Kernel-Based Masking and Zero-Knowledge Folding
A Self-Contained Mathematical Development

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Abstract

We investigate zero-knowledge extensions of KBFold and the representation of masked binary-field ring switching in Boolean-kernel coordinates. This manuscript develops short, fully quantified mathematical statements with explicit proofs, primary-source citations, and dependencies suitable for Lean 4 certification. Chapter 1 establishes the kernel encoding and normalized fold dictionary. Chapter 2 reconstructs sumcheck, masking, FRI-style folding, and ring switching, including an exact high-degree mask-propagation identity. Chapter 3 defines the finite oracle-proof and zero-knowledge security games, proves adaptive linear-mask distribution lemmas and the Reed–Solomon uniqueness bound, and establishes two obstructions: one-query leakage from unmasked codeword oracles and failure of the unmodified final KBFold closure under high-degree masking. Chapter 4 gives a complete field-independent Boolean-kernel algebra, the exact characteristic-dependent multilinearization/folding relation, degree constraints, and a bounded-query masking theorem for one polynomial oracle. Chapter 5 derives the exact masked fold chain, the revised linear final closure, the enlarged Reed–Solomon degree budget, local fold checks, and a precise joint masking criterion; an explicit attack shows why separately uniform messages may still leak through their correlations. Chapter 6 gives the characteristic-two kernel/Frobenius/trace dictionary, an independent anchored residual basis, and a two-challenge bound for an explicitly timed algebraic coefficient-consistency checker. Chapter 7 establishes lossless binary witness packing, the involutive but non-extension-linear transpose, exact masked incoming/outgoing Ring Switching equations, a conditional algebraic false-claim soundness bound and an explicit disclosure attack when the correction is revealed. Chapter 8 reconstructs the actual Diamond–Posen tensor-algebra ring-switching compiler, proves its embedded-field root bound, derives a conditional $2h/|L|$ sumcheck soundness bound, and demonstrates that the unmasked partial tensor message is not zero knowledge. Chapter 9 proves a perfect causal simulator for the single-opening public-message projection of the masked binary-field interface, analyzes conditional mask preimages, derives a multi-opening pad-reuse leakage attack, and isolates an ideal complete-view simulator and a rigorously conditional external-interface composition theorem. Chapter 10 constructs a concrete classical binary-coordinate affine proof using perfectly hiding Pedersen bit commitments and Schnorr OR proofs, with exact honest-verifier simulation, extraction-based binding reductions, and a rigorously conditional outer soundness theorem. It is neither succinct nor quantum-resistant. Chapter 11 replaces the honest-sigma-verifier restriction with sequential one-bit OR proofs and constructs an abort-corrected classical rewinding simulator. It proves perfect expected-time malicious-verifier zero knowledge for the full one-opening binary-field affine transcript, an exact two-run forking and stand-alone early-extraction bound, and identifies the remaining global temporal extraction and KBFold oracle-composition obligations. Chapter 12 constructs a global classical checkpoint-replay extractor for the sequential one-bit affine compiler. It preserves the main transcript and the fresh outer challenges, extracts one chronologically consistent bit registry or yields a discrete-logarithm violation, and proves the explicit false-claim error bound with a separate external authenticated-oracle term.

Chapter 13 specifies a full masked original-field KBFold prover/verifier using authenticated perfectly hiding Pedersen-commitment leaves, committed Sumcheck round coefficients, an exact hidden-offset recurrence and a confidential final degree-one closure. It proves algebraic completeness and the exact online witness-independent distribution of public authenticated shells. The complete proof-transcript simulation and concrete knowledge-extraction security are stated as explicit proof-engine contracts. Chapter 14 proves the final conditional classical one-opening theorem for the fully checked reference protocol: exact completeness, evaluation binding under pre-point NP witness extraction and commitment/hash binding, and a constructive trapdoor-equivocal online ZK simulator under adaptive multi-proof witness indistinguishability. It proves that sparse local folds do not enforce the enlarged RS degree bound and exhibits a quantum false-claim attack against the unchanged Pedersen instantiation. Chapter 15 closes the same manuscript with the final theorem ledger, explicit certification targets and unresolved obligations. No Chapter 16 will be introduced.

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1 Introduction and the exact mathematical problem

1.1 Purpose and scope

The object of this manuscript is an explicit zero-knowledge extension of the evaluation protocol of *KBFold* [Mkh26]. The original KBFold work constructs an encoding of a Boolean-indexed field table using a univariate polynomial in the Boolean-kernel basis. Its normalized fold is exactly restriction of one Boolean variable in the coefficient table. The original paper develops completeness and soundness of its unmasked evaluation protocol and identifies zero knowledge as future work (Section 9, Zero knowledge). In that section the proposed ingredients are a higher-degree random polynomial in the commitment, masking of sumcheck, and masks on the folded words. These ingredients must be analyzed jointly rather than in isolation.

The literature leading to our problem has two related but distinct branches. Multilinear sumcheck and its privacy-preserving variants motivate masking of the intermediate prover polynomials; Libra uses small masking polynomials in a zero-knowledge protocol [Xie+19]. BaseFold connects multilinear evaluation to folding-based commitments [ZCF]. In binary characteristic, Diamond and Posen introduce ring switching for packed Boolean witnesses [DP24]; Kochergin masks its transposition using trace/Frobenius identities [Koc26]; Frigo and shelat use trace-zero algebra to obtain a bit-packing reduction with linear constraints [Fs26]. Chapter 2 will compare the exact protocol statements and assumptions of these works.

Separation of settings. The original KBFold codeword-folding implementation uses an odd-characteristic field with a smooth multiplicative domain. Our algebraic discussion of the ring-switching correction instead takes place in the binary extension $\mathbb{F}_{2^n}$. These are two different ambient fields. Statements from one setting are *not* silently transferred to the other. Every later theorem will declare its field and characteristic as explicit hypotheses.

1.2 Basic finite algebra

Definition 1.1 (Parameters and bit indices)**.** Let $\mathbb{F}$ be a field, let $m \in \mathbb{N}$ with $m \geq 1$, and put $N := 2^m$. Let $I_m := \{0,1\}^m$. Its members are ordered tuples $b = (b_0, \ldots, b_{m-1})$, with $b_i \in \{0,1\} \subseteq \mathbb{F}$. Put

$$\mathrm{ind}_m(b) := \sum_{i=0}^{m-1} b_i 2^i \in \{0, \ldots, N-1\}.$$

For $m \geq 1$ we write $(a, b') \in I_m$ when $a \in \{0,1\}$ is the lowest-index bit and $b' \in I_{m-1}$. We use $I_0 := \{()\}$ and the usual conventions for empty products and sums.

Definition 1.2 (Kernel polynomial and encoding)**.** For $b \in I_m$ define

$$K_b^{(m)}(X) := \prod_{i=0}^{m-1} (X^{2^i} + b_i) \in \mathbb{F}[X]. \tag{1.1}$$

For a table $f : I_m \to \mathbb{F}$, define

$$U_f^{(m)}(X) := \sum_{b \in I_m} f(b) K_b^{(m)}(X). \tag{1.2}$$

For the unique element $() \in I_0$ define $K_{()}^{(0)}(X) := 1$.

Lemma 1.3 (Degree and leading coefficient)**.** *For every $b \in I_m$, $K_b^{(m)}$ is monic and has degree $N-1$.*

Proof. Each factor $X^{2^i} + b_i$ is monic of degree 2^i. The leading term in the product is the product of the leading terms, namely $X^{\sum_{i=0}^{m-1} 2^i}$. Since $\sum_{i=0}^{m-1} 2^i = 2^m - 1 = N - 1$, no other selection of terms has as large a degree. Therefore $K_b^{(m)}$ has degree $N-1$ and leading coefficient 1. □

Lemma 1.4 (Even–odd kernel recursion)**.** *For $b' \in I_{m-1}$, let Y be a formal variable. Then*

$$K_{(0,b')}^{(m)}(X) = X\, K_{b'}^{(m-1)}(X^2), \tag{1.3}$$

$$K_{(1,b')}^{(m)}(X) = (X+1) K_{b'}^{(m-1)}(X^2). \tag{1.4}$$

Proof. Writing $b = (a, b')$, the $i = 0$ factor in (1.1) is $X + a$. For $i \geq 1$ put $j = i - 1$. Then $X^{2^i} = (X^2)^{2^j}$, and the remaining product equals $K_{b'}^{(m-1)}(X^2)$. Taking $a = 0$ and $a = 1$ gives the two formulas. □

Lemma 1.5 (Unique even–odd decomposition)**.** *Let $d \geq 1$ and $P \in \mathbb{F}[X]$ satisfy $\deg P < 2d$, or $P = 0$. There exist unique polynomials $E, O \in \mathbb{F}[Y]$ of degree $< d$ (allowing zero) such that*

$$P(X) = E(X^2) + XO(X^2). \tag{1.5}$$

Proof. Write $P(X) = \sum_{j=0}^{2d-1} a_j X^j$, with zero coefficients beyond the actual degree. Define $E(Y) := \sum_{i=0}^{d-1} a_{2i} Y^i$ and $O(Y) := \sum_{i=0}^{d-1} a_{2i+1} Y^i$. This gives existence. If $E_1(X^2) + XO_1(X^2) = E_2(X^2) + XO_2(X^2)$, comparison of coefficients of every even power X^{2i} gives $E_1 = E_2$; comparison of every odd power X^{2i+1} gives $O_1 = O_2$. This proves uniqueness. □

Proposition 1.6 (Kernel basis theorem)**.** *The family $(K_b^{(m)})_{b \in I_m}$ is an $\mathbb{F}$-basis of the vector space $\mathbb{F}[X]_{<N}$.*

Proof. This vector space has dimension N, and there are $|I_m| = N$ kernel polynomials. It therefore suffices to prove their linear independence. We prove it by induction on m. For $m = 0$ the only polynomial is $K^{(0)}_{()} = 1$, which is linearly independent.

Suppose the assertion holds for $m-1$, and assume that

$$\sum_{b' \in I_{m-1}} a_{0,b'} K^{(m)}_{(0,b')}(X) + \sum_{b' \in I_{m-1}} a_{1,b'} K^{(m)}_{(1,b')}(X) = 0. \tag{1.6}$$

Define $A(Y) := \sum_{b'} a_{0,b'} K^{(m-1)}_{b'}(Y)$ and $B(Y) := \sum_{b'} a_{1,b'} K^{(m-1)}_{b'}(Y)$. Lemma 1.4 turns (1.6) into

$$XA(X^2) + (X+1)B(X^2) = B(X^2) + X(A+B)(X^2) = 0.$$

By uniqueness in Lemma 1.5, $B = 0$ and $A + B = 0$; hence $A = B = 0$. By the induction hypothesis all $a_{0,b'}$ and $a_{1,b'}$ vanish. This proves linear independence for m and completes the induction. □

Corollary 1.7 (Top-coefficient identity)**.** *For every table $f : I_m \to \mathbb{F}$,*

$$[X^{N-1}] U^{(m)}_f = \sum_{b \in I_m} f(b), \tag{1.7}$$

where $[X^j]P$ denotes the coefficient of X^j in P.

Proof. By Lemma 1.3, the coefficient of X^{N-1} in every $K^{(m)}_b$ equals 1. Coefficient extraction is linear, so (1.2) gives $[X^{N-1}] U^{(m)}_f = \sum_b f(b) \cdot 1$. □

1.3 The exact evaluation problem

Definition 1.8 (Equality weights and multilinear extension)**.** For $z = (z_0, \dots, z_{m-1}) \in \mathbb{F}^m$ and $x = (x_0, \dots, x_{m-1}) \in \mathbb{F}^m$ put

$$\mathrm{eq}_m(x, z) := \prod_{i=0}^{m-1} \big((1 - x_i)(1 - z_i) + x_i z_i\big) \in \mathbb{F}. \tag{1.8}$$

For $f : I_m \to \mathbb{F}$ define

$$\widetilde{f}(z) := \sum_{b \in I_m} f(b)\, \mathrm{eq}_m(b, z). \tag{1.9}$$

Lemma 1.9 (Boolean interpolation)**.** *For $b, c \in I_m$, $\mathrm{eq}_m(b, c) = 1$ if $b = c$, and 0 otherwise. Consequently $\widetilde{f}(c) = f(c)$ for every $c \in I_m$.*

Proof. If $b_i = c_i = 0$, the i-th factor of (1.8) equals $(1-0)(1-0)+0 = 1$. If $b_i = c_i = 1$, the factor equals $0 + 1 = 1$. If $b_i \neq c_i$, direct substitution gives 0. Thus the full product is 1 exactly when all coordinates agree. Substituting into (1.9) leaves only the term indexed by $b = c$. □

Definition 1.10 (Evaluation relation)**.** Define the algebraic relation

$$\mathcal{R}_{m,\mathbb{F}} := \left\{ (z, v; f) \in \mathbb{F}^m \times \mathbb{F} \times \mathbb{F}^{I_m} : v = \widetilde{f}(z) \right\}.$$

The public assertion is (z, v); the table f is a witness. In the KBFold opening problem the prover additionally binds an encoding of f to a public commitment or codeword oracle. The exact binding and oracle interfaces will be defined in Chapter 3.

Definition 1.11 (Table restriction)**.** For $r \in \mathbb{F}$ and $f : I_m \to \mathbb{F}$ define $\mathsf{Res}_r(f) : I_{m-1} \to \mathbb{F}$ by

$$\mathsf{Res}_r(f)(b') = (1-r)f(0,b') + rf(1,b'). \tag{1.10}$$

Lemma 1.12 (Multilinear restriction)**.** *For every $r \in \mathbb{F}$ and $y \in \mathbb{F}^{m-1}$,*

$$\widetilde{\mathsf{Res}_r(f)}(y) = \widetilde{f}(r,y). \tag{1.11}$$

Proof. Using the product definition of equality weights, $\mathrm{eq}_m((0,b'),(r,y)) = (1-r)\,\mathrm{eq}_{m-1}(b',y)$ and $\mathrm{eq}_m((1,b'),(r,y)) = r\,\mathrm{eq}_{m-1}(b',y)$. Split the defining sum for $\widetilde{f}(r,y)$ according to the first bit, factor out $\mathrm{eq}_{m-1}(b',y)$, and substitute (1.10). The result is exactly the defining sum for $\widetilde{\mathsf{Res}_r(f)}(y)$. □

Definition 1.13 (Normalized polynomial kernel fold)**.** Let $P \in \mathbb{F}[X]_{<2d}$ and let $P(X) = E(X^2) + XO(X^2)$ be its unique even–odd decomposition. For $r \in \mathbb{F}$ define

$$\mathsf{pfold}_r(P)(Y) := (2r-1)E(Y) + (1-r)O(Y). \tag{1.12}$$

The coefficients are defined over an arbitrary field, including characteristic two; no division is used in this polynomial definition.

Theorem 1.14 (Fold dictionary)**.** *For every $m \geq 1$, table $f : I_m \to \mathbb{F}$, and $r \in \mathbb{F}$,*

$$\mathsf{pfold}_r(U_f^{(m)}) = U_{\mathsf{Res}_r(f)}^{(m-1)}. \tag{1.13}$$

Proof. Write $f_0(b') = f(0,b')$ and $f_1(b') = f(1,b')$, and set $A = U_{f_0}^{(m-1)}$, $B = U_{f_1}^{(m-1)}$. By Lemma 1.4 and the definition of U_f,

$$U_f^{(m)}(X) = XA(X^2) + (X+1)B(X^2) = B(X^2) + X(A+B)(X^2).$$

By Lemma 1.5, its even and odd components are $E = B$ and $O = A+B$, respectively. Definition 1.13 therefore gives

$$\begin{aligned}\mathsf{pfold}_r(U_f^{(m)}) &= (2r-1)B + (1-r)(A+B)\\ &= (1-r)A + \big((2r-1)+(1-r)\big)B\\ &= (1-r)A + rB.\end{aligned}$$

By the linear definition (1.2), this last polynomial equals $U_{(1-r)f_0+rf_1}^{(m-1)}$, and the coefficient table is $\mathsf{Res}_r(f)$ by (1.10). □

Corollary 1.15 (Iterated folds compute the claimed value)**.** *Let $z = (z_0,\dots,z_{m-1}) \in \mathbb{F}^m$. Set $P_0 = U_f^{(m)}$ and define $P_{i+1} = \mathsf{pfold}_{z_i}(P_i)$ for $0 \leq i < m$. Then P_m is a constant polynomial and*

$$P_m = \widetilde{f}(z). \tag{1.14}$$

Proof. For $0 \leq i \leq m$, define the table f_i on I_{m-i} by $f_0 = f$ and $f_{i+1} = \mathsf{Res}_{z_i}(f_i)$. Induction using Theorem 1.14 proves $P_i = U_{f_i}^{(m-i)}$. A second induction using Lemma 1.12 proves that $\widetilde{f}_i(y) = \widetilde{f}(z_0,\dots,z_{i-1},y)$ for every remaining point y. For $i = m$, I_0 has one entry and $U_{f_m}^{(0)} = f_m(())$ is constant. The second identity at $i = m$ gives $f_m(()) = \widetilde{f}(z)$, which is (1.14). □

1.4 Why zero knowledge is a separate requirement

The original KBFold protocol sends a univariate polynomial in each sumcheck round, followed by folded-oracle data and a final table, as specified in the unmasked protocol [Mkh26]. The next lemma demonstrates a concrete leakage channel, without claiming that an abstract binding commitment has any particular hiding property.

Lemma 1.16 (A witness-dependent sumcheck message)**.** *Let $m = 1$, let $z = 0 \in \mathbb{F}$, and consider two tables $f^{(0)}, f^{(1)} : I_1 \to \mathbb{F}$ defined by*

$$f^{(0)}(0) = f^{(0)}(1) = 0, \qquad f^{(1)}(0) = 0, \quad f^{(1)}(1) = 1.$$

Then $\widetilde{f^{(0)}}(z) = \widetilde{f^{(1)}}(z) = 0$, but their honest first-round sumcheck polynomials for the product $\widetilde{f}(X)\,\mathrm{eq}_1(X, z)$ are different: the first is zero and the second is $X(1 - X)$.

Proof. By the definition of the multilinear extension in one variable, $\widetilde{f^{(0)}}(X) = 0$ and $\widetilde{f^{(1)}}(X) = (1-X)\cdot 0 + X \cdot 1 = X$. The weight for $z = 0$ is $\mathrm{eq}_1(X, 0) = 1 - X$. Multiplication gives the claimed polynomials. The polynomial $X(1 - X)$ has coefficient 1 at X and is therefore nonzero over every field. Evaluating either multilinear extension at $X = 0$ gives the same public value 0. □

Remark 1.17 (Interpretation of Lemma 1.16). The first-round message is not determined solely by (z, v). In a model where the original witness is supposed to remain hidden, it therefore requires masking or a separate simulation argument. This lemma does not by itself prove a failure of zero knowledge for an arbitrary concrete commitment: the public commitment and the full transcript must be included in the eventual security game.

Lemma 1.18 (Linearity of kernel masking)**.** *For any tables $f, R : I_m \to \mathbb{F}$ and $r \in \mathbb{F}$,*

$$U^{(m)}_{f+R} = U^{(m)}_f + U^{(m)}_R, \tag{1.15}$$

$$\mathsf{pfold}_r(U^{(m)}_{f+R}) = U^{(m-1)}_{\mathsf{Res}_r(f)} + U^{(m-1)}_{\mathsf{Res}_r(R)}, \tag{1.16}$$

$$\widetilde{f + R}(z) = \widetilde{f}(z) + \widetilde{R}(z). \tag{1.17}$$

Proof. The first identity follows by distributing the finite sum defining U_{f+R}. The second follows by applying the $\mathbb{F}$-linear map pfold_r to the first identity and then applying Theorem 1.14 separately to f and R. The third follows by distributing the finite sum defining the multilinear extension. □

Remark 1.19 (High-degree commitment masking). The original KBFold conclusion also proposes masking U_f by adding $X^N R(X)$ to the committed polynomial. Such a mask changes the degree, and consequently the code and the proof of the fold dictionary must be re-examined on the enlarged polynomial space. Equations (1.15)–(1.17) apply to the separate table mask $f + R$; they do not by themselves establish a folding theorem for $U_f + X^N R$.

1.5 A second setting: binary-field ring switching

Definition 1.20 (Binary extension setting)**.** Let $\ell \geq 1$, put $n := 2^\ell$, and now let $L := \mathbb{F}_{2^n}$. Choose $\alpha \in L$ such that $[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = n$. For $b \in I_\ell$ set $\beta_b := K^{(\ell)}_b(\alpha)$.

Lemma 1.21 (Kernel field basis)**.** *The family $(\beta_b)_{b \in I_\ell}$ is an $\mathbb{F}_2$-basis of L.*

Proof. Proposition 1.6, applied with $\mathbb{F} = \mathbb{F}_2$ and $m = \ell$, says that $(K_b^{(\ell)})_b$ is an $\mathbb{F}_2$-basis of $\mathbb{F}_2[X]_{<n}$. Evaluation at α is $\mathbb{F}_2$-linear. It is injective: a nonzero polynomial of degree $< n$ cannot vanish at α because the minimal polynomial of α has degree n. Its domain and codomain are both n-dimensional over $\mathbb{F}_2$, hence the evaluation map is bijective. The images of the stated basis under this bijection form an $\mathbb{F}_2$-basis of L. □

Definition 1.22 (Coordinates and transposed mask)**.** For $x \in L$ and $u \in I_\ell$ let $d_u(x) \in \mathbb{F}_2$ be the unique coordinate satisfying $x = \sum_u d_u(x)\beta_u$. Let $p = (p_v)_{v \in I_\ell} \in L^{I_\ell}$ be a pad vector, and define

$$c_u^p := \sum_{v \in I_\ell} \beta_v d_u(p_v), \tag{1.18}$$

$$C_p(X) := \sum_{u \in I_\ell} c_u^p K_u^{(\ell)}(X). \tag{1.19}$$

For a challenge $\theta \in L^\ell$ define

$$\Delta_p(\theta) := \sum_{u \in I_\ell} \mathrm{eq}_\ell(u, \theta) c_u^p. \tag{1.20}$$

Finally define the *scalar kernel-fold evaluation* for any $c \in L^{I_\ell}$ by

$$\mathsf{KFold}_\theta(U_c^{(\ell)}) := \sum_{u \in I_\ell} \mathrm{eq}_\ell(u, \theta) c_u.$$

This is well defined by Proposition 1.6.

Proposition 1.23 (Kernel representation of masked transposition)**.** *With the preceding definitions, for every $p \in L^{I_\ell}$ and $\theta \in L^\ell$,*

$$\Delta_p(\theta) = \mathsf{KFold}_\theta(C_p). \tag{1.21}$$

Moreover, if $\eta_u \in L$ denotes the trace-dual basis to β_u, then

$$\Delta_p(\theta) = \sum_{u,v \in I_\ell} \mathrm{eq}_\ell(u, \theta) \beta_v \, \mathrm{Tr}_{L/\mathbb{F}_2}(\eta_u p_v). \tag{1.22}$$

Proof. By the definition of C_p and the scalar kernel-fold evaluation,

$$\mathsf{KFold}_\theta(C_p) = \sum_u \mathrm{eq}_\ell(u, \theta) c_u^p = \Delta_p(\theta),$$

proving (1.21). For the second identity, the trace-dual property implies $d_u(x) = \mathrm{Tr}_{L/\mathbb{F}_2}(\eta_u x)$: expand $x = \sum_v d_v(x)\beta_v$, multiply by η_u, apply trace, and use $\mathrm{Tr}(\eta_u \beta_v) = 1$ for $u = v$ and 0 otherwise. Substitute this coordinate identity in (1.18) and then in (1.20) to obtain (1.22). □

Lemma 1.24 (Characteristic-dependent partition identity)**.** *For every field $\mathbb{F}$ and every $m \geq 1$,*

$$\sum_{b \in I_m} K_b^{(m)}(X) = \prod_{i=0}^{m-1} (2X^{2^i} + 1). \tag{1.23}$$

In particular, if $\mathrm{char}(\mathbb{F}) = 2$, the sum equals 1.

Proof. Expand the sum over bit tuples as a product of sums, one for each index i:

$$\sum_{b \in I_m} \prod_i (X^{2^i} + b_i) = \prod_i ((X^{2^i} + 0) + (X^{2^i} + 1)) = \prod_i (2X^{2^i} + 1).$$

When $\mathrm{char}(\mathbb{F}) = 2$, each factor on the right is 1. □

Definition 1.25 (Binary multilinearization)**.** Let $\mathrm{char}(\mathbb{F}) = 2$, let $m \geq 1$ and $N = 2^m$. For $0 \leq e < N$ write its unique binary expansion $e = \sum_{i=0}^{m-1} e_i 2^i$. Let $\mathbb{F}[Z_0, \ldots, Z_{m-1}]_{\mathrm{ml}}$ be the $\mathbb{F}$-vector space spanned by the squarefree monomials $\prod_i Z_i^{e_i}$. Define the unique $\mathbb{F}$-linear map $\mathscr{M}_m : \mathbb{F}[X]_{<N} \to \mathbb{F}[Z_0, \ldots, Z_{m-1}]_{\mathrm{ml}}$ by

$$\mathscr{M}_m(X^e) := \prod_{i=0}^{m-1} Z_i^{e_i}.$$

We do not assume that $\mathscr{M}_m$ preserves polynomial multiplication.

Lemma 1.26 (Multilinearization of a kernel polynomial)**.** *Under the hypotheses of Definition 1.25, for every $b \in I_m$,*

$$\mathscr{M}_m(K_b^{(m)})(Z) = \prod_{i=0}^{m-1} (Z_i + b_i).$$

Proof. Fix b and write $J = \{0, \ldots, m-1\}$. Applying distributivity to the product defining $K_b^{(m)}$ yields the exact finite expansion

$$K_b^{(m)}(X) = \sum_{S \subseteq J} \left(\prod_{i \notin S} b_i \right) X^{\sum_{i \in S} 2^i}.$$

For each subset S, the integer $\sum_{i \in S} 2^i$ is strictly less than N and has binary digit 1 precisely at the positions in S. By the definition of $\mathscr{M}_m$,

$$\mathscr{M}_m\left(X^{\sum_{i \in S} 2^i}\right) = \prod_{i \in S} Z_i.$$

Applying $\mathbb{F}$-linearity to the preceding finite expansion gives

$$\mathscr{M}_m(K_b^{(m)})(Z) = \sum_{S \subseteq J} \left(\prod_{i \notin S} b_i \right) \left(\prod_{i \in S} Z_i \right).$$

Finally, distributivity of $\prod_{i \in J}(Z_i + b_i)$ gives exactly the same sum over S. Thus the two polynomials agree. No multiplicativity property of $\mathscr{M}_m$ was used. □

Proposition 1.27 (Multilinearization of the complete kernel fold)**.** *Suppose* $\mathrm{char}(\mathbb{F}) = 2$. *For every $P \in \mathbb{F}[X]_{<N}$ and $\theta \in \mathbb{F}^m$,*

$$\mathsf{KFold}_\theta(P) = \mathscr{M}_m(P)(1 + \theta_0, \ldots, 1 + \theta_{m-1}), \tag{1.24}$$

where $\mathsf{KFold}_\theta(P)$ denotes the iterated normalized fold in Corollary 1.15, applied to the unique kernel coefficient table of P.

Proof. By Proposition 1.6, write uniquely $P = \sum_{b \in I_m} c_b K_b^{(m)}$. Corollary 1.15 gives

$$\mathsf{KFold}_\theta(P) = \sum_b c_b \, \mathrm{eq}_m(b, \theta).$$

By $\mathbb{F}$-linearity of $\mathscr{M}_m$ and of evaluation at $1 + \theta$,

$$\mathscr{M}_m(P)(1 + \theta) = \sum_b c_b \mathscr{M}_m(K_b^{(m)})(1 + \theta).$$

Lemma 1.26 identifies each summand as $\prod_i (1 + \theta_i + b_i)$. If $b_i = 0$, this factor is $1 + \theta_i = 1 - \theta_i$, because the characteristic is two. If $b_i = 1$, the factor is $2 + \theta_i = \theta_i$. In both cases this equals $(1 - b_i)(1 - \theta_i) + b_i \theta_i$. Taking the product over i shows $\mathscr{M}_m(K_b^{(m)})(1 + \theta) = \mathrm{eq}_m(b, \theta)$. Substitution in the displayed sum proves (1.24). □

1.6 Formal research obligations

Definition 1.28 (Target zero-knowledge evaluation statement)**.** A full zero-knowledge KBFold extension must specify a public-parameter setup, a commitment and opening relation binding the table f, a distribution for masking randomness, an interactive prover and verifier, and a simulator for the entire verifier view. For each supported public instance (cm, z, v) and witness/opening data tied to f, the intended statements are:

(i) *Completeness*: an honest opening for $v = \widetilde{f}(z)$ is accepted.

(ii) *Soundness*: every strategy proving a false evaluation for a committed witness has explicitly bounded acceptance probability.

(iii) *Zero knowledge*: for a specified verifier class, the real transcript distribution is indistinguishable, in a specified sense, from the simulator's output given only the allowed public instance and public auxiliary information.

The complete probability spaces, quantifiers, and oracle-query bounds are deferred to Chapter 3, where the formal cryptographic interface is introduced.

Logical dependency and status. Lemmas 1.3–1.24 and the kernel-folding dictionary are algebraic theorems proved in this chapter; the original KBFold paper contains corresponding kernel-basis and folding results [Mkh26]. Proposition 1.23 identifies the characteristic-two ring-switching correction in the same language. It does not assert that the original KBFold protocol (over odd characteristic) has already acquired zero knowledge. Chapters 4–7 will develop and audit the binary-field masking protocol, while the later KBFold chapters will treat masked sumcheck polynomials, all folded oracles, the final table, and the enlarged-degree code constraints in the original field.

Lean 4 decomposition. The definitions in this chapter have intentionally small dependencies: bit indexing; polynomial kernel; even–odd split; basis; coefficient extraction; equality weights; restriction; normalized fold; iteration; field evaluation basis; binary-coordinate map; correction identity. The proof of each theorem uses only earlier definitions and statements, with the sole standard finite-field input in Proposition 1.23 being existence of a trace-dual basis in a finite separable extension. The complete theorem/Lean dependency ledger is maintained alongside this manuscript.

2 Historical Background and Related Work

2.1 The algebraic origin: sumcheck

The sumcheck protocol originates in the algebraic work of Lund et al. [Lun+92]. We give a self-contained version to fix the precise leakage surface that motivates the later masking constructions.

Definition 2.1 (Sumcheck instance)**.** Let $\mathbb{F}$ be a finite field of cardinality q, let $m \geq 1$, and let $G \in \mathbb{F}[X_0, \ldots, X_{m-1}]$ satisfy $\deg_{X_i} G \leq d_i$ for prescribed integers $d_i \geq 0$. Define

$$S_G := \sum_{b \in \{0,1\}^m} G(b) \in \mathbb{F}.$$

A sumcheck instance is a claimed value $a \in \mathbb{F}$ for the equality $a = S_G$. We assume that the verifier can evaluate G at one final random point.

Definition 2.2 (Honest partial-sum polynomial)**.** For $0 \le i < m$ and a fixed prefix $r_{<i} = (r_0, \ldots, r_{i-1}) \in \mathbb{F}^i$, define the univariate polynomial

$$h_i(T; r_{<i}) := \sum_{b \in \{0,1\}^{m-i-1}} G(r_0, \ldots, r_{i-1}, T, b).$$

This polynomial has degree at most d_i.

Lemma 2.3 (Partial-sum telescoping)**.** *For $i = 0$,*

$$h_0(0) + h_0(1) = S_G.$$

For $1 \le i < m$ and every prefix $r_{<i}$,

$$h_i(0; r_{<i}) + h_i(1; r_{<i}) = h_{i-1}(r_{i-1}; r_{<i-1}).$$

Proof. For the first identity, partition $\{0,1\}^m$ according to its first bit and use the definition of h_0. For the second, partition the remaining Boolean suffix of $h_{i-1}(r_{i-1}; r_{<i-1})$ according to the i-th bit. The two resulting finite sums are precisely $h_i(0; r_{<i})$ and $h_i(1; r_{<i})$. All equalities are rearrangements of finite sums. □

Definition 2.4 (Sumcheck verifier)**.** Set $a_0 = a$. In round $i = 0, \ldots, m-1$, the prover sends a polynomial $g_i(T)$ of degree at most d_i; the verifier checks

$$g_i(0) + g_i(1) = a_i,$$

rejecting if the equation fails. If it passes, the verifier samples $r_i \xleftarrow{\$} \mathbb{F}$ and sets $a_{i+1} = g_i(r_i)$. After the last round the verifier accepts exactly if

$$a_m = G(r_0, \ldots, r_{m-1}).$$

The prover's round-i polynomial is fixed before the verifier samples r_i.

Lemma 2.5 (Completeness)**.** *If $a = S_G$ and the prover sends $g_i = h_i(\,\cdot\,; r_{<i})$ in every round, the verifier accepts for every challenge sequence.*

Proof. The first test is true by Lemma 2.3. Assume inductively that a_i equals the honest partial sum of round $i-1$ evaluated at r_{i-1} (with $a_0 = S_G$). Lemma 2.3 gives the required endpoint equality for h_i. By the update rule $a_{i+1} = h_i(r_i; r_{<i})$, the induction continues. At $i = m-1$, the Boolean suffix is empty and $h_{m-1}(r_{m-1}; r_{<m-1}) = G(r_0, \ldots, r_{m-1})$, so the final check succeeds. □

Lemma 2.6 (Root bound)**.** *If $A, B \in \mathbb{F}[T]$ are distinct and each has degree at most d, then for uniform $r \in \mathbb{F}$,*

$$\Pr[A(r) = B(r)] \le \min\{1, d/q\}.$$

Proof. The polynomial $A - B$ is nonzero, has degree at most d, and therefore has at most d distinct roots in the field $\mathbb{F}$. Divide the root count by q. □

Lemma 2.7 (Univariate root-count lemma)**.** *Let $\mathbb{F}$ be a field and let $P \in \mathbb{F}[T]$ be nonzero of degree $d \ge 0$. Then P has at most d distinct roots in $\mathbb{F}$.*

Proof. Induct on d. If $d = 0$ then P is a nonzero constant and has no roots. Suppose $d \geq 1$. If P has no roots, the claim is immediate. Otherwise choose one root $a \in \mathbb{F}$. Polynomial division by the monic linear polynomial $T - a$ gives $P(T) = (T - a)Q(T)$ with $\deg Q = d - 1$, since the remainder is the constant $P(a) = 0$. Every root $b \neq a$ of P satisfies $0 = (b - a)Q(b)$, hence $Q(b) = 0$ because $\mathbb{F}$ is a field and $b - a \neq 0$. By induction Q has at most $d - 1$ distinct roots. Thus P has at most $1 + (d - 1) = d$ distinct roots. □

Lemma 2.8 (Multivariate Schwartz–Zippel bound)**.** *Let $s \geq 1$ and let $P \in \mathbb{F}[X_0, \ldots, X_{s-1}]$ be nonzero of total degree at most D. For independent uniform $r_i \xleftarrow{\$} \mathbb{F}$,*

$$\Pr[P(r_0, \ldots, r_{s-1}) = 0] \leq \min\{1, D/|\mathbb{F}|\}.$$

Proof. We prove the bound $D/|\mathbb{F}|$ by induction on s. The case $s = 1$ is Lemma 2.7 divided by $|\mathbb{F}|$. Assume $s > 1$ and expand P in its final variable:

$$P(X_0, \ldots, X_{s-1}) = \sum_{j=0}^{e} A_j(X_0, \ldots, X_{s-2}) X_{s-1}^j, \qquad A_e \neq 0.$$

Every monomial in A_e has degree at most $D - e$, so $\deg A_e \leq D - e$. For a uniform random prefix $r' = (r_0, \ldots, r_{s-2})$, the induction hypothesis gives

$$\Pr[A_e(r') = 0] \leq (D - e)/|\mathbb{F}|.$$

Condition on the event $A_e(r') \neq 0$. Then the specialized univariate polynomial $P(r', T)$ is nonzero and has degree exactly e. By Lemma 2.7, the independent final challenge r_{s-1} is a root with conditional probability at most $e/|\mathbb{F}|$. Split the event $P(r', r_{s-1}) = 0$ according to whether $A_e(r') = 0$. The union/conditional-probability bound yields

$$\Pr[P(r', r_{s-1}) = 0] \leq \frac{D - e}{|\mathbb{F}|} + \frac{e}{|\mathbb{F}|} = \frac{D}{|\mathbb{F}|}.$$

Taking the minimum with 1 finishes the proof. □

Theorem 2.9 (Sumcheck soundness)**.** *If $a \neq S_G$, the acceptance probability of any prover obeying the timing and degree rules in Definition 2.4 is at most*

$$\min\left\{1, \frac{\sum_{i=0}^{m-1} d_i}{q}\right\}.$$

Proof. At the beginning of round zero the asserted value a_0 differs from the correct Boolean sum. More generally, say that round i begins with a *false partial claim* when a_i differs from the correct sum of $G(r_{<i}, b_i, \ldots, b_{m-1})$ over the remaining Boolean variables. Suppose round i begins with such a false partial claim and the verifier does not reject the endpoint check. The honest polynomial $h_i(\,\cdot\,; r_{<i})$ has endpoint sum equal to the correct partial sum by Lemma 2.3. Therefore $g_i \neq h_i$, since equal polynomials would give equal endpoint sums, contradicting the false claim. Conditioned on the full transcript before sampling r_i, these two distinct degree-$\leq d_i$ polynomials are fixed. By Lemma 2.6, with conditional probability at most d_i/q they agree at r_i; outside that event the next partial claim remains false. If it remains false through the final round, the explicit evaluation test rejects. Acceptance therefore entails an agreement event in at least one round. The union bound over these first-agreement events gives the claimed upper bound. □

Remark 2.10 (Why this historical construction is not automatically zero knowledge)*.* The prover's polynomials g_i depend on G, even when the asserted sum a is fixed. Lemma 1.16 supplies an explicit pair of witnesses with the same public evaluation and distinct first-round polynomials. The soundness theorem alone makes no privacy claim.

2.2 Masking a sumcheck: the precise linear algebra

The purpose of zero knowledge is separate from completeness and soundness: a verifier should be able to check the stated relation without learning witness-dependent information beyond the allowed public input. The definitions of full transcript simulation and the oracle-query model will be fixed in Chapter 3. This section proves the elementary masking identities that are used throughout later chapters. Libra [Xie+19] is an important historical example of masking intermediate polynomial messages in an efficient zero-knowledge proof system.

Definition 2.11 (Uniform additive mask)**.** Let V be a finite-dimensional vector space over a finite field $\mathbb{F}$. Let $s \in V$ be fixed and let $p \in V$ be a uniform random variable. The *additively masked message* is $m := s + p$.

Lemma 2.12 (One-time-pad distribution)**.** *For each fixed $s \in V$, the masked message $m = s + p$ is uniformly distributed on V. In particular, its distribution does not depend on s.*

Proof. For any $a \in V$, the event $m = a$ is equivalent to $p = a - s$. Since p is uniform, its probability is $1/|V|$. This is the same for every a and every fixed s. □

Definition 2.13 (Boolean summation functional)**.** Let $m \geq 1$ and let $\mathcal{V}$ be a finite-dimensional $\mathbb{F}$-vector space of polynomials in m variables containing the constant polynomials. Define the $\mathbb{F}$-linear functional

$$\mathscr{S} : \mathcal{V} \to \mathbb{F}, \qquad \mathscr{S}(P) := \sum_{b \in \{0,1\}^m} P(b).$$

Put $\mathcal{Z} := \ker \mathscr{S}$.

Lemma 2.14 (Zero-sum polynomial masks)**.** *For every $G \in \mathcal{V}$ and every $R \in \mathcal{Z}$,*

$$\mathscr{S}(G + R) = \mathscr{S}(G).$$

If R is uniform on the finite vector space $\mathcal{Z}$, then $G + R$ is uniform on the affine coset $G + \mathcal{Z}$. Its distribution depends only on the coset, or equivalently on $\mathscr{S}(G)$ when $\mathscr{S}$ is surjective.

Proof. By linearity, $\mathscr{S}(G + R) = \mathscr{S}(G) + \mathscr{S}(R)$, and the second term is zero. Translation by G is a bijection $\mathcal{Z} \to G + \mathcal{Z}$, which proves uniformity. If $\mathscr{S}$ is surjective, its fibers are precisely the affine cosets of its kernel: $\mathscr{S}(G_1) = \mathscr{S}(G_2)$ iff $G_1 - G_2 \in \mathcal{Z}$. This proves the final assertion. □

Lemma 2.15 (An explicit zero-sum polynomial)**.** *For any field $\mathbb{F}$, any integer $m \geq 1$, and any polynomial $H \in \mathbb{F}[X_1, \ldots, X_{m-1}]$, put*

$$R(X_0, \ldots, X_{m-1}) := (2X_0 - 1)H(X_1, \ldots, X_{m-1}).$$

Then $\mathscr{S}(R) = 0$ whenever R lies in the chosen ambient space $\mathcal{V}$.

Proof. For each fixed $b \in \{0,1\}^{m-1}$, the sum over $x_0 \in \{0,1\}$ is

$$(-1)H(b) + (2 - 1)H(b) = (-1 + 1)H(b) = 0.$$

Summing this identity over all b proves the claim. The calculation holds in every characteristic; in characteristic two the polynomial $2X_0 - 1$ is the constant 1, which still sums to zero over the two Boolean values. □

Lemma 2.16 (First-round masking identity)**.** *Let G and R be multivariate polynomials in m variables, and let*

$$h_0^G(T) = \sum_{b\in\{0,1\}^{m-1}} G(T,b), \qquad h_0^R(T) = \sum_{b\in\{0,1\}^{m-1}} R(T,b).$$

Then the first honest sumcheck message for $G+R$ is exactly

$$h_0^{G+R}(T) = h_0^G(T) + h_0^R(T).$$

Proof. Distribute the finite sum $\sum_b (G(T,b)+R(T,b))$ in $\mathbb{F}[T]$. No probabilistic hypothesis is needed. □

Remark 2.17 (The condition missing from a simple algebraic mask). Preserving $\mathscr{S}(G)$ is only one algebraic requirement. To obtain zero knowledge for a concrete interactive protocol one must ensure that the *joint distribution* of all exposed polynomial coefficients, oracle replies, final tables, and authentication information can be simulated from allowed public data. A uniform mask in $\ker \mathscr{S}$ does not, without a degree-compatible hiding oracle and a simulation theorem, establish zero knowledge. We will not transfer Lemma 2.14 into a full ZK theorem without those hypotheses.

2.3 FRI, BaseFold, and the high-degree masking question

FRI [Ben+18] introduces a fast interactive oracle proof of proximity to Reed–Solomon codes. BaseFold [ZCF23] combines foldable codes with sumcheck to obtain multilinear polynomial commitments. Here we isolate the elementary polynomial and codeword identities used in KBFold, including the exact effect of a high-degree masking term proposed as future work in [Mkh26]. All proofs below are algebraic and are independent of the proximity-soundness theorems of FRI and BaseFold.

Definition 2.18 (Multiplicative-domain evaluation code)**.** Assume $\mathrm{char}(\mathbb{F}) \neq 2$. Let $\Omega \subseteq \mathbb{F}^\times$ be a finite multiplicative subgroup of even order M. Let $1 \le D \le M$. For $P \in \mathbb{F}[X]$ write $\mathrm{ev}_\Omega(P) := (P(x))_{x\in\Omega}$ and set

$$\mathrm{RS}_{\mathbb{F}}(\Omega, D) := \{\mathrm{ev}_\Omega(P) : P \in \mathbb{F}[X],\ \deg P < D\}.$$

The squaring image is $\Omega^2 := \{x^2 : x \in \Omega\}$.

Lemma 2.19 (Two-element square fibers)**.** *For each $y \in \Omega^2$ there are exactly two elements of Ω whose square is y, namely x and $-x$ for any chosen square root $x \in \Omega$.*

Proof. Because Ω has even order, $-1 \in \Omega$. As $\mathrm{char}(\mathbb{F}) \neq 2$ and $x \neq 0$, the points x and $-x$ are distinct and both square to y. If $t^2 = x^2$, then $(t-x)(t+x) = 0$ in the field $\mathbb{F}$, so $t \in \{x, -x\}$. Thus there are no further preimages. □

Definition 2.20 (Even–odd evaluations)**.** For any function $w : \Omega \to \mathbb{F}$ and $y = x^2 \in \Omega^2$, define

$$w_{\mathrm{e}}(y) := \frac{w(x)+w(-x)}{2}, \qquad w_{\mathrm{o}}(y) := \frac{w(x)-w(-x)}{2x}.$$

These expressions are independent of the choice of square root x.

Lemma 2.21 (Well-definedness and polynomial consistency)**.** *The values $w_{\mathrm{e}}(y)$ and $w_{\mathrm{o}}(y)$ in Definition 2.20 are independent of the choice between x and $-x$. If $w = \mathrm{ev}_\Omega(P)$ and $P(X) = E(X^2) + XO(X^2)$, then*

$$w_{\mathrm{e}} = \mathrm{ev}_{\Omega^2}(E), \qquad w_{\mathrm{o}} = \mathrm{ev}_{\Omega^2}(O).$$

Proof. Replacing x by $-x$ leaves the first numerator unchanged. In the second expression both numerator and denominator change sign, hence their quotient is unchanged. For a polynomial P, substitute $P(x) = E(x^2) + xO(x^2)$ and $P(-x) = E(x^2) - xO(x^2)$. Adding gives $2E(x^2)$ and subtracting gives $2xO(x^2)$. Division is legal because $2x \neq 0$. This proves the two identities. □

Definition 2.22 (Normalized word fold)**.** For $r \in \mathbb{F}$ define $\mathsf{wfold}_r(w) : \Omega^2 \to \mathbb{F}$ by

$$\mathsf{wfold}_r(w) := (2r-1)w_{\mathrm{e}} + (1-r)w_{\mathrm{o}}.$$

Proposition 2.23 (Compatibility of polynomial and word folding)**.** *For every polynomial P and every $r \in \mathbb{F}$,*

$$\mathsf{wfold}_r(\mathrm{ev}_\Omega(P)) = \mathrm{ev}_{\Omega^2}(\mathsf{pfold}_r(P)).$$

Proof. Let $P(X) = E(X^2) + XO(X^2)$. By Lemma 2.21, the even and odd parts of the evaluation word are evaluations of E and O. Therefore the left side is the evaluation on Ω^2 of $(2r-1)E + (1-r)O$, which is $\mathsf{pfold}_r(P)$ by Definition 1.13. □

Corollary 2.24 (Degree preservation under a fold)**.** *If P has degree less than $2D$, then $\mathsf{pfold}_r(P)$ has degree less than D, for every $r \in \mathbb{F}$. Consequently, provided $2D \leq |\Omega|$, the word fold sends $\mathrm{RS}_{\mathbb{F}}(\Omega, 2D)$ into $\mathrm{RS}_{\mathbb{F}}(\Omega^2, D)$.*

Proof. The unique even–odd decomposition of a polynomial of degree less than $2D$ has components E, O each of degree less than D (Lemma 1.5). Their linear combination $\mathsf{pfold}_r(P)$ has the same strict degree bound. Apply Proposition 2.23. □

Definition 2.25 (Arbitrary-degree polynomial fold)**.** For a polynomial $P \in \mathbb{F}[X]$ with finite even–odd decomposition $P(X) = E(X^2) + XO(X^2)$, use the same formula

$$\mathsf{pfold}_r(P) := (2r-1)E + (1-r)O.$$

This extends Definition 1.13 to every polynomial; it is independent of a degree bound.

Lemma 2.26 (Even-power shift)**.** *For every polynomial $R \in \mathbb{F}[X]$, every integer $a \geq 0$, and every $r \in \mathbb{F}$,*

$$\mathsf{pfold}_r(X^{2a}R(X))(Y) = Y^a\,\mathsf{pfold}_r(R)(Y).$$

Proof. Write $R(X) = E(X^2) + XO(X^2)$. Then

$$X^{2a}R(X) = (X^2)^a E(X^2) + X(X^2)^a O(X^2).$$

By uniqueness of the even–odd decomposition, its even and odd components are $Y^a E(Y)$ and $Y^a O(Y)$, respectively. Substituting into Definition 2.25 yields

$$(2r-1)Y^a E(Y) + (1-r)Y^a O(Y) = Y^a\big((2r-1)E(Y) + (1-r)O(Y)\big),$$

which is the claimed formula. □

Theorem 2.27 (Propagation of a high-degree mask through kernel folds)**.** *Let $m \geq 1$, let $N = 2^m$, and let $R \in \mathbb{F}[X]$. Fix arbitrary challenges $r_0, \ldots, r_{m-1} \in \mathbb{F}$. Define*

$$R_0 := R, \qquad R_{j+1} := \mathsf{pfold}_{r_j}(R_j) \quad (0 \leq j < m),$$

and separately

$$P_0 := X^N R(X), \qquad P_{j+1} := \mathsf{pfold}_{r_j}(P_j) \quad (0 \leq j < m).$$

Then, for every $0 \leq j \leq m$,

$$\boxed{P_j(X) = X^{2^{m-j}} R_j(X).}$$

Proof. Induct on j. For $j = 0$ the formula is the definition of P_0, since $N = 2^m$. Suppose it holds at some $j < m$. The exponent 2^{m-j} is even; write it as $2a$ with $a = 2^{m-j-1}$. Applying Lemma 2.26 gives

$$P_{j+1} = \mathsf{pfold}_{r_j}(X^{2a}R_j) = X^a \mathsf{pfold}_{r_j}(R_j) = X^{2^{m-(j+1)}} R_{j+1}.$$

This is the induction statement at $j + 1$. Thus the equality holds for every $j \le m$. □

Corollary 2.28 (Folding a masked KBFold polynomial)**.** *For every table $f : I_m \to \mathbb{F}$, every $R \in \mathbb{F}[X]$, and every $0 \le j \le m$, the j-fold polynomial fold of*

$$Q(X) := U_f^{(m)}(X) + X^N R(X)$$

is

$$Q_j(X) = U_{f_j}^{(m-j)}(X) + X^{2^{m-j}} R_j(X),$$

where $f_0 = f$ and $f_{j+1} = \mathsf{Res}_{r_j}(f_j)$, and R_j is as in Theorem 2.27.

Proof. The fold is $\mathbb{F}$-linear by Definition 2.25. Induction therefore separates the fold of $U_f^{(m)}$ from that of $X^N R$. Corollary 1.15 and Theorem 2.27 give the two displayed terms. □

Remark 2.29 (Where the original proof must change). The identity in Corollary 2.28 is a polynomial identity for *any* R. It does not imply that a Reed–Solomon proof with degree bound N can commit to Q: if $R \neq 0$, then $\deg Q \ge N$ in general. The codeword rate, last-round closure, query rules, and proximity analysis must therefore use the *enlarged* degree bound. This is the precise algebraic obstruction recorded as future work in the original KBFold conclusion [Mkh26].

2.4 Diamond–Posen ring switching as an exact transposition

Diamond and Posen [DP24] address commitment and evaluation of Boolean-valued multilinear polynomials using a system that opens packed elements over a large binary extension field. We derive the ring-switching identity at the algebraic level. No hidden correction or masking is used in this subsection.

Definition 2.30 (Packing parameters)**.** Let $\ell \ge 1$, let $n = 2^\ell$, let $L = \mathbb{F}_{2^n}$, and put $I = I_\ell$. Fix any ordered $\mathbb{F}_2$-basis $(\beta_v)_{v \in I}$ of L. For $x \in L$ let $d_u(x) \in \mathbb{F}_2$ be the unique coordinates with $x = \sum_{u \in I} \beta_u d_u(x)$. Let $d \ge 0$, put $J = \{0,1\}^d$, and let $w = (w_{v,j})_{(v,j) \in I \times J}$ be a Boolean witness table, $w_{v,j} \in \mathbb{F}_2$.

Definition 2.31 (Packed witness and evaluation instance)**.** For $j \in J$ let

$$\widehat{w}_j := \sum_{v \in I} \beta_v w_{v,j} \in L.$$

Fix $r_{\mathrm{lo}} \in L^\ell$ and $r_{\mathrm{hi}} \in L^d$, and define the public equality weights

$$q_v := \mathrm{eq}_\ell(v, r_{\mathrm{lo}}), \qquad h_j := \mathrm{eq}_d(j, r_{\mathrm{hi}}).$$

When $d = 0$, J consists of the empty tuple and the empty-product weight is 1. The claimed evaluation is

$$y = \sum_{v \in I} \sum_{j \in J} q_v h_j w_{v,j}.$$

Definition 2.32 (Partial evaluations and binary transposition)**.** Define the honest partial evaluations $s_v^* := \sum_{j\in J} h_j w_{v,j} \in L$. For an arbitrary vector $s = (s_v)_{v\in I} \in L^I$ define its transpose under the chosen binary basis by

$$\mathcal{T}(s)_u := \sum_{v\in I} \beta_v d_u(s_v), \qquad u \in I. \tag{2.1}$$

Lemma 2.33 (Transposition is an involution)**.** *The map $\mathcal{T} : L^I \to L^I$ is $\mathbb{F}_2$-linear and satisfies $\mathcal{T}^2 = \mathrm{id}_{L^I}$.*

Proof. Each coordinate map d_u is $\mathbb{F}_2$-linear, so $\mathcal{T}$ is $\mathbb{F}_2$-linear. Fix an arbitrary vector s. From the definition of coordinates and their uniqueness,

$$d_v(\mathcal{T}(s)_u) = d_v\left(\sum_{a\in I} \beta_a d_u(s_a)\right) = d_u(s_v).$$

It follows that

$$\mathcal{T}(\mathcal{T}(s))_v = \sum_{u\in I} \beta_u d_v(\mathcal{T}(s)_u) = \sum_{u\in I} \beta_u d_u(s_v) = s_v.$$

Thus $\mathcal{T}^2 = \mathrm{id}$, and in particular $\mathcal{T}$ is bijective. □

Lemma 2.34 (Packed transposition identity)**.** *For every $u \in I$,*

$$\mathcal{T}(s^*)_u = \sum_{j\in J} d_u(h_j)\widehat{w}_j. \tag{2.2}$$

Proof. Use Definition 2.32 and expand:

$$\mathcal{T}(s^*)_u = \sum_v \beta_v d_u\left(\sum_j h_j w_{v,j}\right).$$

Because each $w_{v,j} \in \mathbb{F}_2$ and d_u is $\mathbb{F}_2$-linear, $d_u(\sum_j h_j w_{v,j}) = \sum_j d_u(h_j) w_{v,j}$. Interchanging the two finite sums yields

$$\sum_j d_u(h_j)\left(\sum_v \beta_v w_{v,j}\right) = \sum_j d_u(h_j)\widehat{w}_j,$$

which is the assertion. □

Definition 2.35 (Random multilinear batching)**.** For $\theta \in L^\ell$ and $u \in I$, put $\lambda_u(\theta) := \mathrm{eq}_\ell(u, \theta)$. For a candidate vector $s \in L^I$ define

$$B_s(\theta) := \sum_{u\in I} \lambda_u(\theta)\mathcal{T}(s)_u, \qquad a_j(\theta) := \sum_{u\in I} \lambda_u(\theta) d_u(h_j).$$

The batched transposition equation is

$$B_s(\theta) = \sum_{j\in J} a_j(\theta)\widehat{w}_j. \tag{2.3}$$

Lemma 2.36 (Batch correctness)**.** *Equation (2.3) holds for every $\theta \in L^\ell$ when $s = s^*$.*

Proof. Multiply the identity (2.2) by $\lambda_u(\theta)$ and sum over u. Interchange the finite sums and use the definition of $a_j(\theta)$. □

Lemma 2.37 (Nonzero row error yields a nonzero polynomial)**.** *Fix $e = (e_u)_{u \in I} \in L^I$ and define*

$$E_e(\Theta) := \sum_{u \in I} e_u \, \mathrm{eq}_\ell(u, \Theta) \in L[\Theta_0, \ldots, \Theta_{\ell-1}].$$

If $e \neq 0$, then E_e is nonzero and has total degree at most ℓ.

Proof. Each equality polynomial is a product of ℓ affine factors, so the total degree bound holds. Choose u_0 for which $e_{u_0} \neq 0$. At the Boolean point $\Theta = u_0$, Lemma 1.9 gives $E_e(u_0) = e_{u_0} \neq 0$. Hence E_e is not the zero polynomial. □

Theorem 2.38 (Ring-switching batching error)**.** *Let the witness w, public evaluation instance, and candidate partial evaluations s be fixed before a uniform challenge $\theta \xleftarrow{\$} L^\ell$. If $s \neq s^*$, then*

$$\Pr_\theta \left[B_s(\theta) = \sum_j a_j(\theta) \widehat{w}_j \right] \leq \min\{1, \ell/|L|\}.$$

Proof. Because $\mathcal{T}$ is injective by Lemma 2.33, $\mathcal{T}(s) - \mathcal{T}(s^*)$ is a nonzero vector e. By Lemma 2.36, the equation displayed in the theorem holds iff $E_e(\theta) = 0$. By Lemma 2.37, E_e is nonzero of total degree at most ℓ. Lemma 2.8, applied over L to the independent uniform coordinates of θ, gives probability at most $\ell/|L|$. The minimum with 1 is automatic. □

Remark 2.39 (Where commitment enters historically). Ring switching reduces a bit-level multilinear claim to linear relations on packed field elements. Diamond and Posen use a suitable underlying large-field proof system to establish those relations. The algebraic identity and batching theorem above do not prescribe which commitment system implements the final verification.

2.5 Zero-knowledge ring switching and trace-zero bit packing

Kochergin [Koc26] revisits the ring-switching protocol with a zero-knowledge requirement: the field-valued partial evaluations s_v from Section 2.4 must not be sent in the clear. Frigo and shelat [Fs26] give a subsequent general bit-packing approach whose advertised reduction uses only linear constraints and applies to arbitrary bit-linear claims. The results below are small field-algebra statements used to explain these constructions; complete protocol security analyses remain the task of the later chapters.

Definition 2.40 (Uniform pads and masked transposition)**.** Retain the binary extension $L = \mathbb{F}_{2^n}$ and the notation of Section 2.4. Let $p = (p_v)_{v \in I} \in L^I$ be independent uniform pads, chosen before the batching challenge, and let

$$m_v := s_v + p_v, \qquad \Delta_p(\theta) := \sum_{u,v \in I} \lambda_u(\theta) \beta_v d_u(p_v).$$

Lemma 2.41 (Additivity of masked transposition)**.** *For every $s, p \in L^I$ and $\theta \in L^\ell$,*

$$B_{s+p}(\theta) = B_s(\theta) + \Delta_p(\theta).$$

Proof. The coordinate functions d_u are $\mathbb{F}_2$-linear. Therefore

$$\mathcal{T}(s+p)_u = \sum_v \beta_v d_u(s_v + p_v) = \sum_v \beta_v (d_u(s_v) + d_u(p_v)) = \mathcal{T}(s)_u + \mathcal{T}(p)_u.$$

Multiply by $\lambda_u(\theta)$ and sum over u. The second sum is $\Delta_p(\theta)$ by definition. □

Lemma 2.42 (Absolute trace properties)**.** *Let* $L = \mathbb{F}_{2^n}$*. Define* $\mathrm{Tr} : L \to \mathbb{F}_2$ *by*

$$\mathrm{Tr}(x) := \sum_{j=0}^{n-1} x^{2^j}.$$

Then Tr *is* $\mathbb{F}_2$*-linear,* $\mathrm{Tr}(x^2) = \mathrm{Tr}(x)$*, and its image is* $\mathbb{F}_2$*.*

Proof. In characteristic two, $(x+y)^{2^j} = x^{2^j} + y^{2^j}$ for all j, proving additivity; scalar linearity over $\mathbb{F}_2$ is immediate. Since every $x \in L$ satisfies $x^{2^n} = x$,

$$\mathrm{Tr}(x^2) = x^2 + x^{2^2} + \cdots + x^{2^n} = x^2 + \cdots + x^{2^{n-1}} + x = \mathrm{Tr}(x).$$

Also $\mathrm{Tr}(x)^2 = \mathrm{Tr}(x^2) = \mathrm{Tr}(x)$, so $\mathrm{Tr}(x)$ is a root of $T^2 - T = T(T-1)$ and hence belongs to $\mathbb{F}_2$. Finally, the polynomial $X + X^2 + \cdots + X^{2^{n-1}}$ is nonzero and has degree $2^{n-1} < 2^n = |L|$. It cannot vanish at every element of L. Therefore Tr is not the zero map; its image is a nonzero $\mathbb{F}_2$-subspace of $\mathbb{F}_2$, hence equals $\mathbb{F}_2$. □

Lemma 2.43 (Trace-zero characterization)**.** *For the map* $\wp : L \to L$ *defined by* $\wp(t) = t^2 + t$*,*

$$\mathrm{im}\,\wp = \ker \mathrm{Tr}\,.$$

If t *is uniform on* L*, then* $\wp(t)$ *is uniform on* $\ker \mathrm{Tr}$*.*

Proof. In characteristic two, $\wp(t+s) = \wp(t) + \wp(s)$, so $\wp$ is $\mathbb{F}_2$-linear. By Lemma 2.42, $\mathrm{Tr}(\wp(t)) = \mathrm{Tr}(t^2) + \mathrm{Tr}(t) = 0$, proving inclusion of the image in the trace kernel. The polynomial $t^2 + t = t(t+1)$ vanishes precisely at $t \in \{0, 1\}$, so $\ker \wp$ has two elements. Consequently $\mathrm{im}\,\wp$ has $|L|/2$ elements. By surjectivity of Tr in Lemma 2.42, $\ker \mathrm{Tr}$ also has $|L|/2$ elements. The inclusion is thus an equality. Every value in the image has exactly two preimages, and a uniform t therefore induces the uniform distribution on $\ker \mathrm{Tr}$. □

Lemma 2.44 (Existence and uniqueness of the trace-dual basis)**.** *Let* $(\beta_u)_{u \in I}$ *be any* $\mathbb{F}_2$*-basis of* $L = \mathbb{F}_{2^n}$*, indexed by a set* I *of cardinality* n*. There is a unique family* $(\eta_u)_{u \in I} \subseteq L$ *such that*

$$\mathrm{Tr}(\eta_u \beta_v) = \delta_{uv} \qquad \text{for all } u, v \in I.$$

This family is an $\mathbb{F}_2$*-basis of* L*, and for every* $x = \sum_v d_v(x)\beta_v$*,* $d_u(x) = \mathrm{Tr}(\eta_u x)$*.*

Proof. Consider the $\mathbb{F}_2$-linear map

$$\Phi : L \longrightarrow \mathrm{Hom}_{\mathbb{F}_2}(L, \mathbb{F}_2), \qquad \Phi(a)(x) := \mathrm{Tr}(ax).$$

To prove injectivity, let $a \neq 0$. By surjectivity of Tr in Lemma 2.42, choose $y \in L$ with $\mathrm{Tr}(y) = 1$. Since a is invertible, setting $x = a^{-1}y$ gives $\Phi(a)(x) = \mathrm{Tr}(y) = 1$, so $\Phi(a) \neq 0$. Thus $\ker \Phi = \{0\}$. Both the domain and codomain have dimension n over $\mathbb{F}_2$: the codomain has dimension n because an $\mathbb{F}_2$-linear functional is uniquely determined by its n values on a basis. Hence Φ is bijective.

For each $u \in I$, define the coordinate functional $d_u : L \to \mathbb{F}_2$ by $d_u(\sum_v c_v \beta_v) = c_u$. The bijection Φ provides a unique η_u with $\Phi(\eta_u) = d_u$. Evaluating at β_v yields the desired trace pairing. If $\sum_u c_u \eta_u = 0$ with $c_u \in \mathbb{F}_2$, multiply by β_v and apply trace to obtain $c_v = 0$ for all v. Thus (η_u) is linearly independent and hence a basis. Finally $d_u(x) = \Phi(\eta_u)(x) = \mathrm{Tr}(\eta_u x)$ by its definition. □

Lemma 2.45 (Frobenius-linear decomposition of the correction)**.** *Choose the trace-dual field basis* $(\eta_u)_{u\in I}$ *satisfying* $d_u(x) = \mathrm{Tr}(\eta_u x)$*. For* $0 \le j < n$ *define*

$$\mu_j(\theta) := \sum_{u\in I} \lambda_u(\theta)\eta_u^{2^j}, \qquad T_j(\theta, p) := \sum_{v\in I} (\mu_j(\theta)\beta_v)^{2^{n-j}} p_v.$$

Then

$$\Delta_p(\theta) = \sum_{j=0}^{n-1} T_j(\theta, p)^{2^j}.$$

Proof. Expand the trace in Definition 2.40:

$$\begin{aligned}\Delta_p(\theta) &= \sum_{u,v} \lambda_u(\theta)\beta_v \sum_{j=0}^{n-1} (\eta_u p_v)^{2^j} \\ &= \sum_{j=0}^{n-1} \sum_v \beta_v p_v^{2^j} \left(\sum_u \lambda_u(\theta)\eta_u^{2^j} \right) \\ &= \sum_{j=0}^{n-1} \sum_v \mu_j(\theta)\beta_v p_v^{2^j}.\end{aligned}$$

For each j, Frobenius additivity and $x^{2^n} = x$ imply

$$T_j(\theta, p)^{2^j} = \sum_v (\mu_j(\theta)\beta_v)^{2^n} p_v^{2^j} = \sum_v \mu_j(\theta)\beta_v p_v^{2^j}.$$

Substitution yields the identity. □

Lemma 2.46 (Backward Frobenius recurrence)**.** *Let* $T_0, \dots, T_{n-1} \in L$ *be arbitrary. Set* $X_n = 0$ *and recursively*

$$X_j := T_j + X_{j+1}^2, \qquad j = n-1, \dots, 1.$$

Then

$$\sum_{j=0}^{n-1} T_j^{2^j} = T_0 + X_1^2.$$

Proof. From the recurrence, $T_j = X_j + X_{j+1}^2$ for $1 \le j < n$. Raising this identity to 2^j and summing gives

$$\sum_{j=1}^{n-1} T_j^{2^j} = \sum_{j=1}^{n-1} X_j^{2^j} + \sum_{j=1}^{n-1} X_{j+1}^{2^{j+1}}.$$

Terms indexed by $2 \le j \le n-1$ occur twice and cancel in characteristic two. The remaining terms are $X_1^2 + X_n^{2^n} = X_1^2$. Add T_0. □

Lemma 2.47 (Affine verification of a masked recurrence)**.** *Fix* T_j *and* X_j *as in Lemma 2.46. Let* $\rho_j \in L$ *be masks for* $1 \le j < n$ *and put* $Q_j := \rho_j^2$ *and* $Z_j := X_j + \rho_j$*; set* $Z_n = Q_n = 0$*. Then*

$$\begin{aligned} Q_j &= \rho_j^2, \\ T_j &= Z_j + \rho_j + Z_{j+1}^2 + Q_{j+1} \quad (1 \le j < n), \\ \sum_{j=0}^{n-1} T_j^{2^j} &= T_0 + Z_1^2 + Q_1. \end{aligned}$$

Proof. The first relation is the definition of Q_j. Since $X_j = Z_j + \rho_j$,

$$X_{j+1}^2 = (Z_{j+1} + \rho_{j+1})^2 = Z_{j+1}^2 + Q_{j+1}.$$

Substitute both expressions into $T_j = X_j + X_{j+1}^2$ to obtain the second equation. Apply Lemma 2.46 and use $X_1^2 = Z_1^2 + Q_1$ for the third. All the displayed recurrence equations are affine in the hidden committed values once Z_j are public; only the separate square relations $Q_j = \rho_j^2$ are quadratic in hidden values. □

Remark 2.48 (Comparison of the two 2026 constructions). Kochergin [Koc26] uses the preceding Frobenius recurrence to certify the masked transposition with $n-1$ auxiliary squaring constraints. Frigo and shelat [Fs26] instead emphasize the $\mathbb{F}_2$-linearity of trace and the trace-zero characterization, and report a reduction for arbitrary bit-linear claims requiring only linear constraints over a suitable hiding linear commitment. In particular, the trace-zero lemma above is *not* a new invention of the present manuscript.

2.6 The contribution of KBFold and the exact remaining gap

The original KBFold construction [Mkh26] starts from the same kernel family as Chapter 1 and makes its normalized fold equal to restriction of the coefficient table. It then applies this dictionary to Reed–Solomon codewords and interleaves sumcheck with folding. The existing work proves correctness, soundness, evaluation binding, and round-by-round knowledge soundness in its stated regimes. The zero-knowledge extension is explicitly listed as future work; it was not among the results proved in that paper.

Definition 2.49 (Three different mathematical operations)**.** We distinguish:

(i) *Kernel table restriction*: $\mathsf{Res}_r(f)(b') = (1-r)f(0,b') + rf(1,b')$ over an arbitrary field.

(ii) *Binary coordinate transposition*: $\mathcal{T}(s)_u = \sum_v \beta_v d_u(s_v)$ over $L/\mathbb{F}_2$.

(iii) *Zero-knowledge masking*: replacing exposed, witness-dependent messages by masked messages, together with a proof that their algebraic verification relations hold.

None of the three operations is, by definition, a cryptographic proof system.

Lemma 2.50 (Two ways to represent the same binary correction)**.** *In the binary-field setting of Section 2.4, suppose the chosen packing basis is $\beta_u = K_u^{(\ell)}(\alpha)$ for an element $\alpha \in L$ of degree n over $\mathbb{F}_2$. Define*

$$c_u^p := \mathcal{T}(p)_u, \qquad C_p(X) := \sum_{u \in I} c_u^p K_u^{(\ell)}(X).$$

Then, for every $\theta \in L^\ell$,

$$\Delta_p(\theta) = \sum_{u \in I} \mathrm{eq}_\ell(u,\theta) c_u^p = \mathsf{KFold}_\theta(C_p).$$

Proof. By the definition of $\mathcal{T}$, $c_u^p = \sum_v \beta_v d_u(p_v)$. Substitute this expression into Definition 2.40 to obtain the first equality. The second is the defining formula for KFold_θ on the kernel-coefficient table of C_p (Definition 1.22), or equivalently Proposition 1.23. No property of an external commitment is required. □

Proposition 2.51 (The kernel representation does not depend on the masking protocol)**.** *For fixed p, the polynomial C_p in Lemma 2.50 is uniquely determined by its kernel coefficients c_u^p, and its successive normalized folds yield $\Delta_p(\theta)$. Neither the Frobenius recurrence of Lemma 2.46 nor the trace-zero characterization of Lemma 2.43 is needed for that algebraic identity.*

Proof. Uniqueness follows from Proposition 1.6 over L. Corollary 1.15 identifies the successive folds of C_p with the multilinear evaluation of the coefficient table at θ. By Lemma 2.50 this is $\Delta_p(\theta)$. The proof uses only the kernel basis theorem, restriction, and the definition of binary transposition, so the final independence assertion follows from the cited dependencies. □

Remark 2.52 (Characteristic and domain restrictions). The map $x \mapsto x^2 + x$ used in the binary-field masking constructions is a characteristic-two Artin–Schreier map. The original KBFold word fold in Definition 2.20 requires $\mathrm{char}(\mathbb{F}) \neq 2$ and an even-order multiplicative subgroup; its formulas divide by $2x$. One cannot equate these two settings by simply changing notation. A genuine ZK extension of the original KBFold scheme must provide masks and a complete simulation argument over that scheme's field and oracle model.

2.7 Precise comparison and chapter dependencies

For reference, the following comparisons identify the mathematical objects and the *source* of each step; they are not measurements of full proving time.

Construction	Principal algebraic operation	Role in our manuscript
LFKN sumcheck [Lun+92]	Partial-sum polynomials and random evaluation	Exact sumcheck completeness and soundness
Libra [Xie+19]	Zero-knowledge masking of polynomial messages	Motivation for degree-compatible masking
FRI [Ben+18]	Reed–Solomon codeword folding	Oracle proximity and local consistency
BaseFold [ZCF23]	Foldable codes interleaved with sumcheck	Multilinear proof architecture
Diamond–Posen [DP24]	Packed-bit transposition and batching	Binary ring-switching relation
Kochergin [Koc26]	Trace/Frobenius powers and masked recurrence	$n-1$ squaring relations in a ZK reduction
Frigo–shelat [Fs26]	Trace-linearity and trace-zero masking	General linear-constraint bit-packing approach
KBFold [Mkh26]	Boolean-kernel basis and normalized fold dictionary	Original unmasked commitment protocol
Present chapter	Exact kernel mask propagation and correction identities	Starting lemmas for the new development

Definition 2.53 (Later proof obligations)**.** A complete zero-knowledge extension of the original KBFold protocol will require, in addition to its existing soundness analysis:

(i) an explicit distribution of the random polynomial and codeword masks;

(ii) a revised degree and folding invariant, including Theorem 2.27;

(iii) a masked sumcheck protocol with the same public evaluation relation;

(iv) a joint hiding theorem for every exposed round polynomial, oracle response and final message;

(v) completeness and soundness of the modified oracle/query procedure;

(vi) an explicit simulator under a stated verifier-query bound and commitment interface.

These are mathematical conditions to be proved, not assumptions silently inherited from any one of the historical constructions.

Remark 2.54 (Roadmap of the formal development). Chapter 3 defines the exact IOP, commitment and zero-knowledge security games. The subsequent binary-field chapters develop the kernel transposition correction and compare its verification with the two 2026 protocols. The final KBFold chapters use the odd-characteristic fold, enlarged-degree mask propagation, masked sumcheck and oracle simulation. At each step the theorem ledger records the precise Lean declaration and all imported hypotheses.

3 Formal Security Model for Zero-Knowledge KBFold

This chapter supplies the exact definitions and small proof obligations used to distinguish algebraic correctness, oracle proximity soundness, extraction, and zero knowledge. All the objects and probability spaces are finite. The notation for kernel encoding and folding is inherited from Chapters 1 and 2, while the adversarial verifier model is new and is stated independently of a particular implementation of a Merkle tree or a random oracle.

3.1 Finite probability spaces and statistical distance

Every probability calculation in the algebraic IOP model below takes place on a finite sample space. Cryptographic security parameter families will be introduced separately. This separation allows the foundational results to be formalized in Lean 4 without first implementing a computational adversary model.

Definition 3.1 (Finite probability distribution)**.** For a nonempty finite set A, a *probability distribution* on A is a function $\mu : A \to \mathbb{R}_{\geq 0}$ such that

$$\sum_{a \in A} \mu(a) = 1.$$

We write $\mathsf{Dist}(A)$ for this set. For $E \subseteq A$, put $\Pr_\mu[E] := \sum_{a \in E} \mu(a)$. The uniform distribution is $\mathsf{Unif}_A(a) := 1/|A|$. For independent $X \sim \mu$ and $Y \sim \nu$, the joint distribution satisfies $\Pr[X = a, Y = b] = \mu(a)\nu(b)$.

Definition 3.2 (Total variation distance)**.** For $\mu, \nu \in \mathsf{Dist}(A)$ define

$$\Delta_A(\mu, \nu) := \frac{1}{2} \sum_{a \in A} |\mu(a) - \nu(a)|. \tag{3.1}$$

For random variables with distributions μ and ν, we also write $\Delta(X, Y)$.

Lemma 3.3 (Range and symmetry)**.** *For $\mu, \nu \in \mathsf{Dist}(A)$,*

$$0 \leq \Delta_A(\mu, \nu) \leq 1, \qquad \Delta_A(\mu, \nu) = \Delta_A(\nu, \mu),$$

and $\Delta_A(\mu, \nu) = 0$ if and only if $\mu = \nu$.

Proof. Each summand in (3.1) is nonnegative, proving the lower bound. The inequality $|\mu(a) - \nu(a)| \leq \mu(a) + \nu(a)$ follows because both terms are nonnegative; summing gives $\Delta_A \leq (1+1)/2 = 1$. Symmetry follows from $|x - y| = |y - x|$. If the total variation is zero, each nonnegative summand must vanish, so $\mu(a) = \nu(a)$ for all a; the converse is immediate. □

Lemma 3.4 (Triangle inequality)**.** *For $\mu, \nu, \xi \in \mathsf{Dist}(A)$,*

$$\Delta_A(\mu, \xi) \le \Delta_A(\mu, \nu) + \Delta_A(\nu, \xi).$$

Proof. For every a, use the real triangle inequality $|\mu(a) - \xi(a)| \le |\mu(a) - \nu(a)| + |\nu(a) - \xi(a)|$. Sum over A and divide by 2. □

Lemma 3.5 (Event characterization and separating events)**.** *For any $\mu, \nu \in \mathsf{Dist}(A)$ and any $E \subseteq A$,*

$$|\Pr_\mu[E] - \Pr_\nu[E]| \le \Delta_A(\mu, \nu).$$

There exists an $E \subseteq A$ for which equality holds, so

$$\Delta_A(\mu, \nu) = \max_{E \subseteq A} |\Pr_\mu[E] - \Pr_\nu[E]|.$$

Proof. Put $\delta(a) := \mu(a) - \nu(a)$. Since both distributions have mass 1, $\sum_a \delta(a) = 0$. Let $E_+ := \{a : \delta(a) \ge 0\}$ and $E_- := A \setminus E_+$. Then

$$\sum_{a \in E_+} \delta(a) = -\sum_{a \in E_-} \delta(a) = \frac{1}{2} \sum_{a \in A} |\delta(a)| = \Delta_A(\mu, \nu).$$

For any E, the sum $\sum_{a \in E} \delta(a)$ is at most the sum over all nonnegative terms, which is Δ_A, and at least the sum over all negative terms, which is $-\Delta_A$. This proves the inequality. Taking $E = E_+$ gives equality. □

Definition 3.6 (Markov kernel and pushforward)**.** For nonempty finite sets A, B, a *Markov kernel* $K : A \to \mathsf{Dist}(B)$ assigns a distribution $K(a)$ to each $a \in A$. Its action on $\mu \in \mathsf{Dist}(A)$ is

$$(K_* \mu)(b) := \sum_{a \in A} \mu(a) K(a)(b).$$

This is a probability distribution on B.

Lemma 3.7 (Data processing)**.** *For every Markov kernel $K : A \to \mathsf{Dist}(B)$ and $\mu, \nu \in \mathsf{Dist}(A)$,*

$$\Delta_B(K_* \mu, K_* \nu) \le \Delta_A(\mu, \nu).$$

Proof. Expanding the definition and applying the real triangle inequality,

$$\begin{aligned} 2\Delta_B(K_* \mu, K_* \nu) &= \sum_{b \in B} \left| \sum_{a \in A} (\mu(a) - \nu(a)) K(a)(b) \right| \\ &\le \sum_{b \in B} \sum_{a \in A} |\mu(a) - \nu(a)| K(a)(b) \\ &= \sum_{a \in A} |\mu(a) - \nu(a)| \sum_{b \in B} K(a)(b) \\ &= 2\Delta_A(\mu, \nu). \end{aligned}$$

The third equality interchanges finite sums, and the fourth uses the normalization of $K(a)$. □

Corollary 3.8 (Projection does not improve distinguishing)**.** *For a function $\pi : A \to B$, let $\pi_* \mu$ denote the distribution of $\pi(X)$ for $X \sim \mu$. Then*

$$\Delta_B(\pi_* \mu, \pi_* \nu) \le \Delta_A(\mu, \nu).$$

Proof. Apply Lemma 3.7 to the deterministic Markov kernel $K(a)$ concentrated at $\pi(a)$. □

Lemma 3.9 (Convexity under a shared mixing distribution)**.** *Let $p \in \mathsf{Dist}(A)$, and for each $a \in A$ let $\mu_a, \nu_a \in \mathsf{Dist}(B)$. Then*

$$\Delta_B\left(\sum_a p(a)\mu_a, \sum_a p(a)\nu_a\right) \le \sum_a p(a)\Delta_B(\mu_a, \nu_a).$$

Proof. For each b, the absolute difference of the two mixture masses is $|\sum_a p(a)(\mu_a(b) - \nu_a(b))|$, bounded by $\sum_a p(a)|\mu_a(b) - \nu_a(b)|$. Sum over b, interchange the two finite sums, and divide by 2. □

Lemma 3.10 (Stepwise hybrid bound)**.** *Let $A_0, \dots, A_t$ be finite nonempty sets. Let $\mu_0, \nu_0 \in \mathsf{Dist}(A_0)$, and for $1 \le i \le t$ let $K_i, L_i : A_{i-1} \to \mathsf{Dist}(A_i)$ be Markov kernels. Put $\mu_i = (K_i)_*\mu_{i-1}$ and $\nu_i = (L_i)_*\nu_{i-1}$. Suppose that*

$$\Delta_{A_0}(\mu_0, \nu_0) \le \varepsilon_0, \qquad \Delta_{A_i}(K_i(a), L_i(a)) \le \varepsilon_i \quad \forall\, a \in A_{i-1}.$$

Then for $0 \le j \le t$,

$$\Delta_{A_j}(\mu_j, \nu_j) \le \sum_{i=0}^{j} \varepsilon_i.$$

Proof. Induct on j. The base case is the first hypothesis. For the inductive step, insert the intermediate distribution $(K_j)_*\nu_{j-1}$ and use Lemma 3.4:

$$\begin{aligned}\Delta(\mu_j, \nu_j) &\le \Delta((K_j)_*\mu_{j-1}, (K_j)_*\nu_{j-1}) + \Delta((K_j)_*\nu_{j-1}, (L_j)_*\nu_{j-1}) \\ &\le \Delta(\mu_{j-1}, \nu_{j-1}) + \sum_{a \in A_{j-1}} \nu_{j-1}(a)\Delta(K_j(a), L_j(a)).\end{aligned}$$

The second line is Lemma 3.7 followed by Lemma 3.9. The last sum is at most ε_j. Apply the induction hypothesis. □

Remark 3.11 (Cryptographic meaning). Lemma 3.10 is an algebraic tool for constructing a simulator round by round. Its hypotheses require a uniform bound for *every history*, not only for the average transcript of an honest verifier. This difference will matter for adaptive oracle queries.

3.2 Oracle proof systems and binding commitments

Definition 3.12 (Evaluation domain and initial oracle)**.** Let $\mathbb{F}$ be a finite field, let $m \ge 1$, and put $N := 2^m$. Fix a finite set $\mathcal{D} \subseteq \mathbb{F}$ of $M \ge N$ pairwise distinct field elements. A *word* on $\mathcal{D}$ is a function $w : \mathcal{D} \to \mathbb{F}$. Its read-only oracle $\mathcal{O}_w$ answers a query $\xi \in \mathcal{D}$ with $w(\xi)$. The original KBFold setting further requires $\mathrm{char}(\mathbb{F}) \ne 2$ and a smooth multiplicative subgroup $\mathcal{D}$, as in Definition 2.18.

Definition 3.13 (Public-coin IOP transcript)**.** Fix a number of rounds $t \ge 1$ and nonempty finite challenge sets $\mathsf{Ch}_1, \dots, \mathsf{Ch}_t$. In round i, a prover sends a pair

$$m_i = (a_i, \mathcal{O}_{w_i}),$$

where a_i is a finite, explicit message and $w_i : \Lambda_i \to \mathbb{F}$ is a full oracle word on a finite index set Λ_i. An omitted oracle component is interpreted as the unique empty word. The verifier then samples an independent uniform challenge $c_i \in \mathsf{Ch}_i$. A full transcript is the typed tuple

$$\tau = (x, m_1, c_1, \dots, m_t, c_t),$$

where x may contain a read-only initial oracle $\mathcal{O}_{w_0}$. The honest verifier can subsequently query entries of the instance and message oracles, and its acceptance predicate depends only on explicit messages, challenges, and the queried entries.

Definition 3.14 (Causal prover)**.** A deterministic prover is a family of functions $(P_i)_{1\le i\le t}$ such that

$$m_i = P_i(x, \mathsf{wit}, c_1, \dots, c_{i-1}).$$

In particular m_i does not depend on c_i. A randomized prover is a finitely supported distribution over deterministic timing-respecting provers, sampled independently of the verifier challenges.

Lemma 3.15 (The honest transcript distribution is defined)**.** *If every message, challenge, oracle domain, and random tape set is finite, then for each fixed* (x, wit) *the interaction of a randomized prover and the honest public-coin verifier defines a probability distribution on full transcripts.*

Proof. Sample the prover's random tape first and sample each verifier challenge uniformly and independently. Since every prover message is a deterministic function of the already sampled data and the preceding challenges, induction on the round index uniquely determines the full transcript from the sampled tapes. The product of finitely many finite probability spaces is finite. Pushing its distribution forward through the transcript map produces the asserted distribution (Definition 3.6 for deterministic kernels). □

Definition 3.16 (Adaptive verifier and query bound)**.** An *adaptive verifier* V^* is a possibly randomized interactive strategy that may, after receiving an oracle handle, query that handle at valid indices, choose subsequent challenges as functions of its preceding observations, and finally output a bit. It is *Q-query bounded* if every execution makes at most Q read operations across all instance and prover-supplied oracles, counting repeated reads. Its query record is a sequence of triples

$$\mathsf{qry} = ((j_1, \xi_1, a_1), \dots, (j_h, \xi_h, a_h)), \qquad 0 \le h \le Q,$$

with $a_k = w_{j_k}(\xi_k)$; for authenticated commitments the record additionally includes an opening proof π_k. The verifier may not query an oracle handle before that oracle is registered.

Lemma 3.17 (Finite decision-tree representation)**.** *Fix the random tape of a Q-query-bounded adaptive verifier V^*. At every query step, its choice of the next oracle index, the next challenge, or termination is determined by the public input, explicit messages, and the previously received answers. Consequently its behavior can be represented by a decision tree of depth at most Q for oracle-read nodes, with finite branching.*

Proof. For a fixed random tape the verifier is deterministic by definition. At the root its first action is determined by its public input. If that action requests an oracle answer, the finite set of possible field values labels the outgoing edges. On each child, the verifier's next action is determined by the previous action and answer; repeat. No execution has more than Q query nodes, so the resulting tree has bounded query depth. Other finite explicit-message and challenge choices can be included as intermediate nodes. Finiteness of every message and index space ensures finite branching. □

Definition 3.18 (Verifier's view)**.** For a complete execution with adversarial verifier V^*, define its *view* as the tuple consisting of its public input, its own private random tape, every explicit prover message, every challenge, the ordered adaptive query-and-answer record (including any authentication proofs), and its final output bit. The full contents of an unopened oracle are *not* included. Write

$$\mathrm{View}^{P}_{V^*}(x, \mathsf{wit})$$

for the resulting finite random variable.

Definition 3.19 (Abstract oracle commitment)**.** An indexed commitment system with index set $\mathcal{D}$, field $\mathbb{F}$, and security parameter $\lambda_{\mathrm{sec}} \in \mathbb{N}$ is a triple of algorithms

$$\mathsf{Commit}_{\lambda_{\mathrm{sec}}} : \mathbb{F}^{\mathcal{D}} \times \mathsf{R}_{\lambda_{\mathrm{sec}}} \to \mathsf{CM}_{\lambda_{\mathrm{sec}}} \times \mathsf{ST}_{\lambda_{\mathrm{sec}}},$$

$$\mathsf{Open}_{\lambda_{\mathrm{sec}}} : \mathsf{ST}_{\lambda_{\mathrm{sec}}} \times \mathcal{D} \to \mathbb{F} \times \mathsf{Proof}_{\lambda_{\mathrm{sec}}}, \quad \mathsf{Check}_{\lambda_{\mathrm{sec}}} : \mathsf{CM}_{\lambda_{\mathrm{sec}}} \times \mathcal{D} \times \mathbb{F} \times \mathsf{Proof}_{\lambda_{\mathrm{sec}}} \to \{0,1\}.$$

The set $\mathsf{R}_{\lambda_{\mathrm{sec}}}$ contains the commitment random tapes. If $(\mathsf{cm}, \mathsf{st}) = \mathsf{Commit}(w; \rho)$ and $(a, \pi) = \mathsf{Open}(\mathsf{st}, \xi)$, *perfect correctness* is the assertion

$$a = w(\xi) \quad \text{and} \quad \mathsf{Check}(\mathsf{cm}, \xi, a, \pi) = 1$$

for every w, ρ, ξ.

Definition 3.20 (Coordinate binding)**.** For an adversary $\mathcal{A}$ outputting a tuple

$$(\mathsf{cm}, \xi, a, \pi, a', \pi'),$$

define

$$\mathsf{BadBind}(\mathcal{A}) := \{a \neq a' \ \wedge \ \mathsf{Check}(\mathsf{cm}, \xi, a, \pi) = 1 \ \wedge \ \mathsf{Check}(\mathsf{cm}, \xi, a', \pi') = 1\}.$$

The scheme is $(\varepsilon_{\mathrm{bind}}, \mathcal{C})$-coordinate binding if $\Pr[\mathsf{BadBind}(\mathcal{A})] \leq \varepsilon_{\mathrm{bind}}$ for every adversary $\mathcal{A}$ in the specified algorithm class $\mathcal{C}$. All probabilities are over the adversary's and the experiment's randomness.

Definition 3.21 (Standalone commitment hiding)**.** Let $w_0, w_1 \in \mathbb{F}^{\mathcal{D}}$. For independent uniformly sampled commitment random tapes let

$$C_b := \pi_{\mathsf{CM}}(\mathsf{Commit}(w_b; \rho_b)), \qquad b \in \{0,1\}.$$

The commitment is *ε-statistically hiding without openings* if $\Delta(C_0, C_1) \leq \varepsilon$ for every pair w_0, w_1. A computational variant restricts the tests to an explicitly chosen efficient distinguisher class.

Remark 3.22 (A commitment is not a simulator). Coordinate binding says that a root cannot be opened to two different values at the same index; it says nothing about hiding. Conversely, Definition 3.21 does not address leaked values returned by Open. A zero-knowledge KBFold statement must consider the *joint* distribution of the root, the prover's explicit messages, the chosen oracle openings, and the authentication proofs. We will use the full view of Definition 3.18 for that purpose.

Remark 3.23 (Fiat–Shamir and BCS). The original KBFold paper treats the interactive oracle reduction first and then invokes a specific BCS/QROM theorem to obtain a noninteractive knowledge-soundness statement [Mkh26]. Neither a generic random-oracle replacement of the challenges nor salted Merkle commitments alone constitute a zero-knowledge theorem. Any later compilation result must separately state the required simulator and random-oracle security assumptions.

3.3 The evaluation relation and Reed–Solomon proximity

Definition 3.24 (Kernel encoding as a word)**.** Retain the finite field $\mathbb{F}$, integer $m \geq 1$, $N = 2^m$, and evaluation domain $\mathcal{D} \subseteq \mathbb{F}$ with $M := |\mathcal{D}| \geq N$. For each table $f : I_m \to \mathbb{F}$ define

$$\mathrm{Enc}_{\mathcal{D}}(f) : \mathcal{D} \to \mathbb{F}, \qquad \mathrm{Enc}_{\mathcal{D}}(f)(\xi) := U_f^{(m)}(\xi). \tag{3.2}$$

The corresponding Reed–Solomon code is

$$\mathcal{C}_{\mathcal{D},N} := \{\mathrm{Enc}_{\mathcal{D}}(f) : f \in \mathbb{F}^{I_m}\} \subseteq \mathbb{F}^{\mathcal{D}}.$$

Lemma 3.25 (Injectivity of kernel encoding)**.** *If* $\mathrm{Enc}_{\mathcal{D}}(f) = \mathrm{Enc}_{\mathcal{D}}(g)$ *as words, then* $f = g$.

Proof. Let $P := U_f^{(m)} - U_g^{(m)}$. By Lemma 1.3 and the linear definition of U_f, P has degree less than N, or is zero. The equality of words means that P vanishes at every $\xi \in \mathcal{D}$. If P were nonzero, Lemma 2.7 would imply at most $N-1$ roots, contradicting $|\mathcal{D}| = M \geq N$. Thus $P = 0$. By the kernel basis theorem, Proposition 1.6, its coordinates satisfy $f(b) - g(b) = 0$ for every b, hence $f = g$. □

Definition 3.26 (Hamming distance)**.** For two words $w, w' : \mathcal{D} \to \mathbb{F}$ define their Hamming distance by

$$d_H(w, w') := \big|\{\xi \in \mathcal{D} : w(\xi) \neq w'(\xi)\}\big| \in \{0, \ldots, M\}.$$

The relative Hamming distance is $d_H(w, w')/M$.

Lemma 3.27 (Triangle inequality for Hamming distance)**.** *For every three words* $w_1, w_2, w_3 : \mathcal{D} \to \mathbb{F}$,

$$d_H(w_1, w_3) \leq d_H(w_1, w_2) + d_H(w_2, w_3).$$

Proof. If $w_1(\xi) \neq w_3(\xi)$, then at least one of $w_1(\xi) \neq w_2(\xi)$ and $w_2(\xi) \neq w_3(\xi)$ holds; otherwise transitivity of equality would give $w_1(\xi) = w_3(\xi)$. Thus the set of coordinates where w_1 and w_3 differ is contained in the union of the two sets where consecutive words differ. Taking cardinalities and using the union bound for finite sets proves the inequality. □

Theorem 3.28 (Reed–Solomon minimum distance)**.** *For any distinct tables* $f, g : I_m \to \mathbb{F}$,

$$d_H(\mathrm{Enc}_{\mathcal{D}}(f), \mathrm{Enc}_{\mathcal{D}}(g)) \geq M - N + 1. \tag{3.3}$$

Proof. By Proposition 1.6, $f \neq g$ implies $P := U_f^{(m)} - U_g^{(m)} \neq 0$. This polynomial has degree at most $N-1$. The two encoded words agree precisely at positions $\xi \in \mathcal{D}$ with $P(\xi) = 0$. Lemma 2.7 shows that there are at most $N-1$ such positions. Consequently there are at least $M-(N-1) = M-N+1$ differing positions, which is (3.3). □

Definition 3.29 (Exact evaluation relation)**.** An *exact oracle evaluation instance* is $(w; z, v)$ with $w : \mathcal{D} \to \mathbb{F}$, $z \in \mathbb{F}^m$, and $v \in \mathbb{F}$. Its witness relation is

$$\mathcal{R}_{\mathrm{eval}} := \left\{((w; z, v), f) : w = \mathrm{Enc}_{\mathcal{D}}(f) \ \wedge \ v = \widetilde{f}(z)\right\}. \tag{3.4}$$

The semicolon separates the oracle part w from the explicit part (z, v); the verifier may read w only through its oracle.

Definition 3.30 (Proximity evaluation relation)**.** For an integer τ satisfying $0 \leq \tau \leq M$, define

$$\mathcal{R}_\tau := \left\{((w; z, v), f) : d_H(w, \mathrm{Enc}_{\mathcal{D}}(f)) \leq \tau \ \wedge \ v = \widetilde{f}(z)\right\}. \tag{3.5}$$

Its language is

$$\mathcal{L}_\tau := \{(w; z, v) : \exists f, \ ((w; z, v), f) \in \mathcal{R}_\tau\}.$$

The relation $\mathcal{R}_0$ equals $\mathcal{R}_{\mathrm{eval}}$.

Lemma 3.31 (Uniqueness of a nearby codeword)**.** *Assume*

$$2\tau < M - N + 1.$$

For every received word $w : \mathcal{D} \to \mathbb{F}$ *there exists* at most one *table* $f : I_m \to \mathbb{F}$ *such that* $d_H(w, \mathrm{Enc}_{\mathcal{D}}(f)) \leq \tau$.

Proof. Suppose f and g both satisfy the distance condition. By Lemma 3.27,

$$d_H(\mathrm{Enc}_{\mathcal{D}}(f), \mathrm{Enc}_{\mathcal{D}}(g)) \le d_H(\mathrm{Enc}_{\mathcal{D}}(f), w) + d_H(w, \mathrm{Enc}_{\mathcal{D}}(g)) \le 2\tau.$$

If $f \neq g$, Theorem 3.28 gives a contradictory lower bound $M - N + 1 > 2\tau$. Therefore $f = g$. □

Corollary 3.32 (At most one nearby evaluation value)**.** *Under the inequality* $2\tau < M - N + 1$*, for every pair* (w, z) *there exists at most one* $v \in \mathbb{F}$ *such that* $(w; z, v) \in \mathcal{L}_\tau$*.*

Proof. Suppose v and v' both occur. By the definition of $\mathcal{L}_\tau$, there are corresponding tables f, g within distance τ of w. By Lemma 3.31, $f = g$. The relation requires $v = \widetilde{f}(z)$ and $v' = \widetilde{g}(z)$, so $v = v'$. □

Definition 3.33 (Oracle evaluation soundness)**.** For the fixed proximity threshold τ and an oracle IOP $\Pi = (P, V)$, define

$$\mathsf{Sound}_\tau(\Pi) := \sup_{\substack{(w;z,v)\notin\mathcal{L}_\tau \\ P^* \text{ causal}}} \Pr\big[V^{P^*}(w; z, v) \text{ accepts}\big],$$

where the prover's randomness and the honest verifier's challenges are independent, and the supremum over an empty class is defined to be 0. The statement $\mathsf{Sound}_\tau(\Pi) \le \varepsilon$ is *ε-evaluation soundness at proximity τ*.

Remark 3.34 (Scope of the Reed–Solomon lemmas). Theorem 3.28 and Lemma 3.31 establish only code-theoretic uniqueness. They do not prove the soundness of KBFold itself: that requires bounds on the probability of passing its fold and oracle queries. The parent KBFold paper supplies such a soundness analysis in the unique-decoding regime [Mkh26], which must be revisited when the polynomial degree increases under zero-knowledge masking.

3.4 Completeness, soundness, and extraction

Definition 3.35 (Perfect completeness for kernel evaluation)**.** Let $\Pi = (P, V)$ be an oracle IOP for Definition 3.29. It is *perfectly complete* if for every table $f : I_m \to \mathbb{F}$ and every point $z \in \mathbb{F}^m$, with

$$w = \mathrm{Enc}_{\mathcal{D}}(f), \qquad v = \widetilde{f}(z),$$

the honest prover P using witness f makes the honest verifier V accept with probability 1 over all independent prover coins and verifier challenges.

Definition 3.36 (False-claim experiment)**.** Fix $\tau \ge 0$. Define the experiment

$$\mathsf{FalseAccept}_{\Pi,\tau}(w, z, v; P^*).$$

It samples the tapes and public-coin challenges of Π, runs a causal adversarial prover P^* on oracle w and explicit instance (z, v), and outputs 1 exactly when

$$(w; z, v) \notin \mathcal{L}_\tau \quad \wedge \quad V \text{ accepts.}$$

Its success probability is taken over the sampled tapes and challenges, never over an unbound choice of w made *after* challenge sampling.

Lemma 3.37 (Deterministic-prover reduction)**.** *Let E be any event determined by a full IOP transcript. Suppose every deterministic causal prover has probability at most ε of producing a transcript in E. Then every randomized causal prover whose random tape is independent of verifier challenges also has probability at most ε.*

Proof. Let ρ denote the randomized prover tape. Conditioned on ρ, the prover becomes deterministic and causal, so $\Pr[E \mid \rho] \le \varepsilon$. By the finite law of total probability,

$$\Pr[E] = \sum_{\rho} \Pr[\rho] \Pr[E \mid \rho] \le \sum_{\rho} \Pr[\rho]\varepsilon = \varepsilon.$$

□

Definition 3.38 (Doomed partial transcript)**.** For a public-coin IOP with rounds $1, \dots, t$, let $\mathscr{T}_i$ be its set of typed partial transcripts after i challenges; let $\mathscr{T}_0$ consist of the initial instances. A *doomed family* is a tuple of subsets

$$\mathscr{D}_i \subseteq \mathscr{T}_i, \qquad 0 \le i \le t.$$

It is a *round-by-round soundness certificate* with error vector $(\varepsilon_1, \dots, \varepsilon_t)$ for relation $\mathcal{R}_\tau$ if:

(i) every initial instance outside $\mathcal{L}_\tau$ belongs to $\mathscr{D}_0$;

(ii) for each $i = 1, \dots, t$, each $\sigma \in \mathscr{D}_{i-1}$, and every legally timed prover message m_i,

$$\Pr_{c_i \leftarrow \mathsf{Unif}_{\mathsf{Ch}_i}} [(\sigma, m_i, c_i) \notin \mathscr{D}_i] \le \varepsilon_i;$$

(iii) every full transcript in $\mathscr{D}_t$ is rejected by the honest verifier.

All conditions are pointwise in the preceding partial transcript; in (ii) the message is fixed before the fresh challenge.

Lemma 3.39 (First escape is necessary for acceptance)**.** *Suppose $(\mathscr{D}_i)$ satisfies (i) and (iii) in Definition 3.38. For a false instance $(w; z, v) \notin \mathcal{L}_\tau$, every accepted transcript has at least one index $i \in \{1, \dots, t\}$ such that the transcript after $i-1$ challenges belongs to $\mathscr{D}_{i-1}$ but the transcript after i challenges does not belong to $\mathscr{D}_i$.*

Proof. The initial transcript is in $\mathscr{D}_0$ by (i). The accepted full transcript is not in $\mathscr{D}_t$ by contraposition of (iii). Along the finite sequence of membership indicators, there is therefore a first index at which membership changes from true to false. This index has the stated property. □

Theorem 3.40 (Round-by-round implies overall soundness)**.** *If Π admits a doomed family as in Definition 3.38, then*

$$\mathsf{Sound}_\tau(\Pi) \le \min\left\{1, \sum_{i=1}^{t} \varepsilon_i\right\}.$$

Proof. Fix a false instance and a deterministic causal prover. For each i, let E_i be the event that the preceding transcript belongs to $\mathscr{D}_{i-1}$ and the next transcript is outside $\mathscr{D}_i$. Conditional on every fixed prefix of earlier challenges for which the preceding transcript is doomed, the causal prover's message m_i is fixed. Condition (ii) gives conditional probability at most ε_i for E_i; if the preceding

transcript is not doomed, E_i is empty. Averaging over the preceding prefixes yields $\Pr[E_i] \le \varepsilon_i$. By Lemma 3.39, acceptance is contained in $\bigcup_i E_i$. The union bound gives

$$\Pr[\text{accept}] \le \sum_i \Pr[E_i] \le \sum_i \varepsilon_i.$$

Lemma 3.37 extends the bound to randomized provers. Probabilities are at most 1, proving the minimum. □

Definition 3.41 (Round-by-round knowledge witness)**.** Let $(\mathscr{D}_i)$ be a doomed family. An *extractor family* consists of functions

$$\mathsf{Ext}_i : \{(\sigma, m_i) : \sigma \in \mathscr{T}_{i-1},\ m_i \text{ a legal next message}\} \to \mathbb{F}^{I_m}, \quad 1 \le i \le t.$$

It is a knowledge certificate with errors (ε_i) for $\mathcal{R}_\tau$ if:

(i) every initial instance (including true instances) is in $\mathscr{D}_0$;

(ii) every full transcript in $\mathscr{D}_t$ is rejected;

(iii) for all $i, \sigma \in \mathscr{D}_{i-1}$, and legal m_i, if

$$\Pr_{c_i}[(\sigma, m_i, c_i) \notin \mathscr{D}_i] > \varepsilon_i,$$

then $\mathsf{Ext}_i(\sigma, m_i)$ is a valid witness for the instance encoded by σ, in the sense of $\mathcal{R}_\tau$.

For a computational knowledge theorem one must additionally require that the extractors run within an explicitly specified time bound.

Theorem 3.42 (Knowledge certificate implies witness or bounded acceptance)**.** *Assume Definition 3.41 holds. Fix an instance x and a deterministic causal prover P^*. If every pair (σ, m_i) reached with $\sigma \in \mathscr{D}_{i-1}$ satisfies*

$$(x, \mathsf{Ext}_i(\sigma, m_i)) \notin \mathcal{R}_\tau,$$

then the acceptance probability is at most $\sum_{i=1}^{t} \varepsilon_i$. Equivalently, acceptance probability strictly greater than this sum implies that some reachable pair (σ, m_i) has a valid extracted witness.

Proof. Under the stated failure of all reached extraction outputs, the contrapositive of condition (iii) in Definition 3.41 shows that, for every reachable doomed prefix and its next message, the probability of leaving the doomed family is at most ε_i. The initial transcript is doomed by condition (i), and any accepted transcript is non-doomed by condition (ii). The first-escape argument of Lemma 3.39 therefore applies; the conditional union-bound proof of Theorem 3.40 gives the asserted acceptance bound. Its logical contrapositive proves the equivalent formulation. □

Remark 3.43 (Security properties are distinct). Theorems 3.40 and 3.42 concern acceptance and extraction. They place no restriction on information transmitted in the prover's messages or oracle openings. A knowledge-sound proof need not be zero knowledge, and a hiding commitment alone need not yield a knowledge-sound evaluation proof. Zero knowledge is defined separately below.

3.5 Zero knowledge, simulation, and explicit leakage obstructions

Definition 3.44 (Witness-dependent real view)**.** Fix public parameters $(\mathbb{F}, m, \mathcal{D})$, a public evaluation pair (z, v), and a table $f : I_m \to \mathbb{F}$ satisfying $v = \widetilde{f}(z)$. An *evaluation proof construction* consists of a randomized preprocessing algorithm $\mathsf{Prep}(f; \rho)$ outputting an initial public commitment cm, private prover state st, and any committed oracle handles, together with a subsequent interactive proof Π. For a Q-query-bounded verifier V^*, the *real-view distribution* is

$$\mathsf{Real}_{\Pi,V^*}(z, v; f) := \mathrm{Law}\big(\mathrm{View}_{V^*}^{P_f}((\mathsf{cm}; z, v), \mathsf{st})\big),$$

where ρ and the prover and verifier random tapes are freshly and independently sampled, except for any explicitly specified correlations. The view includes the public commitment but not the unopened oracle contents.

Definition 3.45 (Statistical zero knowledge for a fixed evaluation point)**.** Let Q be a nonnegative query bound, and let $\varepsilon \geq 0$. The protocol Π is (Q, ε)-*statistical zero knowledge in the fixed-point commitment-generation game* if for every allowed Q-query-bounded verifier V^* there exists a probabilistic simulator Sim_{V^*}, depending on public parameters and V^* but *not on the witness*, such that for every $z \in \mathbb{F}^m$, $v \in \mathbb{F}$, and every f satisfying $v = \widetilde{f}(z)$,

$$\Delta\big(\mathsf{Real}_{\Pi,V^*}(z, v; f), \mathsf{Sim}_{V^*}(z, v)\big) \leq \varepsilon. \tag{3.6}$$

Perfect zero knowledge is the case $\varepsilon = 0$. The simulator must generate the *complete view*, including the commitment root and all authenticated query responses, not merely the prover's explicit messages.

Remark 3.46 (Order of commitment and evaluation point)*.* Definition 3.45 gives the public pair (z, v) to the simulator before commitment generation. It is deliberately a *fixed-point* security game. A protocol claiming security when the verifier selects z only *after* receiving a commitment must satisfy an *online* version in which the simulator produces a commitment before learning the chosen z, later receives the authorized value v, and answers the verifier's proof and oracle queries consistently. The stronger game is a separate obligation for the final PCS theorem; a proof of Definition 3.45 alone does not imply it.

Definition 3.47 (Computational distinguishing advantage)**.** Let $(\Pi_{\lambda_{\mathrm{sec}}})_{\lambda_{\mathrm{sec}} \geq 1}$ be a family of protocols with finite transcript spaces. A function $\varepsilon : \mathbb{N} \to \mathbb{R}_{\geq 0}$ is *negligible* if for every real-coefficient polynomial p that is positive and unbounded on all sufficiently large integers, there exists n_0 such that $\varepsilon(n) < 1/p(n)$ for all $n \geq n_0$. The computational zero-knowledge variant of Definition 3.45 replaces total variation by the inequality

$$\big|\Pr[D(\mathsf{Real}_{\Pi,V^*}) = 1] - \Pr[D(\mathsf{Sim}_{V^*}) = 1]\big| \leq \varepsilon(\lambda_{\mathrm{sec}})$$

for all distinguishers D in an explicitly chosen polynomial-time class and all allowed witnesses. The security model must specify the representation, running-time bound, and random-oracle access of V^*, Sim and D.

Lemma 3.48 (Two-witness closeness is necessary)**.** *Assume Definition 3.45. For any two tables f_0, f_1 with $\widetilde{f_0}(z) = \widetilde{f_1}(z) = v$ and every allowed V^*,*

$$\Delta\big(\mathsf{Real}_{\Pi,V^*}(z, v; f_0), \mathsf{Real}_{\Pi,V^*}(z, v; f_1)\big) \leq 2\varepsilon.$$

Proof. Let $S = \mathsf{Sim}_{V^*}(z, v)$, the *same* simulator distribution for both witnesses. Lemma 3.4 gives

$$\Delta(\mathsf{Real}(f_0), \mathsf{Real}(f_1)) \le \Delta(\mathsf{Real}(f_0), S) + \Delta(S, \mathsf{Real}(f_1)) \le \varepsilon + \varepsilon = 2\varepsilon.$$

The final inequality is Definition 3.45. □

Lemma 3.49 (A public-message projection remains simulatable)**.** *Under Definition 3.45, for every deterministic projection π of a complete verifier view, the distributions $\pi(\mathsf{Real}_{\Pi,V^*}(z, v; f))$ and $\pi(\mathsf{Sim}_{V^*}(z, v))$ have total variation distance at most ε.*

Proof. Apply Corollary 3.8 to the full-view distributions in (3.6). The projection can select one round polynomial, one oracle answer, a Merkle opening, or any fixed combination of fields of the view. □

Theorem 3.50 (One-query leakage of an unmasked KBFold oracle)**.** *Suppose the oracle supplied to V^* consists of the* unmasked *word* $\mathrm{Enc}_{\mathcal{D}}(f)$*, and the verifier may make at least one read. Assume the public input to be simulated contains only the evaluation pair (z, v) and public parameters, and not the entire oracle as trusted public information. Then for every $m \ge 1$ there exist two tables $f_0 \ne f_1$, a point z, a value v, and a deterministic one-query verifier V^* for which*

$$\widetilde{f_0}(z) = \widetilde{f_1}(z) = v, \qquad \Delta(\mathsf{Real}_{\Pi,V^*}(z, v; f_0), \mathsf{Real}_{\Pi,V^*}(z, v; f_1)) = 1.$$

Consequently such an unmasked-word protocol cannot satisfy Definition 3.45 with $\varepsilon < \frac{1}{2}$.

Proof. Let $z = (0, \dots, 0)$ and $v = 0$. Let f_0 be identically zero. Choose any nonzero Boolean index $b \in I_m$ and let f_1 be the table supported at b with value 1 there and 0 elsewhere. Lemma 1.9 gives $\widetilde{f_0}(z) = f_0(0) = 0$ and $\widetilde{f_1}(z) = f_1(0) = 0$ because $b \ne 0$. By Lemma 3.25, the two encoded words are distinct. Hence there is a coordinate $\xi \in \mathcal{D}$ where $\mathrm{Enc}_{\mathcal{D}}(f_0)(\xi) \ne \mathrm{Enc}_{\mathcal{D}}(f_1)(\xi)$. Construct V^* to query this fixed public coordinate ξ once and record the answer. Under witness f_0, the answer is the fixed value a_0; under witness f_1, it is the distinct fixed value a_1. The event that the recorded answer equals a_0 has probability 1 in the first real view and 0 in the second. Lemma 3.5 implies their distance is at least 1, and Lemma 3.3 implies it is at most 1. Thus it equals 1. Lemma 3.48 would force $1 \le 2\varepsilon$, contradicting $\varepsilon < 1/2$. □

Remark 3.51 (Scope of the obstruction). Theorem 3.50 concerns witness privacy when the allowed public information is the evaluation claim and the commitment is to be simulated. It does *not* assert that a protocol fails a weaker notion relative to an already revealed entire codeword: the entire codeword uniquely determines f by Lemma 3.25, so such a notion would allow that leakage as public input. Likewise, this theorem does not rule out salted and polynomially masked commitments whose opened values have randomized distributions. Those are precisely the mechanisms needed by the intended ZK extension.

Lemma 3.52 (Reusing one pad leaks a difference)**.** *Let V be an additive abelian group, let $s_1, s_2 \in V$ be fixed, and let p be an arbitrary random element of V. If $m_1 = s_1 + p$ and $m_2 = s_2 + p$, then*

$$m_1 - m_2 = s_1 - s_2$$

with probability 1*.*

Proof. By the group laws, $m_1 - m_2 = (s_1 + p) - (s_2 + p) = s_1 - s_2 + (p - p) = s_1 - s_2$ for every value of p. □

Remark 3.53 (Why independent pad sampling is not a cosmetic choice). The marginal distribution of each m_i may be uniform, while their *joint* distribution leaks $s_1 - s_2$ by Lemma 3.52. A simulation proof for KBFold must establish the joint law across all folds and oracle openings, including correlations enforced by the folding equations. Marginal one-time-pad arguments do not suffice.

3.6 Adaptive linear masking and conditional oracle answers

The next lemmas give a criterion for whether a random linear mask actually hides the values seen by an adaptive oracle verifier. Unlike an assertion that one masked element is uniform, these statements describe the joint and conditional distributions of arbitrarily many linear observations.

Definition 3.54 (Linear masking experiment)**.** Let $\mathbb{F}$ be a finite field, let V be a finite-dimensional $\mathbb{F}$-vector space, and let $H \leq V$ be an $\mathbb{F}$-linear subspace. Fix a secret vector $s \in V$. Sample R uniformly from H and publish or expose the *masked vector*

$$Y_s := s + R.$$

For a fixed $\mathbb{F}$-linear map $L : V \to W$ into another finite-dimensional vector space W, the *linear observation* is $L(Y_s)$.

Lemma 3.55 (Uniformity on an affine coset)**.** *The random variable Y_s is uniform on the affine coset $s + H$. For every linear $L : V \to W$, its observation $L(Y_s)$ is uniform on*

$$L(s) + L(H) \subseteq W.$$

Proof. Translation $R \mapsto s + R$ is a bijection $H \to s + H$, so it preserves uniformity. The restriction $L|_H : H \to L(H)$ is a surjective linear map. For any $a \in L(H)$ and any chosen preimage $h_a \in H$, its full fiber is $h_a + \ker(L|_H)$, which has size $|\ker(L|_H)|$ independent of a. Thus $L(R)$ is uniform on $L(H)$, and translation by $L(s)$ proves the second assertion. □

Lemma 3.56 (Exact distinguishing criterion for linear observations)**.** *For fixed linear $L : V \to W$ and two secret vectors $s_0, s_1 \in V$, the distributions $L(Y_{s_0})$ and $L(Y_{s_1})$ satisfy*

$$\Delta(L(Y_{s_0}), L(Y_{s_1})) = \begin{cases} 0, & L(s_0 - s_1) \in L(H), \\ 1, & L(s_0 - s_1) \notin L(H). \end{cases}$$

Proof. By Lemma 3.55, the two variables are uniform respectively on $A_0 = L(s_0) + L(H)$ and $A_1 = L(s_1) + L(H)$. Two cosets of the same additive subgroup are either equal or disjoint. They are equal if and only if $L(s_0) - L(s_1) = L(s_0 - s_1)$ belongs to $L(H)$. In that case the distributions agree, so the distance is zero by Lemma 3.3. Otherwise their supports are disjoint. Taking the event A_0 in Lemma 3.5 shows that their total variation distance is at least 1, and Lemma 3.3 gives the reverse bound. □

Lemma 3.57 (Conditional uniformity after linear queries)**.** *Let $L : V \to W$ be linear, and let $y \in W$ be such that $\Pr[L(Y_s) = y] > 0$. Fix any $r_0 \in H$ satisfying $L(r_0) = y - L(s)$. Then, conditioned on $L(Y_s) = y$, the mask R is uniform on*

$$r_0 + (H \cap \ker L),$$

and the exposed vector Y_s is uniform on

$$s + r_0 + (H \cap \ker L).$$

Proof. The event $L(Y_s) = y$ is equivalent to $R \in H$ and $L(R) = y - L(s)$. By the assumption of positive probability, this fiber is nonempty; choose r_0 in it. For any $r \in H$, $L(r) = L(r_0)$ holds exactly when $r - r_0 \in H \cap \ker L$. Thus the fiber equals $r_0 + (H \cap \ker L)$. The original distribution of R assigns the same probability $1/|H|$ to every member of H, so conditioning on this nonempty fiber assigns equal probability to all its members. Translating by s yields the final statement. □

Corollary 3.58 (One more adaptive linear oracle query)**.** *Retain Lemma 3.57, and let $\varphi : V \to \mathbb{F}$ be another $\mathbb{F}$-linear functional chosen as a function of the observed answer y. Conditional on $L(Y_s) = y$, the next answer $\varphi(Y_s)$ is uniform on*

$$\varphi(s + r_0) + \varphi(H \cap \ker L) \subseteq \mathbb{F}.$$

In particular, its conditional distribution is either a point mass or the uniform distribution on $\mathbb{F}$.

Proof. Apply Lemma 3.55 to the conditional affine mask space in Lemma 3.57, using the linear map φ restricted to $H \cap \ker L$. Its image is an $\mathbb{F}$-linear subspace of the one-dimensional $\mathbb{F}$-vector space $\mathbb{F}$; such a subspace is either $\{0\}$ or all of $\mathbb{F}$. In the former case the translated distribution is a point mass, and in the latter it is uniform on $\mathbb{F}$. □

Remark 3.59 (What must be proved for an adaptive simulator). Corollary 3.58 gives an exact next-answer law *conditional on every realized query history*. It does not by itself give a witness-independent simulator: if the next answer is deterministic, its value could still depend on s. An online simulator must prove that each such deterministic value is determined by the authorized public information, and that each remaining randomized answer is sampled consistently with the earlier answers. The uniform-in-history hypothesis of Lemma 3.10 is the corresponding simulation invariant.

Lemma 3.60 (Evaluation masks at arbitrary distinct points)**.** *Let $\xi_1, \dots, \xi_Q \in \mathbb{F}$ be pairwise distinct, with $Q \geq 1$. The evaluation map*

$$\mathrm{Ev}_Q : \mathbb{F}[X]_{<Q} \to \mathbb{F}^Q, \qquad R \mapsto (R(\xi_1), \dots, R(\xi_Q))$$

is an $\mathbb{F}$-linear bijection.

Proof. Linearity is immediate from polynomial evaluation. For each i put

$$L_i(X) := \prod_{\substack{1 \leq j \leq Q \\ j \neq i}} \frac{X - \xi_j}{\xi_i - \xi_j}.$$

Every denominator is nonzero by pairwise distinctness, and $\deg L_i \leq Q - 1$. Direct substitution gives $L_i(\xi_i) = 1$ and $L_i(\xi_j) = 0$ for $j \neq i$. For any target vector $(y_1, \dots, y_Q)$, the polynomial $R(X) = \sum_{i=1}^{Q} y_i L_i(X)$ has degree less than Q and satisfies $R(\xi_j) = y_j$ for every j, establishing surjectivity. If R maps to zero, it has the Q distinct roots $\xi_1, \dots, \xi_Q$, while its degree is at most $Q - 1$. Lemma 2.7 forces $R = 0$, giving injectivity. □

Corollary 3.61 (Uniform masking of a bounded set of evaluation queries)**.** *Let R be uniform on $\mathbb{F}[X]_{<Q}$ and let $P \in \mathbb{F}[X]$ be fixed. For any fixed Q distinct points $\xi_1, \dots, \xi_Q$, the random vector*

$$(P(\xi_1) + R(\xi_1), \dots, P(\xi_Q) + R(\xi_Q))$$

is uniform on $\mathbb{F}^Q$ and its distribution is independent of P.

Proof. By Lemma 3.60, Ev_Q is a bijection; hence the evaluations of the uniform random polynomial R form a uniform vector in $\mathbb{F}^Q$. Translation by the fixed vector of P evaluations preserves the uniform distribution (Lemma 2.12). □

Remark 3.62 (Why the preceding corollary is only a local hiding claim). For fewer than Q distinct queries, the same conclusion follows by restriction of the uniform Q-vector to the requested coordinates. For *more* than Q queries the image of Ev is a proper linear subspace of $\mathbb{F}^k$ because $\mathbb{F}[X]_{<Q}$ has dimension Q. Lemma 3.56 then identifies which differences of two secrets are visible. This is a dimension condition on a particular linear masking family, not a universal lower bound for every ZK proof system.

3.7 Online evaluation privacy and the exact KBFold proof obligations

Definition 3.63 (Ideal evaluation functionality)**.** For a fixed witness table $f : I_m \to \mathbb{F}$, define the deterministic ideal functionality

$$\mathcal{F}^f_{\mathrm{eval}} : \mathbb{F}^m \to \mathbb{F}, \qquad \mathcal{F}^f_{\mathrm{eval}}(z) := \widetilde{f}(z).$$

Each authorized request at a point z returns exactly the public value $\widetilde{f}(z)$ and reveals no other information by definition. A multi-opening ideal functionality must also specify a maximum number of authorized requests: sufficiently many evaluations determine a multilinear polynomial.

Definition 3.64 (Online real evaluation game)**.** Fix a protocol with a preprocessing commitment phase and a subsequent evaluation-opening phase. In $\mathsf{OnlineReal}_{\Pi,V^*}(f)$:

(i) sample the honest prover's private preprocessing tape; on input f, form the real initial commitment and internal state;

(ii) send the public commitment to V^* *before* the evaluation point is chosen;

(iii) let V^* choose an evaluation point z as a function of its view so far;

(iv) compute the authorized public value $v = \widetilde{f}(z)$ and run the honest evaluation-opening procedure;

(v) permit V^* to make at most Q adaptive authenticated oracle reads, in the order allowed by the protocol;

(vi) output the full verifier view, including the root, z, v, explicit messages, query answers and authentication paths.

The experiment may be iterated for multiple authorized openings if that extension is explicitly included in the parameters.

Definition 3.65 (Online ideal evaluation game)**.** In $\mathsf{OnlineIdeal}^{\mathcal{F}^f_{\mathrm{eval}}}_{\mathsf{Sim},V^*}$, the simulator Sim is not given f. It must first send a simulated public commitment before receiving z. After V^* chooses z, the simulator may query $\mathcal{F}^f_{\mathrm{eval}}(z)$ for the authorized value v, and must then generate the remaining explicit messages and adaptive authenticated oracle replies. Its output is the full verifier view.

Definition 3.66 (Online zero knowledge)**.** For $Q \geq 0$ and $\varepsilon \geq 0$, the protocol is (Q, ε)-*online statistical zero knowledge* if for every Q-query-bounded adversarial verifier V^* there is an *online* simulator Sim_{V^*}, independent of f, such that for every table f,

$$\Delta\left(\mathsf{OnlineReal}_{\Pi,V^*}(f), \mathsf{OnlineIdeal}^{\mathcal{F}^f_{\mathrm{eval}}}_{\mathsf{Sim}_{V^*},V^*}\right) \leq \varepsilon.$$

The computational variant replaces statistical distance with bounded distinguishing advantage, as in Definition 3.47.

Lemma 3.67 (Stepwise online simulation suffices)**.** *Fix a verifier strategy V^* and witness f. Suppose the real and simulated online processes can be represented by finite state spaces $A_0, \ldots, A_t$ (state includes all available query history) and Markov transition kernels K_i, L_i satisfying the hypotheses of Lemma 3.10 with errors $\varepsilon_0, \ldots, \varepsilon_t$. Then the total variation distance of the two final verifier views is at most $\sum_{i=0}^{t} \varepsilon_i$.*

Proof. Apply Lemma 3.10 to the distributions on complete internal states after the last step. The verifier view is a deterministic projection of that internal state, so Corollary 3.8 shows that the distance between projected view distributions is no larger. This gives the stated bound. □

Definition 3.68 (Unmasked KBFold transcript components)**.** Specialize to an odd-characteristic KBFold domain. Let $0 \le s \le m$ be the number of fold rounds, let $N_j := N/2^j$ for $0 \le j \le s$, and let $\mathcal{D}_j := \{x^{2^j} : x \in \mathcal{D}\}$. Let $f_0 = f$, $f_j = \mathsf{Res}_{r_{j-1}}(f_{j-1})$ for $1 \le j \le s$, and let $\chi_0(b) = \mathrm{eq}_m(b, z)$ and $\chi_j = \mathsf{Res}_{r_{j-1}}(\chi_{j-1})$. For $1 \le j \le s$, the honest round polynomial is

$$s_j(T) := \sum_{b \in I_{m-j}} \left((1-T)f_{j-1}(0,b) + T f_{j-1}(1,b)\right)\left((1-T)\chi_{j-1}(0,b) + T\chi_{j-1}(1,b)\right). \tag{3.7}$$

The honest word at fold level j is

$$w_j := \mathrm{ev}_{\mathcal{D}_j}(U_{f_j}^{(m-j)}),$$

and the honest final table is $g := f_s : I_{m-s} \to \mathbb{F}$. The actual view also contains the public commitment to w_0, committed roots for the selected subsequent words, the transmitted polynomials s_j, the final table g, and all authentication openings made in the proof query phase. Each element of this tuple must be included in a full ZK simulator.

Lemma 3.69 (KBFold round-sum equality)**.** *For every $1 \le j \le s$, let $s_j(T)$ be as in (3.7). Then*

$$s_j(0) + s_j(1) = \sum_{a \in I_{m-j+1}} f_{j-1}(a)\chi_{j-1}(a).$$

Proof. At $T = 0$ each first factor in (3.7) is $f_{j-1}(0, b)$ and each second factor is $\chi_{j-1}(0, b)$. At $T = 1$ the corresponding factors are $f_{j-1}(1, b)$ and $\chi_{j-1}(1, b)$. Adding gives the sum over every pair $(0, b)$ and $(1, b)$, which enumerates I_{m-j+1} exactly once. □

Lemma 3.70 (KBFold round-polynomial degree)**.** *For every j and every fixed f_{j-1}, χ_{j-1}, the polynomial $s_j(T)$ in (3.7) has degree at most 2.*

Proof. Each displayed factor is affine in T and therefore has degree at most 1. Their product has degree at most 2. Summing the finitely many products cannot increase the degree beyond 2. □

Theorem 3.71 (A naive high-degree mask does not preserve final closure)**.** *Assume $\mathrm{char}(\mathbb{F}) \neq 2$ and $m \ge 1$. Set $N = 2^m$, let $f : I_m \to \mathbb{F}$ be arbitrary, and use the nonzero high-degree mask $R(X) = 1$ in*

$$Q_0(X) := U_f^{(m)}(X) + X^N.$$

Choose every fold challenge to be $r_i = 0$ for $0 \le i < m$, and define $Q_{i+1} := \mathsf{pfold}_0(Q_i)$. Then

$$\boxed{Q_m(X) = \widetilde{f}(0, \ldots, 0) + (-1)^m X.}$$

In particular Q_m is not a constant polynomial, whereas the original fully folded KBFold closure is constant. Thus the unmodified final closure equation cannot hold for this masking choice.

Proof. By Corollary 2.28, after m rounds

$$Q_m(X) = U_{f_m}^{(0)}(X) + X R_m(X),$$

where $R_0 = 1$ and $R_{i+1} = \mathsf{pfold}_0(R_i)$. For every constant polynomial c, its even–odd decomposition is $E = c$ and $O = 0$, so Definition 2.25 gives $\mathsf{pfold}_0(c) = (2 \cdot 0 - 1)c + (1 - 0)0 = -c$. Induction on i yields $R_i = (-1)^i$, and in particular $R_m = (-1)^m$. Corollary 1.15 gives $U_{f_m}^{(0)} = \widetilde{f}(0, \ldots, 0)$. Substitution proves the displayed formula. Since $(-1)^m$ is a nonzero unit in every field, the coefficient of X is nonzero; hence Q_m has degree 1, not 0. □

Remark 3.72 (Meaning of the closure obstruction). Theorem 3.71 does not rule out the high-degree masking proposal of the original KBFold paper. It proves that *adding the mask while leaving the final consistency check unchanged* is not a correct protocol transformation. A completed construction must either unmask the residual term in a zero-knowledge manner or replace the final check by a degree-aware relation that includes the residual R_s.

Definition 3.73 (ZK-KBFold certification obligations)**.** For a claimed ZK extension of the actual KBFold protocol, the following statements will be proved as separate theorems rather than combined as an assumption:

(i) **Fold invariant:** the masked word and polynomial folds agree over the enlarged degree space for every round.

(ii) **Sumcheck invariant:** the masked round polynomials preserve the authorized evaluation claim and each verifier endpoint check.

(iii) **Closure invariant:** the verifier's last-round test correctly handles the residual polynomial mask (Theorem 3.71).

(iv) **Soundness:** the revised code rate and enlarged degree bound preserve an explicitly quantified evaluation/proximity soundness statement.

(v) **Adaptive view simulation:** the *joint* root/round-polynomial/oracle-opening/final-table distribution is simulated with bounded query access, using conditional laws as in Lemma 3.57.

(vi) **Compiler interface:** any proposed noninteractive or QROM variant has its own, separately stated simulator and proof of security; the original KBFold post-quantum soundness theorem is not a zero-knowledge theorem.

Remark 3.74 (Formalization roadmap). Chapter 3 has defined the typed finite distributions, security games, and elementary oracle/masking lemmas. In Lean, these can be formalized using `Fintype`, `PMF` or `Measure` on finite types, `Polynomial`, `Submodule`, and finite linear maps. A complete computer-checked proof still requires the explicit implementation and semantics of the prover, verifier, simulator, and authenticated opening interface. No cryptographic zero-knowledge theorem for KBFold is asserted by the definitions of this chapter.

4 The Boolean-Kernel Basis: Complete Algebraic Theory

We develop the Boolean-kernel representation with all indices, maps, fields, and degree conventions explicit. Each result is split into elementary steps suitable for future Lean 4 certification. The original KBFold paper proves the kernel basis, Kronecker transform, low-degree criterion, and fold dictionary [Mkh26]; we reconstruct those results with complete proofs, then derive algebraic consequences directly relevant to degree-aware masking and bounded-query simulation. Throughout this chapter, $\mathbb{F}$ is an arbitrary field unless a lemma explicitly assumes a finite field or specifies the characteristic.

4.1 Bit indexing, kernels, and coefficients

The statements in this section do not require a finite field or an assumption on the characteristic. They rederive the structural facts of the original KBFold basis [Mkh26] with complete proofs and explicit indexing conventions.

Definition 4.1 (Bit sets and binary exponents)**.** Let $\mathbb{F}$ be an arbitrary field and let $m \geq 0$ be an integer. Write $J_m := \{0, \ldots, m-1\}$, $I_m := \{0,1\}^m$, and $N := 2^m$. For $S \subseteq J_m$, define

$$e(S) := \sum_{i \in S} 2^i \in \{0, \ldots, N-1\}.$$

For $b \in I_m$, set $\mathrm{supp}(b) := \{i : b_i = 1\}$, $Z(b) := J_m \setminus \mathrm{supp}(b)$, and $\mathrm{wt}(b) := |\mathrm{supp}(b)|$. The empty set J_0 and empty tuple $I_0 = \{()\}$ have their usual meanings.

Lemma 4.2 (Binary-subset bijection)**.** *The function $e : \mathcal{P}(J_m) \to \{0, \ldots, N-1\}$ is a bijection. Its inverse sends the binary expansion $a = \sum_{i=0}^{m-1} a_i 2^i$ to $\{i : a_i = 1\}$.*

Proof. For every S, $0 \leq e(S) \leq \sum_{i=0}^{m-1} 2^i = 2^m - 1$. If $e(S) = e(T)$ and $S \neq T$, choose the largest j in the symmetric difference. The contribution of 2^j to $e(S) - e(T)$ has absolute value 2^j, whereas the sum of all contributions from smaller indices has absolute value at most $\sum_{i=0}^{j-1} 2^i = 2^j - 1$. Equality is impossible. Thus e is injective. Domain and codomain both have cardinality 2^m, so it is bijective. The inverse formula follows directly from the definition of e. □

Definition 4.3 (General kernel and Boolean kernel)**.** For $y = (y_i)_{i \in J_m} \in \mathbb{F}^m$ define

$$K_y^{(m)}(X) := \prod_{i=0}^{m-1} (X^{2^i} + y_i) \in \mathbb{F}[X].$$

For $b \in I_m \subseteq \mathbb{F}^m$ write $K_b^{(m)} := K_y^{(m)}|_{y=b}$. We put $K_{()}^{(0)} = 1$. For a table $f : I_m \to \mathbb{F}$, set

$$U_f^{(m)}(X) := \sum_{b \in I_m} f(b) K_b^{(m)}(X).$$

Lemma 4.4 (Explicit product expansion)**.** *For every $y \in \mathbb{F}^m$,*

$$K_y^{(m)}(X) = \sum_{S \subseteq J_m} \left(\prod_{i \in J_m \setminus S} y_i \right) X^{e(S)}. \tag{4.1}$$

Proof. Apply distributivity to the product defining $K_y^{(m)}$. For each factor indexed by i, choose either the monomial X^{2^i} or the constant y_i. Let S be the set of indices where the monomial is chosen. The resulting term is exactly $\left(\prod_{i \notin S} y_i\right) X^{\sum_{i \in S} 2^i}$. Every choice corresponds to precisely one subset S, so summing the terms gives (4.1). □

Lemma 4.5 (Coefficient formula)**.** *For $S \subseteq J_m$ and $y \in \mathbb{F}^m$,*

$$[X^{e(S)}] K_y^{(m)} = \prod_{i \notin S} y_i.$$

For Boolean $b \in I_m$, this coefficient equals 1 precisely when $Z(b) \subseteq S$ and equals 0 otherwise.

Proof. Lemma 4.2 shows that the exponents $e(T)$ in (4.1) are pairwise distinct. Thus the only term of exponent $e(S)$ is the one indexed by S, proving the first formula. If $y = b$ is Boolean, the product equals 1 if all factors indexed by $J_m \setminus S$ are 1, and is zero if any of those factors is 0. The condition is $(J_m \setminus S) \cap Z(b) = \varnothing$, which is equivalent to $Z(b) \subseteq S$. □

Corollary 4.6 (Degree, leading term, and constant term)**.** *For each $b \in I_m$, $K_b^{(m)}$ is monic of degree $N-1$ and*

$$[X^0]K_b^{(m)} = \prod_{i \in J_m} b_i, \qquad [X^{N-1}]K_b^{(m)} = 1.$$

The same degree and leading coefficient assertion holds for $K_y^{(m)}$ with arbitrary $y \in \mathbb{F}^m$.

Proof. The greatest exponent occurring in (4.1) is $e(J_m) = N-1$, and its coefficient is the empty product 1. The exponent $e(\varnothing) = 0$ has coefficient $\prod_i y_i$. These statements hold also for $m = 0$, where $K_{()}^{(0)} = 1$. □

Lemma 4.7 (Block factorization)**.** *For integers $0 \le t \le m$, $c \in I_t$, and $h \in I_{m-t}$, write (c,h) for the concatenation of the t low-index bits of c with the $m-t$ high-index bits of h. Then*

$$K_{(c,h)}^{(m)}(X) = K_c^{(t)}(X)\, K_h^{(m-t)}(X^{2^t}). \tag{4.2}$$

Proof. Split the defining product into factors with indices $i < t$ and $i \ge t$. The low-index product is $K_c^{(t)}(X)$. In the remaining product put $i = t+j$, so that $X^{2^i} = (X^{2^t})^{2^j}$. The high-index product is therefore $K_h^{(m-t)}(X^{2^t})$. Multiply the two expressions. □

Corollary 4.8 (Lowest-bit kernel recursion)**.** *For $m \ge 1$ and $b' \in I_{m-1}$,*

$$K_{(0,b')}^{(m)}(X) = XK_{b'}^{(m-1)}(X^2), \qquad K_{(1,b')}^{(m)}(X) = (X+1)K_{b'}^{(m-1)}(X^2).$$

Proof. Take $t = 1$ in Lemma 4.7 and note that $K_0^{(1)}(X) = X$ and $K_1^{(1)}(X) = X+1$. □

4.2 Kronecker transform and the basis theorem

Definition 4.9 (Coefficient vectors and kernel coordinates)**.** For $P \in \mathbb{F}[X]_{<N}$ define its monomial coefficient vector $a \in \mathbb{F}^{I_m}$ by $a(b) := [X^{e(\mathrm{supp}(b))}]P$. For any table $f \in \mathbb{F}^{I_m}$, its encoded polynomial is $U_f^{(m)}$. Define the $N \times N$ matrix $\mathbf{K}_m$, with rows indexed by $a \in I_m$ and columns by $b \in I_m$, by

$$\mathbf{K}_m[a,b] := [X^{e(\mathrm{supp}(a))}]K_b^{(m)}.$$

Definition 4.10 (Bitwise tensor product of matrices)**.** For matrices $A_i \in \mathrm{Mat}_{2\times 2}(\mathbb{F})$ indexed by $i \in J_m$, define $\bigotimes_{i=0}^{m-1} A_i$ to be the $I_m \times I_m$ matrix with entries

$$\left(\bigotimes_{i=0}^{m-1} A_i\right)[a,b] := \prod_{i=0}^{m-1} A_i[a_i, b_i].$$

For $m = 0$ this is the 1×1 identity matrix.

Lemma 4.11 (Tensor multiplication rule)**.** *For each $i \in J_m$ let A_i, B_i be 2×2 matrices over $\mathbb{F}$. Then*

$$\left(\bigotimes_i A_i\right)\left(\bigotimes_i B_i\right) = \bigotimes_i (A_i B_i).$$

Proof. Fix $a, b \in I_m$. The $[a, b]$ entry of the matrix product on the left is

$$\sum_{c \in I_m} \prod_{i \in J_m} A_i[a_i, c_i] B_i[c_i, b_i].$$

The finite sum ranges over all independent binary choices $c_i \in \{0, 1\}$. Distributivity therefore rewrites it as

$$\prod_{i \in J_m} \left(\sum_{u \in \{0,1\}} A_i[a_i, u] B_i[u, b_i] \right) = \prod_{i \in J_m} (A_i B_i)[a_i, b_i],$$

which is the $[a, b]$ entry of the right-hand side. Equality of all entries proves the identity. For $m = 0$, both sides are the 1×1 identity. □

Lemma 4.12 (One-bit coefficient transform)**.** *Put*

$$C = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad D = \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then $CD = DC = I_2$. The vector of monomial coefficients of $f_0 K_0^{(1)} + f_1 K_1^{(1)}$ is $C(f_0, f_1)^{\mathsf{T}} = (f_1, f_0 + f_1)^{\mathsf{T}}$, and its inverse is $(f_0, f_1) = (a_1 - a_0, a_0)$.

Proof. Since $K_0^{(1)} = X$ and $K_1^{(1)} = X + 1$, we have $f_0 K_0^{(1)} + f_1 K_1^{(1)} = f_1 + (f_0 + f_1)X$, giving the formula for C. Direct matrix multiplication shows $CD = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right) = DC$ over any field, since $-1 + 1 = 0$. Therefore $D(a_0, a_1)^{\mathsf{T}} = (a_1 - a_0, a_0)^{\mathsf{T}}$ is the inverse coefficient map. □

Proposition 4.13 (Kronecker coefficient matrix)**.** *With C as in Lemma 4.12,*

$$\boxed{\mathbf{K}_m = C^{\otimes m}.}$$

Proof. The coefficient formula in Lemma 4.5 gives

$$\mathbf{K}_m[a, b] = \prod_{i : a_i = 0} b_i.$$

But $C[0, b_i] = b_i$ and $C[1, b_i] = 1$, so the same quantity equals $\prod_{i \in J_m} C[a_i, b_i]$, which is the defining entry of $C^{\otimes m}$. This proves equality entry by entry, including $m = 0$. □

Theorem 4.14 (Boolean-kernel basis and inverse transform)**.** *For every field $\mathbb{F}$ and $m \geq 0$, the family $\{K_b^{(m)} : b \in I_m\}$ is an $\mathbb{F}$-basis of $\mathbb{F}[X]_{<N}$. Its coefficient matrix and inverse are*

$$\mathbf{K}_m = C^{\otimes m}, \qquad \mathbf{K}_m^{-1} = D^{\otimes m}.$$

Proof. By Lemma 4.12, $CD = DC = I_2$. Applying the mixed-product rule in Lemma 4.11 to m identical factors gives

$$C^{\otimes m} D^{\otimes m} = (CD)^{\otimes m} = I_{I_m}, \quad D^{\otimes m} C^{\otimes m} = I_{I_m}.$$

Thus the coefficient matrix is invertible. Its N columns are the monomial coefficient vectors of the N kernel polynomials. Invertibility implies linear independence of these column vectors. The monomials $1, X, \ldots, X^{N-1}$ form an N-element basis of $\mathbb{F}[X]_{<N}$, so its dimension is N. Therefore an independent set of N kernel polynomials is a basis. No restriction on the characteristic was used. □

Corollary 4.15 (Unique kernel coordinates)**.** *For each $P \in \mathbb{F}[X]_{<N}$ there exists a unique $f : I_m \to \mathbb{F}$ with $P = U_f^{(m)}$, and if a is its monomial coefficient vector, then*

$$a = \mathbf{K}_m f, \qquad f = \mathbf{K}_m^{-1} a.$$

Proof. The existence and uniqueness are the defining properties of a basis in Theorem 4.14. Passing to monomial coefficient vectors gives $a = \mathbf{K}_m f$. Applying its explicit inverse gives the second equality. □

4.3 Möbius inversion, constants, and fast transforms

Lemma 4.16 (Forward transform in subset form)**.** *Let $f : I_m \to \mathbb{F}$, set $P := U_f^{(m)}$, and let $a_S := [X^{e(S)}]P$ for $S \subseteq J_m$. Then*

$$\boxed{a_S = \sum_{\substack{b \in I_m \\ Z(b) \subseteq S}} f(b).} \tag{4.3}$$

Proof. By Definition 4.3, $P = \sum_b f(b) K_b^{(m)}$. Coefficient extraction is $\mathbb{F}$-linear. Thus

$$a_S = \sum_b f(b) [X^{e(S)}] K_b^{(m)}.$$

Lemma 4.5 says the bracket is 1 precisely when $Z(b) \subseteq S$ and 0 otherwise. Removing the zero summands gives (4.3). □

Lemma 4.17 (Monomial expansion in the kernel basis)**.** *For every $S \subseteq J_m$,*

$$\boxed{X^{e(S)} = \sum_{\substack{b \in I_m \\ S \subseteq Z(b)}} (-1)^{|Z(b) \setminus S|} K_b^{(m)}(X).} \tag{4.4}$$

Proof. For each $i \in S$ use $X^{2^i} = X^{2^i} + 0$. For each $i \notin S$, use the identity

$$1 = (X^{2^i} + 1) - (X^{2^i} + 0).$$

Multiplying these identities gives

$$X^{e(S)} = \prod_{i \in S} (X^{2^i} + 0) \prod_{i \notin S} \left[(X^{2^i} + 1) - (X^{2^i} + 0) \right].$$

Expand the second product by distributivity. Each term is indexed by a Boolean vector b with $b_i = 0$ for all $i \in S$; equivalently $S \subseteq Z(b)$. Its factors, before their signs, multiply to $K_b^{(m)}$. A negative sign is incurred for every index $i \notin S$ where the term $-(X^{2^i} + 0)$ was chosen, i.e. precisely for the indices in $Z(b) \setminus S$. Its coefficient is therefore $(-1)^{|Z(b) \setminus S|}$. Summing these terms proves the equality. This calculation works in every characteristic; in characteristic two all signs reduce to $+1$. □

Proposition 4.18 (Inverse Möbius formula)**.** *Let $P(X) = \sum_{S \subseteq J_m} a_S X^{e(S)}$ be arbitrary of degree $< N$. Its unique kernel coordinates f satisfy*

$$\boxed{f(b) = \sum_{S \subseteq Z(b)} (-1)^{|Z(b) \setminus S|} a_S.} \tag{4.5}$$

Proof. Substitute the monomial expansion (4.4) into $P = \sum_S a_S X^{e(S)}$ and interchange the two finite sums:

$$P = \sum_{b \in I_m} \left(\sum_{S \subseteq Z(b)} (-1)^{|Z(b) \setminus S|} a_S \right) K_b^{(m)}.$$

By uniqueness of kernel coordinates (Corollary 4.15), the coefficient in parentheses equals $f(b)$. □

Lemma 4.19 (One-coordinate transform algorithms)**.** *For a fixed coordinate i, apply to every pair of table entries with equal other bits the update*

$$(x_0, x_1) \longmapsto (x_1, x_0 + x_1).$$

Composing these updates over all $i \in J_m$ sends a kernel-coordinate table f to the monomial coefficient table a of $U_f^{(m)}$. Reversing each update as

$$(y_0, y_1) \longmapsto (y_1 - y_0, y_0)$$

recovers f from a.

Proof. By Lemma 4.12, the forward pair update is multiplication by C and the reverse is multiplication by $D = C^{-1}$. When performed on coordinate i while fixing the other bits, the update is multiplication by the tensor matrix that has C at coordinate i and I_2 elsewhere. The mixed-product rule of Lemma 4.11 shows that multiplying these matrices once for each coordinate yields $C^{\otimes m} = \mathbf{K}_m$. The reverse operations similarly yield $D^{\otimes m} = \mathbf{K}_m^{-1}$, proving both claims. □

Corollary 4.20 (Exact transform cost)**.** *For $m \geq 1$, either transform in Lemma 4.19 takes exactly $m2^{m-1} = mN/2$ additions or subtractions in $\mathbb{F}$ and no nontrivial field multiplications. For $m = 0$, both transforms are the identity.*

Proof. For each coordinate i, there are exactly $2^{m-1} = N/2$ disjoint pairs, since the other $m - 1$ bits can be chosen freely. Each forward pair operation copies one entry and computes one sum; each reverse pair copies one entry and computes one difference. Thus each coordinate costs $N/2$ additions or subtractions, and there are m coordinates. Multiplication by 0, 1, or -1 is not counted as a nontrivial field multiplication. When $m = 0$ there are no coordinate updates. □

Corollary 4.21 (Highest and constant coefficients)**.** *For every table $f : I_m \to \mathbb{F}$,*

$$[X^{N-1}]U_f^{(m)} = \sum_{b \in I_m} f(b), \qquad [X^0]U_f^{(m)} = f(1, \ldots, 1).$$

Proof. In Lemma 4.16, take $S = J_m$; then every $Z(b)$ is a subset of S, so all $f(b)$ appear. For the constant term take $S = \varnothing$; the condition $Z(b) \subseteq \varnothing$ forces $Z(b) = \varnothing$, hence $b = (1, \ldots, 1)$, leaving only this term. □

Proposition 4.22 (Kernel coordinates of a constant)**.** *For every $c \in \mathbb{F}$,*

$$\boxed{c = \sum_{b \in I_m} c(-1)^{m - \mathrm{wt}(b)} K_b^{(m)}(X).} \tag{4.6}$$

Proof. Apply Lemma 4.17 with $S = \varnothing$ to $X^{e(\varnothing)} = X^0 = 1$. Its sum runs over all b, and its sign is $(-1)^{|Z(b)|} = (-1)^{m - \mathrm{wt}(b)}$. Multiply the resulting polynomial identity by c. □

Proposition 4.23 (Kernel partition product)**.** *For every $m \geq 0$ and every field $\mathbb{F}$,*

$$\boxed{\sum_{b \in I_m} K_b^{(m)}(X) = \prod_{i=0}^{m-1} (2X^{2^i} + 1).} \tag{4.7}$$

If $\operatorname{char} \mathbb{F} = 2$*, the left side equals* 1*; if* $\operatorname{char} \mathbb{F} \neq 2$ *and* $m \geq 1$*, the right side is a nonconstant polynomial of degree* $N - 1$*.*

Proof. Since b_i ranges independently over 0 and 1, repeated distributivity factors the sum as

$$\sum_{(b_i)_i} \prod_i (X^{2^i} + b_i) = \prod_i ((X^{2^i} + 0) + (X^{2^i} + 1)) = \prod_i (2X^{2^i} + 1).$$

In characteristic two, every factor is 1. Otherwise every factor has leading term $2X^{2^i}$ with nonzero coefficient, so their product has degree $\sum_i 2^i = N - 1 > 0$. For $m = 0$ both sides are 1. □

4.4 Degree constraints in kernel coordinates

Definition 4.24 (Coefficient blocks)**.** Fix integers $0 \leq t \leq m$ and set $B := 2^t$, $H := 2^{m-t}$, so $N = BH$. For any $P \in \mathbb{F}[X]_{<N}$, its B-coefficient blocks are the polynomials $P_j \in \mathbb{F}[X]_{<B}$, $0 \leq j < H$, uniquely determined by

$$P(X) = \sum_{j=0}^{H-1} X^{Bj} P_j(X).$$

Lemma 4.25 (Existence and uniqueness of coefficient blocks)**.** *The decomposition in Definition 4.24 exists and is unique for every $P \in \mathbb{F}[X]_{<N}$.*

Proof. Write $P(X) = \sum_{a=0}^{N-1} p_a X^a$. Every a has a unique quotient–remainder decomposition $a = Bj + i$ with $0 \leq i < B$ and $0 \leq j < H$. Define $P_j(X) := \sum_{i=0}^{B-1} p_{Bj+i} X^i$. Then $\sum_j X^{Bj} P_j(X) = \sum_{j,i} p_{Bj+i} X^{Bj+i} = P(X)$. Conversely, comparing coefficients of the pairwise distinct powers X^{Bj+i} determines all coefficients of every P_j, proving uniqueness. □

Lemma 4.26 (Block degree criterion)**.** *With the preceding notation,* $\deg P < B$ *(including $P = 0$) if and only if $P_j = 0$ for all $1 \leq j < H$.*

Proof. The terms of $X^{Bj} P_j$ have degrees in the interval $[Bj, B(j+1) - 1]$. These intervals are pairwise disjoint. If $P_j \neq 0$ for some $j \geq 1$, the polynomial P has a nonzero coefficient of degree at least B, so $\deg P \not< B$. Conversely, if all blocks with $j \geq 1$ vanish, then $P = P_0$ with $\deg P_0 < B$ by definition. □

Definition 4.27 (High-bit kernel polynomials)**.** For a table $f : I_m \to \mathbb{F}$ and each $c \in I_t$, define

$$Y_c(Z) := \sum_{h \in I_{m-t}} f(c, h) K_h^{(m-t)}(Z) \in \mathbb{F}[Z]_{<H}.$$

If $Y_c(Z) = \sum_{j=0}^{H-1} y_{c,j} Z^j$, define

$$A_j(X) := \sum_{c \in I_t} y_{c,j} K_c^{(t)}(X) \in \mathbb{F}[X]_{<B}.$$

Lemma 4.28 (Kernel blocks equal monomial blocks)**.** *For the polynomials of Definition 4.27,*

$$U_f^{(m)}(X) = \sum_{c \in I_t} K_c^{(t)}(X) Y_c(X^B) = \sum_{j=0}^{H-1} X^{Bj} A_j(X).$$

Proof. Split the finite sum defining U_f according to the unique low/high decomposition $b = (c, h)$. Lemma 4.7 gives

$$U_f = \sum_c \sum_h f(c,h) K_c^{(t)}(X) K_h^{(m-t)}(X^B) = \sum_c K_c^{(t)}(X) Y_c(X^B).$$

Substitute $Y_c(X^B) = \sum_j y_{c,j} X^{Bj}$, interchange the two finite sums, and use the definition of A_j. By Corollary 4.6, each $K_c^{(t)}$ has degree $B-1$, so $A_j \in \mathbb{F}[X]_{<B}$ and the displayed sum is precisely the unique block decomposition. □

Lemma 4.29 (Low degree iff each high kernel polynomial is constant)**.** *With the preceding notation,*

$$\deg U_f^{(m)} < B \quad \Longleftrightarrow \quad \text{every } Y_c(Z) \text{ is constant.}$$

Proof. By Lemma 4.26 and Lemma 4.28, $\deg U_f < B$ is equivalent to $A_j = 0$ for every $j \geq 1$. By the basis theorem (Theorem 4.14) with t variables, each $A_j = \sum_c y_{c,j} K_c^{(t)}$ vanishes iff every $y_{c,j} = 0$. Thus all A_j with $j \geq 1$ vanish iff all coefficients of all Y_c of positive degree vanish, exactly when every Y_c is constant. □

Theorem 4.30 (Complete low-degree characterization)**.** *Let $0 \leq t \leq m$ and let $f : I_m \to \mathbb{F}$. Put*

$$U_f^{(m)} = \sum_{b \in I_m} f(b) K_b^{(m)}.$$

Then

$$\boxed{\deg U_f^{(m)} < 2^t} \tag{4.8}$$

if and only if there exists a unique table $\varphi : I_t \to \mathbb{F}$ satisfying

$$\boxed{f(c,h) = (-1)^{(m-t)-\mathrm{wt}(h)} \varphi(c) \qquad (c \in I_t,\ h \in I_{m-t}).} \tag{4.9}$$

In this case,

$$U_f^{(m)} = U_\varphi^{(t)}, \qquad \varphi(c) = f(c, 1, \ldots, 1).$$

Proof. Put $B = 2^t$, $H = 2^{m-t}$ and use Definition 4.27. By Lemma 4.29, $\deg U_f < B$ holds if and only if for every c, $Y_c(Z) = \varphi(c)$ for some scalar $\varphi(c)$. Proposition 4.22, applied with $m - t$ variables, gives the kernel expansion

$$\varphi(c) = \sum_{h \in I_{m-t}} (-1)^{(m-t)-\mathrm{wt}(h)} \varphi(c) K_h^{(m-t)}(Z).$$

The definition $Y_c = \sum_h f(c,h) K_h^{(m-t)}$ and uniqueness of kernel coordinates (Corollary 4.15) imply that $Y_c = \varphi(c)$ iff the coefficients satisfy (4.9) for every h. This proves the equivalence. The bit vector $h = (1, \ldots, 1)$ has weight $m-t$, so its sign in (4.9) is $+1$. Hence $\varphi(c) = f(c, 1, \ldots, 1)$ is forced and unique. Finally, substituting $Y_c = \varphi(c)$ into Lemma 4.28 gives $U_f^{(m)} = \sum_c K_c^{(t)} \varphi(c) = U_\varphi^{(t)}$. □

Corollary 4.31 (Dimension of a kernel low-degree subspace)**.** *The set of coefficient tables*

$$\mathcal{H}_t := \{f \in \mathbb{F}^{I_m} : \deg U_f^{(m)} < 2^t\}$$

is an $\mathbb{F}$-linear subspace of dimension 2^t.

Proof. By Theorem 4.30, $\mathcal{H}_t$ consists exactly of the tables obtained by applying the linear map $\varphi \mapsto ((c,h) \mapsto (-1)^{m-t-\mathrm{wt}(h)}\varphi(c))$ to $\mathbb{F}^{I_t}$. This map is injective because $\varphi(c)$ is recovered from $f(c,1,\dots,1)$. Its image is $\mathcal{H}_t$. Since $\mathbb{F}^{I_t}$ has dimension $|I_t| = 2^t$, the same holds for $\mathcal{H}_t$. □

Remark 4.32 (Characteristic-two simplification). When $\mathrm{char}\,\mathbb{F} = 2$, every sign in (4.9) equals $+1$, so low-degree kernel tables are constant along each high-bit fiber: $f(c,h) = \varphi(c)$. This simplification fails in general over the odd-characteristic fields used by the original KBFold protocol.

4.5 Evaluation on a curve in multilinear coordinates

Definition 4.33 (Multilinear equality weights)**.** For $b \in I_m$ and $z = (z_i)_{i\in J_m} \in \mathbb{F}^m$, define

$$\lambda_b(z) := \prod_{i=0}^{m-1} ((1-b_i)(1-z_i) + b_i z_i), \qquad \widetilde{f}(z) := \sum_{b\in I_m} f(b)\lambda_b(z).$$

For $m = 0$, $\lambda_{()}(()) = 1$ and $\widetilde{f}(()) = f(())$.

Lemma 4.34 (Kernel evaluation in factored form)**.** *For $b \in I_m$ and $\zeta \in \mathbb{F}$, write $t_i := \zeta^{2^i}$. Then*

$$K_b^{(m)}(\zeta) = \prod_{i=0}^{m-1} ((1-b_i)t_i + b_i(t_i+1)). \tag{4.10}$$

Proof. For $b_i = 0$ the i-th factor on the right is t_i, and for $b_i = 1$ it is $t_i + 1$. Both equal $t_i + b_i = \zeta^{2^i} + b_i$, the corresponding factor in the definition of $K_b^{(m)}(\zeta)$. Multiplication over i gives the equality. □

Definition 4.35 (Evaluation curve and scaling)**.** Fix $\zeta \in \mathbb{F}$. For every $i \in J_m$ put

$$t_i(\zeta) := \zeta^{2^i}, \qquad \gamma_i(\zeta) := 2t_i(\zeta) + 1, \qquad \Gamma_m(\zeta) := \prod_{i\in J_m} \gamma_i(\zeta).$$

When $\Gamma_m(\zeta) \neq 0$, define

$$\psi_i(\zeta) := \frac{t_i(\zeta)+1}{\gamma_i(\zeta)}, \qquad \psi(\zeta) := (\psi_0(\zeta),\dots,\psi_{m-1}(\zeta)).$$

Lemma 4.36 (One-factor scaled equality identity)**.** *Suppose $\gamma_i(\zeta) \neq 0$. Then for each $b_i \in \{0,1\}$,*

$$t_i(\zeta) + b_i = \gamma_i(\zeta)((1-b_i)(1-\psi_i(\zeta)) + b_i\psi_i(\zeta)).$$

Proof. For $b_i = 1$, the right side is $\gamma_i\psi_i = t_i + 1$ by definition of ψ_i. For $b_i = 0$, the right side is $\gamma_i(1-\psi_i) = \gamma_i - (t_i+1) = (2t_i+1) - (t_i+1) = t_i$. Both equal $t_i + b_i$. The two Boolean cases exhaust all possibilities. □

Theorem 4.37 (Scaled multilinear evaluation)**.** *For any field $\mathbb{F}$, any table $f : I_m \to \mathbb{F}$, and every $\zeta \in \mathbb{F}$ with $\Gamma_m(\zeta) \neq 0$,*

$$\boxed{U_f^{(m)}(\zeta) = \Gamma_m(\zeta)\,\widetilde{f}(\psi(\zeta)).} \tag{4.11}$$

Proof. Since a product in a field is nonzero precisely when all factors are nonzero, the assumption $\Gamma_m(\zeta) \neq 0$ implies $\gamma_i(\zeta) \neq 0$ for every i. Apply Lemma 4.36 to every factor in $K_b^{(m)}(\zeta) = \prod_i (t_i + b_i)$ to obtain

$$K_b^{(m)}(\zeta) = \left(\prod_i \gamma_i(\zeta)\right) \prod_i ((1-b_i)(1-\psi_i(\zeta)) + b_i\psi_i(\zeta)) = \Gamma_m(\zeta)\lambda_b(\psi(\zeta)).$$

Multiply this identity by $f(b)$ and sum over $b \in I_m$. The left side is $U_f^{(m)}(\zeta)$ by definition and the right side is $\Gamma_m(\zeta)\widetilde{f}(\psi(\zeta))$ by Definition 4.33. □

Proposition 4.38 (Exceptional-point bound)**.** *Assume $m \geq 1$ and $\operatorname{char} \mathbb{F} \neq 2$. Then $\Gamma_m \in \mathbb{F}[X]$ is nonzero of degree $N-1$. Consequently*

$$\left|\{\zeta \in \mathbb{F} : \Gamma_m(\zeta) = 0\}\right| \leq N - 1.$$

When $\mathbb{F}$ is finite and $|\mathbb{F}| > N-1$, at least $|\mathbb{F}| - (N-1)$ points admit the evaluation formula (4.11).

Proof. Each polynomial $2X^{2^i} + 1$ is nonzero with leading coefficient $2 \neq 0$ and degree 2^i. Their product Γ_m has nonzero leading coefficient 2^m and degree $\sum_i 2^i = N - 1$. A nonzero degree-d polynomial has at most d roots by Lemma 2.7. For a finite field, subtracting the number of exceptional roots from the field cardinality gives the final lower bound. □

Corollary 4.39 (Characteristic-two evaluation curve)**.** *If $\operatorname{char} \mathbb{F} = 2$, then $\Gamma_m(\zeta) = 1$ for every ζ, and*

$$\boxed{U_f^{(m)}(\zeta) = \widetilde{f}(\zeta + 1, \zeta^2 + 1, \ldots, \zeta^{2^{m-1}} + 1).} \tag{4.12}$$

No evaluation points are exceptional.

Proof. In characteristic two, $2t_i + 1 = 1$ for every i, so $\Gamma_m = 1$ and $\psi_i = (t_i + 1)/1 = \zeta^{2^i} + 1$. Substitute these identities into Theorem 4.37. The proof also covers $m = 0$ with the empty-product convention. □

Remark 4.40 (No division on exceptional points). When $\Gamma_m(\zeta) = 0$ in odd characteristic, the rational expression defining ψ is not available. The universal polynomial identity (4.10) remains valid at that point; one must use it instead of dividing by a vanishing γ_i. This distinction is essential for a formalized evaluation algorithm.

4.6 Folds on monomials and the general multilinearization functional

Definition 4.41 (Even–odd polynomial operators)**.** For every polynomial $P(X) = \sum_{j \geq 0} p_j X^j \in \mathbb{F}[X]$, define

$$E(P)(Y) := \sum_{j \geq 0} p_{2j} Y^j, \qquad O(P)(Y) := \sum_{j \geq 0} p_{2j+1} Y^j.$$

Thus $P(X) = E(P)(X^2) + XO(P)(X^2)$. Both operators are $\mathbb{F}$-linear. For $r \in \mathbb{F}$ put

$$\mathsf{pfold}_r(P) := (2r-1)E(P) + (1-r)O(P).$$

Lemma 4.42 (Fold of a single monomial)**.** *For every integer $j \geq 0$ and $r \in \mathbb{F}$,*

$$\mathsf{pfold}_r(X^{2j}) = (2r-1)Y^j, \qquad \mathsf{pfold}_r(X^{2j+1}) = (1-r)Y^j.$$

Proof. For $P(X) = X^{2j}$, its even component is $E(P) = Y^j$ and its odd component is 0 by the coefficient definition; substitute into Definition 4.41. For $P(X) = X^{2j+1}$, the even component is 0 and the odd component is Y^j. Substitute once more. □

Definition 4.43 (Complete normalized fold)**.** For $P \in \mathbb{F}[X]_{<N}$ and $r = (r_0, \ldots, r_{m-1}) \in \mathbb{F}^m$, define polynomials $P_0 := P$ and $P_{i+1} := \mathsf{pfold}_{r_i}(P_i)$ for $0 \le i < m$. The polynomial P_m is constant. Define $\mathsf{KFold}_r(P) \in \mathbb{F}$ to be its constant coefficient. For $m = 0$ this is the constant value of P.

Lemma 4.44 (Degree halving)**.** *If* $\deg P < 2d$*, then* $\deg \mathsf{pfold}_r(P) < d$*, including the zero polynomial. In particular* $\mathsf{KFold}_r(P)$ *is well defined.*

Proof. When all nonzero coefficients of P have indices strictly less than $2d$, both $E(P)$ and $O(P)$ have all coefficients at degrees strictly less than d by Definition 4.41. Their linear combination $\mathsf{pfold}_r(P)$ has no term of degree $\ge d$. Applying this with $d = 2^{m-i-1}$ at every step shows that P_m has degree < 1, i.e. is constant or zero. □

Lemma 4.45 (Iterated fold of one monomial)**.** *For* $S \subseteq J_m$ *and* $r \in \mathbb{F}^m$*,*

$$\boxed{\mathsf{KFold}_r(X^{e(S)}) = \left(\prod_{i \in J_m \setminus S} (2r_i - 1)\right)\left(\prod_{i \in S}(1 - r_i)\right).} \tag{4.13}$$

Proof. We induct on m. At $m = 0$, $S = \varnothing$, $e(S) = 0$ and the fold is $X^0 = 1$, while both products are empty and equal 1. Suppose $m \ge 1$. Write $e(S) = b_0 + 2j$, where $b_0 = 1$ if $0 \in S$ and 0 otherwise, and $j = \sum_{i \in S, i > 0} 2^{i-1}$. By Lemma 4.42, the first fold yields $(2r_0 - 1)Y^j$ if $b_0 = 0$, or $(1 - r_0)Y^j$ if $b_0 = 1$. All remaining folds are $\mathbb{F}$-linear, so this scalar factors out. Apply the induction hypothesis to Y^j and the remaining challenge coordinates $(r_1, \ldots, r_{m-1})$. Multiplying the scalar from the first fold by the inductively obtained factors gives exactly the two products in (4.13). □

Definition 4.46 (Binary-exponent multilinearization)**.** Let $\mathscr{V}_m := \mathbb{F}[Z_0, \ldots, Z_{m-1}]_{\mathrm{ml}}$ be the $\mathbb{F}$-linear span of squarefree monomials. Define the unique $\mathbb{F}$-linear map $\mathscr{M}_m : \mathbb{F}[X]_{<N} \to \mathscr{V}_m$ by

$$\mathscr{M}_m(X^{e(S)}) = \prod_{i \in S} Z_i.$$

For $P(X) = \sum_{S \subseteq J_m} a_S X^{e(S)}$ we consequently have $\mathscr{M}_m(P)(Z) = \sum_S a_S \prod_{i \in S} Z_i$. No multiplicativity of $\mathscr{M}_m$ is assumed.

Lemma 4.47 (Multilinearization is a vector-space isomorphism)**.** *The map* $\mathscr{M}_m$ *is an* $\mathbb{F}$*-linear isomorphism* $\mathbb{F}[X]_{<N} \to \mathscr{V}_m$*.*

Proof. By Lemma 4.2, the monomials $X^{e(S)}$ for $S \subseteq J_m$ form the usual basis $1, X, \ldots, X^{N-1}$. The squarefree monomials $\prod_{i \in S} Z_i$ form an N-element basis of $\mathscr{V}_m$ by its definition. The map $\mathscr{M}_m$ sends the first basis bijectively onto the second; a linear map doing so has a linear inverse determined by the reverse basis correspondence. □

Lemma 4.48 (Multilinearization of kernels, every characteristic)**.** *For every* $b \in I_m$ *and every field* $\mathbb{F}$*,*

$$\boxed{\mathscr{M}_m(K_b^{(m)})(Z) = \prod_{i \in J_m} (Z_i + b_i).} \tag{4.14}$$

Proof. By Lemma 4.4, $K_b^{(m)} = \sum_{S \subseteq J_m} (\prod_{i \notin S} b_i) X^{e(S)}$. Apply the $\mathbb{F}$-linearity of $\mathscr{M}_m$ and then its defining formula on each monomial:

$$\mathscr{M}_m(K_b^{(m)})(Z) = \sum_{S \subseteq J_m} \left(\prod_{i \notin S} b_i \right) \left(\prod_{i \in S} Z_i \right).$$

By distributivity, the right-hand side is precisely the expansion of $\prod_i (Z_i + b_i)$. This proof explicitly expands the product; it does not use, or assert, that $\mathscr{M}_m$ is a ring homomorphism. □

Definition 4.49 (Tensor evaluation functional)**.** For $r \in \mathbb{F}^m$ define an $\mathbb{F}$-linear map $\Lambda_r : \mathscr{V}_m \to \mathbb{F}$ on squarefree monomials by

$$\Lambda_r \left(\prod_{i \in S} Z_i \right) := \prod_{i \notin S} (2r_i - 1) \prod_{i \in S} (1 - r_i).$$

This is well defined because the squarefree monomials form a basis.

Theorem 4.50 (Complete fold as a tensor functional)**.** *For every field $\mathbb{F}$, every $m \geq 0$, $P \in \mathbb{F}[X]_{<N}$, and $r \in \mathbb{F}^m$,*

$$\boxed{\mathsf{KFold}_r(P) = \Lambda_r(\mathscr{M}_m(P)).} \tag{4.15}$$

Proof. Both sides define $\mathbb{F}$-linear functionals in P: the left side is a composition of $\mathbb{F}$-linear folds by Definition 4.41, and the right side is a composition of the $\mathbb{F}$-linear maps $\mathscr{M}_m$ and Λ_r. It suffices to check equality on the basis monomials $X^{e(S)}$. The left side on such a monomial is given explicitly in Lemma 4.45. The right side is $\Lambda_r(\prod_{i \in S} Z_i)$ by Definition 4.46, which equals the same product by Definition 4.49. Thus the functionals agree on a basis and therefore on every P. □

Corollary 4.51 (Ordinary evaluation when all weights are invertible)**.** *If $2r_i - 1 \neq 0$ for every i, then*

$$\mathsf{KFold}_r(P) = \left(\prod_i (2r_i - 1) \right) \mathscr{M}_m(P) \left(\frac{1 - r_0}{2r_0 - 1}, \ldots, \frac{1 - r_{m-1}}{2r_{m-1} - 1} \right). \tag{4.16}$$

Proof. Apply Theorem 4.50 and expand P in monomials. In the summand indexed by S, each factor with $i \in S$ is $(1 - r_i)$; since $2r_i - 1$ is invertible, this equals $(2r_i - 1) \cdot (1 - r_i)/(2r_i - 1)$. Factoring $\prod_i (2r_i - 1)$ from all summands leaves the evaluation of $\mathscr{M}_m(P)$ at the displayed point. □

Corollary 4.52 (Characteristic-two fold identity)**.** *If* $\operatorname{char} \mathbb{F} = 2$*, then $2r_i - 1 = 1$ and $1 - r_i = 1 + r_i$ for every i, and hence*

$$\boxed{\mathsf{KFold}_r(P) = \mathscr{M}_m(P)(1 + r_0, \ldots, 1 + r_{m-1}).} \tag{4.17}$$

Proof. In characteristic two the hypotheses of Corollary 4.51 hold automatically because $2r_i - 1 = -1 = 1$. Substituting $2r_i - 1 = 1$ and $1 - r_i = 1 + r_i$ into (4.16) yields the statement. □

Remark 4.53 (Critical characteristic distinction). The identity (4.17) is special to characteristic two. In odd characteristic the correct universal identity is the weighted tensor functional (4.15); one must not evaluate $\mathscr{M}_m(P)$ at $1 + r$ without the appropriate weights and reparametrization.

4.7 Classical FRI, field extension, and degree-aware masking

Definition 4.54 (Classical even–odd FRI fold)**.** For $P \in \mathbb{F}[X]$ and $\theta \in \mathbb{F}$, put

$$\mathsf{cfold}_\theta(P) := E(P) + \theta O(P),$$

where E and O are the operators of Definition 4.41.

Lemma 4.55 (Classical fold of the kernel coordinate pair)**.** *Let $m \geq 1$ and write a table $f : I_m \to \mathbb{F}$ as $f_0(b') := f(0, b')$ and $f_1(b') := f(1, b')$. Then*

$$\mathsf{cfold}_\theta(U_f^{(m)}) = U_{\theta f_0 + (1+\theta) f_1}^{(m-1)}.$$

Proof. By Corollary 4.8,

$$U_f^{(m)}(X) = X U_{f_0}^{(m-1)}(X^2) + (X+1) U_{f_1}^{(m-1)}(X^2).$$

Collecting the even and odd terms gives $E(U_f) = U_{f_1}^{(m-1)}$ and $O(U_f) = U_{f_0}^{(m-1)} + U_{f_1}^{(m-1)}$. Thus

$$\mathsf{cfold}_\theta(U_f) = U_{f_1} + \theta(U_{f_0} + U_{f_1}) = U_{\theta f_0 + (1+\theta) f_1}$$

by linearity of $f \mapsto U_f$. □

Lemma 4.56 (Classical versus normalized fold)**.** *Let $\theta \in \mathbb{F}$ satisfy $1 + 2\theta \neq 0$, and set*

$$r := \frac{1+\theta}{1+2\theta}.$$

Then for every polynomial $P \in \mathbb{F}[X]$,

$$\boxed{\mathsf{cfold}_\theta(P) = (1+2\theta)\mathsf{pfold}_r(P).} \tag{4.18}$$

Conversely, for $r \in \mathbb{F}$ with $2r - 1 \neq 0$, setting $\theta := (1-r)/(2r-1)$ gives

$$\mathsf{pfold}_r(P) = (2r-1)\mathsf{cfold}_\theta(P).$$

Proof. For the first identity,

$$2r - 1 = \frac{2 + 2\theta - (1+2\theta)}{1+2\theta} = \frac{1}{1+2\theta}, \qquad 1 - r = \frac{\theta}{1+2\theta}.$$

Therefore $(1+2\theta)((2r-1)E(P) + (1-r)O(P)) = E(P) + \theta O(P) = \mathsf{cfold}_\theta(P)$. For the converse, substituting $\theta = (1-r)/(2r-1)$ into $(2r-1)(E + \theta O)$ gives $(2r-1)E + (1-r)O = \mathsf{pfold}_r(P)$. Every division is legal by the explicitly stated nonvanishing hypotheses. □

Lemma 4.57 (One classical fold preserves evaluation on a square)**.** *For every $P \in \mathbb{F}[X]$, every $\zeta \in \mathbb{F}$, and $\theta = \zeta$,*

$$P(\zeta) = \mathsf{cfold}_\zeta(P)(\zeta^2).$$

Proof. By Definition 4.41, $P(\zeta) = E(P)(\zeta^2) + \zeta O(P)(\zeta^2)$. The right side is exactly $\mathsf{cfold}_\zeta(P)(\zeta^2)$ by Definition 4.54. □

Theorem 4.58 (Classical folding computes univariate evaluation)**.** *Fix $\zeta \in \mathbb{F}$ and $P \in \mathbb{F}[X]_{<N}$. Define $P_0 = P$ and $P_{i+1} := \mathsf{cfold}_{\zeta^{2^i}}(P_i)$ for $0 \le i < m$. Then P_m is constant and*

$$\boxed{P_m = P(\zeta).}$$

Proof. The classical fold maps polynomials of degree less than $2d$ to polynomials of degree less than d, because E and O have that degree bound by Definition 4.41. Therefore P_m is constant. Applying Lemma 4.57 with P_i and evaluation point ζ^{2^i} shows

$$P_i(\zeta^{2^i}) = P_{i+1}(\zeta^{2^{i+1}})$$

for every $0 \le i < m$. Chaining these m equalities yields $P(\zeta) = P_m(\zeta^{2^m}) = P_m$, since P_m is constant. □

Lemma 4.59 (Naturality under field extensions)**.** *Let $\iota : \mathbb{F} \hookrightarrow \mathbb{K}$ be an injective field homomorphism. Extend it coefficientwise to polynomials and pointwise to tables. Then*

$$\iota(K_b^{(m)}) = K_b^{(m)}, \qquad \iota(U_f^{(m)}) = U_{\iota \circ f}^{(m)},$$

and for every $P \in \mathbb{F}[X]$ and $r \in \mathbb{F}$,

$$\iota(\mathsf{pfold}_r(P)) = \mathsf{pfold}_{\iota(r)}(\iota(P)).$$

Consequently the kernel-coordinate transforms and complete normalized folds commute with ι.

Proof. Every coefficient of $K_b^{(m)}$ is an integer expression in 0 and 1, fixed by the homomorphism ι, proving the first equality. Apply ι to the finite sum defining U_f and use preservation of addition and multiplication to obtain the second. Even–odd coefficient extraction commutes with ι because ι acts coefficientwise: $E(\iota(P)) = \iota(E(P))$ and $O(\iota(P)) = \iota(O(P))$. Applying ι to $(2r-1)E(P)+(1-r)O(P)$ therefore gives the third equality. Repeating it round by round proves the corresponding complete fold identity. Uniqueness of kernel coordinates from Theorem 4.14 gives the transform compatibility. □

Lemma 4.60 (Degree of a high-degree masked kernel polynomial)**.** *Let $m \ge 1$, $N = 2^m$, $f : I_m \to \mathbb{F}$, and $R \in \mathbb{F}[X]$ be nonzero of degree $d \ge 0$. Then*

$$\boxed{\deg(U_f^{(m)} + X^N R) = N + d.} \tag{4.19}$$

Proof. Corollary 4.6 gives $\deg U_f^{(m)} \le N-1$, unless the polynomial vanishes. Because R is nonzero of degree d, the polynomial $X^N R$ has degree exactly $N+d$ with the same nonzero leading coefficient as R. There is no term of degree $N+d$ in U_f, so the top coefficient of $X^N R$ cannot cancel in the sum. The sum has degree $N+d$. □

Corollary 4.61 (The code dimension must change)**.** *Suppose $\mathcal{D} \subseteq \mathbb{F}$ has M distinct elements and $M \ge N+d+1$, with $R \ne 0$ of degree d. Then the evaluation word of $U_f + X^N R$ does not lie in $\mathrm{RS}_{\mathbb{F}}(\mathcal{D}, N)$. It belongs to $\mathrm{RS}_{\mathbb{F}}(\mathcal{D}, N+d+1)$.*

Proof. The second claim follows from Lemma 4.60: the polynomial has degree $N+d < N+d+1$. Suppose its evaluation word also lay in $\mathrm{RS}_{\mathbb{F}}(\mathcal{D}, N)$. Then some polynomial T of degree less than N would agree with $U_f + X^N R$ at every point of $\mathcal{D}$. Their difference is nonzero of degree $N+d$, because the leading term of $X^N R$ cannot cancel with terms of degree less than N. This difference has at least $M \ge N+d+1$ distinct roots, contradicting Lemma 2.7. Hence the word is outside the original degree-N code. □

4.8 Kernel-mask subspaces and bounded-query hiding

Throughout this subsection, $\mathbb{F}$ is a *finite* field.

Definition 4.62 (Random low-degree kernel mask)**.** Let $\mathbb{F}$ be a finite field. Fix $0 \le t \le m$ and let $B = 2^t$. Let $\mathcal{H}_t \subseteq \mathbb{F}^{I_m}$ be the coefficient subspace of Corollary 4.31. Sample a random table H uniformly from $\mathcal{H}_t$, and let $R := U_H^{(m)}$ be its encoded polynomial, which satisfies $\deg R < B$. For an arbitrary fixed $P \in \mathbb{F}[X]$, the *high-degree masked polynomial* is

$$Q(X) := P(X) + X^N R(X), \qquad N = 2^m.$$

Lemma 4.63 (Uniformity of the mask polynomial)**.** *Under Definition 4.62, the random polynomial R is uniform on the B-dimensional vector space $\mathbb{F}[X]_{<B}$.*

Proof. Theorem 4.30 identifies $\mathcal{H}_t$ with the tables $\varphi : I_t \to \mathbb{F}$ via an injective linear map onto $\mathcal{H}_t$. The same theorem identifies $U_H^{(m)}$ with $U_\varphi^{(t)}$. By the basis theorem (Theorem 4.14) with t variables, the map $\varphi \mapsto U_\varphi^{(t)}$ is a linear bijection from $\mathbb{F}^{I_t}$ to $\mathbb{F}[X]_{<B}$. The composition $H \mapsto R$ is therefore a bijection between the finite sets $\mathcal{H}_t$ and $\mathbb{F}[X]_{<B}$. A bijection sends the uniform distribution to the uniform distribution. □

Lemma 4.64 (Uniform answers at prescribed nonzero points)**.** *Let $1 \le q \le B$, and let $\xi_1, \dots, \xi_q \in \mathbb{F}^\times$ be pairwise distinct. With R as in Definition 4.62, for any fixed $P \in \mathbb{F}[X]$ the random vector*

$$(Q(\xi_1), \dots, Q(\xi_q)) \in \mathbb{F}^q, \qquad Q = P + X^N R,$$

is uniform on $\mathbb{F}^q$ and independent of the choice of P.

Proof. By Lemma 4.63, R is uniform on $\mathbb{F}[X]_{<B}$. Let $\mathrm{Ev}_q : \mathbb{F}[X]_{<B} \to \mathbb{F}^q$ be evaluation at the q specified points. Since $q \le B$, the Lagrange polynomials of Lemma 3.60 have degree $< q \le B$, and prove that Ev_q is surjective. As an $\mathbb{F}$-linear surjection between finite vector spaces, all its fibers are cosets of the same kernel and have equal cardinality. Consequently the evaluation vector $(R(\xi_1), \dots, R(\xi_q))$ is uniform on $\mathbb{F}^q$. Every ξ_j is nonzero, so multiplication of coordinate j by ξ_j^N is invertible. The diagonal map $(a_j)_j \mapsto (\xi_j^N a_j)_j$ is a bijection of $\mathbb{F}^q$, and adding the fixed vector $(P(\xi_j))_j$ is another bijection. Their composition sends the uniform vector of R-evaluations to a uniform vector of Q-evaluations, independently of P. □

Theorem 4.65 (Adaptive bounded-query hiding for one masked polynomial)**.** *Let R be sampled as in Definition 4.62 with $B = 2^t$. Consider a possibly randomized verifier whose random tape is independent of R and which adaptively queries the oracle $Q = P + X^N R$ at at most B distinct nonzero field points, with repeat queries allowed but returning their previous value. The strategy for selecting the next query depends only on public parameters, the verifier's tape, and the earlier query-and-answer pairs. If the verifier receives no other P-dependent messages, the distribution of its complete query-and-answer record is independent of the fixed secret polynomial P.*

Proof. Condition first on a fixed random tape of the verifier, so the query strategy is deterministic. We prove by induction on the number q of distinct queried points that the conditional law of the next *fresh* answer, given any realized history of $q < B$ distinct queries, is uniform on $\mathbb{F}$ and independent of P. The base $q = 0$ is Lemma 4.64 with one point. For the induction step, fix a history of q pairwise distinct nonzero indices $\xi_1, \dots, \xi_q$ and answers $a_1, \dots, a_q$ with positive probability. The verifier's deterministic strategy fixes its next fresh point ξ_{q+1} from this history. Since it is distinct

from previous points and $q+1 \le B$, Lemma 4.64 applied to this fixed $(q+1)$-tuple states that the joint vector of answers is uniform on $\mathbb{F}^{q+1}$. Thus

$$\Pr[Q(\xi_{q+1}) = a \mid Q(\xi_j) = a_j \ (1 \le j \le q)] = \frac{|\mathbb{F}|^{-(q+1)}}{|\mathbb{F}|^{-q}} = |\mathbb{F}|^{-1},$$

for each $a \in \mathbb{F}$, independently of P. On a repeated query the answer is deterministically the previously observed value, again independent of P conditional on the history. By induction on query steps, the distributions of the entire records coincide for all choices of P. Finally average over the verifier's independent random tape; the mixture is the same for every P. □

Theorem 4.66 (Mask-query and code-rate tradeoff)**.** *Let $\mathbb{F}$ be a finite field, let $m \ge 1$, let $N = 2^m$, and let $0 \le t \le m$ with $B = 2^t$. Let $\mathcal{D} \subseteq \mathbb{F}^\times$ have $M \ge N + B$ pairwise distinct elements. For a fixed secret table $f : I_m \to \mathbb{F}$, sample R uniformly from $\mathbb{F}[X]_{<B}$ and set*

$$Q(X) := U_f^{(m)}(X) + X^N R(X).$$

Then the following three statements hold:

(i) $\mathrm{ev}_{\mathcal{D}}(Q) \in \mathrm{RS}_{\mathbb{F}}(\mathcal{D}, N+B)$ *for every R.*

(ii) *A verifier making at most B adaptive distinct queries to this one oracle, at nonzero domain points and without receiving other witness-dependent messages, obtains a query-and-answer distribution independent of f.*

(iii) *The code $\mathrm{RS}_{\mathbb{F}}(\mathcal{D}, N+B)$ has minimum distance at least $M-(N+B)+1$, and a received word can be within Hamming distance τ of at most one codeword whenever $2\tau < M-(N+B)+1$.*

Proof. For (i), $\deg U_f^{(m)} \le N-1$ and $\deg R < B$. Therefore $\deg(X^N R) < N+B$, with the zero polynomial also allowed. The same degree bound holds for their sum Q, which by definition belongs to the evaluation code of dimension bound $N+B$. For (ii), Lemma 4.63 shows that uniform kernel masks from $\mathcal{H}_t$ are equivalent to uniform polynomials of degree $< B$, so Theorem 4.65 applies verbatim to the oracle Q on the nonzero domain $\mathcal{D}$. For (iii), two distinct polynomials of degree $< N+B$ differ by a nonzero polynomial of degree at most $N+B-1$. The root-count lemma (Lemma 2.7) shows that their evaluation words agree in at most $N+B-1$ of the M positions. Thus their Hamming distance is at least $M-(N+B)+1$. If two codewords both had distance at most τ from one received word, the Hamming triangle inequality (Lemma 3.27) would bound their mutual distance by 2τ, contrary to the strict hypothesis. □

Remark 4.67 (Exact scope of the bounded-query theorem). Theorem 4.65 concerns *only* direct oracle evaluations of a single high-degree masked polynomial at nonzero points. It does not prove zero knowledge of KBFold: the original protocol reveals additional sumcheck round polynomials, correlated folded words, a last-round table, and authenticated commitment openings. Those additional messages can invalidate the theorem's hypotheses. Moreover, to use the masked word in a proximity protocol the Reed–Solomon degree and final closure must be modified as proved in Corollary 4.61 and Theorem 3.71.

4.9 Formal status and next algebraic dependencies

The foundations above depend only on field arithmetic, polynomial coefficients, finite sums, and finite-dimensional linear algebra, except for the explicitly finite-field masking statements in Section 4.8. The tensor fold in Theorem 4.50 is valid in every characteristic; the ordinary evaluation formula of Corollary 4.52 is the characteristic-two specialization. These are not interchangeable in the odd-characteristic KBFold protocol. In the next chapter, the typed mathematical statements on multilinearization, folding, and masks can be imported as standalone lemmas rather than reproved inside a cryptographic security argument. The dedicated theorem ledger accompanying the source provides a proposed Lean name for each declaration. No new Lean machine-checked proof is asserted here.

5 Multilinearization, Kernel Folding, and Mask Propagation

This chapter develops the algebraic folding invariants needed for a zero-knowledge extension of KBFold. Its starting point is the Boolean-kernel encoding

$$U_f^{(m)}(X) = \sum_{b \in I_m} f(b) K_b^{(m)}(X)$$

from Chapters 1 and 4, and the normalized odd-characteristic word fold in the original KBFold scheme [Mkh26]. We separate four different claims: that the coefficient table is restricted correctly; that polynomial and word folds commute; that an added high-degree mask propagates with a controlled degree budget; and that the *joint* masked oracle answers satisfy a precise linear-algebraic hiding condition. None of these algebraic statements alone asserts a complete zero-knowledge simulator.

5.1 Prefix restrictions and multilinear evaluation

Throughout this chapter, $\mathbb{F}$ is a field, $m \geq 1$ is an integer, $N = 2^m$, and $I_k = \{0,1\}^k$ for every $0 \leq k \leq m$. The first bit is the lowest-index bit, as in Definition 1.1. The variables of a table are ordered, and no permutation of their indices is implicit.

Definition 5.1 (Prefix equality weight)**.** For $0 \leq j \leq m$, $a \in I_j$ and $r = (r_0, \ldots, r_{j-1}) \in \mathbb{F}^j$, define

$$\lambda_a^{(j)}(r) := \prod_{i=0}^{j-1} \big((1-a_i)(1-r_i) + a_i r_i\big) \in \mathbb{F}.$$

For $j = 0$, the unique weight is the empty product 1.

Lemma 5.2 (First-bit factorization)**.** *For $j < m$, any $a = (a_0, \ldots, a_{j-1}) \in I_j$, $e \in \{0,1\}$ and $r' = (r_0, \ldots, r_{j-1}) \in \mathbb{F}^j$,*

$$\lambda_{(a,e)}^{(j+1)}(r', t) = \lambda_a^{(j)}(r')\big((1-e)(1-t) + et\big).$$

Proof. By Definition 5.1, the left side is a product indexed by $i = 0, \ldots, j$. Separate the factor with $i = j$ from the product with $i < j$. The latter is $\lambda_a^{(j)}(r')$ and the former is the displayed affine factor. □

Definition 5.3 (Simultaneous prefix restriction)**.** For $0 \le j \le m$, $r \in \mathbb{F}^j$, and $f : I_m \to \mathbb{F}$, define

$$\mathsf{Res}_r^{[j]}(f) : I_{m-j} \to \mathbb{F}, \qquad \mathsf{Res}_r^{[j]}(f)(b) := \sum_{a \in I_j} \lambda_a^{(j)}(r)\, f(a,b).$$

Here $(a,b) \in I_m$ denotes concatenation in the original bit order.

Lemma 5.4 (Zero-prefix identity)**.** *For every table $f : I_m \to \mathbb{F}$,*

$$\mathsf{Res}_{()}^{[0]}(f) = f.$$

Proof. The set I_0 contains only the empty tuple $()$, whose equality weight equals 1. In the defining sum, $((), b) = b$. Thus $\mathsf{Res}_{()}^{[0]}(f)(b) = f(b)$ for every b. □

Lemma 5.5 (Prefix extension by one restriction)**.** *For $0 \le j < m$, $r \in \mathbb{F}^j$, and $t \in \mathbb{F}$,*

$$\mathsf{Res}_{(r,t)}^{[j+1]}(f) = \mathsf{Res}_t(\mathsf{Res}_r^{[j]}(f)).$$

Proof. Fix $b \in I_{m-j-1}$ and expand the left side:

$$\mathsf{Res}_{(r,t)}^{[j+1]}(f)(b) = \sum_{a \in I_j} \sum_{e \in \{0,1\}} \lambda_{(a,e)}^{(j+1)}(r,t) f(a,e,b).$$

By Lemma 5.2, the summand equals $\lambda_a^{(j)}(r)((1-e)(1-t) + et) f(a,e,b)$. Separate the two values of e:

$$(1-t) \sum_{a \in I_j} \lambda_a^{(j)}(r) f(a,0,b) + t \sum_{a \in I_j} \lambda_a^{(j)}(r) f(a,1,b).$$

The two sums are $\mathsf{Res}_r^{[j]}(f)(0,b)$ and $\mathsf{Res}_r^{[j]}(f)(1,b)$, respectively. Their affine combination is $\mathsf{Res}_t(\mathsf{Res}_r^{[j]}(f))(b)$ by Definition 1.11. Since b was arbitrary, the table identity follows. □

Lemma 5.6 (Linearity of prefix restriction)**.** *For any $f, g : I_m \to \mathbb{F}$, $c \in \mathbb{F}$, and every prefix $r \in \mathbb{F}^j$,*

$$\mathsf{Res}_r^{[j]}(f + cg) = \mathsf{Res}_r^{[j]}(f) + c\,\mathsf{Res}_r^{[j]}(g).$$

Proof. At each remaining bit tuple b, distribute the sum $\sum_a \lambda_a^{(j)}(r)(f(a,b) + cg(a,b))$, and take the scalar c outside the finite sum. □

Lemma 5.7 (Partial multilinear-evaluation identity)**.** *For every $0 \le j \le m$, $r \in \mathbb{F}^j$, and $y \in \mathbb{F}^{m-j}$,*

$$\widetilde{\mathsf{Res}_r^{[j]}(f)}(y) = \widetilde{f}(r,y).$$

Proof. By the definition of multilinear extension, the left side is

$$\sum_{b \in I_{m-j}} \mathsf{Res}_r^{[j]}(f)(b) \lambda_b^{(m-j)}(y).$$

Substitute Definition 5.3 and interchange the two finite sums:

$$\sum_{a \in I_j} \sum_{b \in I_{m-j}} f(a,b) \lambda_a^{(j)}(r) \lambda_b^{(m-j)}(y).$$

By splitting the product in Definition 1.8, $\lambda_{(a,b)}^{(m)}(r,y) = \lambda_a^{(j)}(r)\lambda_b^{(m-j)}(y)$. The map $(a,b) \mapsto (a,b)$ is a bijection $I_j \times I_{m-j} \to I_m$; thus the displayed expression equals $\sum_{c \in I_m} f(c) \lambda_c^{(m)}(r,y) = \widetilde{f}(r,y)$. □

Corollary 5.8 (Complete prefix restriction)**.** *For $r \in \mathbb{F}^m$ the table $\mathsf{Res}^{[m]}_r(f)$ has one entry and*

$$\mathsf{Res}^{[m]}_r(f)(()) = \widetilde{f}(r).$$

Proof. Apply Lemma 5.7 with $j = m$ and the unique empty tuple of $\mathbb{F}^0$; the multilinear extension of a one-entry table is its only entry. □

5.2 Iterated polynomial folds and the tensor functional

We use the even–odd decomposition of Lemma 1.5 and the normalized fold of Definition 2.25. All results in this subsection hold over *any* field unless the characteristic is explicitly restricted.

Definition 5.9 (Bitwise fold weights)**.** For $r \in \mathbb{F}$ define

$$A_0(r) := 2r - 1, \qquad A_1(r) := 1 - r.$$

For $j \geq 0$, $e \in \mathbb{N}$, and $r = (r_0, \ldots, r_{j-1}) \in \mathbb{F}^j$, define

$$W_e^{[j]}(r) := \prod_{i=0}^{j-1} A_{e_i}(r_i),$$

where $e_i \in \{0, 1\}$ is the i-th binary digit of e. For $j = 0$, the product is 1.

Lemma 5.10 (Monomial fold, even exponent)**.** *For every $a \in \mathbb{N}$ and $r \in \mathbb{F}$,*

$$\mathsf{pfold}_r(X^{2a}) = (2r - 1)X^a = A_0(r)X^a.$$

Proof. The even–odd decomposition of X^{2a} is $E(Y) = Y^a$, $O(Y) = 0$. Substituting into $\mathsf{pfold}_r(P) = (2r - 1)E + (1 - r)O$ proves the identity. □

Lemma 5.11 (Monomial fold, odd exponent)**.** *For every $a \in \mathbb{N}$ and $r \in \mathbb{F}$,*

$$\mathsf{pfold}_r(X^{2a+1}) = (1 - r)X^a = A_1(r)X^a.$$

Proof. The even–odd decomposition of X^{2a+1} is $E(Y) = 0$, $O(Y) = Y^a$. Substitute into the same definition. □

Definition 5.12 (Iterated normalized polynomial fold)**.** For a polynomial $P \in \mathbb{F}[X]$ and $r \in \mathbb{F}^j$, define

$$\mathsf{PFold}^{[0]}_{()}(P) := P, \qquad \mathsf{PFold}^{[j+1]}_{(r,t)}(P) := \mathsf{pfold}_t\left(\mathsf{PFold}^{[j]}_r(P)\right).$$

Each step is defined on *every* polynomial; no degree bound is required to define the operator.

Lemma 5.13 (Iterated fold of a monomial)**.** *For every $e \in \mathbb{N}$, $0 \leq j$, and $r \in \mathbb{F}^j$,*

$$\mathsf{PFold}^{[j]}_r(X^e) = W_e^{[j]}(r)X^{\lfloor e/2^j \rfloor}.$$

Proof. Induct on j. When $j = 0$, the weight is the empty product 1 and $\lfloor e/1 \rfloor = e$, so the formula is the definition. Assume it holds for j, and write $a = \lfloor e/2^j \rfloor$. By linearity of pfold_t and the induction hypothesis,

$$\mathsf{PFold}^{[j+1]}_{(r,t)}(X^e) = W_e^{[j]}(r)\mathsf{pfold}_t(X^a).$$

By Lemmas 5.10 and 5.11,

$$\mathsf{pfold}_t(X^a) = A_{a \bmod 2}(t)X^{\lfloor a/2 \rfloor}.$$

The least significant bit of $a = \lfloor e/2^j \rfloor$ is precisely e_j; also $\lfloor a/2 \rfloor = \lfloor e/2^{j+1} \rfloor$. Hence the expression equals $W_e^{[j]}(r)A_{e_j}(t)X^{\lfloor e/2^{j+1} \rfloor} = W_e^{[j+1]}(r, t)X^{\lfloor e/2^{j+1} \rfloor}$. This proves the induction step. □

Theorem 5.14 (Explicit polynomial-fold formula)**.** *For any polynomial* $P(X) = \sum_{e=0}^{d} p_e X^e \in \mathbb{F}[X]$ *and every* $r \in \mathbb{F}^j$*,*

$$\boxed{\mathsf{PFold}_r^{[j]}(P)(Y) = \sum_{e=0}^{d} p_e \, W_e^{[j]}(r) \, Y^{\lfloor e/2^j \rfloor}.}$$

Proof. By Definition 2.25, pfold_t is linear in the polynomial argument, since the even and odd coefficient-extraction maps are linear. Compositions of linear maps are linear, so $\mathsf{PFold}_r^{[j]}$ is linear. Apply it to the finite monomial expansion of P and use Lemma 5.13 on each term. □

Corollary 5.15 (Degree-halving bound)**.** *If* $D \geq 1$ *and* $\deg P < D$ *(allowing* $P = 0$*), then*

$$\deg \mathsf{PFold}_r^{[j]}(P) < \left\lceil \frac{D}{2^j} \right\rceil$$

or the iterated fold is zero.

Proof. Every exponent in the sum of Theorem 5.14 satisfies $e \leq D - 1$. Thus every nonzero monomial in the result has exponent at most $\lfloor (D-1)/2^j \rfloor$. The integer identity $\lfloor (D-1)/2^j \rfloor + 1 = \lceil D/2^j \rceil$ gives the asserted strict degree bound; cancellation can only lower degree. □

Corollary 5.16 (Complete tensor functional)**.** *For* $N = 2^m$*,* $P \in \mathbb{F}[X]_{<N}$ *with monomial coefficients* p_e*, and* $r \in \mathbb{F}^m$*,*

$$\boxed{\mathsf{PFold}_r^{[m]}(P) = \sum_{e=0}^{N-1} p_e \prod_{i=0}^{m-1} \begin{cases} 2r_i - 1, & e_i = 0, \\ 1 - r_i, & e_i = 1. \end{cases}}$$

The result is a constant polynomial.

Proof. Apply Theorem 5.14 with $j = m$. Every $e < N = 2^m$ satisfies $\lfloor e/2^m \rfloor = 0$, so all powers of the output variable are $Y^0 = 1$. Expand the definition of the weights. □

Definition 5.17 (Binary-exponent multilinearization)**.** For $P = \sum_{e=0}^{N-1} p_e X^e$, define

$$\mathscr{M}_m(P)(Z_0, \ldots, Z_{m-1}) := \sum_{e=0}^{N-1} p_e \prod_{i=0}^{m-1} Z_i^{e_i}.$$

This is the $\mathbb{F}$-linear isomorphism between $\mathbb{F}[X]_{<N}$ and the space of multilinear polynomials already introduced in Chapter 4.

Lemma 5.18 (Characteristic-two specialization)**.** *If* $\mathrm{char}(\mathbb{F}) = 2$*, then for every* $P \in \mathbb{F}[X]_{<N}$ *and* $r \in \mathbb{F}^m$*,*

$$\mathsf{PFold}_r^{[m]}(P) = \mathscr{M}_m(P)(1 + r_0, \ldots, 1 + r_{m-1}).$$

Proof. When $2 = 0$, $2r_i - 1 = -1 = 1$ and $1 - r_i = 1 + r_i$. Therefore the weight for a binary exponent e in Corollary 5.16 is $\prod_{i:e_i=1}(1 + r_i)$; factors with $e_i = 0$ equal 1. This is exactly the coefficient of p_e in the evaluation of Definition 5.17 at $Z_i = 1 + r_i$. Equality follows by summing over e. □

Proposition 5.19 (Failure of the unshifted characteristic-two identity in odd characteristic)**.** *Assume* $\mathrm{char}(\mathbb{F}) \neq 2$*. For* $m = 1$*,* $P(X) = 1$*, and* $r = 0$*,*

$$\mathsf{PFold}_{(0)}^{[1]}(P) = -1 \neq 1 = \mathscr{M}_1(P)(1).$$

Proof. Since $P = 1$ is even with odd component zero, $\mathsf{pfold}_0(1) = (2 \cdot 0 - 1) \cdot 1 = -1$. By definition $\mathscr{M}_1(1)$ is the constant polynomial 1. If $-1 = 1$ in $\mathbb{F}$, then $2 = 0$, contrary to the hypothesis. Hence the displayed inequality. □

Remark 5.20 (No hidden ring-homomorphism step). The map $\mathscr{M}_m$ is linear but not multiplicative: products in $\mathbb{F}[X]$ may have degrees outside its domain, and even within that domain ordinary multiplication does not generally agree with multiplication of multilinear forms. Lemma 5.18 follows from the explicit monomial formula, not from an assumed multiplicativity property.

5.3 The normalized fold dictionary as a commuting diagram

This subsection relates prefix restriction and polynomial folding for every intermediate level. The result is the algebraic input to the oracle-level proofs, not yet a claim about an authenticated proof transcript. The structural identity originates in KBFold [Mkh26].

Lemma 5.21 (First-bit kernel decomposition). *For $f : I_m \to \mathbb{F}$, $m \geq 1$, write $f_0(b) := f(0, b)$, $f_1(b) := f(1, b)$ for $b \in I_{m-1}$. Then*

$$U_f^{(m)}(X) = U_{f_1}^{(m-1)}(X^2) + X(U_{f_0}^{(m-1)} + U_{f_1}^{(m-1)})(X^2).$$

Proof. Split the defining finite sum for $U_f^{(m)}$ into terms indexed by $(0, b)$ and $(1, b)$. Lemma 1.4 identifies the two kernel terms. Substituting those identities and collecting the terms gives

$$U_f^{(m)}(X) = XU_{f_0}^{(m-1)}(X^2) + (X + 1)U_{f_1}^{(m-1)}(X^2).$$

Distribute the second term and collect the constant and X parts to obtain the stated formula. □

Lemma 5.22 (Single-fold intertwining). *For every $f : I_m \to \mathbb{F}$ and $r \in \mathbb{F}$,*

$$\mathsf{pfold}_r(U_f^{(m)}) = U_{\mathsf{Res}_r(f)}^{(m-1)}.$$

Proof. By Lemma 5.21, the unique even–odd components of U_f are $E = U_{f_1}^{(m-1)}$ and $O = U_{f_0}^{(m-1)} + U_{f_1}^{(m-1)}$. Hence

$$\begin{aligned}\mathsf{pfold}_r(U_f^{(m)}) &= (2r - 1)U_{f_1}^{(m-1)} + (1 - r)(U_{f_0}^{(m-1)} + U_{f_1}^{(m-1)}) \\ &= (1 - r)U_{f_0}^{(m-1)} + ((2r - 1) + (1 - r))U_{f_1}^{(m-1)} \\ &= (1 - r)U_{f_0}^{(m-1)} + rU_{f_1}^{(m-1)}.\end{aligned}$$

Linearity of the kernel encoding makes the last expression $U_{(1-r)f_0+rf_1}^{(m-1)} = U_{\mathsf{Res}_r(f)}^{(m-1)}$. □

Theorem 5.23 (Iterated folding–restriction dictionary). *For every $0 \leq j \leq m$, $r \in \mathbb{F}^j$, and $f : I_m \to \mathbb{F}$,*

$$\boxed{\mathsf{PFold}_r^{[j]}(U_f^{(m)}) = U_{\mathsf{Res}_r^{[j]}(f)}^{(m-j)}.}$$

Proof. Induct on j. For $j = 0$, the left side is $U_f^{(m)}$ and the right side equals the same polynomial by Lemma 5.4. Suppose the identity holds for $j < m$. For $t \in \mathbb{F}$,

$$\begin{aligned}\mathsf{PFold}_{(r,t)}^{[j+1]}(U_f^{(m)}) &= \mathsf{pfold}_t\left(\mathsf{PFold}_r^{[j]}(U_f^{(m)})\right)\\ &= \mathsf{pfold}_t\left(U^{(m-j)}_{\mathsf{Res}_r^{[j]}(f)}\right)\\ &= U^{(m-j-1)}_{\mathsf{Res}_t(\mathsf{Res}_r^{[j]}(f))}\\ &= U^{(m-j-1)}_{\mathsf{Res}_{(r,t)}^{[j+1]}(f)}.\end{aligned}$$

The first step uses Definition 5.12, the second is the induction hypothesis, the third is Lemma 5.22, and the last is Lemma 5.5. This completes the induction. □

Corollary 5.24 (Complete fold equals multilinear evaluation)**.** *For every $f : I_m \to \mathbb{F}$ and $r \in \mathbb{F}^m$,*

$$\boxed{\mathsf{PFold}_r^{[m]}(U_f^{(m)}) = \widetilde{f}(r).}$$

Proof. Apply Theorem 5.23 with $j = m$. The right side is the encoding of the one-entry table $\mathsf{Res}_r^{[m]}(f)$, hence equals its sole entry because $K_{()}^{(0)} = 1$. That entry is $\widetilde{f}(r)$ by Corollary 5.8. □

Corollary 5.25 (Two descriptions of the complete fold)**.** *For any polynomial $P \in \mathbb{F}[X]_{<N}$, write uniquely $P = \sum_b f(b) K_b^{(m)}$ and also $P = \sum_{e=0}^{N-1} p_e X^e$. Then for each $r \in \mathbb{F}^m$,*

$$\sum_b f(b)\,\mathrm{eq}_m(b, r) = \sum_{e=0}^{N-1} p_e W_e^{[m]}(r).$$

Proof. By Corollary 5.24, the left side is $\mathsf{PFold}_r^{[m]}(P)$. By Corollary 5.16, the right side is the same complete normalized fold. Equality follows by transitivity, using uniqueness of the two polynomial expansions from Proposition 1.6. □

Lemma 5.26 (Composition of prefix restrictions)**.** *Let $j, k \geq 0$ with $j + k \leq m$, and let $r \in \mathbb{F}^j$, $s \in \mathbb{F}^k$. Then*

$$\mathsf{Res}_s^{[k]}(\mathsf{Res}_r^{[j]}(f)) = \mathsf{Res}_{(r,s)}^{[j+k]}(f).$$

Proof. For each $b \in I_{m-j-k}$, expand the left side as

$$\sum_{c \in I_k} \lambda_c^{(k)}(s) \left(\sum_{a \in I_j} \lambda_a^{(j)}(r) f(a, c, b) \right).$$

Interchange the two finite sums and use $\lambda_{(a,c)}^{(j+k)}(r, s) = \lambda_a^{(j)}(r)\lambda_c^{(k)}(s)$, which follows by splitting the defining product across the first j and next k coordinates. Reindex the sum via the bijection $I_j \times I_k \to I_{j+k}$. The result is $\sum_{d \in I_{j+k}} \lambda_d^{(j+k)}(r, s) f(d, b)$, the right-hand side. □

Corollary 5.27 (Compatibility of nested polynomial folds)**.** *For $j+k \leq m$ and a kernel polynomial $U_f^{(m)}$,*

$$\mathsf{PFold}_s^{[k]}\left(\mathsf{PFold}_r^{[j]}(U_f^{(m)})\right) = \mathsf{PFold}_{(r,s)}^{[j+k]}(U_f^{(m)}).$$

Proof. The left side equals $U^{(m-j-k)}_{\mathsf{Res}_s^{[k]}(\mathsf{Res}_r^{[j]}(f))}$ by two applications of Theorem 5.23. Lemma 5.26 turns its coefficient table into $\mathsf{Res}_{(r,s)}^{[j+k]}(f)$, whose encoding is the right side by Theorem 5.23. □

5.4 A degree-aware module of masked kernel polynomials

The original KBFold paper proposes adding a random high-degree term $X^N R$ to its kernel encoding as part of a zero-knowledge extension [Mkh26]. The following statements explain exactly how this term transforms under the same normalized folds. No privacy or soundness claim is made without an explicit verifier game.

Definition 5.28 (Mask degree and extended encoding)**.** Fix $1 \le B \le N$. Let $\mathcal{R}_B := \mathbb{F}[X]_{<B}$ and let $\mathcal{T}_m := \mathbb{F}^{I_m}$. Define

$$\mathscr{Q}_{m,B} : \mathcal{T}_m \times \mathcal{R}_B \longrightarrow \mathbb{F}[X]_{<N+B}, \qquad \mathscr{Q}_{m,B}(f, R) := U_f^{(m)} + X^N R.$$

Both arguments are field vectors, and the map is linear in the pair.

Lemma 5.29 (No overlap between table and high-degree masks)**.** *If $P \in \mathbb{F}[X]_{<N}$ and $R \in \mathbb{F}[X]_{<B}$ satisfy $P + X^N R = 0$, then $P = 0$ and $R = 0$.*

Proof. For every exponent $e < N$, the coefficient of X^e in $X^N R$ equals zero. Coefficient comparison in $P + X^N R = 0$ therefore shows that all coefficients of P vanish, so $P = 0$. The equality reduces to $X^N R = 0$ in the polynomial ring $\mathbb{F}[X]$. Since $\mathbb{F}$ is a field, $\mathbb{F}[X]$ has no zero divisors and $X^N \neq 0$, so $R = 0$. □

Proposition 5.30 (Extended encoding is a linear isomorphism)**.** *The map $\mathscr{Q}_{m,B}$ is an $\mathbb{F}$-linear bijection from $\mathcal{T}_m \times \mathcal{R}_B$ onto $\mathbb{F}[X]_{<N+B}$.*

Proof. Linearity follows from linearity of $f \mapsto U_f^{(m)}$ and of multiplication by X^N. If $\mathscr{Q}_{m,B}(f, R) = 0$, Lemma 5.29 gives $R = 0$ and $U_f^{(m)} = 0$; the kernel basis theorem implies $f = 0$. Thus the map is injective. For surjectivity, let $Q(X) = \sum_{e=0}^{N+B-1} q_e X^e$. Define $P(X) = \sum_{e=0}^{N-1} q_e X^e$ and $R(X) = \sum_{e=0}^{B-1} q_{N+e} X^e$. Then $Q = P + X^N R$ by coefficient comparison. The kernel basis theorem provides a table f with $P = U_f^{(m)}$, whence $Q = \mathscr{Q}_{m,B}(f, R)$. □

Lemma 5.31 (Fold of an even monomial shift)**.** *For every $a \in \mathbb{N}$, $S \in \mathbb{F}[X]$, and $r \in \mathbb{F}$,*

$$\mathsf{pfold}_r(X^{2a} S) = X^a \mathsf{pfold}_r(S).$$

Proof. Write $S(X) = E(X^2) + XO(X^2)$. Then $X^{2a} S(X) = (X^2)^a E(X^2) + X(X^2)^a O(X^2)$. By uniqueness of the even–odd decomposition, the even and odd parts of the shifted polynomial are respectively $Y^a E(Y)$ and $Y^a O(Y)$. Its normalized fold is therefore $(2r-1)Y^a E + (1-r)Y^a O = Y^a \mathsf{pfold}_r(S)$. □

Definition 5.32 (Masked-table and residual-mask sequences)**.** Fix a vector $r = (r_0, \ldots, r_{m-1}) \in \mathbb{F}^m$. Define, for $0 \le j \le m$,

$$f_j := \mathsf{Res}^{[j]}_{(r_0,\ldots,r_{j-1})}(f), \qquad R_j := \mathsf{PFold}^{[j]}_{(r_0,\ldots,r_{j-1})}(R),$$

and

$$Q_j := \mathsf{PFold}^{[j]}_{(r_0,\ldots,r_{j-1})}(\mathscr{Q}_{m,B}(f, R)).$$

Theorem 5.33 (Masked folding invariant at every level)**.** *For each $0 \le j \le m$,*

$$\boxed{Q_j(X) = U_{f_j}^{(m-j)}(X) + X^{2^{m-j}} R_j(X).}$$

Proof. For $j = 0$, $Q_0 = \mathscr{Q}_{m,B}(f, R) = U_f^{(m)} + X^N R$, which is the assertion. Suppose the formula holds at $j < m$. By linearity of pfold_{r_j},

$$Q_{j+1} = \mathsf{pfold}_{r_j}(U_{f_j}^{(m-j)}) + \mathsf{pfold}_{r_j}(X^{2^{m-j}} R_j).$$

Lemma 5.22 identifies the first term as $U^{(m-j-1)}_{\mathsf{Res}_{r_j}(f_j)} = U^{(m-j-1)}_{f_{j+1}}$ by Lemma 5.5. Since $j < m$, the exponent 2^{m-j} is even. Applying Lemma 5.31 with $a = 2^{m-j-1}$ transforms the second term into $X^{2^{m-j-1}} \mathsf{pfold}_{r_j}(R_j) = X^{2^{m-(j+1)}} R_{j+1}$. This proves the induction step. □

Lemma 5.34 (Degree of the residual mask)**.** *For each* $0 \le j \le m$,

$$\deg R_j < B_j, \qquad B_j := \left\lceil \frac{B}{2^j} \right\rceil,$$

where the zero polynomial is allowed.

Proof. By definition R_j is the j-fold polynomial fold of a polynomial of degree less than B. Apply Corollary 5.15 with $D = B$. □

Corollary 5.35 (Exact upper degree budget)**.** *For* $N_j = 2^{m-j}$ *and* $D_j := N_j + B_j$,

$$\deg Q_j < D_j \qquad (0 \le j \le m).$$

Proof. The kernel encoding $U_{f_j}^{(m-j)}$ has degree less than N_j by Lemma 1.3. By Lemma 5.34, the shifted mask $X^{N_j} R_j$ has degree less than $N_j + B_j$. The sum of two polynomials of degrees below D_j also has degree below D_j, proving the claim. □

Lemma 5.36 (Residual mask is constant at the final level)**.** *Under* $1 \le B \le N = 2^m$, R_m *has degree less than* 1, *hence is a constant polynomial.*

Proof. From Lemma 5.34, $\deg R_m < \lceil B/N \rceil = 1$ because $0 < B \le N$. Thus R_m is a constant or is the zero polynomial. □

Theorem 5.37 (Linear final closure for high-degree masked KBFold)**.** *For every* $f : I_m \to \mathbb{F}$, *every mask* $R \in \mathbb{F}[X]_{<B}$ *with* $1 \le B \le N$, *and every* $r \in \mathbb{F}^m$, *there exists a unique scalar* $\rho_r(R) \in \mathbb{F}$ *such that*

$$\boxed{Q_m(X) = \widetilde{f}(r) + \rho_r(R) X.}$$

More precisely,

$$\rho_r(R) = \mathsf{PFold}_r^{[m]}(R) = \sum_{e=0}^{B-1} [X^e] R \prod_{i=0}^{m-1} A_{e_i}(r_i).$$

Proof. Apply Theorem 5.33 with $j = m$: $Q_m = U_{f_m}^{(0)} + X R_m$. The first polynomial is constant and equals $\widetilde{f}(r)$ by Corollary 5.8. By Lemma 5.36 there is a unique scalar $\rho_r(R)$ such that $R_m = \rho_r(R)$; uniqueness follows from uniqueness of coefficients in $\mathbb{F}[X]$. Since $R_m = \mathsf{PFold}_r^{[m]}(R)$ by definition, Corollary 5.16, with missing coefficients of R filled by zero, gives the displayed formula. □

Corollary 5.38 (Constant-term closure identity)**.** *For* $B \le N$,

$$\boxed{[X^0] Q_m = \widetilde{f}(r).}$$

Proof. Theorem 5.37 expresses Q_m as a constant plus $\rho_r(R)X$. The coefficient of X^0 in the second summand is zero. □

Remark 5.39 (What this corrects). Chapter 3 proved that an unchanged *constant-polynomial* closure test rejects certain honestly masked executions. Theorem 5.37 supplies the correct algebraic replacement: a *linear* final polynomial whose constant coefficient equals the folded evaluation. This resolves that algebraic closure obstruction but does not supply a hiding proof for the final authentication paths or the whole interactive transcript.

5.5 Reed–Solomon words, local folds, and degree budgets

For this subsection assume $\mathbb{F}$ has odd characteristic and $\Omega_0 \leq \mathbb{F}^\times$ is a multiplicative subgroup of order $M = 2^{m+s}$, with $s \geq 1$. Define $\Omega_j := \{x^{2^j} : x \in \Omega_0\}$ for $0 \leq j \leq m$.

Lemma 5.40 (Domain sizes)**.** *For every $0 \leq j \leq m$,*

$$|\Omega_j| = M_j := \frac{M}{2^j}, \qquad \Omega_{j+1} = \Omega_j^2 \quad (j < m).$$

Each Ω_j has even order.

Proof. A finite multiplicative subgroup of a field is cyclic. In a cyclic group of order $M = 2^{m+s}$, the homomorphism $x \mapsto x^{2^j}$ has kernel of order $\gcd(M, 2^j) = 2^j$ for $j \leq m$; hence its image has order $M/2^j$. Taking the image again under squaring gives $\Omega_{j+1} = \Omega_j^2$. Because $s \geq 1$ and $j \leq m$, we have $M_j \geq 2^s \geq 2$, so each order is even. □

Definition 5.41 (Word fold on a smooth domain)**.** Let $w : \Omega_j \to \mathbb{F}$ with $j < m$. For $y = x^2 \in \Omega_{j+1}$ set

$$w_{\mathrm{e}}(y) = \frac{w(x) + w(-x)}{2}, \qquad w_{\mathrm{o}}(y) = \frac{w(x) - w(-x)}{2x},$$

and

$$\mathsf{WFold}_{r,j}(w)(y) = (2r-1)w_{\mathrm{e}}(y) + (1-r)w_{\mathrm{o}}(y).$$

Lemma 5.42 (Word fold is well defined)**.** *The values in Definition 5.41 are independent of the choice of $x \in \Omega_j$ with $x^2 = y$.*

Proof. By Lemma 5.40, the domain Ω_j has even order, hence $-1 \in \Omega_j$. Since the field has odd characteristic, $x \neq -x$ and both square to y. Conversely $u^2 = x^2$ implies $(u-x)(u+x) = 0$, so $u = \pm x$. Replacing x with $-x$ fixes the numerator of w_{e}. It changes the signs of both the numerator and denominator of w_{o}, leaving that quotient unchanged. Thus both components and their linear combination are well defined. □

Lemma 5.43 (Evaluations split into even and odd parts)**.** *Let $P(X) = E(X^2) + XO(X^2)$ and let $w(x) = P(x)$ on Ω_j. Then for every $y \in \Omega_{j+1}$,*

$$w_{\mathrm{e}}(y) = E(y), \qquad w_{\mathrm{o}}(y) = O(y).$$

Proof. Choose $x \in \Omega_j$ with $x^2 = y$. Then $P(x) = E(y) + xO(y)$ and $P(-x) = E(y) - xO(y)$. Their sum is $2E(y)$ and their difference is $2xO(y)$. Both 2 and x are invertible in the field, giving the two equalities. □

Theorem 5.44 (Word/polynomial fold commutation)**.** *For every $P \in \mathbb{F}[X]$, $r \in \mathbb{F}$, and $0 \le j < m$,*

$$\boxed{\mathsf{WFold}_{r,j}(\mathrm{ev}_{\Omega_j}(P)) = \mathrm{ev}_{\Omega_{j+1}}(\mathsf{pfold}_r(P)).}$$

Proof. Let $P = E(X^2) + XO(X^2)$. By Lemma 5.43, the two components of the evaluation word at $y \in \Omega_{j+1}$ equal $E(y)$ and $O(y)$. Their normalized linear combination is $(2r-1)E(y)+(1-r)O(y) = \mathsf{pfold}_r(P)(y)$. This holds at every coordinate y, proving equality of words. □

Corollary 5.45 (Masked oracle fold chain)**.** *Let Q_j be the polynomials in Definition 5.32, and define $w_j := \mathrm{ev}_{\Omega_j}(Q_j)$. Then*

$$w_{j+1} = \mathsf{WFold}_{r_j,j}(w_j) \qquad (0 \le j < m).$$

Proof. The definition of iterated polynomial folds gives $Q_{j+1} = \mathsf{pfold}_{r_j}(Q_j)$. Apply Theorem 5.44 with $P = Q_j$. □

Definition 5.46 (Enlarged degree budgets)**.** For $0 \le j \le m$ put

$$N_j := 2^{m-j}, \qquad B_j := \left\lceil B/2^j \right\rceil, \qquad D_j := N_j + B_j, \qquad M_j := M/2^j.$$

The corresponding degree-bounded evaluation code is $\mathrm{RS}_{\mathbb{F}}(\Omega_j, D_j)$.

Lemma 5.47 (Budget recurrence)**.** *For $0 \le j < m$,*

$$\left\lceil \frac{D_j}{2} \right\rceil = D_{j+1}.$$

Proof. Since $N_j = 2N_{j+1}$,

$$\left\lceil \frac{D_j}{2} \right\rceil = N_{j+1} + \left\lceil \frac{B_j}{2} \right\rceil.$$

For every positive integer B and every $j \ge 0$, $\lceil \lceil B/2^j \rceil /2 \rceil = \lceil B/2^{j+1} \rceil$, as follows directly by writing $B = 2^{j+1}q + t$ with $0 \le t < 2^{j+1}$ and checking the cases $t = 0$ and $t > 0$. The second summand is therefore B_{j+1}, which proves the formula. □

Theorem 5.48 (Degree-correct masked RS chain)**.** *Assume $1 \le B \le N$ and $N + B \le M$. For every masked encoding $Q_0 = U_f^{(m)} + X^N R$ and every challenge sequence $r \in \mathbb{F}^m$,*

$$w_j \in \mathrm{RS}_{\mathbb{F}}(\Omega_j, D_j), \qquad D_j \le M_j,$$

and

$$\mathsf{WFold}_{r_j,j} : \mathrm{RS}_{\mathbb{F}}(\Omega_j, D_j) \longrightarrow \mathrm{RS}_{\mathbb{F}}(\Omega_{j+1}, D_{j+1})$$

for $j < m$.

Proof. Corollary 5.35 gives $\deg Q_j < D_j$, so w_j belongs to the stated code. Since $N + B \le M$, we have

$$D_j = N/2^j + \lceil B/2^j \rceil = \left\lceil \frac{N+B}{2^j} \right\rceil \le M/2^j = M_j,$$

where divisibility of N and M by 2^j is used. For any polynomial P of degree $< D_j$, Corollary 5.15 gives $\deg \mathsf{pfold}_{r_j}(P) < \lceil D_j/2 \rceil = D_{j+1}$. Theorem 5.44 identifies the word fold of its evaluation with the evaluation of that polynomial, which belongs to the next RS code. □

Corollary 5.49 (Exact rate transition for a power-of-two mask budget)**.** *Let $B = 2^t$ with $0 \le t \le m$, and put $\rho = N/M$. Then the relative degree budget D_j/M_j equals*

$$\frac{D_j}{M_j} = \begin{cases} \dfrac{N+B}{M}, & 0 \le j \le t, \\ \dfrac{N+2^j}{M}, & t < j \le m. \end{cases}$$

In particular $D_m = 2$ and $D_m/M_m = 2\rho$.

Proof. For $j \le t$, $B_j = \lceil 2^t/2^j \rceil = 2^{t-j}$, so $D_j = (N+B)/2^j$ and division by $M/2^j$ gives $(N+B)/M$. For $j > t$, $0 < 2^t/2^j < 1$, hence $B_j = 1$, so $D_j = N/2^j + 1$, and division gives $(N+2^j)/M$. At $j = m$, $N_m = 1$ and $B_m = 1$, so $D_m = 2$; since $M_m = M/N$, its rate is $2N/M = 2\rho$. □

Remark 5.50 (Quantified effect on proximity analysis). The original KBFold proof uses constant-rate RS chains of degree N_j. A high-degree mask replaces them by $D_j = N_j + B_j$. Theorem 5.48 proves code membership and correctness of the folds, but does not prove a proximity soundness theorem against adversarial *non-codewords*. Every argument using code distance or correlated agreement must use the actual D_j/M_j at the corresponding level.

5.6 Local fold checks and a revised final closure

We now give exact local algebraic verification statements for the enlarged-degree codes of Section 5.5. They assume that the queried words *are evaluations of degree-bounded polynomials*; proving this property for maliciously committed words requires the separate proximity analysis of KBFold [Mkh26].

Definition 5.51 (Fold-consistency residual)**.** Fix $0 \le j < m$, a challenge $r \in \mathbb{F}$, and two polynomials $P \in \mathbb{F}[X]_{<D_j}$ and $Q \in \mathbb{F}[X]_{<D_{j+1}}$. Define

$$E_{P,Q,r}(Y) := Q(Y) - \mathsf{pfold}_r(P)(Y).$$

For $y = x^2 \in \Omega_{j+1}$, the *local fold test* is the equation

$$Q(y) = (2r-1)\frac{P(x)+P(-x)}{2} + (1-r)\frac{P(x)-P(-x)}{2x}.$$

Lemma 5.52 (Local test equals a residual root)**.** *With the notation above, the local test at y passes if and only if $E_{P,Q,r}(y) = 0$.*

Proof. By Lemma 5.43, the even and odd expressions on the right side of the local test evaluate the even and odd parts of P at y. Their normalized combination equals $\mathsf{pfold}_r(P)(y)$ by Definition 2.25. Thus the local test is $Q(y) = \mathsf{pfold}_r(P)(y)$, equivalent to $E_{P,Q,r}(y) = 0$. □

Lemma 5.53 (Residual degree)**.** *The polynomial $E_{P,Q,r}$ has degree less than D_{j+1} or is zero.*

Proof. By assumption $\deg Q < D_{j+1}$. By Corollary 5.15 and Lemma 5.47, $\deg \mathsf{pfold}_r(P) < \lceil D_j/2 \rceil = D_{j+1}$. A difference of two polynomials of degree less than D_{j+1} has degree less than D_{j+1} unless it vanishes. □

Theorem 5.54 (Single-query local fold detection)**.** *Under $D_{j+1} \le M_{j+1}$, if $Q \neq \mathsf{pfold}_r(P)$ and y is uniform on Ω_{j+1}, then*

$$\Pr_y[\textit{local fold test passes}] \le \frac{D_{j+1}-1}{M_{j+1}}.$$

For $k \ge 1$ independent uniform $y_1, \dots, y_k \in \Omega_{j+1}$, the probability that all k tests pass is at most

$$\left(\frac{D_{j+1}-1}{M_{j+1}}\right)^k.$$

Proof. Since $Q \neq \mathsf{pfold}_r(P)$, the residual polynomial $E_{P,Q,r}$ is nonzero. By Lemma 5.53, its degree is at most $D_{j+1}-1$. The univariate root-count lemma, Lemma 2.7, gives at most $D_{j+1}-1$ roots in the entire field, hence at most that many in Ω_{j+1}. By Lemma 5.52, acceptance at uniform y occurs on exactly those roots, proving the first bound. Conditional on the fixed polynomials, the k query points are independent and their acceptance events depend only on the individual query values; therefore the probability that all pass is the product of the individual probabilities. Apply the first bound to every factor. □

Definition 5.55 (Linear final closure predicate)**.** Let $v \in \mathbb{F}$ be the claimed *folded* scalar at a fixed challenge vector $r \in \mathbb{F}^m$. For a final polynomial $Q \in \mathbb{F}[X]_{<2}$, define

$$\mathsf{Close}_v(Q) \quad \Longleftrightarrow \quad [X^0]Q = v.$$

The original KBFold constant-closure predicate is the stronger condition $Q = v$ as a constant polynomial. These are different predicates for a nonzero linear coefficient.

Lemma 5.56 (Honest masked closure)**.** *For every table $f : I_m \to \mathbb{F}$ and mask R of degree $< B \le N$, the final polynomial Q_m satisfies $\mathsf{Close}_{\widetilde{f}(r)}(Q_m)$ for every $r \in \mathbb{F}^m$.*

Proof. By Theorem 5.37, $Q_m = \widetilde{f}(r) + \rho_r(R)X$. Its coefficient of X^0 is $\widetilde{f}(r)$, proving the predicate. □

Lemma 5.57 (Two evaluations determine the constant of a line)**.** *Let $Q(X) = a + bX \in \mathbb{F}[X]_{<2}$ and let $x, y \in \mathbb{F}$ be distinct. Then*

$$a = \frac{yQ(x) - xQ(y)}{y-x}, \qquad b = \frac{Q(y)-Q(x)}{y-x}.$$

Proof. Expanding $yQ(x) - xQ(y)$ gives $y(a+bx) - x(a+by) = (y-x)a + (yx-xy)b = (y-x)a$. Because $x \neq y$, the scalar $y-x$ is nonzero and invertible, so division yields the formula for a. Likewise $Q(y) - Q(x) = b(y-x)$, yielding the formula for b. □

Corollary 5.58 (Two-point linear closure check)**.** *Assume $Q \in \mathbb{F}[X]_{<2}$ and choose distinct $x, y \in \Omega_m$. Then $\mathsf{Close}_v(Q)$ holds exactly when*

$$yQ(x) - xQ(y) = v(y-x).$$

Proof. By Lemma 5.57, the left side equals $(y-x)[X^0]Q$. Since $y - x \neq 0$, its equality with $v(y-x)$ is equivalent to $[X^0]Q = v$. □

Remark 5.59 (Protocol-level scope). The test of Corollary 5.58 is algebraically correct only once the final word is known to be an evaluation of a degree-< 2 polynomial. Two unauthenticated values of an arbitrary word do not certify its degree. A full ZK-KBFold protocol must enforce the degree bound and folding consistency, and then hide or correctly simulate all requested Merkle openings. The final folded scalar $\widetilde{f}(r)$ is generally *not* the initially claimed $\widetilde{f}(z)$; the original sumcheck links these values.

5.7 Correlated queries across folded oracles

The polynomials $Q_0, \dots, Q_m$ share the *same* random mask R. Consequently their oracle responses are correlated. We now give the exact joint distribution for any fixed set of observations. This extends the single-oracle masking criterion of Chapter 4, but it is not yet an online simulation theorem for a malicious adaptive verifier.

Definition 5.60 (Fixed fold-query schedule)**.** Retain the fields and domains of Section 5.5. Fix a complete challenge vector $r \in \mathbb{F}^m$ and an integer $k \geq 0$. A *fixed query schedule* is a sequence

$$\Sigma = ((j_1, \xi_1), \dots, (j_k, \xi_k)), \qquad 0 \leq j_h \leq m, \quad \xi_h \in \Omega_{j_h}.$$

The schedule is fixed independently of the mask R; it may depend on the already fixed challenge vector r.

Definition 5.61 (Witness and mask observation maps)**.** For such a fixed r and Σ, define the maps

$$\mathcal{S}_{r,\Sigma} : \mathbb{F}^{I_m} \to \mathbb{F}^{k+1}, \qquad \mathcal{T}_{r,\Sigma} : \mathbb{F}[X]_{<B} \to \mathbb{F}^{k+1}$$

by

$$\mathcal{S}_{r,\Sigma}(f) := \left(U_{f_{j_h}}^{(m-j_h)}(\xi_h)\right)_{h=1}^{k} \,\|\, (0),$$

$$\mathcal{T}_{r,\Sigma}(R) := \left(\xi_h^{2^{m-j_h}} R_{j_h}(\xi_h)\right)_{h=1}^{k} \,\|\, (\rho_r(R)),$$

where $\|$ denotes concatenation with one final coordinate, f_j and R_j are as in Definition 5.32, and $\rho_r(R)$ is the scalar in Theorem 5.37.

Lemma 5.62 (Both observation maps are linear)**.** *The maps $\mathcal{S}_{r,\Sigma}$ and $\mathcal{T}_{r,\Sigma}$ are $\mathbb{F}$-linear.*

Proof. The map $f \mapsto f_j$ is linear by Lemma 5.6; $f_j \mapsto U_{f_j}^{(m-j)}$ is linear by the finite-sum definition; evaluation at a fixed ξ_h is linear. Thus each of the first k coordinates of $\mathcal{S}_{r,\Sigma}$ is linear, and the final zero coordinate is linear. For $\mathcal{T}_{r,\Sigma}$, each fold map pfold_{r_i} is linear, so $R \mapsto R_{j_h}$ is linear. Evaluation at ξ_h and multiplication by the fixed scalar $\xi_h^{2^{m-j_h}}$ are linear. Finally $R \mapsto \rho_r(R) = \mathsf{PFold}_r^{[m]}(R)$ is linear by the same fold-linearity argument. Hence all coordinates are linear. □

Lemma 5.63 (Exact observation decomposition)**.** *Let Q_j be the masked fold polynomials. Define the observed vector*

$$\mathcal{O}_{r,\Sigma}(f, R) := (Q_{j_h}(\xi_h))_{h=1}^{k} \,\|\, (\rho_r(R)).$$

Then

$$\boxed{\mathcal{O}_{r,\Sigma}(f, R) = \mathcal{S}_{r,\Sigma}(f) + \mathcal{T}_{r,\Sigma}(R).}$$

Proof. For each query (j_h, ξ_h), Theorem 5.33 gives

$$Q_{j_h}(\xi_h) = U_{f_{j_h}}^{(m-j_h)}(\xi_h) + \xi_h^{2^{m-j_h}} R_{j_h}(\xi_h).$$

These are exactly the h-th coordinates of $\mathcal{S}_{r,\Sigma}(f)$ and $\mathcal{T}_{r,\Sigma}(R)$. The last coordinate of the first vector is 0, and the last coordinate of the second is $\rho_r(R)$, which also equals the last coordinate of $\mathcal{O}_{r,\Sigma}(f, R)$. Thus all $k+1$ coordinates agree. □

Theorem 5.64 (Exact joint fixed-schedule masking criterion)**.** *Assume $\mathbb{F}$ is finite and R is uniform on the vector space $\mathbb{F}[X]_{<B}$. For fixed r and Σ, the vector $\mathcal{O}_{r,\Sigma}(f,R)$ is uniform on the affine coset*

$$\mathcal{S}_{r,\Sigma}(f) + \operatorname{im} \mathcal{T}_{r,\Sigma} \subseteq \mathbb{F}^{k+1}.$$

For any two tables $f, g : I_m \to \mathbb{F}$,

$$\boxed{\Delta\left(\mathcal{O}_{r,\Sigma}(f,R), \mathcal{O}_{r,\Sigma}(g,R)\right) = \begin{cases} 0, & \mathcal{S}_{r,\Sigma}(f-g) \in \operatorname{im} \mathcal{T}_{r,\Sigma}, \\ 1, & \textit{otherwise.} \end{cases}}$$

Proof. By Lemma 5.62, $\mathcal{T}_{r,\Sigma}$ is linear. Lemma 5.63 expresses the random vector as the sum of a fixed vector $\mathcal{S}_{r,\Sigma}(f)$ and the image of a uniform random element of the mask space. Lemma 3.55 therefore gives the uniform coset distribution. Apply Lemma 3.56 with $V = \mathbb{F}^{k+1}$, $H = \operatorname{im} \mathcal{T}_{r,\Sigma}$, and secret vectors $\mathcal{S}_{r,\Sigma}(f)$ and $\mathcal{S}_{r,\Sigma}(g)$. Their difference equals $\mathcal{S}_{r,\Sigma}(f-g)$ by linearity, proving both cases. □

Corollary 5.65 (Sufficient condition for witness-independent fixed-schedule answers)**.** *If $\mathcal{T}_{r,\Sigma}$ is surjective onto $\mathbb{F}^{k+1}$, the joint observed vector is uniform on $\mathbb{F}^{k+1}$ for every witness f, hence independent of f.*

Proof. Surjectivity gives $\operatorname{im} \mathcal{T}_{r,\Sigma} = \mathbb{F}^{k+1}$. Its unique affine coset is the whole target space. Apply the uniform-coset conclusion of Theorem 5.64. □

Lemma 5.66 (Dimension obstruction to surjectivity)**.** *If $\mathcal{T}_{r,\Sigma}$ is surjective, then*

$$k + 1 \le B.$$

Proof. The source $\mathbb{F}[X]_{<B}$ has dimension B over $\mathbb{F}$. The target $\mathbb{F}^{k+1}$ has dimension $k+1$. The image of a linear map has dimension at most that of its source. Surjectivity would give $k + 1 = \dim \operatorname{im} \mathcal{T}_{r,\Sigma} \le B$. □

Theorem 5.67 (One-query attack on a reused degree-zero mask)**.** *Assume $\mathbb{F}$ is a finite field of odd characteristic and $\Omega_0 \le \mathbb{F}^\times$ is a multiplicative subgroup of order at least 4, containing the point 1. Let $m = 1$, $N = 2$, $B = 1$, and fix the fold challenge $r = 0$. Consider the tables $f^{(0)}, f^{(1)} : I_1 \to \mathbb{F}$ given by*

$$f^{(0)}(0) = f^{(0)}(1) = 0, \qquad f^{(1)}(0) = 0, \quad f^{(1)}(1) = 1.$$

Both satisfy the same *folded evaluation $\widetilde{f^{(0)}}(0) = \widetilde{f^{(1)}}(0) = 0$. Let $R = c$ be uniform on $\mathbb{F}[X]_{<1}$, so c is a uniform scalar. For the schedule containing the initial oracle query $Q_0(1)$ and the final slope $\rho_0(R)$, the joint observations of the two tables have total variation distance exactly 1.*

Proof. By the kernel definition for $m = 1$, $K_0(X) = X$ and $K_1(X) = X + 1$. Thus $U_{f^{(0)}} = 0$ and $U_{f^{(1)}} = X + 1$. The initial masked polynomial is $Q_0^{(0)}(X) = cX^2$ for the first table and $Q_0^{(1)}(X) = X + 1 + cX^2$ for the second. Evaluating at $X = 1$ gives, respectively, $Q_0^{(0)}(1) = c$ and $Q_0^{(1)}(1) = 2 + c$.

The normalized fold of the constant mask $R = c$ at challenge 0 is $\mathsf{pfold}_0(c) = (2 \cdot 0 - 1)c = -c$, by Lemma 5.10 with exponent zero. Therefore the final folded line has slope $\rho_0(R) = -c$ for both witnesses. For the observed pair $(a, \rho) = (Q_0(1), \rho_0(R))$, the verifier computes the deterministic statistic $a + \rho$. It equals $c - c = 0$ under $f^{(0)}$, and $(2 + c) - c = 2$ under $f^{(1)}$. Because $\operatorname{char}(\mathbb{F}) \neq 2$, these two values are distinct. The supports of the two observed-pair distributions are therefore disjoint. By Lemma 3.5, their total variation distance is 1. The common public folded scalar is zero by Boolean interpolation, Lemma 1.9. □

Corollary 5.68 (Separate uniform marginals are insufficient)**.** *In Theorem 5.67, each of $Q_0(1)$ and $\rho_0(R)$ is individually uniform on $\mathbb{F}$ for both witnesses, although their joint distributions are perfectly distinguishable.*

Proof. The variable c is uniform on $\mathbb{F}$. Each initial opening has the form $c+a$ for a fixed $a \in \{0, 2\}$, so it is uniform by Lemma 2.12. The slope is $-c$ and negation is a bijection of $\mathbb{F}$, so the slope is uniform as well. Theorem 5.67 proves the perfect distinguishability of the joint pair. □

Remark 5.69 (Final line coefficients are correlated with earlier openings)*.* The last observation $\rho_r(R)$ is a linear functional of the *same* mask that randomizes the earlier oracle values. Even if each marginal query is uniform, the joint distribution may distinguish two witnesses unless $\mathcal{S}_{r,\Sigma}(f-g)$ lies in the mask image. Theorem 5.64 is therefore the exact fixed-schedule proof obligation for a candidate simulator. When a verifier chooses later queries after seeing earlier answers, the schedule is no longer fixed; one must use the conditional-coset description in Lemma 3.57 for every adaptive history.

Remark 5.70 (Not a complete zero-knowledge theorem)*.* The observations here do not include commitment roots, Merkle authentication paths, the sumcheck round-polynomial messages, or adaptively selected challenge/query schedules. No claim that $\mathcal{T}_{r,\Sigma}$ is always surjective is made. Consequently the chapter supplies a necessary-and-sufficient linear-algebraic test for a *specified* fixed schedule, not zero knowledge for the complete KBFold protocol.

5.8 Main theorem chain and formal scope

For $N = 2^m$, $1 \le B \le N$, and $R \in \mathbb{F}[X]_{<B}$, define the high-degree masked encoding $Q_0 = U_f^{(m)} + X^N R$. The proven chain is

$$\boxed{Q_j = U^{(m-j)}_{\mathsf{Res}^{[j]}_{r_{<j}}(f)} + X^{N/2^j}\,\mathsf{PFold}^{[j]}_{r_{<j}}(R),}$$

$$\boxed{\deg Q_j < \frac{N}{2^j} + \left\lceil \frac{B}{2^j} \right\rceil,} \qquad \boxed{Q_m(X) = \widetilde{f}(r) + \rho_r(R)X.}$$

When $\mathbb{F}$ has odd characteristic and the code domain is a suitable power-of-two multiplicative group, the same polynomial identities propagate to the committed evaluation words and yield degree-correct local fold checks. For fixed challenge vectors and query schedules, the exact hiding condition is the coset-membership criterion of Theorem 5.64. Theorem 5.67 gives an explicit reused-mask attack: the initial masked opening and the final line slope each have uniform marginal distributions, but their joint observation reveals a secret-dependent value.

Lean certification boundary. Each result above has a self-contained algebraic proof, with all field and indexing hypotheses explicit. The proposed Lean targets are collected in the accompanying Chapter 5 theorem map. A proof that the full KBFold IOP is zero knowledge still requires a masked sumcheck, a simulator for adaptive authenticated openings, and soundness for malicious words in the enlarged Reed–Solomon code; these are not inferred from polynomial identities.

6 Characteristic-Two Specializations and Kernel Consistency

This chapter develops the characteristic-two algebra required to express and verify the masked correction of binary-field ring switching in Boolean-kernel coordinates. The distinction from the

original KBFold protocol is essential: KBFold's smooth multiplicative-domain word fold assumes odd characteristic, while the trace-zero and Artin–Schreier identities here assume characteristic two. We use the original Boolean-kernel basis and normalized polynomial folds established in Chapters 1, 4, and 5 [Mkh26]. The trace/Frobenius techniques should be compared with Kochergin [Koc26] and Frigo–shelat [Fs26]; we do not claim their trace-zero masking principle as new.

Every theorem is split into elementary lemmas with full proofs. The principal new algebraic statements of this chapter are the kernel-coordinate affine trace identity, the independent anchored residual basis, and a conditional two-challenge coefficient-consistency bound. The final result is *not* a zero-knowledge theorem for the whole Ring Switching reduction or for the original KBFold PCS.

6.1 Frobenius as a field and polynomial operation

Throughout this chapter, $\mathbb{F}$ denotes a field of characteristic 2. Unless explicitly assumed finite, the statements do not require that $\mathbb{F}$ have finitely many elements. For integers $j \geq 0$ we write $q_j := 2^j$.

Lemma 6.1 (Characteristic-two squaring)**.** *For all $a, b \in \mathbb{F}$, $(a+b)^2 = a^2 + b^2$.*

Proof. Distributivity gives $(a+b)^2 = a^2 + ab + ba + b^2$. Since $\mathbb{F}$ is commutative and $2 = 0$, the middle terms sum to $2ab = 0$. □

Lemma 6.2 (Iterated Frobenius additivity)**.** *For all $a, b \in \mathbb{F}$ and $j \geq 0$,*

$$(a+b)^{2^j} = a^{2^j} + b^{2^j}.$$

Proof. Induct on j. At $j = 0$ the identity is $a + b = a + b$. If it holds at j, square both sides and apply Lemma 6.1:

$$(a+b)^{2^{j+1}} = \left(a^{2^j} + b^{2^j}\right)^2 = a^{2^{j+1}} + b^{2^{j+1}}.$$

□

Definition 6.3 (Frobenius on scalars)**.** For $j \geq 0$ define $\sigma_j : \mathbb{F} \to \mathbb{F}$ by $\sigma_j(a) = a^{2^j}$.

Lemma 6.4 (Frobenius endomorphism)**.** *For $j \geq 0$, σ_j preserves $0, 1$, addition, and multiplication, and satisfies $\sigma_{i+j} = \sigma_i \circ \sigma_j$ for all $i, j \geq 0$.*

Proof. The identities for $0, 1$ are immediate. Additivity is Lemma 6.2. Multiplicativity follows from the exponent rule $(ab)^{2^j} = a^{2^j} b^{2^j}$ in a commutative field. Finally $\sigma_i(\sigma_j(a)) = (a^{2^j})^{2^i} = a^{2^{i+j}} = \sigma_{i+j}(a)$. □

Lemma 6.5 (Frobenius is injective)**.** *For every $j \geq 0$, σ_j is injective.*

Proof. If $\sigma_j(a) = \sigma_j(b)$, then by Lemma 6.2, $\sigma_j(a+b) = 0$. Because a field has no zero divisors, $x^{2^j} = 0$ implies $x = 0$; therefore $a + b = 0$, or $a = b$ in characteristic two. □

Corollary 6.6 (Frobenius permutes a finite field)**.** *If $\mathbb{F}$ is finite, every σ_j is bijective, and sends a uniform $\mathbb{F}$-valued variable to another uniform variable.*

Proof. An injective map from a finite set to itself is surjective, proving bijectivity. A bijection preserves the uniform distribution because every output has exactly one preimage. □

Definition 6.7 (Coefficient Frobenius of a polynomial)**.** For $P(X) = \sum_{e=0}^{d} p_e X^e \in \mathbb{F}[X]$, define

$$P^{\sigma_j}(X) := \sum_{e=0}^{d} p_e^{2^j} X^e.$$

This notation acts on coefficients and does not replace X by X^{2^j}.

Lemma 6.8 (Polynomial Frobenius identity)**.** *For every $P \in \mathbb{F}[X]$ and $j \geq 0$,*

$$\boxed{P(X)^{2^j} = P^{\sigma_j}(X^{2^j}).}$$

Proof. Write $P = \sum_e p_e X^e$. Repeated application of Lemma 6.1 in the polynomial ring $\mathbb{F}[X]$ gives $P(X)^{2^j} = \sum_e (p_e X^e)^{2^j}$. Exponent rules yield $\sum_e p_e^{2^j} X^{e2^j}$, which equals $\sum_e p_e^{2^j} (X^{2^j})^e = P^{\sigma_j}(X^{2^j})$ by definition. □

Lemma 6.9 (Coefficient Frobenius and products)**.** *For $P, Q \in \mathbb{F}[X]$ and $j \geq 0$,*

$$(P+Q)^{\sigma_j} = P^{\sigma_j} + Q^{\sigma_j}, \qquad (PQ)^{\sigma_j} = P^{\sigma_j} Q^{\sigma_j}.$$

Proof. The first identity follows coefficientwise from Lemma 6.2. The coefficient of X^e in PQ is $\sum_{a+b=e} p_a q_b$; applying σ_j and using its additivity and multiplicativity gives $\sum_{a+b=e} p_a^{2^j} q_b^{2^j}$, which is the coefficient of X^e in $P^{\sigma_j} Q^{\sigma_j}$. □

Remark 6.10 (Formal distinction). The identity in Lemma 6.8 is an equality of ordinary polynomials. The operation $P \mapsto P^{\sigma_j}$ and the substitution $P \mapsto P(X^{2^j})$ have different types in Lean and must not be conflated.

6.2 Frobenius stability and the Boolean kernel partition

Fix $\ell \geq 1$ and put $n := 2^\ell$. Let $I_\ell = \{0,1\}^\ell$, and let $K_b(X) = \prod_{i=0}^{\ell-1} (X^{2^i} + b_i)$ denote the Boolean kernel from Definition 1.2.

Lemma 6.11 (Binary coefficients of a kernel)**.** *For every $b \in I_\ell$, the polynomial K_b has all coefficients in the prime subfield $\mathbb{F}_2 \subseteq \mathbb{F}$.*

Proof. Every factor $X^{2^i} + b_i$ has coefficients in $\mathbb{F}_2$ because $b_i \in \{0,1\}$. A finite product of polynomials with coefficients in a subfield again has coefficients in that subfield, by the finite-sum formula for multiplication coefficients. □

Lemma 6.12 (Kernel coefficient Frobenius invariance)**.** *For $b \in I_\ell$ and $j \geq 0$,*

$$K_b^{\sigma_j}(X) = K_b(X).$$

Proof. By Lemma 6.11, each coefficient is 0 or 1. Both are fixed by σ_j, so the coefficientwise Frobenius operation leaves the polynomial unchanged. □

Theorem 6.13 (Kernel Frobenius substitution)**.** *For every $b \in I_\ell$ and $j \geq 0$,*

$$\boxed{K_b(X)^{2^j} = K_b(X^{2^j}).}$$

Proof. Lemma 6.8 gives $K_b(X)^{2^j} = K_b^{\sigma_j}(X^{2^j})$. By Lemma 6.12 the latter equals $K_b(X^{2^j})$. □

Definition 6.14 (Kernel coefficient Frobenius)**.** For a coefficient table $c : I_\ell \to \mathbb{F}$ and $j \geq 0$, define $c^{\sigma_j}(b) := c(b)^{2^j}$ and $U_c(X) := \sum_{b \in I_\ell} c(b) K_b(X)$.

Theorem 6.15 (Frobenius covariance of kernel encoding)**.** *For every coefficient table c and $j \geq 0$,*

$$\boxed{U_c(X)^{2^j} = U_{c^{\sigma_j}}(X^{2^j}).}$$

Proof. Repeatedly apply Lemma 6.1 to the finite sum defining U_c:

$$U_c(X)^{2^j} = \sum_b c(b)^{2^j} K_b(X)^{2^j}.$$

Theorem 6.13 turns each kernel factor into $K_b(X^{2^j})$, giving $\sum_b c^{\sigma_j}(b) K_b(X^{2^j})$, which is $U_{c^{\sigma_j}}(X^{2^j})$. □

Lemma 6.16 (Kernel partition sum)**.**

$$\boxed{\sum_{b \in I_\ell} K_b(X) = 1.}$$

Proof. Distribute the sum over all bit strings:

$$\sum_{b \in I_\ell} \prod_{i=0}^{\ell-1} (X^{2^i} + b_i) = \prod_{i=0}^{\ell-1} ((X^{2^i} + 0) + (X^{2^i} + 1)).$$

Inside each parenthesis, $X^{2^i} + X^{2^i} = 0$ in characteristic two, leaving the scalar 1. The product of ℓ copies of 1 is 1. □

Corollary 6.17 (All-one kernel coordinates of the constant polynomial)**.** *Let $\mathbf{1} : I_\ell \to \mathbb{F}$ be the constant table with value 1. Then $U_{\mathbf{1}}(X) = 1$. Moreover the coordinates of 1 in the kernel basis are uniquely all equal to 1.*

Proof. By definition $U_{\mathbf{1}} = \sum_b K_b = 1$, using Lemma 6.16. Uniqueness follows from the kernel basis theorem, Proposition 1.6. □

Lemma 6.18 (Two complementary lowest-bit kernels)**.** *For every $b' \in I_{\ell-1}$,*

$$K_{(0,b')}(X) + K_{(1,b')}(X) = K_{b'}^{(\ell-1)}(X^2).$$

Proof. By Lemma 1.4, $K_{(0,b')} = X K_{b'}^{(\ell-1)}(X^2)$ and $K_{(1,b')} = (X+1) K_{b'}^{(\ell-1)}(X^2)$. Add and factor out the common polynomial. The scalar $X + (X+1)$ equals 1 because $X + X = 0$. □

Remark 6.19 (One important characteristic restriction). Lemma 6.16 is false over a general odd-characteristic field. The exact all-characteristic identity is Proposition 4.23. We will use $\sum_b K_b = 1$ to identify the degeneracy of an *unanchored* residual polynomial, so the characteristic assumption cannot be dropped.

6.3 Multilinearization and the exact kernel-fold functional

We retain $\ell \geq 1$, $n = 2^\ell$, and a characteristic-two field $\mathbb{F}$. For $0 \leq e < n$ write its unique binary expansion as $e = \sum_{i=0}^{\ell-1} e_i 2^i$ with $e_i \in \{0, 1\}$.

Definition 6.20 (Binary-exponent multilinearization)**.** Let $\mathscr{ML}_\ell$ be the $\mathbb{F}$-vector space of polynomials multilinear in $Z_0, \ldots, Z_{\ell-1}$. Define the $\mathbb{F}$-linear map

$$\mathscr{M}_\ell : \mathbb{F}[X]_{<n} \longrightarrow \mathscr{ML}_\ell, \qquad \mathscr{M}_\ell\Big(\sum_{e=0}^{n-1} a_e X^e\Big) := \sum_{e=0}^{n-1} a_e \prod_{i=0}^{\ell-1} Z_i^{e_i}.$$

Lemma 6.21 (Multilinearization is a vector-space isomorphism)**.** *The map $\mathscr{M}_\ell$ is an $\mathbb{F}$-linear bijection.*

Proof. Linearity is part of its definition. The monomials X^e, $0 \leq e < n$, form a basis of $\mathbb{F}[X]_{<n}$. The monomials $\prod_i Z_i^{e_i}$ indexed by the binary digits of $0 \leq e < n$ form a basis of $\mathscr{ML}_\ell$. The binary expansion gives a bijection between these two bases. The map sends each element of the first basis to the corresponding element of the second, hence is an isomorphism. □

Lemma 6.22 (Explicit subset expansion of a kernel)**.** *For $b \in I_\ell$,*

$$K_b(X) = \sum_{S \subseteq \{0,\ldots,\ell-1\}} \left(\prod_{i \notin S} b_i\right) X^{\sum_{i \in S} 2^i}.$$

Proof. In the product $K_b = \prod_i (X^{2^i} + b_i)$, a term is obtained by selecting X^{2^i} from a subset S of factors and b_i from the complementary factors. The resulting term has scalar coefficient $\prod_{i \notin S} b_i$ and monomial factor $X^{\sum_{i \in S} 2^i}$. Every subset occurs exactly once in the full distributive expansion, establishing the equality. □

Lemma 6.23 (Multilinearization of every kernel basis element)**.** *For every $b \in I_\ell$,*

$$\boxed{\mathscr{M}_\ell(K_b)(Z) = \prod_{i=0}^{\ell-1} (Z_i + b_i).}$$

Proof. By Lemma 6.22 and linearity,

$$\mathscr{M}_\ell(K_b) = \sum_S \left(\prod_{i \notin S} b_i\right) \mathscr{M}_\ell\left(X^{\sum_{i \in S} 2^i}\right).$$

The binary digits of $\sum_{i \in S} 2^i$ equal 1 precisely for $i \in S$, so the last factor is $\prod_{i \in S} Z_i$. Therefore the sum equals $\sum_S (\prod_{i \notin S} b_i)(\prod_{i \in S} Z_i)$. The distributive expansion of $\prod_i (Z_i + b_i)$ is the same sum. This argument uses only linearity of $\mathscr{M}_\ell$, not multiplicativity. □

Definition 6.24 (Characteristic-two equality weights)**.** For $b \in I_\ell$ and $\theta = (\theta_0, \ldots, \theta_{\ell-1}) \in \mathbb{F}^\ell$, define

$$\lambda_b(\theta) := \prod_{i=0}^{\ell-1} (1 + \theta_i + b_i).$$

For $P = \sum_b c_b K_b$ with its unique kernel coefficient table, define

$$D_P(\theta) := \sum_b c_b \lambda_b(\theta).$$

Lemma 6.25 (Equality weights agree)**.** *For all $b \in I_\ell$ and $\theta \in \mathbb{F}^\ell$, $\lambda_b(\theta) = \mathrm{eq}_\ell(b, \theta)$.*

Proof. For each coordinate i, if $b_i = 0$ then $1 + \theta_i + b_i = 1 + \theta_i = 1 - \theta_i$. If $b_i = 1$ then $1 + \theta_i + b_i = \theta_i$, because $1 + 1 = 0$. These are precisely the two cases of $(1 - b_i)(1 - \theta_i) + b_i\theta_i$. Multiply the coordinatewise identities. □

Lemma 6.26 (Boolean weights partition unity)**.** *For every $\theta \in \mathbb{F}^\ell$, $\sum_{b \in I_\ell} \lambda_b(\theta) = 1$.*

Proof. Factor the sum over bit strings into a product of two-term sums:

$$\sum_b \lambda_b(\theta) = \prod_{i=0}^{\ell-1} ((1 + \theta_i + 0) + (1 + \theta_i + 1)) = \prod_i ((1 + \theta_i) + \theta_i) = 1.$$

□

Theorem 6.27 (Complete kernel fold equals multilinearized evaluation)**.** *For every $P \in \mathbb{F}[X]_{<n}$ and $\theta \in \mathbb{F}^\ell$,*

$$\boxed{\mathsf{PFold}_\theta^{[\ell]}(P) = D_P(\theta) = \mathscr{M}_\ell(P)(1 + \theta_0, \ldots, 1 + \theta_{\ell-1}).}$$

Proof. By Proposition 1.6, uniquely write $P = \sum_b c_b K_b$. The fold dictionary, Corollary 5.24, identifies its complete fold with $\sum_b c_b \,\mathrm{eq}_\ell(b, \theta)$, which equals $D_P(\theta)$ by Lemma 6.25. For the second equality, apply the linear map $\mathscr{M}_\ell$ to the kernel expansion of P and evaluate:

$$\mathscr{M}_\ell(P)(1 + \theta) = \sum_b c_b \,\mathscr{M}_\ell(K_b)(1 + \theta).$$

Lemma 6.23 evaluates each summand as $\prod_i (1 + \theta_i + b_i) = \lambda_b(\theta)$. Thus the sum is precisely $D_P(\theta)$. □

Definition 6.28 (Partial multilinearization)**.** For $0 \le j \le \ell$, define

$$\mathscr{M}_{\ell,j}(P)(Z_0, \ldots, Z_{j-1}, Y) := \sum_{e=0}^{n-1} a_e \left(\prod_{i=0}^{j-1} Z_i^{e_i} \right) Y^{\lfloor e/2^j \rfloor}$$

for $P(X) = \sum_{e=0}^{n-1} a_e X^e$.

Lemma 6.29 (Partial folding formula)**.** *For $0 \le j \le \ell$ and $r \in \mathbb{F}^j$,*

$$\boxed{\mathsf{PFold}_r^{[j]}(P)(Y) = \mathscr{M}_{\ell,j}(P)(1 + r_0, \ldots, 1 + r_{j-1}, Y).}$$

Proof. The explicit fold formula of Theorem 5.14 gives a summand $a_e(\prod_{i<j} A_{e_i}(r_i))Y^{\lfloor e/2^j \rfloor}$. In characteristic two, $A_0(r_i) = 2r_i - 1 = 1$ and $A_1(r_i) = 1 - r_i = 1 + r_i$. Thus $\prod_{i<j} A_{e_i}(r_i) = \prod_{i<j}(1 + r_i)^{e_i}$. Substitution produces exactly the evaluation of Definition 6.28. □

Lemma 6.30 (Frobenius covariance of equality weights)**.** *For $b \in I_\ell$, $\theta \in \mathbb{F}^\ell$, and $j \ge 0$,*

$$\lambda_b(\theta)^{2^j} = \lambda_b(\theta_0^{2^j}, \ldots, \theta_{\ell-1}^{2^j}).$$

Proof. Apply σ_j to the product $\prod_i (1 + \theta_i + b_i)$, using the ring-homomorphism property of Lemma 6.4. Since 1 and $b_i \in \mathbb{F}_2$ are fixed by σ_j, each factor becomes $1 + \theta_i^{2^j} + b_i$. Their product is the equality weight on the right. □

Corollary 6.31 (Frobenius covariance of kernel-fold values)**.** *For $c : I_\ell \to \mathbb{F}$, $\theta \in \mathbb{F}^\ell$, and $j \geq 0$,*

$$\boxed{D_{U_c}(\theta)^{2^j} = D_{U_{c^{\sigma_j}}}(\theta^{\sigma_j}).}$$

Proof. Raise the finite sum $D_{U_c}(\theta) = \sum_b c(b)\lambda_b(\theta)$ to its 2^j-th power. By Lemma 6.2 and multiplicativity, it becomes $\sum_b c(b)^{2^j}\lambda_b(\theta)^{2^j}$. Apply Lemma 6.30 and Definition 6.14. □

Remark 6.32 (Nonmultiplicativity warning). The multilinearization map is *not* a ring homomorphism, even where both sides happen to be defined. For $\ell \geq 2$, $\mathscr{M}_\ell(X^2) = Z_1$ while $\mathscr{M}_\ell(X)\mathscr{M}_\ell(X) = Z_0^2$. The polynomials Z_1 and Z_0^2 are distinct. All foregoing kernel identities were proved from expansions and the linearity of $\mathscr{M}_\ell$.

6.4 Finite binary extensions, trace, and Artin–Schreier masks

For the rest of the chapter, specialize to $L = \mathbb{F}_{2^n}$ with $n = 2^\ell$ and $\ell \geq 1$. The statements that mention uniformity use the finite uniform distribution from Definition 3.1. We write $q := |L| = 2^n$.

Lemma 6.33 (Frobenius has period n)**.** *For every $a \in L$, $a^{2^n} = a$.*

Proof. If $a = 0$, the identity is immediate. Suppose $a \neq 0$. Multiplication by a is a bijection of $L^\times$, whose cardinality is $q - 1$. Define $P := \prod_{x \in L^\times} x$, a nonzero element of L. Reindexing the product under the bijection $x \mapsto ax$ gives

$$P = \prod_{x \in L^\times} (ax) = a^{q-1} \prod_{x \in L^\times} x = a^{q-1} P.$$

Cancel the nonzero element P, obtaining $a^{q-1} = 1$. Multiplying by a yields $a^q = a$, with $q = 2^n$. □

Definition 6.34 (Absolute trace)**.** For $x \in L$ define

$$\mathrm{Tr}(x) := \sum_{j=0}^{n-1} x^{2^j}.$$

Initially the sum is taken in L; the next lemma proves its value lies in the prime subfield $\mathbb{F}_2$.

Lemma 6.35 (Trace is additive and Frobenius-invariant)**.** *For every $x, y \in L$,*

$$\mathrm{Tr}(x + y) = \mathrm{Tr}(x) + \mathrm{Tr}(y), \qquad \mathrm{Tr}(x^2) = \mathrm{Tr}(x).$$

In particular, Tr *is $\mathbb{F}_2$-linear.*

Proof. The first equality follows by applying Lemma 6.2 to each term of the finite sum. For the second, write

$$\mathrm{Tr}(x^2) = x^2 + x^{2^2} + \cdots + x^{2^n}.$$

By Lemma 6.33, the last term is x. Thus the sum is a cyclic permutation of the terms in $\mathrm{Tr}(x)$. Linearity for the scalar 0 is immediate, and for scalar 1 it is the identity $\mathrm{Tr}(1x) = \mathrm{Tr}(x)$. □

Lemma 6.36 (Trace takes binary values)**.** *For every $x \in L$, $\mathrm{Tr}(x) \in \mathbb{F}_2 = \{0, 1\}$.*

Proof. By Lemma 6.35, $\mathrm{Tr}(x)^2 = \mathrm{Tr}(x^2) = \mathrm{Tr}(x)$. Hence $\mathrm{Tr}(x)$ is a root of $T^2 + T = T(T + 1)$. A field has no zero divisors, so the only possibilities are $T = 0$ and $T = 1$, both in $\mathbb{F}_2$. □

Lemma 6.37 (Trace is surjective)**.** *The linear map* $\mathrm{Tr} : L \to \mathbb{F}_2$ *is nonzero and therefore surjective.*

Proof. Consider the polynomial $H(X) = X + X^2 + \cdots + X^{2^{n-1}} \in L[X]$. It is nonzero: its coefficient of X is 1, and all higher exponents are distinct. Its degree 2^{n-1} is strictly less than $q = 2^n$. If $H(x) = 0$ for all $x \in L$, this nonzero polynomial would have q distinct roots, contradicting Lemma 2.7. Thus some x satisfies $\mathrm{Tr}(x) = H(x) \neq 0$. By Lemma 6.36, that value equals 1. Therefore the image of Tr contains both 0 and 1. □

Corollary 6.38 (Sizes of the two trace fibers)**.** *The sets* $\ker \mathrm{Tr}$ *and* $\{x \in L : \mathrm{Tr}(x) = 1\}$ *each have* $q/2$ *elements.*

Proof. Choose $a \in L$ with $\mathrm{Tr}(a) = 1$ by Lemma 6.37. Translation $x \mapsto x + a$ is a bijection and, by trace additivity, sends elements of trace zero exactly to elements of trace one and vice versa. The two fibers partition L, and hence have equal cardinality $q/2$. □

Definition 6.39 (Artin–Schreier operator)**.** Define $\wp : L \to L$ by

$$\wp(t) := t^2 + t.$$

Lemma 6.40 (Artin–Schreier is binary-linear)**.** *For all* $s, t \in L$, $\wp(s + t) = \wp(s) + \wp(t)$.

Proof. Expand $(s + t)^2 + s + t$ and apply Lemma 6.1. The result is $(s^2 + s) + (t^2 + t)$. □

Lemma 6.41 (Artin–Schreier kernel)**.** $\ker \wp = \{0, 1\}$.

Proof. $\wp(t) = t^2 + t = t(t + 1)$. This product is zero exactly when $t = 0$ or $t + 1 = 0$, the latter being $t = 1$ in characteristic two. Both values satisfy the equation. □

Lemma 6.42 (Every Artin–Schreier value has trace zero)**.** *For every* $t \in L$, $\mathrm{Tr}(\wp(t)) = 0$.

Proof. By trace additivity and Frobenius invariance, $\mathrm{Tr}(t^2 + t) = \mathrm{Tr}(t^2) + \mathrm{Tr}(t) = \mathrm{Tr}(t) + \mathrm{Tr}(t) = 0$. □

Theorem 6.43 (Artin–Schreier characterizes trace zero)**.**

$$\boxed{\operatorname{im} \wp = \ker \mathrm{Tr}\,.}$$

Every element of $\ker \mathrm{Tr}$ *has exactly two preimages under* $\wp$.

Proof. Lemma 6.42 gives $\operatorname{im} \wp \subseteq \ker \mathrm{Tr}$. By Lemma 6.40, the nonempty fiber above any $y = \wp(t)$ is the coset $t + \ker \wp$: indeed $\wp(s) = \wp(t)$ is equivalent to $\wp(s + t) = 0$ and hence $s + t \in \ker \wp$. By Lemma 6.41, every nonempty fiber contains exactly two elements. Thus $|\operatorname{im} \wp| = q/2$. Corollary 6.38 shows $|\ker \mathrm{Tr}| = q/2$. The inclusion between equal-sized finite sets is an equality. □

Corollary 6.44 (Uniform trace-zero masking)**.** *If* T *is uniform on* L*, then* $\wp(T)$ *is uniform on* $\ker \mathrm{Tr}$.

Proof. By Theorem 6.43, every trace-zero element has exactly two preimages. Therefore its probability as the output $\wp(T)$ is $2/q = 1/(q/2)$, independent of the chosen trace-zero element. □

Theorem 6.45 (Perfect hiding of an individual trace-zero shift)**.** *Fix* $v \in \ker \mathrm{Tr}$ *and let* T *be uniform on* L*. Then*

$$U := v + \wp(T)$$

is uniform on $\ker \mathrm{Tr}$*, independently of* v.

Proof. Corollary 6.44 makes $\wp(T)$ uniform on the additive subgroup $\ker \mathrm{Tr}$. Since v lies in that subgroup, translation $y \mapsto v + y$ is a bijection of $\ker \mathrm{Tr}$, so U is uniform there. Its distribution is therefore independent of the fixed value v. □

Remark 6.46 (No joint-ZK conclusion). Theorem 6.45 concerns one public field element. If the same T is subsequently used in another published equation, the joint transcript can have correlations. The complete verification protocol must analyze those correlations, as illustrated by the reused-mask attack of Theorem 5.67.

6.5 Trace-dual coordinates and affine verification of kernel corrections

All vector indices in this subsection belong to I_ℓ and all field elements belong to $L = \mathbb{F}_{2^n}$, $n = 2^\ell$. The characteristic-two coordinate construction is related to binary-field ring switching [DP24] and to its masked versions [Koc26; Fs26].

Lemma 6.47 (Existence of an element of full binary degree). *There exists $\alpha \in L = \mathbb{F}_{2^n}$ such that $[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = n$.*

Proof. For any $a \in L$, the tower law applied to $\mathbb{F}_2 \subseteq \mathbb{F}_2(a) \subseteq L$ shows that $d := [\mathbb{F}_2(a) : \mathbb{F}_2]$ divides n. If every a had $d < n$, then, because $n = 2^\ell$, every such d would divide $n/2$. The field $\mathbb{F}_2(a)$ has exactly 2^d elements. Repeating the product-permutation proof of Lemma 6.33 in this finite field gives $a^{2^d} = a$. Iterating Frobenius then yields $a^{2^{n/2}} = a$ for every $a \in L$. This would make all 2^n elements of L roots of $X^{2^{n/2}} + X$, a nonzero polynomial of degree $2^{n/2} < 2^n$. It contradicts the root-count lemma, Lemma 2.7. Therefore some α has degree n. □

Definition 6.48 (Evaluation basis). Choose $\alpha \in L$ of degree n over $\mathbb{F}_2$, whose existence is guaranteed by Lemma 6.47, and set

$$\beta_u := K_u(\alpha), \qquad u \in I_\ell.$$

Lemma 6.49 (Kernel values form a binary field basis). *The family $(\beta_u)_{u \in I_\ell}$ is an $\mathbb{F}_2$-basis of L.*

Proof. The polynomials $(K_u)_u$ form an $\mathbb{F}_2$-basis of $\mathbb{F}_2[X]_{<n}$ by Proposition 1.6. The evaluation map $P \mapsto P(\alpha)$ is injective on that space: if a nonzero polynomial of degree $< n$ vanished at α, it would contradict the minimality of the degree-n minimal polynomial. Both domain and codomain have dimension n over $\mathbb{F}_2$. Thus the map is an isomorphism, and the images of the kernel basis form a basis of L. □

Definition 6.50 (Binary coordinate maps). For each $u \in I_\ell$, define the unique $\mathbb{F}_2$-linear coordinate map $d_u : L \to \mathbb{F}_2$ by

$$x = \sum_{u \in I_\ell} \beta_u d_u(x) \quad \text{for every } x \in L.$$

Lemma 6.51 (Coordinate extraction). *For $u, v \in I_\ell$, $d_u(\beta_v) = 1$ if $u = v$ and 0 otherwise; each d_u is additive.*

Proof. The kernel-basis expansion of β_v has coefficient 1 at index v and 0 elsewhere; uniqueness of coordinates gives the first assertion. If $x = \sum_v \beta_v a_v$ and $y = \sum_v \beta_v b_v$, with $a_v, b_v \in \mathbb{F}_2$, then $x + y = \sum_v \beta_v (a_v + b_v)$. Uniqueness gives $d_u(x + y) = a_u + b_u = d_u(x) + d_u(y)$. □

Lemma 6.52 (Trace pairing is nondegenerate). *For every nonzero $a \in L$ there exists $x \in L$ with $\mathrm{Tr}(ax) = 1$.*

Proof. Choose $y \in L$ with $\mathrm{Tr}(y) = 1$, using Lemma 6.37. As $a \neq 0$, a^{-1} exists. Put $x = a^{-1}y$; then $\mathrm{Tr}(ax) = \mathrm{Tr}(y) = 1$. □

Theorem 6.53 (Existence and uniqueness of trace-dual coordinates)**.** *There is a unique family* $(\eta_u)_{u \in I_\ell}$ *in* L *such that*

$$\boxed{\mathrm{Tr}(\eta_u \beta_v) = \delta_{uv}.}$$

For every $x \in L$*,*

$$\boxed{d_u(x) = \mathrm{Tr}(\eta_u x).}$$

Proof. Define the $\mathbb{F}_2$-linear map $\Phi : L \to \mathrm{Hom}_{\mathbb{F}_2}(L, \mathbb{F}_2)$ by $\Phi(a)(x) = \mathrm{Tr}(ax)$. Linearity follows from Lemma 6.35. Lemma 6.52 proves that Φ is injective. Its domain has dimension n over $\mathbb{F}_2$; its codomain also has dimension n, since a linear functional is uniquely determined by its values on the n basis elements β_v. Hence Φ is bijective. For each coordinate functional d_u, choose the unique $\eta_u = \Phi^{-1}(d_u)$. Then $\mathrm{Tr}(\eta_u \beta_v) = d_u(\beta_v) = \delta_{uv}$ by Lemma 6.51, and $\mathrm{Tr}(\eta_u x) = d_u(x)$ for all x by construction. Uniqueness of each η_u follows from injectivity of Φ. □

Definition 6.54 (Binary transpose and kernel correction)**.** For $p = (p_v)_{v \in I_\ell} \in L^{I_\ell}$ define

$$c_u^p := \sum_{v \in I_\ell} \beta_v d_u(p_v), \qquad C_p(X) := \sum_{u \in I_\ell} c_u^p K_u(X).$$

For $\theta \in L^\ell$ define

$$\Delta_p(\theta) := \sum_{u \in I_\ell} \lambda_u(\theta) c_u^p = D_{C_p}(\theta).$$

Lemma 6.55 (Trace expression for the correction)**.** *For every* $p \in L^{I_\ell}$ *and* $\theta \in L^\ell$*,*

$$\boxed{\Delta_p(\theta) = \sum_{u,v \in I_\ell} \lambda_u(\theta) \beta_v \, \mathrm{Tr}(\eta_u p_v).}$$

Proof. Expand $\Delta_p(\theta)$ using Definition 6.54:

$$\Delta_p(\theta) = \sum_u \lambda_u(\theta) \sum_v \beta_v d_u(p_v).$$

Interchange the two finite sums and replace $d_u(p_v)$ by $\mathrm{Tr}(\eta_u p_v)$ using Theorem 6.53. □

Definition 6.56 (Public affine coefficients)**.** For $\theta \in L^\ell$ and $u, v, k \in I_\ell$ put

$$H_{v,k}(\theta) := \sum_{u \in I_\ell} d_k(\beta_v \lambda_u(\theta)) \eta_u.$$

For a candidate coefficient table $c : I_\ell \to L$ define

$$D_c(\theta) := \sum_u \lambda_u(\theta) c_u$$

and

$$V_k(p, c, \theta) := \eta_k D_c(\theta) + \sum_{v \in I_\ell} H_{v,k}(\theta) p_v.$$

The coefficients $H_{v,k}(\theta)$ depend only on public parameters and θ, not on the hidden pads p_v.

Lemma 6.57 (Affine transpose trace identity)**.** *For every p, θ, k,*

$$\mathrm{Tr}\left(\sum_v H_{v,k}(\theta) p_v\right) = d_k(\Delta_p(\theta)).$$

Proof. Expand the definition of H and use $\mathbb{F}_2$-linearity of trace:

$$\mathrm{Tr}\left(\sum_v H_{v,k} p_v\right) = \sum_{u,v} d_k(\beta_v \lambda_u(\theta))\ \mathrm{Tr}(\eta_u p_v).$$

Each factor $\mathrm{Tr}(\eta_u p_v)$ lies in $\mathbb{F}_2$. Using additivity and $\mathbb{F}_2$-linearity of d_k, the last sum equals

$$d_k\left(\sum_{u,v} \beta_v \lambda_u(\theta)\,\mathrm{Tr}(\eta_u p_v)\right).$$

The field element inside d_k equals $\Delta_p(\theta)$ by Lemma 6.55. □

Lemma 6.58 (Trace coordinates of affine residuals)**.** *For all p, c, θ and k,*

$$\boxed{\mathrm{Tr}(V_k(p, c, \theta)) = d_k(D_c(\theta) + \Delta_p(\theta)).}$$

Proof. By the definition of V_k and trace additivity,

$$\mathrm{Tr}(V_k) = \mathrm{Tr}(\eta_k D_c(\theta)) + \mathrm{Tr}\left(\sum_v H_{v,k} p_v\right).$$

The first term equals $d_k(D_c(\theta))$ by Theorem 6.53; the second equals $d_k(\Delta_p(\theta))$ by Lemma 6.57. Additivity of d_k combines them into the claimed formula. □

Theorem 6.59 (Exact affine trace consistency)**.** *For fixed p, c, θ,*

$$\boxed{D_c(\theta) = \Delta_p(\theta) \iff \mathrm{Tr}(V_k(p, c, \theta)) = 0 \quad \forall k \in I_\ell.}$$

Proof. In characteristic two the equality $D_c(\theta) = \Delta_p(\theta)$ is equivalent to $D_c(\theta) + \Delta_p(\theta) = 0$. Lemma 6.58 identifies $\mathrm{Tr}(V_k)$ with the coordinate $d_k(D_c + \Delta_p)$. A field element is zero exactly when every coordinate in the basis (β_k) is zero, by Definition 6.50. This proves both directions. □

Lemma 6.60 (Mismatch gives a nonzero multilinear polynomial)**.** *If $c \neq c^p$ as tables, then*

$$E_{c,p}(\Theta) := D_c(\Theta) + \Delta_p(\Theta) = \sum_{u \in I_\ell} (c_u + c^p_u) \lambda_u(\Theta)$$

is a nonzero polynomial in $L[\Theta_0, \ldots, \Theta_{\ell-1}]$ of total degree at most ℓ.

Proof. Each λ_u is a product of ℓ affine factors, giving the degree bound. Choose u_0 with $c_{u_0} \neq c^p_{u_0}$. Evaluate at the Boolean point $\Theta = u_0$. By Lemma 1.9 and Lemma 6.25, $E_{c,p}(u_0) = c_{u_0} + c^p_{u_0} \neq 0$. Hence the polynomial is nonzero. □

Corollary 6.61 (Random-challenge detection of a coefficient mismatch)**.** *If $c \neq c^p$ is fixed before uniform $\theta \in L^\ell$ is sampled, then*

$$\Pr_\theta[D_c(\theta) = \Delta_p(\theta)] \leq \min\{1, \ell/q\}, \qquad q = 2^n.$$

Proof. By Lemma 6.60, $E_{c,p}$ is nonzero of total degree at most ℓ. The event on the left is $E_{c,p}(\theta) = 0$. Apply the multivariate Schwartz–Zippel bound proved in Lemma 2.8. The fixed-before-challenge hypothesis ensures the polynomial is independent of the uniform θ. □

Remark 6.62 (Meaning of this part of the construction). The coefficients $H_{v,k}(\theta)$ turn the $\mathbb{F}_2$-linear coordinate relation into *affine equations over the extension field*. This is an algebraic ingredient of our proposed kernel-coordinate checker; it does not by itself establish hiding of a committed witness or the soundness of the complete ring-switching protocol.

6.6 Public trace-zero messages and their algebraic residuals

Fix p, c, θ as in Section 6.5. The trace-zero mask of Frigo–shelat [Fs26] is a relevant prior algebraic mechanism; the following results make its conditional distributions explicit for our particular kernel-correction coefficients.

Definition 6.63 (Artin–Schreier-masked public values)**.** For a mask vector $t = (t_k)_{k \in I_\ell} \in L^{I_\ell}$ define

$$U_k(p, c, \theta, t) := V_k(p, c, \theta) + t_k^2 + t_k.$$

When t is random, its coordinates are sampled independently and uniformly from L unless specified otherwise.

Lemma 6.64 (Masking preserves the trace of each residual)**.** *For every $k \in I_\ell$,*

$$\mathrm{Tr}(U_k) = \mathrm{Tr}(V_k) = d_k(D_c(\theta) + \Delta_p(\theta)).$$

Proof. By trace additivity and Lemma 6.42, $\mathrm{Tr}(U_k) = \mathrm{Tr}(V_k) + \mathrm{Tr}(t_k^2 + t_k) = \mathrm{Tr}(V_k)$. Apply Lemma 6.58 to obtain the second equality. □

Corollary 6.65 (Consistency iff all public traces vanish)**.** *For fixed p, c, θ, t,*

$$D_c(\theta) = \Delta_p(\theta) \quad \Longleftrightarrow \quad \mathrm{Tr}(U_k) = 0 \quad \forall k \in I_\ell.$$

Proof. Combine Lemma 6.64 with Theorem 6.59. □

Theorem 6.66 (Joint uniformity of independently masked traces)**.** *Suppose $D_c(\theta) = \Delta_p(\theta)$. If the coordinates t_k are independent uniform samples in L, then the vector $(U_k)_{k \in I_\ell}$ is uniform on $(\ker \mathrm{Tr})^{I_\ell}$. Its distribution does not depend on the fixed values p, c, θ once the consistency equality holds.*

Proof. By Theorem 6.59, every V_k has trace zero. The random variable $U_k = V_k + \wp(t_k)$ is therefore uniform on $\ker \mathrm{Tr}$ by Theorem 6.45. The independent masks t_k are transformed separately to produce each U_k; hence the U_k are independent. Their joint law is the product of the uniform distributions, which is the uniform distribution on the Cartesian product $(\ker \mathrm{Tr})^{I_\ell}$. This product law is independent of the fixed shifts V_k, and therefore of fixed p, c, θ satisfying consistency. □

Definition 6.67 (Algebraic consistency residual)**.** For arbitrary public values $U = (U_k)_{k \in I_\ell} \in L^{I_\ell}$ and an arbitrary mask vector t, set

$$e_k := U_k + V_k(p, c, \theta) + t_k^2 + t_k.$$

Thus the honest masked value of Definition 6.63 satisfies $e_k = 0$.

Lemma 6.68 (Trace of an algebraic residual)**.** *For every k,*

$$\mathrm{Tr}(e_k) = \mathrm{Tr}(U_k) + d_k(D_c(\theta) + \Delta_p(\theta)).$$

Proof. By trace additivity, $\mathrm{Tr}(e_k) = \mathrm{Tr}(U_k) + \mathrm{Tr}(V_k) + \mathrm{Tr}(\wp(t_k))$. The last summand is zero by Lemma 6.42 and the middle summand is $d_k(D_c + \Delta_p)$ by Lemma 6.58. □

Lemma 6.69 (Trace checks force a nonzero residual under mismatch)**.** *If $D_c(\theta) \neq \Delta_p(\theta)$ and $\mathrm{Tr}(U_k) = 0$ for all k, then $e_k \neq 0$ for at least one k.*

Proof. Since $D_c(\theta) + \Delta_p(\theta) \neq 0$, at least one binary coordinate $d_k(D_c(\theta) + \Delta_p(\theta))$ equals 1, by Definition 6.50. Lemma 6.68 and the trace-zero assumptions give $\mathrm{Tr}(e_k) = 1$ for this k. As $\mathrm{Tr}(0) = 0$, necessarily $e_k \neq 0$. □

Remark 6.70 (Conditional, not full-transcript hiding). Theorem 6.66 asserts uniformity of the public vector U for a *fixed* consistent instance, assuming independent uniform masks. It does not prove that a later public linear combination of the same masks t_k, combined with U, remains witness-independent. Such correlations must be checked in the full protocol simulation game defined in Chapter 3.

6.7 The unanchored dependence and the anchored kernel family

The identity $\sum_b K_b = 1$ creates an exact linear dependence between the constant polynomial and the kernel basis. This makes a naive residual polynomial identity test unsound unless the additional scalar residual is distinguished algebraically. We derive the minimal-degree anchoring mechanism; a full protocol using it will appear later.

Lemma 6.71 (Unanchored residual dependence)**.** *Let $\delta \in L$ be nonzero and set $e_b := \delta$ for every $b \in I_\ell$. Then*

$$\delta + \sum_{b \in I_\ell} e_b K_b(X) = 0$$

as a polynomial, even though $(\delta, e) \neq 0$.

Proof. By Lemma 6.16, $\sum_b K_b = 1$. Therefore the left side is $\delta + \delta \sum_b K_b = \delta + \delta = 0$ in characteristic two. The first residual component δ is nonzero by assumption. □

Definition 6.72 (Anchored Boolean kernels)**.** Let $0_\ell = (0, \ldots, 0) \in I_\ell$. For $b \in I_\ell$ define

$$W_b(X) := \begin{cases} K_{0_\ell}(X) + X^n, & b = 0_\ell, \\ K_b(X), & b \neq 0_\ell. \end{cases}$$

Every W_b is a polynomial with coefficients in $\mathbb{F}_2$.

Lemma 6.73 (Anchored family degree)**.** $\deg W_{0_\ell} = n$*, while* $\deg W_b = n - 1$ *for every $b \neq 0_\ell$.*

Proof. By Lemma 1.3, each kernel has degree $n - 1$. Therefore the added monomial X^n is the unique degree-n contribution to W_{0_ℓ}, with nonzero coefficient 1, so its degree is n. For $b \neq 0_\ell$, $W_b = K_b$, which has degree $n - 1$. □

Lemma 6.74 (Sum of anchored kernels)**.**

$$\sum_{b \in I_\ell} W_b(X) = 1 + X^n.$$

Proof. By definition only the term W_{0_ℓ} differs from K_{0_ℓ}, by the summand X^n. Thus $\sum_b W_b = \sum_b K_b + X^n = 1 + X^n$ by Lemma 6.16. □

Theorem 6.75 (Independence of the constant and all anchored kernels)**.** *The $n+1$ polynomials*

$$\boxed{\{1\} \cup \{W_b : b \in I_\ell\}}$$

are linearly independent over L in $L[X]_{\le n}$.

Proof. Suppose

$$\delta + \sum_{b \in I_\ell} e_b W_b(X) = 0 \qquad (\delta, e_b \in L).$$

The coefficient of X^n in the left side is exactly e_{0_ℓ} by Lemma 6.73. Hence $e_{0_\ell} = 0$. The remaining polynomial identity is $\delta + \sum_{b \neq 0_\ell} e_b K_b = 0$. Using $1 = \sum_b K_b$ from Lemma 6.16, rewrite it as

$$\delta K_{0_\ell} + \sum_{b \neq 0_\ell} (\delta + e_b) K_b = 0.$$

By the kernel basis theorem, Proposition 1.6, all kernel coefficients vanish. In particular $\delta = 0$, and then $\delta + e_b = 0$ gives every $e_b = 0$. Thus the anchored family together with 1 is independent. □

Corollary 6.76 (Anchored family is a basis in degree at most n)**.** *The $n+1$ polynomials from Theorem 6.75 form an L-basis of $L[X]_{\le n}$.*

Proof. The monomials $1, X, \ldots, X^n$ form a basis of $L[X]_{\le n}$, which therefore has dimension $n+1$. The displayed family consists of $n+1$ linearly independent vectors in this space, hence is a basis. □

Proposition 6.77 (Degree-n anchoring is dimensionally minimal)**.** *No family of $n+1$ polynomials, all of degree $< n$, can be linearly independent over L.*

Proof. The space $L[X]_{<n}$ has basis $1, X, \ldots, X^{n-1}$ and therefore dimension n. A vector space of dimension n cannot contain $n+1$ linearly independent vectors. □

Lemma 6.78 (Anchored kernels commute with Frobenius evaluation)**.** *For every $b \in I_\ell$ and $z \in L$,*

$$W_b(z)^2 = W_b(z^2).$$

Proof. Every coefficient of W_b lies in $\mathbb{F}_2$ by Definition 6.72. The coefficient Frobenius of W_b is therefore W_b. Apply Lemma 6.8 with $j = 1$ and evaluate at z: $W_b(z)^2 = W_b^{\sigma_1}(z^2) = W_b(z^2)$. □

Definition 6.79 (Anchored residual polynomial)**.** For $\delta \in L$ and $e = (e_b)_{b \in I_\ell} \in L^{I_\ell}$, define

$$F_{\delta,e}(X) := \delta + \sum_{b \in I_\ell} e_b W_b(X).$$

Lemma 6.80 (Nonzero residual gives a nonzero polynomial)**.** *If $(\delta, e) \neq (0, 0)$, then $F_{\delta,e}$ is a nonzero polynomial of degree at most n.*

Proof. Each W_b has degree at most n, so the linear combination has degree at most n. If it vanished identically, the linear independence of $\{1, W_b\}_b$ proved in Theorem 6.75 would force $\delta = 0$ and $e_b = 0$ for all b, contradicting the hypothesis. □

Theorem 6.81 (Root bound for an anchored residual)**.** *Fix* $(\delta, e) \neq (0,0)$ *independently of a uniform challenge* $z \xleftarrow{\$} L$*. Then*

$$\boxed{\Pr_z\left[\delta + \sum_b e_b W_b(z)^2 = 0\right] \leq \frac{n}{q}.}$$

Proof. By Lemma 6.78, the tested expression is $F_{\delta,e}(z^2)$. By Corollary 6.6, the map $z \mapsto z^2$ is a bijection of the finite field L, so z^2 is uniform on L. By Lemma 6.80, $F_{\delta,e}$ is nonzero of degree at most n. The univariate root bound of Lemma 2.7 gives at most n roots. Divide by q. □

Remark 6.82 (Challenge timing is necessary). The bound is conditional on (δ, e) being fixed independently of the fresh challenge z. If the coefficients are chosen after seeing z, the prover could select a nonzero polynomial that vanishes at that particular point. Thus the commit-then-challenge order is part of the soundness hypothesis, not a cosmetic presentation detail.

6.8 An algebraic two-challenge consistency checker

We now combine the preceding lemmas into a small *algebraic checking theorem*. This is the kernel-coordinate consistency component of the proposed binary-field construction. The theorem does not assert zero knowledge for the complete IOP or its commitment scheme.

Definition 6.83 (Affine consistency checks)**.** Let the secret values p, c, t, r, s have types

$$p, c, t \in L^{I_\ell}, \qquad r, s \in L.$$

After a challenge $\theta \in L^\ell$, let $V_k = V_k(p, c, \theta)$ be as in Definition 6.56, and let $U = (U_k)_k \in L^{I_\ell}$ be public. After a second challenge $z \in L$, a public scalar $A \in L$ is supplied. The *algebraic consistency checker* accepts precisely if

$$\mathrm{Tr}(U_k) = 0 \quad \text{for all } k \in I_\ell, \tag{6.1}$$

$$A + r + \sum_k W_k(z) t_k = 0, \tag{6.2}$$

$$A^2 + s + \sum_k W_k(z)^2 (U_k + V_k + t_k) = 0. \tag{6.3}$$

The last two equations are affine in the committed secret variables p, c, t, r, s once θ, z, U, A are fixed: A^2 and $W_k(z)^2$ are public scalars.

Definition 6.84 (Scalar and vector checking residuals)**.** Put

$$\delta := s + r^2, \qquad e_k := U_k + V_k + t_k^2 + t_k$$

and form the anchored residual polynomial $F_{\delta,e}$ of Definition 6.79.

Lemma 6.85 (First equation determines the public scalar)**.** *For fixed* r, t, z*, equation* (6.2) *holds if and only if*

$$A = r + \sum_k W_k(z) t_k.$$

Proof. Move the other terms to the right side of equation (6.2). In characteristic two, additive inverses equal the original values, so subtraction and addition coincide. The resulting formula is both necessary and sufficient. □

Lemma 6.86 (The two affine checks imply an anchored residual root)**.** *If equations* (6.2) *and* (6.3) *both hold, then*

$$\boxed{\delta + \sum_k e_k W_k(z)^2 = 0.}$$

Proof. By Lemma 6.85, $A = r + \sum_k W_k(z)t_k$. Square this identity. Characteristic-two additivity of squares (Lemma 6.1) gives

$$A^2 = r^2 + \sum_k W_k(z)^2 t_k^2.$$

Substitute it into equation (6.3):

$$0 = r^2 + s + \sum_k W_k(z)^2 (t_k^2 + U_k + V_k + t_k).$$

The first two terms are $\delta = s + r^2$, and the parenthesis is exactly e_k by Definition 6.84. This is the claimed identity. □

Lemma 6.87 (Honest trace check)**.** *Assume $c = c^p$ and choose $U_k := V_k + \wp(t_k)$. Then the trace equations* (6.1) *hold for every θ and every t.*

Proof. By the definition of c^p and Definition 6.54, $D_c(\theta) = \Delta_p(\theta)$ for every θ. Corollary 6.65 then yields $\mathrm{Tr}(U_k) = 0$ for all k. □

Lemma 6.88 (Honest affine checks)**.** *Assume $c = c^p$, $s = r^2$, and $U_k = V_k + \wp(t_k)$ for every k. For every θ, z, choose*

$$A := r + \sum_k W_k(z)t_k.$$

Then the two affine equations (6.2) *and* (6.3) *hold.*

Proof. The first equation holds by definition of A, since $A + A = 0$ in characteristic two. The second equation follows from $U_k + V_k + t_k = t_k^2$:

$$A^2 + s + \sum_k W_k(z)^2 (U_k + V_k + t_k) = A^2 + r^2 + \sum_k W_k(z)^2 t_k^2.$$

By the same squaring calculation used in Lemma 6.86, $A^2 = r^2 + \sum_k W_k(z)^2 t_k^2$. The displayed sum therefore consists of two identical copies of each term and equals zero. □

Theorem 6.89 (Perfect algebraic completeness)**.** *If $c = c^p$, $s = r^2$, $U_k = V_k + \wp(t_k)$, and $A = r + \sum_k W_k(z)t_k$, the checker accepts for every choice of the challenges (θ, z).*

Proof. Lemma 6.87 proves the trace tests, and Lemma 6.88 proves the two remaining affine equations. Thus all tests in Definition 6.83 hold. □

Theorem 6.90 (Conditional anchored-check soundness)**.** *Fix θ, p, c, t, r, s, U before a uniform $z \xleftarrow{\$} L$ is sampled. Suppose the trace checks hold and $D_c(\theta) \neq \Delta_p(\theta)$. Then, even if A is chosen after seeing z, the probability that both affine checks pass is at most n/q.*

Proof. Because the traces of all U_k vanish while $D_c(\theta) \neq \Delta_p(\theta)$, Lemma 6.69 shows that some e_k is nonzero. Thus $(\delta, e) \neq (0, 0)$. The residual values depend only on θ, p, c, t, r, s, U, which are fixed before z, so the hypotheses of Theorem 6.81 apply. If both affine checks pass, Lemma 6.86 forces $\delta + \sum_k e_k W_k(z)^2 = 0$. The anchored-root bound gives probability at most n/q, independently of how A is selected after the challenge. □

Theorem 6.91 (Two-challenge coefficient-consistency bound)**.** *Fix $p, c \in L^{I_\ell}$ with $c \neq c^p$ before sampling uniform $\theta \in L^\ell$. After seeing θ, an arbitrary strategy may select t, r, s, U (and must satisfy all trace checks), but these values must be fixed* before *an independent uniform $z \xleftarrow{\$} L$ is sampled. The strategy may choose A after seeing z. Then the probability that all checks in Definition 6.83 accept is at most*

$$\boxed{\min\left\{1, \frac{\ell + n}{q}\right\}, \qquad q = 2^n.}$$

Proof. Let E be the event $D_c(\theta) = \Delta_p(\theta)$. Since $c \neq c^p$ was fixed before θ, Corollary 6.61 gives $\Pr[E] \leq \ell/q$. On the complementary event E^c, condition on any particular θ and on the strategy's chosen values t, r, s, U (and its private random tape), provided the public trace checks pass. These values are fixed before the fresh challenge z, so Theorem 6.90 bounds the conditional probability of passing both affine checks by n/q. If the trace checks fail, the acceptance probability is zero. Averaging over these conditional cases yields

$$\Pr[\text{accept}] \leq \Pr[E] + \Pr[\text{accept} \cap E^c] \leq \frac{\ell}{q} + \frac{n}{q}.$$

Probabilities are also bounded by 1. This proves the stated minimum. □

Lemma 6.92 (No even-order multiplicative subgroup in a binary field)**.** *Every subgroup of $L^\times$ has odd order. In particular, the odd-characteristic pairing $x \leftrightarrow -x$ used in the original KBFold word fold has no distinct paired points in $L^\times$.*

Proof. The group $L^\times$ has order $|L| - 1 = 2^n - 1$, an odd integer. Let $H \leq L^\times$. The distinct left cosets aH partition $L^\times$, and each has cardinality $|H|$ because multiplication by a is a bijection. Consequently $|L^\times| = k|H|$ for some integer k. As $|L^\times| = 2^n - 1$ is odd, $|H|$ is odd. Furthermore $-x = x$ for every $x \in L$ because the characteristic is two. Thus x and $-x$ cannot form two distinct points. □

Remark 6.93 (Precise boundary of the result). Theorems 6.89 and 6.91 concern only the *algebraic coefficient-consistency checker*. They assume the stated timing and do not incorporate a concrete commitment-binding error, the remaining ring-switching equations, the simulator for the joint public messages (U, A), or any proof of security for the odd-characteristic KBFold IOP. The final lemma also proves that the binary checker cannot simply reuse the $\pm x$ multiplicative-domain fold. These issues will be addressed in the appropriate later chapters.

6.9 The theorem chain and its formal boundary

For $L = \mathbb{F}_{2^n}$ with $n = 2^\ell$ and $c, p \in L^{I_\ell}$, the algebraic chain is

$$\boxed{\Delta_p(\theta) = D_{C_p}(\theta) = \mathsf{PFold}_\theta^{[\ell]}(C_p) = \mathscr{M}_\ell(C_p)(1 + \theta).}$$

The correction can be verified through traces of affine expressions over L:

$$\boxed{D_c(\theta) = \Delta_p(\theta) \iff \operatorname{Tr}(V_k(p, c, \theta)) = 0 \quad \forall k.}$$

The Artin–Schreier map $\wp(t) = t^2 + t$ has image $\ker \operatorname{Tr}$, and independent uniform masks make the *individual public trace vector* uniformly distributed over $(\ker \operatorname{Tr})^{I_\ell}$ for a fixed consistent instance. The anchored family

$$W_b = K_b + \mathbf{1}_{\{b=0_\ell\}} X^n$$

satisfies that $\{1, W_b : b \in I_\ell\}$ is a basis of $L[X]_{\le n}$. With secret coefficients and messages fixed before the relevant fresh challenges, the two-challenge consistency checker has mismatch-acceptance probability at most

$$\boxed{\min\{1, (\ell + n)/2^n\}.}$$

This bound concerns algebraic coefficient consistency alone. The complete security claim requires the remaining ring-switching equations, binding commitments, a joint-transcript simulator, and the composition analysis developed in later chapters. No new Lean certificate is asserted here; each result is mapped to a proposed Lean declaration in the chapter ledger.

7 Binary-Field Packing and Transposition

This chapter constructs the exact algebraic interface between a Boolean witness table, its binary field-packed columns, and a collection of extension-field affine relations. It builds on the characteristic-two kernel, trace-dual basis, and anchored consistency checker of Chapter 6; it rederives the transposition equations of Diamond–Posen [DP24] in the chosen Boolean-kernel coordinates and distinguishes them from the separate trace-zero masking mechanisms of Kochergin [Koc26] and Frigo–shelat [Fs26]. Every claim has a declared field and challenge order, and all algebraic statements are proved in small steps suitable for later Lean 4 formalization.

7.1 Binary witness tables and lossless field packing

Fix integers $\ell \ge 1$ and $h \ge 0$, and put

$$n := 2^\ell, \qquad L := \mathbb{F}_{2^n}, \qquad q := |L| = 2^n, \qquad I := I_\ell = \{0,1\}^\ell, \qquad J := I_h = \{0,1\}^h.$$

Choose $\alpha \in L$ of degree n over $\mathbb{F}_2$ and use the kernel evaluation basis $\beta_v = K_v^{(\ell)}(\alpha)$, $v \in I$, and trace-dual basis η_v from Section 6.5. The equality weights are the polynomials eq_d from Chapter 1, interpreted as the constant 1 when $d = 0$. The packed-ring-switching construction and its history are described in [DP24]. We derive each required identity rather than importing a protocol-level correctness statement.

Definition 7.1 (Boolean witness and its field embedding)**.** A *Boolean witness* is a function $w : I \times J \to \mathbb{F}_2$. We identify the prime subfield $\mathbb{F}_2 \subseteq L$ with $\{0,1\} \subseteq L$ when forming products or sums in L. For $v \in I$ and $j \in J$ write $w_{v,j} := w(v,j)$.

Definition 7.2 (Packing a Boolean column)**.** For $a = (a_v)_{v \in I} \in \mathbb{F}_2^I$ define

$$\mathrm{Pack}(a) := \sum_{v \in I} \beta_v a_v \in L.$$

For $x \in L$ define $\mathrm{Unpack}(x) := (d_v(x))_{v \in I} \in \mathbb{F}_2^I$, where d_v are the unique coordinates in the basis (β_v).

Lemma 7.3 (Unpack after Pack)**.** *For every $a \in \mathbb{F}_2^I$, $\mathrm{Unpack}(\mathrm{Pack}(a)) = a$.*

Proof. The element $\mathrm{Pack}(a)$ is by definition $\sum_v \beta_v a_v$. By uniqueness of coordinates in the basis (β_v), $d_u(\mathrm{Pack}(a)) = a_u$ for every $u \in I$. These equalities are the equality of the two vectors. □

Lemma 7.4 (Pack after Unpack)**.** *For every $x \in L$, $\mathrm{Pack}(\mathrm{Unpack}(x)) = x$.*

Proof. The defining coordinate expansion is $x = \sum_v \beta_v d_v(x)$. Its right side is $\mathrm{Pack}(\mathrm{Unpack}(x))$. □

Lemma 7.5 (Packing is binary-linear)**.** *For $a, b \in \mathbb{F}_2^I$ and $e \in \mathbb{F}_2$, $\mathrm{Pack}(a + eb) = \mathrm{Pack}(a) + e\,\mathrm{Pack}(b)$. The map* Unpack *is also $\mathbb{F}_2$-linear.*

Proof. Distribute the finite sum $\sum_v \beta_v(a_v + eb_v)$, using $e \in \mathbb{F}_2$. Each coordinate map d_v is $\mathbb{F}_2$-linear by Lemma 6.51, so the coordinatewise map Unpack is. □

Theorem 7.6 (Lossless Boolean packing)**.** $\mathrm{Pack} : \mathbb{F}_2^I \to L$ *and* $\mathrm{Unpack} : L \to \mathbb{F}_2^I$ *are mutually inverse $\mathbb{F}_2$-linear isomorphisms.*

Proof. Lemmas 7.3 and 7.4 show that the maps are mutually inverse. Lemma 7.5 proves their $\mathbb{F}_2$-linearity. □

Definition 7.7 (Packed witness and its column coordinates)**.** For each $j \in J$ define the packed field element

$$\widehat{w}_j := \mathrm{Pack}((w_{v,j})_{v \in I}) = \sum_{v \in I} \beta_v w_{v,j}.$$

Corollary 7.8 (Recovering every Boolean witness bit)**.** *For all $v \in I$ and $j \in J$,*

$$\boxed{d_v(\widehat{w}_j) = w_{v,j}.}$$

Proof. Apply Lemma 7.3 to the column vector $(w_{v,j})_{v \in I}$ and compare its v-th coordinate. □

Remark 7.9 (What packing does not prove). Every field element $x \in L$ has a unique vector of *binary* coordinates in this fixed basis. Thus the equality $x = \mathrm{Pack}(\mathrm{Unpack}(x))$ does not require a separate Booleanity constraint. However it does not bind an independently committed array of bits to x: if both representations are provided independently, their equality is an additional relation that must be checked.

Definition 7.10 (Multilinear evaluation weights)**.** Choose two public evaluation points $r_{\mathrm{lo}} \in L^\ell$ and $r_{\mathrm{hi}} \in L^h$, and define

$$q_v := \mathrm{eq}_\ell(v, r_{\mathrm{lo}}), \qquad h_j := \mathrm{eq}_h(j, r_{\mathrm{hi}}).$$

For $h = 0$ the unique weight is $h_{()} = 1$. Define the true evaluation of a Boolean table by

$$y_w := \sum_{v \in I} \sum_{j \in J} q_v h_j w_{v,j} \in L.$$

Lemma 7.11 (Two-axis equality-weight factorization)**.** *For all $(v, j) \in I \times J$,*

$$\mathrm{eq}_{\ell+h}((v, j), (r_{\mathrm{lo}}, r_{\mathrm{hi}})) = q_v h_j.$$

Proof. The equality polynomial is the product of one affine factor per coordinate. Its first ℓ factors involve only (v, r_{lo}) and multiply to q_v; its final h factors involve only (j, r_{hi}) and multiply to h_j. Splitting the finite product gives the identity. □

Corollary 7.12 (Packed witness is the original Boolean multilinear table)**.** *Let $\widetilde{w} : L^{\ell+h} \to L$ be the multilinear extension of the field-embedded Boolean table w. Then*

$$\boxed{\widetilde{w}(r_{\mathrm{lo}}, r_{\mathrm{hi}}) = y_w.}$$

Proof. Expand the multilinear extension as $\sum_{(v,j) \in I \times J} w_{v,j}\,\mathrm{eq}_{\ell+h}((v, j), (r_{\mathrm{lo}}, r_{\mathrm{hi}}))$. Substitute the factorization of Lemma 7.11. The resulting sum is the definition of y_w. □

Lemma 7.13 (Partition of evaluation weights)**.** $\sum_{v \in I} q_v = 1$ *and* $\sum_{j \in J} h_j = 1$.

Proof. For each coordinate, the equality weights for Boolean values 0 and 1 sum to $(1 - t) + t = 1$. Distributing the finite sum over all bit strings as a product of these two-term sums gives 1 for each axis. When $h = 0$, the second sum has one term 1. □

7.2 Matrix transposition over the binary prime field

The transpose below is an $\mathbb{F}_2$-linear operation on a vector of n elements of L. It is generally *not* an L-linear transformation; this is the source of the nontrivial ring-switching correction in the subsequent sections.

Definition 7.14 (Binary coordinate matrix)**.** For $s = (s_v)_{v\in I} \in L^I$ define its binary coordinate matrix

$$M(s) = (M_{u,v}(s))_{u,v\in I}, \qquad M_{u,v}(s) := d_u(s_v) \in \mathbb{F}_2.$$

The first index labels a binary basis coordinate; the second labels an entry of the original field vector.

Lemma 7.15 (Recovering a field vector from its matrix)**.** *For every $s \in L^I$ and every $v \in I$,*

$$s_v = \sum_{u\in I} \beta_u M_{u,v}(s).$$

Proof. By Definition 7.14, $M_{u,v}(s) = d_u(s_v)$. The displayed sum is therefore the unique basis expansion of s_v, namely $\sum_u \beta_u d_u(s_v) = s_v$. □

Definition 7.16 (Binary coordinate transpose)**.** For $s \in L^I$ define $\mathcal{T}(s) \in L^I$ by

$$\boxed{\mathcal{T}(s)_u := \sum_{v\in I} \beta_v d_u(s_v) = \sum_v \beta_v M_{u,v}(s).}$$

Lemma 7.17 (Transpose is binary-linear)**.** *For every $s, t \in L^I$ and $e \in \mathbb{F}_2$,*

$$\mathcal{T}(s + et) = \mathcal{T}(s) + e\mathcal{T}(t).$$

Proof. For each u, expand $\mathcal{T}(s + et)_u = \sum_v \beta_v d_u(s_v + et_v)$. Linearity of d_u over $\mathbb{F}_2$ gives $d_u(s_v) + ed_u(t_v)$. Distribute the sum and factor out the scalar e. The resulting coordinates equal those of $\mathcal{T}(s) + e\mathcal{T}(t)$. □

Lemma 7.18 (Coordinates of a transposed vector)**.** *For all $u, v \in I$ and $s \in L^I$,*

$$\boxed{d_v(\mathcal{T}(s)_u) = d_u(s_v).}$$

Proof. Use Definition 7.16 and the coordinate property $d_v(\beta_a) = \delta_{va}$ (Lemma 6.51):

$$d_v(\mathcal{T}(s)_u) = \sum_{a\in I} d_u(s_a) d_v(\beta_a) = d_u(s_v).$$

The first equality uses $\mathbb{F}_2$-linearity of d_v and that $d_u(s_a) \in \mathbb{F}_2$. □

Theorem 7.19 (Transposition is an involution)**.** *For every $s \in L^I$,*

$$\boxed{\mathcal{T}(\mathcal{T}(s)) = s.}$$

Proof. For each $v \in I$, expand the second transpose:

$$\mathcal{T}(\mathcal{T}(s))_v = \sum_{u\in I} \beta_u d_v(\mathcal{T}(s)_u).$$

Lemma 7.18 replaces the last coordinate with $d_u(s_v)$. The sum becomes $\sum_u \beta_u d_u(s_v)$, which equals s_v by Lemma 7.15. This holds for all v. □

Corollary 7.20 (Transpose is a bijective binary-linear map)**.** *$\mathcal{T} : L^I \to L^I$ is an $\mathbb{F}_2$-linear isomorphism with inverse $\mathcal{T}$.*

Proof. Lemma 7.17 gives linearity; Theorem 7.19 makes the map its own two-sided inverse. □

Lemma 7.21 (The coordinate matrices are transposes)**.** *For every $s \in L^I$ and all $u, v \in I$,*

$$\boxed{M_{u,v}(\mathcal{T}(s)) = M_{v,u}(s).}$$

Proof. The left side is $d_u(\mathcal{T}(s)_v)$ by Definition 7.14. Lemma 7.18 turns it into $d_v(s_u)$, which is the right side. □

Lemma 7.22 (Nonzero field elements have some nonzero coordinate)**.** *For every $x \in L$,*

$$x \neq 0 \quad \iff \quad \exists u \in I,\ d_u(x) = 1.$$

Proof. If every coordinate vanishes, the expansion $x = \sum_u \beta_u d_u(x)$ is zero. Contrapositively, $x \neq 0$ implies some coordinate is nonzero, and an element of $\mathbb{F}_2$ is nonzero exactly when it is 1. Conversely the zero element has all coordinates zero by $\mathbb{F}_2$-linearity of the maps d_u. □

Proposition 7.23 (Transposition is not extension-field-linear)**.** *Because $n = 2^\ell \geq 2$, the operation $\mathcal{T} : L^I \to L^I$ is not L-linear.*

Proof. Suppose toward a contradiction that $\mathcal{T}$ is L-linear. Choose $v \in I$, and let $e_v \in L^I$ be the vector equal to 1 at v and 0 elsewhere. Since $1 \neq 0$, the preceding lemma gives an index u with $d_u(1) = 1$. For any $a \in L$, the definition of $\mathcal{T}$ gives $\mathcal{T}(ae_v)_u = \beta_v d_u(a)$ and $\mathcal{T}(e_v)_u = \beta_v d_u(1) = \beta_v$. The assumed L-linearity would imply $\beta_v d_u(a) = a\beta_v$. Because $\beta_v \neq 0$, division gives $d_u(a) = a$ for every $a \in L$. But $d_u(a) \in \mathbb{F}_2$ and $|L| = 2^n > 2$, so choosing $a \in L \setminus \mathbb{F}_2$ is a contradiction. □

Lemma 7.24 (Uniformity under binary transposition)**.** *If S is uniform on L^I, then $\mathcal{T}(S)$ is uniform on L^I.*

Proof. By Corollary 7.20, $\mathcal{T}$ is a bijection of the finite set L^I. Each output has exactly one preimage, hence the uniform input law is mapped to the uniform output law. □

Remark 7.25 (Connection with field-linear proof systems). A proof system verifying only affine equations over L cannot directly replace $\mathcal{T}(m)$ by a linear combination of hidden field entries m_v: the coefficient extraction d_u is only $\mathbb{F}_2$-linear. Our protocol therefore reveals masked field entries m_v and enforces the hidden correction using affine trace coordinates, rather than pretending that $\mathcal{T}$ is L-linear.

7.3 Exact transposed evaluation identities

Definition 7.26 (Partial evaluations of the original Boolean witness)**.** For the Boolean table w and public high-variable weights h_j from Definition 7.10, define

$$s^*_v := \sum_{j \in J} h_j w_{v,j} \in L, \qquad v \in I.$$

Write $s^* = (s^*_v)_{v \in I} \in L^I$.

Lemma 7.27 (Partial evaluations recover the claimed value)**.** *For q_v and y_w as in Definition 7.10,*

$$\boxed{y_w = \sum_{v \in I} q_v s_v^*.}$$

Proof. Substitute the definition of s_v^* and distribute the two finite sums:

$$\sum_v q_v s_v^* = \sum_v q_v \left(\sum_j h_j w_{v,j} \right) = \sum_{v,j} q_v h_j w_{v,j}.$$

The last sum equals y_w by definition. □

Lemma 7.28 (Coordinate of a Boolean partial sum)**.** *For every $u, v \in I$,*

$$\boxed{d_u(s_v^*) = \sum_{j \in J} d_u(h_j) w_{v,j},}$$

where the sum on the right is taken in $\mathbb{F}_2$.

Proof. By Definition 7.26, $s_v^* = \sum_j h_j w_{v,j}$. The coordinate map d_u is additive and $\mathbb{F}_2$-linear; as $w_{v,j} \in \mathbb{F}_2$, $d_u(h_j w_{v,j}) = d_u(h_j) w_{v,j}$. Distribute d_u over the finite sum. □

Theorem 7.29 (Packed transposition identity)**.** *For every $u \in I$,*

$$\boxed{\mathcal{T}(s^*)_u = \sum_{j \in J} d_u(h_j) \widehat{w}_j.}$$

Proof. Expand the transpose and apply Lemma 7.28:

$$\mathcal{T}(s^*)_u = \sum_v \beta_v d_u(s_v^*) = \sum_v \beta_v \sum_j d_u(h_j) w_{v,j}.$$

Interchange the finite sums and take the binary scalar $d_u(h_j)$ outside the inner sum:

$$\mathcal{T}(s^*)_u = \sum_j d_u(h_j) \left(\sum_v \beta_v w_{v,j} \right).$$

The term in parentheses is $\widehat{w}_j$ by Definition 7.7. □

Definition 7.30 (Random batching weights)**.** For a public challenge $\theta \in L^\ell$, put

$$\lambda_u(\theta) := \mathrm{eq}_\ell(u, \theta)$$

and define

$$a_j(\theta) := \sum_{u \in I} \lambda_u(\theta) d_u(h_j) \in L, \qquad B_x(\theta) := \sum_{u \in I} \lambda_u(\theta) \mathcal{T}(x)_u \in L$$

for any $x \in L^I$.

Lemma 7.31 (Linearity of batching over the binary prime field)**.** *For $x, y \in L^I$,*

$$B_{x+y}(\theta) = B_x(\theta) + B_y(\theta).$$

Proof. By Lemma 7.17, $\mathcal{T}(x+y) = \mathcal{T}(x)+\mathcal{T}(y)$. Substitute into $\sum_u \lambda_u(\theta)\mathcal{T}(x+y)_u$, then distribute the sum. □

Theorem 7.32 (Packed evaluation batching identity)**.** *For every $\theta \in L^\ell$,*

$$\boxed{B_{s^*}(\theta) = \sum_{j\in J} a_j(\theta)\widehat{w}_j.}$$

Proof. Multiply Theorem 7.29 by $\lambda_u(\theta)$ and sum over u:

$$B_{s^*}(\theta) = \sum_u \lambda_u(\theta) \sum_j d_u(h_j)\widehat{w}_j.$$

Interchange the two finite sums and factor out $\widehat{w}_j$:

$$B_{s^*}(\theta) = \sum_j \left(\sum_u \lambda_u(\theta) d_u(h_j) \right) \widehat{w}_j.$$

The inner sum is $a_j(\theta)$. □

Definition 7.33 (Transposition discrepancy polynomial)**.** For $x \in L^I$ define

$$E_x(\Theta) := \sum_{u\in I} \mathcal{T}(x)_u \operatorname{eq}_\ell(u, \Theta) \in L[\Theta_0, \ldots, \Theta_{\ell-1}].$$

Lemma 7.34 (Nonzero vector gives a nonzero discrepancy)**.** *If $x \neq 0$, then E_x is nonzero and has total degree at most ℓ.*

Proof. The invertibility of $\mathcal{T}$ from Theorem 7.19 implies $\mathcal{T}(x) \neq 0$. Choose u with $\mathcal{T}(x)_u \neq 0$. Evaluate the discrepancy at the Boolean challenge $\Theta = u$. Boolean interpolation (Lemma 1.9) gives $E_x(u) = \mathcal{T}(x)_u \neq 0$. Thus E_x is not identically zero. Each summand is a scalar multiple of a product of ℓ affine factors, so its total degree is at most ℓ; summing cannot increase it. □

Theorem 7.35 (Unmasked batching soundness)**.** *Fix $x \in L^I$, $x \neq 0$, before an independent uniform $\theta \in L^\ell$. Then*

$$\boxed{\Pr_\theta[B_x(\theta) = 0] \leq \min\{1, \ell/q\}.}$$

Proof. By Definition 7.33, $B_x(\theta) = E_x(\theta)$. Lemma 7.34 makes E_x a nonzero polynomial of total degree at most ℓ. Apply the multivariate Schwartz–Zippel bound, Lemma 2.8, to the independent uniform coordinates of θ. □

Definition 7.36 (Kernel representation of a transposed vector)**.** For $x \in L^I$, define

$$S_x(X) := \sum_{u\in I} \mathcal{T}(x)_u K_u^{(\ell)}(X) \in L[X]_{<n}.$$

Lemma 7.37 (Kernel-fold representation of batching)**.** *For every $x \in L^I$ and $\theta \in L^\ell$,*

$$\boxed{B_x(\theta) = \mathsf{PFold}_\theta^{[\ell]}(S_x).}$$

Proof. By the complete kernel folding dictionary, Corollary 5.24,

$$\mathsf{PFold}_\theta^{[\ell]}(S_x) = \sum_u \mathrm{eq}_\ell(u,\theta)\mathcal{T}(x)_u.$$

The sum is exactly $B_x(\theta)$ by Definition 7.30. □

Remark 7.38 (Index and field constraints). The packed identity is valid because the *witness entries* $w_{v,j}$ belong to $\mathbb{F}_2$. The high-axis weights h_j and low-axis weights q_v may be arbitrary elements of L. We never move h_j through the coordinate map as an extension-field scalar: in general $d_u(h_j x) \neq h_j d_u(x)$. Instead we apply d_u to h_j and use the Boolean scalar $w_{v,j}$.

7.4 Masking partial evaluations and both affine interfaces

The raw partial evaluations s_v^* can depend on the witness. The protocol instead exposes $m_v = s_v^* + p_v$ and checks the correction through hidden kernel coordinates. The following identities are algebraic, not yet a complete zero-knowledge simulation result.

Definition 7.39 (Field pads and masked partial evaluations)**.** Let $p = (p_v)_{v\in I} \in L^I$ be a pad vector and define

$$\boxed{m_v := s_v^* + p_v, \qquad v \in I.}$$

In distributional statements the coordinates of p are independent uniform elements of L.

Lemma 7.40 (Uniform masked partial evaluations)**.** *For every fixed witness and high-axis evaluation point, if p is uniform on L^I then $m = s^* + p$ is uniform on L^I, independently of the witness.*

Proof. For each $a \in L^I$, the event $m = a$ holds precisely when $p = a + s^*$, since subtraction equals addition in characteristic two. Translation by s^* is a bijection of L^I, so $\Pr[m = a] = q^{-n}$, independently of s^*. □

Definition 7.41 (Transposed pad coefficients)**.** Set

$$c_u^p := \mathcal{T}(p)_u = \sum_{v\in I} \beta_v d_u(p_v), \qquad C_p(X) := \sum_{u\in I} c_u^p K_u^{(\ell)}(X).$$

For $\theta \in L^\ell$ define

$$\Delta_p(\theta) := B_p(\theta) = \sum_{u\in I} \mathrm{eq}_\ell(u,\theta) c_u^p.$$

Lemma 7.42 (Pad correction equals its complete kernel fold)**.** *For every p, θ,*

$$\boxed{\Delta_p(\theta) = \mathsf{PFold}_\theta^{[\ell]}(C_p) = \mathscr{M}_\ell(C_p)(1+\theta).}$$

Proof. The kernel coefficients of C_p are c^p. Corollary 5.24 therefore gives the first equality by the definition of Δ_p. The second equality is the characteristic-two identity of Theorem 6.27. □

Lemma 7.43 (Transpose of a masked vector)**.** *If $m = s^* + p$, then*

$$\boxed{\mathcal{T}(m) = \mathcal{T}(s^*) + c^p.}$$

Proof. By $\mathbb{F}_2$-linearity of $\mathcal{T}$, $\mathcal{T}(s^* + p) = \mathcal{T}(s^*) + \mathcal{T}(p)$. The last term equals c^p by definition. □

Corollary 7.44 (Masked batching with the true correction)**.** *For all θ,*

$$\boxed{B_m(\theta) + \Delta_p(\theta) = \sum_{j \in J} a_j(\theta)\widehat{w}_j.}$$

Proof. The preceding lemma implies $B_m = B_{s^*} + B_p$ by distributivity of the finite sum defining B. Adding $\Delta_p = B_p$ cancels two copies of B_p in characteristic two. Theorem 7.32 then identifies the result with the right side. □

Definition 7.45 (Candidate correction and outgoing interface)**.** A committed candidate $c = (c_u)_{u \in I} \in L^I$ is fixed before the batching challenge. Define

$$D_c(\theta) := \sum_{u \in I} \mathrm{eq}_\ell(u, \theta) c_u.$$

The intended consistency condition is $c = c^p$. The *outgoing affine equation* is

$$\boxed{\sum_{j \in J} a_j(\theta)\widehat{w}_j = B_m(\theta) + D_c(\theta).} \tag{OUT}$$

After the public m and challenge θ are fixed, B_m is publicly computable and the equation is affine in hidden field values $(\widehat{w}_j, c_u)$.

Lemma 7.46 (Outgoing equation with consistent correction)**.** *If $c = c^p$, equation* (OUT) *holds for every θ.*

Proof. The consistency condition gives $D_c = \Delta_p$. Substitute in Corollary 7.44. □

Definition 7.47 (Incoming masked scalar)**.** Let $y_{\mathrm{pub}} \in L$ be a public evaluation claim and let $p_{\mathrm{in}} \in L$ be an independent hidden pad. Define

$$y' := y_{\mathrm{pub}} + p_{\mathrm{in}}.$$

Lemma 7.48 (Incoming masked scalar is uniform)**.** *For fixed y_{pub}, a uniform p_{in} makes y' uniform on L.*

Proof. For every $a \in L$, the equation $y' = a$ has exactly one pad solution $p_{\mathrm{in}} = a + y_{\mathrm{pub}}$. The uniform distribution assigns it probability $1/q$. □

Lemma 7.49 (Joint uniformity of basic public masks)**.** *For fixed $w, y_{\mathrm{pub}}, r_{\mathrm{hi}}$, if p and p_{in} are independent uniform samples, then (m, y') is uniform on $L^I \times L$.*

Proof. For a target pair (a, b), the equations $(m, y') = (a, b)$ uniquely determine $p = a + s^*$ and $p_{\mathrm{in}} = b + y_{\mathrm{pub}}$. The independent uniform masks give probability $q^{-n}q^{-1} = q^{-(n+1)}$ for each target pair. □

Definition 7.50 (Incoming affine relation)**.** The incoming equation is

$$\boxed{\sum_{v \in I} q_v m_v + y' = \sum_{v \in I} q_v p_v + p_{\mathrm{in}}.} \tag{IN}$$

It is affine in the hidden field values p_v, p_{in} once public messages and evaluation weights are fixed.

Lemma 7.51 (Incoming elimination of both masks)**.** *Under* $y' = y_{\mathrm{pub}} + p_{\mathrm{in}}$*, equation* (IN) *is equivalent to*

$$\boxed{y_{\mathrm{pub}} = \sum_{v \in I} q_v (m_v + p_v).}$$

Proof. Substitute $y_{\mathrm{pub}} + p_{\mathrm{in}}$ for y' in the incoming equation. Add p_{in} to both sides to cancel it, then add $\sum_v q_v m_v$ to both sides. Distributivity yields the displayed equality. Every operation is reversible in the additive group of L, establishing equivalence. □

Theorem 7.52 (Perfect correctness of both masked interfaces)**.** *Assume* $y_{\mathrm{pub}} = y_w$*,* $m = s^* + p$*,* $c = c^p$*, and* $y' = y_{\mathrm{pub}} + p_{\mathrm{in}}$*. Then both* (IN) *and* (OUT) *hold for every* θ*.*

Proof. Since $m + p = s^*$, Lemma 7.27 gives $\sum_v q_v (m_v + p_v) = \sum_v q_v s^*_v = y_w = y_{\mathrm{pub}}$. Lemma 7.51 proves the incoming equation. Lemma 7.46 proves the outgoing equation. □

Remark 7.53 (Privacy boundary). The pair (m, y') is individually and jointly uniform in the stated setting, but the complete protocol view also includes commitments, outgoing constraints, trace-zero messages and later affine checks. Their joint distribution requires an independent zero-knowledge proof.

7.5 Elimination, random batching, and algebraic composition soundness

The following analysis fixes the hidden witness and all algebraic values that enter the outgoing equation *before* the first random batching challenge. This is necessary to apply Schwartz–Zippel: if the prover could change the witness or the masked vector after seeing θ, it could manufacture a root of the discrepancy polynomial.

Definition 7.54 (The three public-value constraints)**.** For public w-dependent packed values $(\widehat{w}_j)_j$, public masked messages $m \in L^I$, $y' \in L$, a public claimed evaluation y_{pub}, and hidden p, p_{in}, c, consider:

$$y' = y_{\mathrm{pub}} + p_{\mathrm{in}}, \tag{CLAIM}$$

$$\sum_v q_v m_v + y' = \sum_v q_v p_v + p_{\mathrm{in}}, \tag{IN}$$

$$\sum_j a_j(\theta)\widehat{w}_j = B_m(\theta) + D_c(\theta). \tag{OUT}$$

The first two are affine relations in hidden extension-field values; the third is affine in hidden packed witness elements and c after m, θ are public. The further constraint $c = \mathcal{T}(p)$ is not L-affine; it will be checked by the characteristic-two algebraic checker from Chapter 6.

Definition 7.55 (Partial-evaluation and correction errors)**.** For arbitrary m, p, c, define

$$e := m + s^* + p \in L^I, \qquad \kappa := c + c^p \in L^I.$$

Because the characteristic is two, $e = 0$ is equivalent to $m = s^* + p$, and $\kappa = 0$ to $c = c^p$.

Lemma 7.56 (Exact outgoing discrepancy decomposition)**.** *For every* $\theta \in L^\ell$*,*

$$\boxed{B_m(\theta) + D_c(\theta) + \sum_j a_j(\theta)\widehat{w}_j = B_e(\theta) + D_\kappa(\theta).}$$

Proof. By the correct packed identity Theorem 7.32, $\sum_j a_j \widehat{w}_j = B_{s^*}$. By the definition $e = m + s^* + p$, we have $m = e + s^* + p$. Since B is additive, $B_m = B_e + B_{s^*} + B_p$. Substitution into the left side cancels the two copies of B_{s^*}. The remaining expression is $B_e + B_p + D_c$. By Definition 7.41, $B_p = D_{c^p}$, and linearity of D in its coefficient table gives $D_{c^p} + D_c = D_{c^p + c} = D_\kappa$. □

Lemma 7.57 (A correct correction isolates the partial-evaluation error)**.** *If $c = c^p$, then the outgoing equation* (OUT) *holds exactly when $B_e(\theta) = 0$.*

Proof. The assumption gives $\kappa = 0$, so $D_\kappa = 0$ by its defining sum. Lemma 7.56 shows that the sum of the two sides of the outgoing equation is $B_e(\theta)$. In characteristic two, two field elements are equal precisely when their sum is zero. □

Lemma 7.58 (A false claim and the incoming relations force a row error)**.** *Assume* CLAIM *and* IN*, and suppose $y_{\text{pub}} \neq y_w$. Then $e \neq 0$.*

Proof. Suppose instead $e = 0$. By Definition 7.55, $m = s^* + p$. The claim and incoming equations imply $y_{\text{pub}} = \sum_v q_v(m_v + p_v)$ by Lemma 7.51. Substituting $m + p = s^*$ gives $y_{\text{pub}} = \sum_v q_v s^*_v$, which equals y_w by Lemma 7.27. This contradicts the false-claim hypothesis. □

Theorem 7.59 (False-claim soundness with exact correction)**.** *Fix $w, m, p, c, y', p_{\text{in}}, y_{\text{pub}}, r_{\text{lo}}, r_{\text{hi}}$ before uniform $\theta \in L^\ell$. Suppose $c = c^p$ and $y_{\text{pub}} \neq y_w$. Then*

$$\Pr_\theta[\mathsf{CLAIM} \wedge \mathsf{IN} \wedge \mathsf{OUT}] \leq \min\{1, \ell/q\}.$$

Proof. If either the claim or incoming equation fails, the acceptance event is empty. Otherwise Lemma 7.58 implies that the fixed vector $e = m + s^* + p$ is nonzero. Lemma 7.57 makes the outgoing equation equivalent to $B_e(\theta) = 0$. Because e is fixed before θ, Theorem 7.35 bounds that event by ℓ/q. □

Definition 7.60 (Combined algebraic acceptance event)**.** Let $\mathsf{Check}_6(\theta, z)$ denote acceptance of the algebraic consistency checker in Definition 6.83, instantiated with the same hidden pad vector p, candidate coefficient table c, and public challenge θ. It uses auxiliary hidden vectors/scalars t, ρ, σ and published values U, A subject to the timing in Chapter 6. Define

$$\mathsf{Accept}_7 := \mathsf{CLAIM} \wedge \mathsf{IN} \wedge \mathsf{OUT} \wedge \mathsf{Check}_6(\theta, z).$$

Theorem 7.61 (Complete algebraic soundness of packed Ring Switching)**.** *Fix w, its packed column elements $(\widehat{w}_j)$, m, p, c, $y_{\text{pub}}, y', p_{\text{in}}$, and the evaluation points $r_{\text{lo}}, r_{\text{hi}}$ before an independent uniform $\theta \in L^\ell$. After θ, an adversary may choose the auxiliary hidden values and public trace-zero vector of the Chapter 6 checker, but all those choices are fixed before an independent uniform $z \in L$; it may choose the final public scalar A after seeing z. Suppose $y_{\text{pub}} \neq y_w$. Then*

$$\boxed{\Pr_{\theta,z}[\mathsf{Accept}_7] \leq \min\left\{1, \frac{\ell + n}{q}\right\}, \qquad n = 2^\ell,\ q = 2^n.}$$

The bound is for the algebraic constraints*, not for a concrete externally instantiated commitment or affine-proof system.*

Proof. Because p, c are fixed before θ, precisely one of the two cases below holds before challenges are drawn.

If $c \neq c^p$, acceptance of Accept_7 entails acceptance of the Chapter 6 checker. Theorem 6.91 bounds this probability by $\min\{1, (\ell+n)/q\}$ under the stated timing, regardless of the additional incoming and outgoing constraints.

If $c = c^p$, the Chapter 6 checker can only reduce the probability of acceptance. Therefore $\Pr[\mathsf{Accept}_7]$ is at most the probability of satisfying CLAIM, IN and OUT. By Theorem 7.59, this is at most $\min\{1, \ell/q\}$.

The maximum of the bounds of the two mutually exclusive cases is $\min\{1, (\ell+n)/q\}$. This proves the theorem. □

Theorem 7.62 (Complete algebraic correctness of packed Ring Switching)**.** *Fix a Boolean witness w, its exact packed columns $\widehat{w}_j$, and the true public claim $y_{\mathrm{pub}} = y_w$. Choose any $p, p_{\mathrm{in}} \in L$ of the declared vector/scalar types, set $m = s^* + p$, $c = c^p$ and $y' = y_{\mathrm{pub}} + p_{\mathrm{in}}$. Choose the Chapter 6 auxiliary secrets with its honest square relation and the public messages as in Theorem 6.89. Then Accept_7 holds for every θ, z.*

Proof. The honest masked interfaces satisfy IN and OUT by Theorem 7.52; the defining choice of y' gives CLAIM. Theorem 6.89 proves that the Chapter 6 checker accepts every challenge for the honest choices of its auxiliary secrets and public messages. Thus every conjunct in Accept_7 is true. □

Remark 7.63 (What the bound excludes). The theorem assumes exact enforcement of the claimed field-affine relations, and that the packed columns, witnesses, pads, and correction table cannot be changed after seeing θ. In an actual system, a commitment or auxiliary proof protocol must bind those values and prove each affine relation with its own soundness bound. Our theorem does not claim that the resulting IOP has a zero-knowledge simulator, nor does it prove the odd-characteristic KBFold proximity theorem.

7.6 Affine-proof interfaces, trace coordinates, and privacy limits

Definition 7.64 (Available affine constraints)**.** An *extension-field affine proof interface* for L allows a verifier to specify public coefficients $(a_i)_i \in L^t$ and a public right-hand side $b \in L$, and to require committed hidden values $(x_i)_i$ to satisfy

$$\sum_i a_i x_i = b$$

without exposing those values. The algebraic interface does not itself assert binding, knowledge soundness, or zero knowledge for the implementation.

Lemma 7.65 (Incoming constraints are extension-field affine)**.** *For public $m, y', q_v, y_{\mathrm{pub}}$ the relations* CLAIM *and* IN *are extension-field affine equations in hidden $(p_v)_v, p_{\mathrm{in}}$.*

Proof. The claim relation is $p_{\mathrm{in}} = y' + y_{\mathrm{pub}}$ in characteristic two, so it is affine. The incoming relation is $\sum_v q_v p_v + p_{\mathrm{in}} = \sum_v q_v m_v + y'$, with public right side and public coefficients q_v. Both have exactly the form in Definition 7.64. □

Lemma 7.66 (Outgoing constraints are extension-field affine)**.** *For public θ, m and weights h_j, the outgoing equation is extension-field affine in hidden $(\widehat{w}_j)_j$ and $c = (c_u)_u$.*

Proof. Rewrite (OUT) as $\sum_j a_j(\theta)\widehat{w}_j + \sum_u \mathrm{eq}_\ell(u, \theta) c_u = B_m(\theta)$. Every coefficient is publicly computable from θ, h_j, and the public right side $B_m(\theta)$ is computed by transposing the public vector m. No coordinate extraction of a hidden field element occurs in this equation, proving affinity. □

Definition 7.67 (Affine correction coefficients)**.** Use the trace-dual basis η_u and the public coefficients from Chapter 6:

$$H_{v,k}(\theta) = \sum_{u \in I} d_k(\beta_v \operatorname{eq}_\ell(u, \theta))\eta_u.$$

Define

$$V_k(p, c, \theta) = \eta_k D_c(\theta) + \sum_{v \in I} H_{v,k}(\theta) p_v.$$

Lemma 7.68 (Every correction expression is extension-field affine)**.** *For fixed public θ, the expression $V_k(p, c, \theta)$ is L-linear in the hidden values $(p_v)_v$ and $(c_u)_u$.*

Proof. Substitute $D_c(\theta) = \sum_u \lambda_u(\theta) c_u$. Then

$$V_k = \sum_u \eta_k \lambda_u(\theta) c_u + \sum_v H_{v,k}(\theta) p_v.$$

All coefficients multiplying hidden variables depend only on public parameters and θ. There is no constant term, so the expression is in fact homogeneous L-linear. □

Theorem 7.69 (Trace-coordinate interface for candidate correction)**.** *For every p, c, θ,*

$$\boxed{D_c(\theta) = \Delta_p(\theta) \quad \Longleftrightarrow \quad \operatorname{Tr}(V_k(p, c, \theta)) = 0 \quad \forall k \in I.}$$

Proof. By Theorem 6.53, $d_k(x) = \operatorname{Tr}(\eta_k x)$. Lemma 6.57 shows $\operatorname{Tr}(\sum_v H_{v,k} p_v) = d_k(\Delta_p)$. Therefore $\operatorname{Tr}(V_k) = d_k(D_c + \Delta_p)$, which is Lemma 6.58. Every coordinate vanishes if and only if the field element vanishes, by the basis expansion. In characteristic two $D_c + \Delta_p = 0$ is equivalent to $D_c = \Delta_p$. □

Lemma 7.70 (A revealed transposed pad reveals the original pad)**.** *If the entire field vector $c^p = \mathcal{T}(p)$ is revealed, then the pad vector p is uniquely recovered by*

$$\boxed{p = \mathcal{T}(c^p).}$$

Proof. Apply $\mathcal{T}$ to the identity $c^p = \mathcal{T}(p)$, then use Theorem 7.19. □

Theorem 7.71 (Revealing the correction destroys masking)**.** *Suppose both $m = s^* + p$ and $c^p = \mathcal{T}(p)$ are sent in the clear. Then every unmasked partial evaluation s^*_v is deterministically recoverable:*

$$\boxed{s^* = m + \mathcal{T}(c^p).}$$

Proof. Lemma 7.70 recovers $p = \mathcal{T}(c^p)$. As $m = s^* + p$, adding p to both sides yields $s^* = m + p = m + \mathcal{T}(c^p)$. Thus the unmasked vector can be computed from the two exposed vectors. □

Corollary 7.72 (Masked-value uniformity alone does not imply privacy)**.** *The marginal uniformity result for m in Lemma 7.40 does not imply witness privacy for the joint message pair (m, c^p).*

Proof. For any fixed s^*, the joint pair (m, c^p) deterministically reveals s^* by Theorem 7.71. Choose two distinct possible partial-evaluation vectors s^*_0, s^*_1. The events that the reconstruction equals s^*_0 distinguish their joint message distributions with probabilities 1 and 0, respectively. Thus the pair need not be witness independent, despite each m being uniform. □

Remark 7.73 (Public parameters and hidden values). All $H_{v,k}(\theta)$ and $a_j(\theta)$ may be computed by the verifier from public inputs; the vectors p, c must remain hidden behind a commitment or an appropriate zero-knowledge affine proof. Whether such a proof can simulate its commitments, openings and messages is a separate cryptographic statement. The current chapter proves only the algebraic affine interface.

Remark 7.74 (Independent paper contributions). The matrix transposition and packed bit-linear reduction originate in the Ring Switching line of work [DP24]. The trace/Frobenius approach to hiding has prior treatments [Koc26; Fs26]. Our kernel-coordinate identities use the basis from KBFold [Mkh26]. No novelty claim about the underlying trace-zero principle is made here.

7.7 Exact composition boundary and implementation obligations

Definition 7.75 (Ideal versus implemented acceptance)**.** Let $\mathcal{A}_{\text{ideal}}$ be the algebraic acceptance event of Definition 7.60 for fixed committed values. Let $\mathcal{A}_{\text{impl}}$ denote acceptance by a proposed implemented PCS/IOP using an auxiliary commitment and affine-proof system. Call $\mathcal{B}$ the event that the implementation accepts even though some supposedly enforced commitment equality, claimed bit packing, or field-affine relation is invalid on the fixed underlying witness. Assume the implementation has the *faithful acceptance* property

$$\mathcal{A}_{\text{impl}} \cap \mathcal{B}^c \subseteq \mathcal{A}_{\text{ideal}}.$$

Lemma 7.76 (Bad-execution decomposition)**.** *Under Definition 7.75,*

$$\mathcal{A}_{\text{impl}} \subseteq \mathcal{A}_{\text{ideal}} \cup \mathcal{B}.$$

Proof. Every execution either belongs to $\mathcal{B}$ or to its complement. In the first case it belongs to the union on the right. In the second case, faithful acceptance places it in $\mathcal{A}_{\text{ideal}}$. This exhausts all executions. □

Proposition 7.77 (Conditional soundness under an external failure bound)**.** *Consider a false evaluation claim under the pre-challenge binding and strategy-timing hypotheses of Theorem 7.61. If an implementation admits an ideal coupling as in Definition 7.75 and the probability of the bad event is at most ε_{ext}, then*

$$\boxed{\Pr[\mathcal{A}_{\text{impl}}] \leq \min\left\{1, \frac{\ell + n}{q} + \varepsilon_{\text{ext}}\right\}.}$$

Proof. By Lemma 7.76 and the probability union bound,

$$\Pr[\mathcal{A}_{\text{impl}}] \leq \Pr[\mathcal{A}_{\text{ideal}}] + \Pr[\mathcal{B}].$$

Theorem 7.61 bounds the first term by $(\ell + n)/q$. The external coupling assumption bounds the second by ε_{ext}. Cap the sum by the universal upper bound 1 for a probability. □

Definition 7.78 (Formal verification obligations)**.** To turn the preceding identities into a complete zero-knowledge Ring Switching IOP and then reuse them safely as a component of another proof system, one needs separate formal statements that establish:

(i) the exact binding relation between the independently committed Boolean witness and packed elements $\widehat{w}_j$;

(ii) the commitment-before-θ timing for $w, m, p, c, y', p_{\text{in}}$ and the pre-z timing for the checker residuals;

(iii) exact or knowledge-sound implementation of the three affine interfaces and the Chapter 6 checker;

(iv) the joint zero-knowledge simulation of m, y', U, A together with commitments and authenticated openings;

(v) a separate analysis for any multiple-opening and Fiat–Shamir/QROM compilation; and

(vi) if combined with the original KBFold PCS, the odd-characteristic folding, degree and proximity arguments of Chapters 3–5 rather than an unjustified characteristic-two substitution.

Remark 7.79 (Where Chapter 7 ends). We have proved lossless packing, the exact involutive transpose, the outgoing packed-evaluation identity, incoming/outgoing masked equations, algebraic completeness and a quantified false-claim bound in an idealized affine interface. We have *not* proved cryptographic zero knowledge or a new PCS theorem. A future simulator must cover the full conditional distribution of every committed and published message, not just the uniform marginals of m and y'.

7.8 Main theorem chain

Fix $n = 2^\ell$, $L = \mathbb{F}_{2^n}$, and a Boolean witness $w : I_\ell \times I_h \to \mathbb{F}_2$. The exact identities established above are

$$\boxed{\text{Unpack} \circ \text{Pack} = \text{id}, \qquad \mathcal{T}^2 = \text{id},}$$

$$\boxed{\mathcal{T}(s^*)_u = \sum_{j \in J} d_u(h_j)\widehat{w}_j, \qquad B_{s^*}(\theta) = \sum_j a_j(\theta)\widehat{w}_j,}$$

and, for $m = s^* + p$ with $c = c^p = \mathcal{T}(p)$,

$$\boxed{y_{\text{pub}} = \sum_v q_v(m_v + p_v), \qquad \sum_j a_j(\theta)\widehat{w}_j = B_m(\theta) + D_c(\theta).}$$

The first incoming equation is equivalent to the two concrete affine equations CLAIM and IN. The second is the outgoing affine equation OUT, and the non-affine identity $c = \mathcal{T}(p)$ is checked by the anchored trace-coordinate checker of Chapter 6.

When a false evaluation $y_{\text{pub}} \neq y_w$ is bound before the independent challenges, the combined *ideal algebraic* checks obey

$$\boxed{\Pr[\mathsf{Accept}_7] \leq \min\{1, (\ell + n)/2^n\}.}$$

This is a conditional algebraic result, not a complete cryptographic theorem. The chapter also proves that publishing (m, c^p) reveals s^* exactly, even though the marginal distribution of m is uniform. The remaining obligations are external commitment binding, authenticated affine proofs, an online joint-view simulator, and composition with the original odd-characteristic KBFold PCS.

8 The Diamond–Posen Ring-Switching Compiler

We now reconstruct the *original* Diamond–Posen ring-switching reduction from a small-field multilinear evaluation to a committed large-field multilinear evaluation [DP24, Section 4]. Its defining intermediate object is a tensor algebra $A = L \otimes_K L$ with two distinct field embeddings. This is not equivalent to treating binary coordinate transposition as an extension-field-linear map. We specialize to $K = \mathbb{F}_2$ and $[L : K] = 2^\ell$ to connect the construction with Chapters 6–7; the original compiler applies to more general finite extensions of power-of-two degree.

We first prove the ring and column identities, then the tensor sumcheck reduction and its degree-two root lemma, and finally a conditional composition theorem and a precise disclosure theorem. The manuscript distinguishes every algebraic proof from claims requiring an authenticated commitment, a proximity theorem, or a full zero-knowledge simulator.

8.1 The tensor algebra and its two scalar embeddings

The original Diamond–Posen ring-switching compiler is formulated over $A = L \otimes_K L$, not merely over a matrix transpose. We reconstruct the specialization $K = \mathbb{F}_2$ of Construction 4.1 in [DP24]. Write $n = 2^\ell$, $L = \mathbb{F}_{2^n}$, and use the binary basis $(\beta_v)_{v \in I_\ell}$ from Chapter 6.

Definition 8.1 (Two embeddings)**.** Let $A := L \otimes_{\mathbb{F}_2} L$ be the tensor product with multiplication $(a \otimes b)(c \otimes d) = ac \otimes bd$ and identity $1 \otimes 1$. Define

$$\varphi_0(a) := a \otimes 1, \qquad \varphi_1(a) := 1 \otimes a.$$

For $x = \sum_i a_i \otimes b_i$ define the multiplication map $\mu(x) := \sum_i a_i b_i \in L$.

Lemma 8.2 (The multiplication map is well defined)**.** *The map $\mu : A \to L$ is an $\mathbb{F}_2$-linear map, and $\mu(\varphi_0(a)) = \mu(\varphi_1(a)) = a$.*

Proof. The map $(a, b) \mapsto ab$ is additive in each argument and balanced over $\mathbb{F}_2$. By the universal property of the tensor product it induces a unique $\mathbb{F}_2$-linear map μ with $\mu(a \otimes b) = ab$. Evaluating at $a \otimes 1$ and $1 \otimes a$ gives the two identities. □

Lemma 8.3 (Two commuting field embeddings)**.** *Both $\varphi_0, \varphi_1 : L \to A$ are injective unital ring homomorphisms, and $\varphi_0(a)\varphi_1(b) = a \otimes b$.*

Proof. Additivity, multiplicativity and preservation of the unit follow from the defining tensor relations and multiplication rule. The formula for the product is $(a \otimes 1)(1 \otimes b) = a \otimes b$. If $\varphi_i(a) = 0$, Lemma 8.2 implies $a = \mu(\varphi_i(a)) = 0$, so both maps are injective. The two images commute because A is a commutative algebra. □

Definition 8.4 (Column coordinates in the tensor algebra)**.** For $x \in A$ define $\mathrm{Col}(x)_v \in L$ as the unique coefficients satisfying

$$\boxed{x = \sum_{v \in I_\ell} \varphi_0(\mathrm{Col}(x)_v)\varphi_1(\beta_v).}$$

Lemma 8.5 (Existence and uniqueness of tensor columns)**.** *The family $(\varphi_1(\beta_v))_v$ is an L-basis of A with scalar action through φ_0, so Definition 8.4 is well defined.*

Proof. Every $b \in L$ has a unique expansion $b = \sum_v d_v(b)\beta_v$ with $d_v(b) \in \mathbb{F}_2$. For a pure tensor,

$$a \otimes b = \sum_v d_v(b)\,(a \otimes \beta_v) = \sum_v \varphi_0(a d_v(b))\varphi_1(\beta_v).$$

Every tensor is a finite sum of pure tensors, so the proposed family spans A over the left L-action. For independence, let $\sum_v a_v \otimes \beta_v = 0$. For each u, apply the well-defined linear map $\mathrm{id}_L \otimes d_u : A \to L \otimes_{\mathbb{F}_2} \mathbb{F}_2 \simeq L$; the result is $\sum_v a_v d_u(\beta_v) = a_u = 0$. All coefficients vanish. □

Lemma 8.6 (Columns of the two embeddings)**.** *For all $a, b \in L$,*

$$\mathrm{Col}(\varphi_0(a))_v = a\, d_v(1), \qquad \mathrm{Col}(\varphi_1(b))_v = d_v(b).$$

In particular $\mathrm{Col}(\varphi_1(b))$ *has entries in* $\mathbb{F}_2$.

Proof. Use the basis expansions $1 = \sum_v d_v(1)\beta_v$ and $b = \sum_v d_v(b)\beta_v$ in the second tensor factor, then compare the unique left-L coefficients from Lemma 8.5. □

Lemma 8.7 (Left scalar multiplication acts columnwise)**.** *For all $a \in L$ and $x \in A$,*

$$\mathrm{Col}(\varphi_0(a)x)_v = a\ \mathrm{Col}(x)_v.$$

Proof. Insert the unique column expansion of x, multiply each term by $\varphi_0(a)$, use multiplicativity of φ_0, and compare coefficients in the left-L basis. □

Definition 8.8 (Weighted column contraction)**.** For $q = (q_v)_v \in L^{I_\ell}$ define

$$C_q : A \to L, \qquad C_q(x) := \sum_v q_v\, \mathrm{Col}(x)_v.$$

Lemma 8.9 (Contraction is linear in the left scalar action)**.** *For $a \in L$ and $x, y \in A$, $C_q(\varphi_0(a)x + y) = aC_q(x) + C_q(y)$.*

Proof. Every coordinate map Col is additive, and Lemma 8.7 shows compatibility with the left scalar action. Multiply by q_v and sum over v. □

Lemma 8.10 (Ordinary multiplication loses tensor information)**.** *The map $\mu : A \to L$ is not injective when $n > 1$. More precisely, for any $a \in L \setminus \mathbb{F}_2$ the tensor $\varphi_0(a) + \varphi_1(a)$ is nonzero but has image zero under μ.*

Proof. By Lemma 8.2, $\mu(\varphi_0(a)+\varphi_1(a)) = a+a = 0$. Suppose the tensor were zero. Its v-th column would satisfy $a\, d_v(1) + d_v(a) = 0$ for every v, by Lemma 8.6. As $1 \neq 0$, some coordinate $d_v(1) = 1$. For that v the equality gives $a = d_v(a) \in \mathbb{F}_2$, a contradiction. □

Remark 8.11 (Provenance and scope). The two embeddings and the distinction between columns and the multiplication map are essential to Diamond–Posen's actual ring-switching Construction 4.1. The binary transposition map $\mathcal{T}$ of Chapter 7 is a related coordinate operation, but it is not a substitute for the entire tensor-algebra sumcheck.

8.2 Packed polynomial and the tensor-valued evaluation

Fix a Boolean witness $w : I_\ell \times I_h \to \mathbb{F}_2$, with $h \geq 1$, and the evaluation points $r_{\mathrm{lo}} \in L^\ell$, $r_{\mathrm{hi}} \in L^h$ from Chapter 7. Write $q_v = \mathrm{eq}_\ell(v, r_{\mathrm{lo}})$, $h_j = \mathrm{eq}_h(j, r_{\mathrm{hi}})$.

Definition 8.12 (Packed multilinear polynomial)**.** The packed table is $t'(j) := \sum_v \beta_v w_{v,j} = \widehat{w}_j$, $j \in I_h$. Its multilinear extension is

$$T'(X) = \sum_{j \in I_h} \widehat{w}_j \, \mathrm{eq}_h(j, X) \in L[X_0, \ldots, X_{h-1}]_{\mathrm{mult}}.$$

Let $\varphi_1(T')$ denote the polynomial obtained by applying φ_1 to each coefficient of T'.

Lemma 8.13 (Tensor expansion on Boolean columns)**.** *For every $j \in I_h$,*

$$\varphi_1(\widehat{w}_j) = \sum_v \varphi_0(w_{v,j}) \varphi_1(\beta_v).$$

Proof. By definition $\widehat{w}_j = \sum_v \beta_v w_{v,j}$. Apply the additive ring homomorphism φ_1. Because $w_{v,j} \in \mathbb{F}_2$, the scalar embeddings agree: $\varphi_1(w_{v,j}) = \varphi_0(w_{v,j})$. Rearranging the commutative products gives the displayed expansion. □

Definition 8.14 (Tensor partial evaluation)**.** Put

$$S := \varphi_1(T')(\varphi_0(r_{\mathrm{hi},0}), \ldots, \varphi_0(r_{\mathrm{hi},h-1})) \in A.$$

Lemma 8.15 (Evaluation of equality weights through the left embedding)**.** *For each $j \in I_h$,*

$$\mathrm{eq}_h(j, \varphi_0(r_{\mathrm{hi}})) = \varphi_0(h_j).$$

Proof. Each equality weight is a product of factors of the form $(1 - r_i)$ or r_i, selected by the Boolean index j_i. A unital ring homomorphism preserves addition, products, and the elements $0, 1$. Applying φ_0 coordinatewise to the defining product thus gives the same result as evaluating at the embedded point. □

Theorem 8.16 (Columns of the tensor evaluation)**.** *For every $v \in I_\ell$,*

$$\boxed{\mathrm{Col}(S)_v = \sum_{j \in I_h} h_j w_{v,j} = s^*_v.}$$

Proof. Multilinear interpolation and Lemma 8.15 give

$$S = \sum_j \varphi_1(\widehat{w}_j) \varphi_0(h_j).$$

Insert Lemma 8.13:

$$S = \sum_{j,v} \varphi_0(w_{v,j} h_j) \varphi_1(\beta_v).$$

Interchange the finite sums, use additivity of φ_0, and obtain

$$S = \sum_v \varphi_0 \left(\sum_j h_j w_{v,j} \right) \varphi_1(\beta_v).$$

Uniqueness of column coefficients in Lemma 8.5 shows that the v-th coefficient is $\sum_j h_j w_{v,j}$, precisely the partial evaluation s^*_v defined in Chapter 7. □

Theorem 8.17 (Original evaluation is a column contraction)**.** *The true evaluation of the original Boolean multilinear table is*

$$\boxed{y_w = C_q(S).}$$

Proof. By Theorem 8.16, $C_q(S) = \sum_v q_v s_v^*$. Lemma 7.27 identifies this finite sum with y_w. □

Lemma 8.18 (Multiplication recovers only the packed high-axis evaluation)**.**

$$\mu(S) = T'(r_{\mathrm{hi}}) = \sum_v \beta_v s_v^*.$$

Proof. Apply the multiplication map μ to the displayed expansion of S in the proof of Theorem 8.16. As $\mu(\varphi_0(a)\varphi_1(b)) = ab$, the image is $\sum_v \beta_v s_v^*$. On the other hand multilinear interpolation gives $T'(r_{\mathrm{hi}}) = \sum_j h_j \widehat{w}_j = \sum_{j,v} h_j \beta_v w_{v,j} = \sum_v \beta_v s_v^*$. □

Definition 8.19 (Sumcheck summand over A)**.** Define the A-coefficient polynomial

$$H(X) := \varphi_1(T')(X)\ \mathrm{eq}_h(\varphi_0(r_{\mathrm{hi}}), X) \in A[X_0, \ldots, X_{h-1}],$$

where the multivariate equality polynomial uses the usual affine factors in each variable.

Lemma 8.20 (Individual variable degree of the sumcheck summand)**.** *The degree of H in each variable X_i is at most 2.*

Proof. The polynomial T' is multilinear, so its coefficientwise image under φ_1 has degree at most 1 in each variable. The equality polynomial is also multilinear, hence has individual variable degree at most 1. In a commutative polynomial ring, degrees of products in any fixed variable are bounded above by the sum of the degrees of the factors. Therefore $\deg_{X_i} H \le 2$. □

Theorem 8.21 (Tensor Boolean sum equals its partial evaluation)**.**

$$\boxed{\sum_{j \in I_h} H(j) = S.}$$

Proof. For Boolean j, multilinear interpolation and Lemma 8.15 give

$$\varphi_1(T')(j) = \varphi_1(\widehat{w}_j), \qquad \mathrm{eq}_h(\varphi_0(r_{\mathrm{hi}}), j) = \varphi_0(h_j).$$

Consequently,

$$\sum_j H(j) = \sum_j \varphi_1(\widehat{w}_j)\varphi_0(h_j) = S,$$

where the last equality is the expansion in Theorem 8.16. □

Lemma 8.22 (Reduction to a large-field evaluation at a common challenge)**.** *For $z \in L^h$,*

$$\boxed{H(\varphi_1(z)) = \varphi_1(T'(z))\ \mathrm{eq}_h(\varphi_0(r_{\mathrm{hi}}), \varphi_1(z)).}$$

Proof. Evaluate Definition 8.19 at the tuple of elements $\varphi_1(z_i)$. Because φ_1 is a ring homomorphism, substituting $\varphi_1(z_i)$ into the coefficientwise image $\varphi_1(T')$ equals $\varphi_1(T'(z))$. The second factor is already the displayed public equality polynomial at that tuple. □

Remark 8.23 (Why the tensor is essential). The ordinary packed value $T'(r_{\mathrm{hi}})$ equals $\mu(S)$; it does not preserve the independent columns of S. The verifier needs these columns to apply its arbitrary low-axis weights q. The noninjectivity of μ from Lemma 8.10 explains why evaluating only T' at r_{hi} cannot replace the tensor phase of the original compiler.

8.3 Soundness of sumcheck over a tensor algebra

The ring A is not a field, so the ordinary polynomial root bound over an integral domain cannot be invoked without a proof. The following elementary lemma is the precise ring-theoretic substitute needed for Diamond–Posen's sumcheck (cf. [DP24, Lemma 4.5]).

Lemma 8.24 (Factor theorem for a commutative ring)**.** *Let R be a commutative ring with identity, $P \in R[X]$ and $a \in R$. There exist $Q \in R[X]$ and a constant remainder $c \in R$ with*

$$P(X) = (X - a)Q(X) + c, \qquad c = P(a).$$

If $P \neq 0$ and $\deg P = d \geq 1$, then $\deg Q = d - 1$.

Proof. For every exponent $j \geq 1$,

$$X^j - a^j = (X - a) \sum_{i=0}^{j-1} X^{j-1-i} a^i,$$

as follows by distributing and canceling successive terms in the finite sum. Write

$$P(X) = \sum_{j=0}^{d} p_j X^j.$$

Subtracting $P(a)$ gives $(X - a)$ multiplied by

$$\sum_{j=1}^{d} p_j \sum_{i=0}^{j-1} X^{j-1-i} a^i.$$

This supplies Q and establishes $c = P(a)$. If $p_d \neq 0$, the coefficient of X^{d-1} in Q is exactly p_d: only the term $j = d, i = 0$ contributes to that degree. Thus $\deg Q = d - 1$. □

Lemma 8.25 (Units from the embedded challenge field)**.** *For distinct $a, b \in L$, the element $\varphi_1(a) - \varphi_1(b)$ is a unit of A.*

Proof. The difference equals $\varphi_1(a - b)$, by additivity of φ_1. Since $a - b \neq 0$, its inverse $(a - b)^{-1}$ exists in L. Multiplicativity and preservation of the unit show $\varphi_1(a - b)\varphi_1((a - b)^{-1}) = 1$. Hence the difference is invertible in A. □

Theorem 8.26 (Root bound inside the embedded field)**.** *Let $P \in A[X]$ be nonzero with degree at most d, where $d \geq 0$. Then*

$$\boxed{|\{z \in L : P(\varphi_1(z)) = 0\}| \leq d.}$$

Proof. Induct on $\deg P$. If $\deg P = 0$, P is a nonzero constant and has no roots. Suppose $\deg P = D \geq 1$. If it has no root of the form $\varphi_1(z)$, there is nothing to prove. Otherwise choose one such root a. Lemma 8.24 gives $P(X) = (X - \varphi_1(a))Q(X)$ with $\deg Q = D - 1$. For any other root $b \neq a$,

$$0 = P(\varphi_1(b)) = (\varphi_1(b) - \varphi_1(a))Q(\varphi_1(b)).$$

The first factor is a unit by Lemma 8.25, so $Q(\varphi_1(b)) = 0$. The inductive hypothesis gives at most $D - 1$ such other roots. Including a, the total is at most $D \leq d$. No assumption that A is an integral domain was used. □

Corollary 8.27 (Uniform challenge root probability)**.** *For $P \neq 0$ of degree at most d chosen independently of a fresh uniform $z \in L$,*

$$\Pr_z[P(\varphi_1(z)) = 0] \leq d/q.$$

Proof. Theorem 8.26 allows at most d values of z out of the q equally likely field elements. □

Definition 8.28 (True partial-sum polynomial)**.** For each $0 \leq i < h$ and each previously sampled challenge prefix $z_{<i} = (z_0, \ldots, z_{i-1}) \in L^i$, define

$$G_i(X) := \sum_{b \in I_{h-i-1}} H(\varphi_1(z_0), \ldots, \varphi_1(z_{i-1}), X, b) \in A[X].$$

The notation for $i = 0$ means the empty prefix.

Lemma 8.29 (True sumcheck degree and endpoint identity)**.** *For each i, the polynomial G_i has degree at most 2, and*

$$G_i(0) + G_i(1) = \sum_{b \in I_{h-i}} H(\varphi_1(z_{<i}), b).$$

Proof. By Lemma 8.20, each variable of H has degree at most 2. Specializing earlier variables and later Boolean variables, then summing, cannot increase the degree in X. The right-hand finite sum splits uniquely according to its first bit 0 or 1; its two parts are $G_i(0)$ and $G_i(1)$, respectively. □

Definition 8.30 (Degree-two tensor sumcheck)**.** Given public initial claim $S_0 \in A$, let $s_0 = S_0$. For $i = 0, \ldots, h-1$: the prover sends a polynomial $g_i(X) \in A[X]$ of degree at most 2 *before* the verifier samples $z_i \in L$ uniformly. The verifier requires

$$s_i = g_i(0) + g_i(1),$$

then sets $s_{i+1} := g_i(\varphi_1(z_i))$. At the end it checks $s_h = H(\varphi_1(z_0), \ldots, \varphi_1(z_{h-1}))$, provided the target evaluation can be obtained from the underlying large-field evaluation oracle.

Lemma 8.31 (A false running claim forces a nonzero discrepancy)**.** *Fix a legal round i and a previously sampled prefix for which*

$$s_i \neq \sum_{b \in I_{h-i}} H(\varphi_1(z_{<i}), b).$$

If the endpoint condition $s_i = g_i(0) + g_i(1)$ passes, then the polynomial $g_i - G_i \in A[X]$ is nonzero and has degree at most 2.

Proof. Lemma 8.29 shows that the Boolean endpoint sum of G_i is the true running sum. If $g_i = G_i$, the accepted endpoint condition would imply that the claimed s_i equals this true sum, contrary to hypothesis. Thus $g_i - G_i \neq 0$. Both polynomials have degree at most 2 by the protocol and Lemma 8.29; their difference has degree at most 2 or is zero, and we have excluded the latter. □

Lemma 8.32 (One-round false-to-true transition bound)**.** *Under the hypotheses of Lemma 8.31, conditioned on the fixed prefix and the prover message g_i,*

$$\Pr_{z_i}\left[s_{i+1} = \sum_{b \in I_{h-i-1}} H(\varphi_1(z_{\leq i}), b)\right] \leq 2/q.$$

Proof. The true next running claim is $G_i(\varphi_1(z_i))$, by Definition 8.28. The new claimed value is $g_i(\varphi_1(z_i))$. Their equality means that the nonzero degree-at-most-two polynomial $g_i - G_i$ vanishes at $\varphi_1(z_i)$. The challenge is independent of the fixed prefix and polynomial, so Corollary 8.27 bounds that probability by $2/q$. □

Theorem 8.33 (Tensor sumcheck soundness)**.** *Suppose $S_0 \neq \sum_{b \in I_h} H(b)$ and the prover's messages are fixed before their respective fresh challenges. Then the probability that all degree-two sumcheck endpoint conditions pass* and *the final true equality $s_h = H(\varphi_1(z))$ holds is at most*

$$\boxed{\min\{1, 2h/q\}.}$$

This statement holds for arbitrary adaptive, randomized causal provers.

Proof. The initial running claim is false by hypothesis and the definition of s_0. If every endpoint condition passes and the final true equality holds, there must be a first round at which the false running claim becomes true. At every fixed prefix with a false claim and a passing endpoint check, Lemma 8.32 bounds the probability of that transition by $2/q$, conditional on the already chosen prover message. A conditional union bound over the h rounds gives probability at most $2h/q$. For a randomized prover, condition on its private random tape and average using the finite law of total probability, as in Lemma 3.37. The result is also at most 1. □

8.4 The original ring-switching evaluation reduction

We state an algebraically explicit specialization of Diamond–Posen's Construction 4.1 [DP24]. Its large-field evaluation protocol is represented by a black-box committed-oracle interface, not silently replaced by our Chapter 6 trace checker.

Definition 8.34 (Underlying large-field evaluation interface)**.** An *underlying large-field evaluation protocol* consists of algorithms

$$\mathsf{Commit}'(T'), \quad \mathsf{Prove}'(C', z, u; T'), \quad \mathsf{Verify}'(C', z, u, \pi').$$

Here T' is a multilinear polynomial over L, C' is the commitment to its Boolean coefficient table, $z \in L^h$, and $u \in L$. Perfect correctness means every honest opening for $u = T'(z)$ is accepted. For a *fixed, bound* T' we assume the following evaluation soundness property: for any causal adversary and any choice of the claimed (z, u) arising from the higher-level protocol,

$$\Pr[\mathsf{Verify}' \text{ accepts } \wedge \ u \neq T'(z)] \leq \varepsilon_{\mathrm{eval}}.$$

This is an assumption on the external protocol, not a theorem of the tensor algebra.

Definition 8.35 (Ring-switching prover and verifier)**.** The public instance is $(C', r_{\mathrm{lo}}, r_{\mathrm{hi}}, y_{\mathrm{pub}})$; the prover holds the Boolean table w that packs into the committed T'. The ring-switching reduction runs as follows.

(i) **Commit:** The prover packs w into the table of T' and invokes $\mathsf{Commit}'(T')$ to obtain C'. The commitment and its bound table are fixed before the interactive evaluation.

(ii) **Tensor partial message:** The prover forms $S = \varphi_1(T')(\varphi_0(r_{\mathrm{hi}}))$ and sends an alleged value $S_0 \in A$.

(iii) **Original-claim check:** The verifier requires

$$\boxed{y_{\mathrm{pub}} = C_q(S_0).}$$

This step reads every tensor column but does not reveal the packed polynomial.

(iv) **Ring sumcheck:** Run Definition 8.30 starting with $s_0 = S_0$ for the degree-two polynomial $H(X) = \varphi_1(T')(X)\,\mathrm{eq}_h(\varphi_0(r_{\mathrm{hi}}), X)$. For every $i < h$, the prover sends $g_i \in A[X]$ of degree at most 2; the verifier checks $s_i = g_i(0) + g_i(1)$, then samples $z_i \leftarrow L$ and puts $s_{i+1} = g_i(\varphi_1(z_i))$.

(v) **Reduction to the large field:** The prover sends an alleged $u \in L$ for $T'(z)$, where $z = (z_0, \ldots, z_{h-1})$, and the verifier requires

$$\boxed{s_h = \varphi_1(u)\ \mathrm{eq}_h(\varphi_0(r_{\mathrm{hi}}), \varphi_1(z)).}$$

(vi) **Delegated evaluation:** The parties execute the large-field evaluation protocol and accept only if $\mathsf{Verify}'(C', z, u, \pi') = 1$.

The verifier accepts precisely when all the displayed checks and the external proof pass.

Lemma 8.36 (All verifier quantities are typed)**.** *Every equality checked in Definition 8.35 is between elements of the same ring or field, and the final evaluation request to the black box is over L, not over the tensor algebra A.*

Proof. The element S_0 and each $s_i, g_i(0), g_i(1), g_i(\varphi_1(z_i))$ belong to A. The contraction $C_q(S_0)$ is in L, matching $y_{\mathrm{pub}} \in L$. The terminal expression is a product of two elements of A, matching $s_h \in A$. Finally u and $T'(z)$ belong to L, and C' is a commitment to an L-coefficient table. □

Lemma 8.37 (The public terminal equality factor is computable)**.** *The element $\mathrm{eq}_h(\varphi_0(r_{\mathrm{hi}}), \varphi_1(z))$ is computable by the verifier from public values without accessing the witness table.*

Proof. All coordinates of r_{hi} and the freshly sampled z are public at the final step. Both embeddings φ_0, φ_1 and the equality polynomial are specified by the public field representation. Evaluating their finite product requires no private coefficient of T'. □

Definition 8.38 (Committed-word timing invariant)**.** A run is *causal and bound* if: T' is fixed by its commitment before the evaluation interaction; S_0 is sent before the sumcheck challenges; each g_i is fixed before z_i; the final u may depend on the sampled z but must be verified against the already bound T'; and the external evaluation protocol satisfies the assumed error bound of Definition 8.34.

Remark 8.39 (Protocol name and attribution). This is the *unmasked* reduction of Diamond–Posen, specialized to $K = \mathbb{F}_2$, extension L/K of degree $n = 2^\ell$, and h remaining variables. It is distinct from the *masked* field-affine variant developed in Chapter 7. In particular, the original prover sends S_0 in the clear, whereas the masking program aims to prevent disclosure of its witness-dependent columns.

8.5 Completeness and composition soundness of the compiler

Lemma 8.40 (Honest initial contraction)**.** *If the public claim is true, $y_{\mathrm{pub}} = y_w$, and the prover sends the correct tensor message $S_0 = S$, the initial contraction check passes.*

Proof. By Theorem 8.17, $C_q(S) = y_w$. Substitute $S_0 = S$ and $y_{\mathrm{pub}} = y_w$. □

Lemma 8.41 (Honest sumcheck telescoping)**.** *If $S_0 = S = \sum_{b \in I_h} H(b)$ and the prover sends the true partial-sum polynomial $g_i = G_i$ at every round, then all endpoint checks pass and*

$$s_h = H(\varphi_1(z)).$$

Proof. Induct on the number i of completed rounds. Initially $s_0 = S$ equals the full Boolean sum by Theorem 8.21. Suppose s_i equals the true Boolean sum over the remaining variables at challenge prefix $z_{<i}$. The endpoint identity of Lemma 8.29 gives $s_i = G_i(0) + G_i(1) = g_i(0) + g_i(1)$, so the verifier's check passes. It then sets $s_{i+1} = G_i(\varphi_1(z_i))$, which by the definition of G_i is exactly the sum over the remaining $h - i - 1$ Boolean variables. This proves the induction invariant. At $i = h$, no variables remain, and the sole term is $H(\varphi_1(z_0), \ldots, \varphi_1(z_{h-1}))$. □

Lemma 8.42 (Honest terminal check)**.** *If the sumcheck is honest and the prover sends $u = T'(z)$, the terminal tensor equality in Definition 8.35 passes.*

Proof. Lemma 8.41 gives $s_h = H(\varphi_1(z))$. Lemma 8.22 identifies this element with $\varphi_1(T'(z))$ times the public equality factor. Replacing $T'(z)$ by u proves the terminal check. □

Theorem 8.43 (Perfect completeness of the unmasked compiler)**.** *Assume the underlying large-field evaluation protocol has perfect completeness in the sense of Definition 8.34. For an honest Boolean witness w and true public evaluation claim $y_{\mathrm{pub}} = y_w$, an honest execution of Definition 8.35 is accepted with probability* 1*.*

Proof. The prover constructs the packed polynomial and commits to it. The initial contraction passes by Lemma 8.40. Every sumcheck round passes by Lemma 8.41. The terminal equality passes by Lemma 8.42. The final large-field claim $u = T'(z)$ is true, so the delegated evaluation proof passes by its assumed perfect completeness. All checks pass for every choice of verifier challenges. □

Definition 8.44 (Unpacking an arbitrary committed polynomial)**.** For any multilinear $T' \in L[X_0, \ldots, X_{h-1}]_{\mathrm{mult}}$, define its underlying Boolean table

$$\mathrm{Dec}(T')(v, j) := d_v(T'(j)) \in \mathbb{F}_2.$$

Let $w^{T'}$ denote this table.

Lemma 8.45 (Packing after extraction gives the original packed polynomial)**.** *For each $j \in I_h$,*

$$\sum_v \beta_v w^{T'}_{v,j} = T'(j).$$

Therefore the packed multilinear extension of $w^{T'}$ is exactly T'.

Proof. The first equality is the unique binary basis expansion $T'(j) = \sum_v \beta_v d_v(T'(j))$. Both T' and the packed multilinear extension of $w^{T'}$ are multilinear polynomials over L with identical values at all Boolean vertices $j \in I_h$. The uniqueness of multilinear interpolation (from Chapter 1) therefore makes them identical. □

Lemma 8.46 (False original claim implies false tensor initial sum)**.** *Fix a multilinear packed polynomial T' and its extracted Boolean table $w^{T'}$. If $y_{\mathrm{pub}} \neq \widetilde{w^{T'}}(r_{\mathrm{lo}}, r_{\mathrm{hi}})$ and the initial contraction check $y_{\mathrm{pub}} = C_q(S_0)$ passes, then*

$$S_0 \neq \sum_{j \in I_h} H(j).$$

Proof. By Lemma 8.45, T' is the exact packing of $w^{T'}$. Theorem 8.21 identifies the true tensor sum with S. Theorems 8.16 and 8.17, applied to the extracted table, give $C_q(\sum_j H(j)) = \widetilde{w^{T'}}(r_{\mathrm{lo}}, r_{\mathrm{hi}})$. If S_0 were that true sum, the passing initial contraction would force y_{pub} to equal the extracted evaluation, contrary to hypothesis. □

Lemma 8.47 (A correct delegated opening implies the true terminal tensor value)**.** *Suppose the delegated claimed value satisfies $u = T'(z)$. If the terminal equality*

$$s_h = \varphi_1(u)\, \mathrm{eq}_h(\varphi_0(r_{\mathrm{hi}}), \varphi_1(z))$$

passes, then $s_h = H(\varphi_1(z))$.

Proof. Substitute $u = T'(z)$ into the terminal equality and apply Lemma 8.22, which identifies the product with $H(\varphi_1(z))$. □

Theorem 8.48 (False-claim soundness with a bound underlying large-field oracle)**.** *Fix a committed multilinear T' and its extracted Boolean table $w^{T'}$ before any interactive challenges. Assume the underlying evaluation protocol satisfies Definition 8.34 with error $\varepsilon_{\mathrm{eval}}$. If $y_{\mathrm{pub}} \neq \widetilde{w^{T'}}(r_{\mathrm{lo}}, r_{\mathrm{hi}})$, then for every randomized causal prover*

$$\boxed{\Pr[\text{ring switch verifier accepts}] \leq \min\left\{1, \frac{2h}{q} + \varepsilon_{\mathrm{eval}}\right\}.}$$

Proof. Let E denote the event that the verifier accepts its delegated large-field evaluation proof even though $u \neq T'(z)$. The underlying soundness assumption gives $\Pr[E] \leq \varepsilon_{\mathrm{eval}}$. We now condition on the complement of E and analyze the ring-switching checks.

If the initial contraction fails, the verifier rejects. Otherwise Lemma 8.46 proves that the initial ring-sumcheck claim is false. On a run in E^c where the underlying verification accepts, the final claimed opening satisfies $u = T'(z)$. If the terminal tensor equality passes, Lemma 8.47 then forces the final sumcheck claim to be the *true* value $H(\varphi_1(z))$. Theorem 8.33 bounds the probability of that event, together with all sumcheck checks passing, by $2h/q$. Thus acceptance is contained in the union of E and the false-to-true sumcheck event. The union bound gives $\Pr[\text{accept}] \leq \varepsilon_{\mathrm{eval}} + 2h/q$. Cap by 1 to finish. □

Corollary 8.49 (Ideal large-field oracle error)**.** *If the delegated evaluation is perfectly sound ($\varepsilon_{\mathrm{eval}} = 0$), then a false original evaluation claim survives the tensor ring-switching reduction with probability at most $\min\{1, 2h/q\}$.*

Proof. Set $\varepsilon_{\mathrm{eval}} = 0$ in Theorem 8.48. □

Remark 8.50 (Difference from a full IOP knowledge theorem). Theorem 8.48 assumes that the committed polynomial T' has already been bound. It does not reprove the emulator/extractor theorem for the actual vector-oracle model of Diamond–Posen, nor does it account for all implementation-level binding and query errors. The original authors give their own compiler security argument in Theorem 4.4 of [DP24]. This chapter isolates a quantitative algebraic soundness interface for later formalization.

8.6 Information disclosure and its relationship to masked kernel coordinates

The tensor compiler above is *not* zero knowledge. Its first prover message is the collection of all partial evaluations s_v^*, represented as an element of A. This is precisely the leakage surface addressed by later work [Koc26; Fs26].

Lemma 8.51 (The first message reveals every partial evaluation)**.** *In an honest execution, the verifier can calculate*

$$\boxed{s_v^* = \mathrm{Col}(S_0)_v}$$

for every $v \in I_\ell$ using only its first received ring-switching message.

Proof. For an honest prover $S_0 = S$. Theorem 8.16 identifies the v-th tensor column of S with the desired partial evaluation s_v^*. The column operation is explicitly available from the public tensor-field basis, so the verifier can evaluate it. □

Theorem 8.52 (Perfect two-witness distinction from the first message)**.** *For every $\ell \geq 1$ and $h \geq 1$, there exist two distinct Boolean witness tables $w^{(0)}, w^{(1)}$ and a public evaluation point $(r_{\mathrm{lo}}, r_{\mathrm{hi}})$ such that both tables have the same claimed evaluation $y_{\mathrm{pub}} = 0$, but the distributions of the honest first tensor messages S_0 have total variation distance 1. Thus the unmasked tensor protocol does not satisfy statistical zero knowledge in the evaluation-only public-input game of Chapter 3.*

Proof. Choose $r_{\mathrm{lo}} = 0_\ell$ and $r_{\mathrm{hi}} = 0_h$, both Boolean evaluation points. Let $w^{(0)}$ be identically zero. Choose a nonzero Boolean index $v_* \in I_\ell$ and let $w^{(1)}$ be zero everywhere except at $(v_*, 0_h)$, where it equals 1. At the Boolean point $(0_\ell, 0_h)$, both multilinear extensions evaluate to the witness entry $w^{(i)}_{0_\ell, 0_h} = 0$, so both public claims agree.

For $w^{(0)}$, every partial evaluation s_v^* is zero. For $w^{(1)}$, the high-axis weights at $r_{\mathrm{hi}} = 0_h$ equal $\mathrm{eq}_h(j, 0_h) = \delta_{j,0_h}$. Therefore the partial vector has entry $s^*_{v_*} = 1$ and every other entry zero. By Theorem 8.16, the honest first messages have different v_*-th columns: 0 and 1, respectively. The deterministic test asking whether the v_*-th column equals zero outputs 1 with probability 1 for the first witness and 0 for the second. Lemma 3.5 therefore makes their total variation distance equal to 1. □

Definition 8.53 (Masked field-column interface)**.** To avoid exposing s_v^* directly, the field-affine interface of Chapter 7 uses the public vector

$$m_v := s_v^* + p_v$$

for hidden independent uniform pads $p_v \in L$. It also introduces the hidden transposed correction $c^p = \mathcal{T}(p)$ and checks the contraction $D_c(\theta) = \Delta_p(\theta)$ using the trace-coordinate checker of Chapter 6.

Lemma 8.54 (The basic masked column view does not distinguish)**.** *For any two fixed Boolean witness tables, the distributions of their basic public masked vectors m from Definition 8.53 are identical and uniform on L^I.*

Proof. For either fixed witness, the partial vector s^* is fixed. Lemma 7.40 proves that independent uniform $p \in L^I$ makes $s^* + p$ uniform on L^I. This distribution is the same for both witnesses. □

Lemma 8.55 (Revealing the full hidden correction reverses the mask)**.** *If the verifier obtains both $m = s^* + p$ and the full transposed pad vector $c^p = \mathcal{T}(p)$, then it can reconstruct all of s^* exactly.*

Proof. Theorem 7.19 gives $p = \mathcal{T}(c^p)$. Consequently $s^* = m + p = m + \mathcal{T}(c^p)$, as already proved in Theorem 7.71. □

Proposition 8.56 (The two protocol routes are distinct)**.** *The original unmasked compiler of Definition 8.35 and the masked algebraic checker of Chapters 6–7 have different verifier interfaces:*

(i) *The former sends an A-valued partial evaluation S_0 and executes an h-round degree-two sumcheck over A with challenges in $\varphi_1(L)$.*

(ii) *The latter exposes uniform $m \in L^I$, keeps p, c hidden, and uses a random kernel contraction and an anchored trace-zero consistency checker over L with challenges (θ, z).*

Theorem 8.48 does not automatically imply soundness of the latter; Theorem 7.61 does not automatically imply zero knowledge or proximity soundness of the former.

Proof. The first distinction follows directly by comparing the typed messages and checks in Definitions 8.35 and 8.53. The two soundness theorems quantify different acceptance events and impose different timing and underlying-oracle hypotheses. Neither theorem's hypotheses entail the other's, so no implication is asserted without a new reduction proof. □

Remark 8.57 (Prior-work attribution). Diamond–Posen introduce the tensor-algebra ring-switching compiler [DP24]. Diamond's zero-knowledge large-field PCS note explicitly leaves ring-switching outside its scope [Dia25]. Kochergin and Frigo–shelat subsequently give masking approaches for binary-field ring switching [Koc26; Fs26]. Our use of the Boolean kernel basis as a representation of the correction has been established algebraically; a claim that the complete cryptographic construction is new or secure requires a separate comparison and a full zero-knowledge simulator.

8.7 Assumptions, efficiencies, and formalization boundaries

Definition 8.58 (The three proof-system layers)**.** The development distinguishes:

(i) *Algebraic reduction:* exact packing, the tensor evaluation S, and the sumcheck polynomial H;

(ii) *Committed evaluation:* a fixed packed polynomial T', a binding representation C', and an underlying large-field evaluation proof;

(iii) *Zero-knowledge implementation:* commitments, masked messages and a simulator producing the full adaptive verifier view.

Only the first layer is proved unconditionally in this chapter. The second has an explicit conditional soundness theorem; the third remains an obligation.

Lemma 8.59 (No increase in the packed Boolean table size)**.** *The original Boolean table has $2^{\ell+h} = n\,2^h$ binary entries. The packed table has 2^h elements of L, each represented by n binary coordinates. Thus both data representations contain exactly $n\,2^h$ binary coordinates.*

Proof. The set I_ℓ has cardinality $2^\ell = n$ and I_h has cardinality 2^h, so the original table has $n2^h$ entries. The packed table has one element of L for each $j \in I_h$. Since $[L : \mathbb{F}_2] = n$, every field element has exactly n binary basis coordinates, and hence the packed table has $n2^h$ such coordinates. □

Lemma 8.60 (Message types and sumcheck round count)**.** *The tensor reduction sends one initial A-element S_0, then exactly h univariate A-polynomials of degree at most 2, and one final field element $u \in L$ before invoking the delegated large-field evaluation protocol.*

Proof. The number and type of messages are specified in Definition 8.35: one initial tensor message, one polynomial in each iteration $i = 0, \dots, h-1$, and one claimed evaluation in the last reduction step. The degree bound follows from Definition 8.30. □

Remark 8.61 (Why this is not the masked protocol). The initial tensor message S_0 reveals the partial table s^* by Lemma 8.51. The masking proposal in Chapters 6–7 changes the public messages and proof constraints; it cannot inherit the original sumcheck theorem merely by reusing the name ring switching. Its own proof needs an end-to-end simulation of (m, y', U, A) and all commitment/opening messages, with every challenge timing and correlation checked.

Definition 8.62 (Chapter 8 Lean certification contract)**.** To certify the exact compiler in Lean 4, the formal model will provide:

(i) the finite extension $L/\mathbb{F}_2$, two L-module structures on A, tensor product and its universal map;

(ii) binary packing/unpacking and the polynomial T';

(iii) the typed column functional and tensor equality interpolation;

(iv) the polynomial ring $A[X]$ and its root bound restricted to embedded units;

(v) causal sumcheck messages, independent challenge sampling, and its soundness event; and

(vi) a separate abstract large-field commitment interface with an explicitly parameterized evaluation error.

No Lean declaration is claimed certified until its code has been compiled and checked.

Remark 8.63 (History and non-novelty). The *unmasked* compiler and its tensor-sumcheck soundness come from Diamond–Posen [DP24, Construction 4.1, Theorems 4.2 and 4.4]. The proof of the embedded-field root bound corresponds to their Lemma 4.5. The present chapter reorganizes those established arguments as small dependency-ordered proofs, specialized to the Boolean-kernel basis already used in this manuscript. It is not presented as a new ring-switching construction.

8.8 The theorem chain and its exact scope

Put $T'(j) = \sum_v \beta_v w_{v,j}$ and

$$S = \varphi_1(T')(\varphi_0(r_{\mathrm{hi}})), \qquad H(X) = \varphi_1(T')(X)\,\mathrm{eq}_h(\varphi_0(r_{\mathrm{hi}}), X).$$

The original Boolean evaluation and tensor sumcheck identities are

$$\boxed{y_w = C_q(S), \qquad \sum_{j \in I_h} H(j) = S, \qquad \deg_{X_i} H \le 2.}$$

A nonzero degree-d polynomial in $A[X]$ has at most d zeros in the embedded subfield $\varphi_1(L)$, since distinct embedded field values differ by units in A. Consequently, under a fixed committed packed polynomial and causal prover timing,

$$\boxed{\Pr[\text{false original claim accepted}] \le \min\{1, 2h/q + \varepsilon_{\mathrm{eval}}\}.}$$

This is a conditional algebraic statement depending on the soundness of the delegated large-field evaluation protocol. The first unmasked tensor message reveals every s_v^* and is perfectly distinguishable for some equal-evaluation witnesses. Thus the unmasked protocol does not establish zero knowledge.

Chapter 7's hidden-pad correction is a *different* verification interface. It must independently satisfy soundness, commitment binding and full adaptive-view simulation before being claimed as a zero-knowledge implementation.

9 Zero-Knowledge Ring Switching: Public Simulation and Proof Interfaces

This chapter formalizes the masking interface built in Chapters 6–7 and distinguishes three progressively stronger statements: *perfect privacy of the public field messages*, *perfect complete-view privacy under an opaque ideal proof functionality*, and *conditional real-world zero knowledge* requiring a separately proved external-interface simulator. The order matters. One cannot infer that a cryptographic protocol is zero knowledge merely because each individual field message is uniform. The original unmasked tensor compiler is treated in Chapter 8 [DP24]; the ZK ring-switching literature is discussed in [Koc26; Fs26]. We make no claim that our specific binary-field kernel encoding is a drop-in zero-knowledge extension of the existing odd-characteristic KBFold PCS.

9.1 Security target, message schedule, and witness types

We retain $L = \mathbb{F}_{2^n}$, $n = 2^\ell$, $q = 2^n$, $I = I_\ell$, and the notation of Chapters 6–7. The binary witness $w : I \times I_h \to \mathbb{F}_2$ has packed columns $\widehat{w}_j = \sum_v \beta_v w_{v,j}$ and true claimed evaluation $y_w = \sum_{v,j} q_v h_j w_{v,j}$. We do not identify the masked ring-switching extension with the original unmasked tensor-sumcheck of Diamond–Posen [DP24]. Our public-message protocol is a kernel-coordinate realization of a candidate masked affine interface; trace-zero masking also appears in Frigo–shelat [Fs26] and the earlier ring-switching discussion of Kochergin [Koc26].

Definition 9.1 (Admissible single-opening instance)**.** An admissible instance consists of public $(r_{\mathrm{lo}}, r_{\mathrm{hi}}, y)$, a private Boolean table w, and exact packed columns $\widehat{w}$, with $y = \widetilde{w}(r_{\mathrm{lo}}, r_{\mathrm{hi}})$. The evaluation points are fixed for the duration of this one-opening experiment. The public value y is *authorized* information.

Definition 9.2 (Pre-challenge hidden state)**.** For one honest opening, sample independently

$$p \xleftarrow{\$} L^I, \qquad p_{\mathrm{in}} \xleftarrow{\$} L,$$

and form

$$s_v^* = \sum_j h_j w_{v,j}, \quad m = s^* + p, \quad y' = y + p_{\mathrm{in}}, \quad c = c^p = \mathcal{T}(p).$$

The mask vector p and correction c must be bound before the first consistency challenge θ. The public pre-challenge messages are (m, y').

Definition 9.3 (Post-challenge hidden state)**.** After $\theta \in L^\ell$ has been received, sample independently

$$t = (t_k)_{k \in I} \xleftarrow{\$} L^I, \qquad \rho \xleftarrow{\$} L, \qquad \sigma := \rho^2.$$

Write

$$V_k = V_k(p, c, \theta), \qquad U_k := V_k + t_k^2 + t_k.$$

Publish $U = (U_k)_k$ and bind (t, ρ, σ) before the second challenge z. After receiving $z \in L$, publish

$$A := \rho + \sum_k W_k(z) t_k.$$

The scalar ρ in this chapter is the auxiliary mask denoted r in Chapter 6; the distinct notation avoids confusion with the public evaluation point $(r_{\mathrm{lo}}, r_{\mathrm{hi}})$.

Definition 9.4 (Public-message schedule)**.** The *public core* is the ordered transcript

$$\mathsf{Core} := (m, y'; \theta; U; z; A).$$

A causal verifier may select θ after seeing (m, y') and z after seeing (m, y', θ, U), using its private randomness. The underlying commitments, their roots and authenticated openings, and the auxiliary proofs are *not* part of Core; they belong to the complete transcript defined later.

Lemma 9.5 (Perfect consistency of an honest correction)**.** *Under Definition 9.2, $D_c(\theta) = \Delta_p(\theta)$ for every $\theta \in L^\ell$, and therefore every V_k has trace zero.*

Proof. The equality $c = c^p$ is part of the honest definition. By Definition 6.54, the two kernel evaluations D_c and $\Delta_p = D_{c^p}$ coincide identically. Theorem 6.59 implies $\mathrm{Tr}(V_k) = 0$ for every coordinate k, independently of θ. □

Lemma 9.6 (All three checker equations hold for honest values)**.** *For every θ, z, the honest values of Definitions 9.2 and 9.3 pass all tests in Definition 6.83, with (r, s) there interpreted as (ρ, σ) here.*

Proof. By Lemma 9.5, $D_c(\theta) = \Delta_p(\theta)$. The honest values also satisfy $\sigma = \rho^2$, $U_k = V_k + t_k^2 + t_k$ and $A = \rho + \sum_k W_k(z) t_k$ by construction. These are exactly the hypotheses of Theorem 6.89, which proves every checker equation. □

Lemma 9.7 (Complete algebraic correctness of the masked interfaces)**.** *For every admissible instance and honest state as above,* CLAIM, IN *and* OUT *from Chapter 7 hold for all θ.*

Proof. The admissible-instance condition gives $y = y_w$. By definition $m = s^* + p$, $c = c^p$ and $y' = y + p_{\mathrm{in}}$. These are the hypotheses of Theorem 7.52, which proves all three relations. □

Remark 9.8 (Challenge order and single-opening scope). A malicious verifier's choices need not be uniform for the privacy theorems of this chapter. Uniform independent challenges are, however, required for the quantitative soundness estimates of Chapters 6–7. The present distributional analysis uses fresh $(p, p_{\mathrm{in}}, t, \rho)$ for a *single* opening; it must not be silently extended to pad reuse across openings.

9.2 Exact public-core distribution and a causal simulator

Lemma 9.9 (Masked input pair is jointly uniform)**.** *For every admissible instance and every fixed witness w, (m, y') is uniform on $L^I \times L$; this law is independent of w.*

Proof. The fresh independent masks (p, p_{in}) are uniform on $L^I \times L$. For fixed s^* and y, the map $(p, p_{\mathrm{in}}) \mapsto (s^* + p, y + p_{\mathrm{in}})$ is a translation and hence a bijection of the finite additive group. Every output pair has one preimage and therefore probability $q^{-(n+1)}$, independently of w. □

Lemma 9.10 (Conditional trace-zero uniformity)**.** *Fix any realized public prefix (m, y', θ) with positive probability in an honest execution, and fix any admissible witness w. Conditional on this prefix, the vector U is uniform on*

$$H^I, \qquad H := \ker(\mathrm{Tr} : L \to \mathbb{F}_2).$$

This conditional law is independent of the witness and of the prefix.

Proof. The equality $m = s^* + p$ determines $p = m + s^*$ once m and the fixed witness are given. Likewise $y' = y + p_{\mathrm{in}}$ determines p_{in}. Therefore $c = \mathcal{T}(p)$ and the $V_k(p, c, \theta)$ are fixed under the conditioning. By Lemma 9.5, every V_k lies in the trace-zero subspace H. The vector t, sampled only after θ, is independent uniform on L^I, even under this conditioning. For each k, Theorem 6.45 shows that $V_k + t_k^2 + t_k$ is uniform on H, independent of the fixed V_k. Independence of the t_k makes the coordinates of U independent. Consequently U is uniform on H^I. Because this law does not depend on the fixed prefix or w, it remains the same after conditioning only on that prefix and averaging over any remaining private randomness. □

Lemma 9.11 (Conditional uniformity of the final scalar)**.** *Fix any realized history (m, y', θ, U, z) with positive probability, and fix an admissible witness w. Then the honest final scalar A is uniform on L, and its conditional law is independent of w and the history.*

Proof. The scalar ρ is sampled independently and uniformly after θ. It is independent of p, p_{in}, t, the public messages derived from them, and the verifier's private random tape. The challenge z may be an arbitrary causal function of the already exposed history and this verifier tape, but it has no access to ρ. Thus, even after conditioning on (m, y', θ, U, z) and on any fixed value of t, the variable ρ remains uniform on L. The expression $A = \rho + \sum_k W_k(z) t_k$ is a translation of ρ by a fixed element under this finer conditioning and hence is uniform. Averaging over the conditional values of t keeps the uniform law. Its value is therefore independent of both the witness and public history. □

Definition 9.12 (Causal ideal public-core simulator)**.** For a fixed admissible public evaluation instance, the simulator $\mathsf{Sim}_{\mathrm{core}}$ has no witness and interacts with an arbitrary causal verifier as follows:

(i) Independently sample $\bar{m} \xleftarrow{\$} L^I$ and $\bar{y}' \xleftarrow{\$} L$, and publish them.

(ii) Receive the verifier's chosen $\theta \in L^\ell$.

(iii) Independently sample $\bar{U}$ uniformly in H^I and publish it.

(iv) Receive the verifier's chosen $z \in L$.

(v) Independently sample $\bar{A}$ uniformly in L and publish it.

All verifier choices may depend on the public history and on its own private random tape.

Theorem 9.13 (Perfect public-core zero knowledge for one opening)**.** *Let V^* be any causal verifier making arbitrary (not necessarily uniform) choices of θ and z in the domains above. For any two admissible witnesses having the same public instance, the complete joint distributions of the public core and the verifier's private random tape are identical. More precisely the honest public-core interaction has* exactly *the same law as $\mathsf{Sim}_{\mathrm{core}}$ interacting with V^*:*

$$\boxed{\mathsf{Law}(\mathsf{Core}^{P_w, V^*}) = \mathsf{Law}(\mathsf{Core}^{\mathsf{Sim}_{\mathrm{core}}, V^*}).}$$

This is a perfect simulation theorem for the public-message projection only*, not for commitment roots, authenticated queries, or auxiliary proof transcripts.*

Proof. Include the verifier's random tape in the experiment state and fix its sampled value for this proof. The verifier's next challenge is then a (possibly history-dependent) deterministic function of the messages so far. Initially Lemma 9.9 gives exactly the simulator's joint uniform distribution for (m, y'). Given any such realized pair, the verifier selects the same conditional distribution of θ in both processes (the same deterministic value under the fixed tape). Lemma 9.10 shows that the conditional distribution of the next message U is exactly uniform on H^I, matching the simulator independently of the preceding history. The verifier therefore selects z with the same conditional law in both processes. Finally Lemma 9.11 makes A uniform on L conditional on the full history, matching the simulator. The probability of each complete finite transcript factors into these identical successive conditional probabilities. Thus the full joint laws, including the fixed verifier tape, coincide. Average over the verifier tape to obtain the theorem. □

Corollary 9.14 (Conditional product law for fixed challenges)**.** *For fixed nonadaptive θ, z, the honest public messages (m, y', U, A) have the product-uniform distribution*

$$\boxed{\mathrm{Unif}(L^I) \otimes \mathrm{Unif}(L) \otimes \mathrm{Unif}(H^I) \otimes \mathrm{Unif}(L).}$$

Proof. In Theorem 9.13, take the verifier to output the two fixed challenges. The simulator draws the four factors independently and uniformly by Definition 9.12. Equality of laws gives the displayed product distribution. □

Remark 9.15 (The mask ρ is essential to this proof). Without an independent uniform ρ, the scalar $A = \sum_k W_k(z) t_k$ would be a function of the same t used to generate U. The conditional uniformity of A would no longer follow. This would require a new joint distribution analysis rather than relying on Theorem 6.66.

9.3 Conditional witness distributions and the limits of public simulation

Lemma 9.16 (Two solutions of a trace-zero equation)**.** *For every $a \in H = \ker \mathrm{Tr}$ and every solution $t_0 \in L$ of $t_0^2 + t_0 = a$, the complete solution set is $\{t_0, t_0 + 1\}$. The two values are distinct.*

Proof. Theorem 6.43 guarantees at least one solution t_0. If t is another, add the two equations: $(t + t_0)^2 + (t + t_0) = 0$ by Lemma 6.40. Lemma 6.41 implies $t + t_0 \in \{0, 1\}$, so $t = t_0$ or $t = t_0 + 1$. Conversely both values satisfy the equation because $1^2 + 1 = 0$. The values differ since $1 \neq 0$. □

Lemma 9.17 (Uniform conditional mask preimages)**.** *Fix $V \in H$ and sample t uniformly from L. For any $u \in H$, conditioned on $u = V + t^2 + t$, the variable t is uniform on the two-element set of roots of $t^2 + t = u + V$.*

Proof. The event is equivalent to $t^2 + t = u + V \in H$. By Lemma 9.16, it contains exactly two equally likely values of the original uniform sample t. Conditioning on this nonempty event gives probability $1/2$ to each. □

Lemma 9.18 (Conditioned vector mask is a product of two-point fibers)**.** *Fix an admissible witness, a public prefix (m, y', θ), and a realized trace-zero vector U. Under the honest execution, conditional on that data, the vector t is uniform on the Cartesian product*

$$\prod_{k \in I} \{t_{0,k}, t_{0,k} + 1\}, \qquad t_{0,k}^2 + t_{0,k} = U_k + V_k(p, c, \theta),$$

and each of its 2^n points is equally likely.

Proof. Given the fixed witness and m, the pad $p = m + s^*$ and the correction $c = \mathcal{T}(p)$ are determined, and hence so is each V_k. The original coordinates t_k are independent uniform variables and each equation $U_k = V_k + t_k^2 + t_k$ restricts only t_k. Lemma 9.17 gives a uniform two-point conditional law per coordinate. Because the conditioning event factors over the coordinates, these conditional laws remain independent, giving the stated product distribution. There are $|I| = n$ binary choices, hence 2^n vectors. □

Lemma 9.19 (Conditional reconstruction of the last scalar mask)**.** *For fixed (t, z, A) in an honest execution, the value of ρ is uniquely determined:*

$$\boxed{\rho = A + \sum_k W_k(z) t_k, \qquad \sigma = \rho^2.}$$

Proof. The honest last message satisfies $A = \rho + \sum_k W_k(z) t_k$. Adding the field sum to both sides, and using characteristic two, gives the displayed expression. The definition of the honest σ then gives $\sigma = \rho^2$. Uniqueness follows because these are deterministic field operations. □

Proposition 9.20 (Algebraic completion for a fixed witness and a public core)**.** *Fix an admissible private witness and any public core (m, y', θ, U, z, A) satisfying $\mathrm{Tr}(U_k) = 0$ for all k. There exist hidden values $(p, p_{\mathrm{in}}, c, t, \rho, \sigma)$ satisfying all honest algebraic relations in Definitions 9.2 and 9.3. Exactly 2^n choices of the vector t are possible; for each, all other hidden variables are uniquely determined.*

Proof. The equations for the pre-challenge data force $p = m + s^*$ and $p_{\mathrm{in}} = y' + y$, and then $c = \mathcal{T}(p)$. The trace consistency lemma (Lemma 9.5) gives $\mathrm{Tr}(V_k) = 0$ for this p, c, θ. Hence each $U_k + V_k$ has trace zero and Lemma 9.16 gives exactly two solutions of $t_k^2 + t_k = U_k + V_k$. Independent selection across n coordinates gives 2^n distinct choices of t. For each choice, Lemma 9.19 determines ρ and σ uniquely, and all defining equations then hold. Conversely every valid completion must obey the same forced equations, so the count is exact. □

Remark 9.21 (The crucial witness-dependence). The existence and count in Proposition 9.20 do *not* yield a witness-free simulator for the hidden values. The forced pad $p = m + s^*$ depends on the private partial vector s^*. A simulator receiving only the public evaluation claim does not know that vector. A hiding commitment and a zero-knowledge proof of the affine relations must therefore permit transcript simulation without extracting or revealing these hidden values. Ordinary binding or marginal hiding, by themselves, are insufficient.

Lemma 9.22 (Revealing the hidden pad breaks the single-opening privacy)**.** *If a verifier receives m and p in the clear, it can recover the unmasked partial vector s^* by $s^* = m + p$.*

Proof. By definition $m = s^* + p$. Adding p to both sides gives $m + p = s^* + p + p = s^*$ because $2p = 0$. □

Corollary 9.23 (Revealing the transposed pad has the same effect)**.** *If a verifier receives (m, c) in the clear with $c = \mathcal{T}(p)$, it can recover $s^* = m + \mathcal{T}(c)$.*

Proof. The transpose is an involution by Theorem 7.19, so $\mathcal{T}(c) = p$. Apply Lemma 9.22. □

9.4 Field-affine verification, timing, and residual checking

Definition 9.24 (Complete ideal affine verification)**.** For a fixed public instance and public core, the ideal checker receives hidden bound values $(\widehat{w}_j)_j, p, p_{\mathrm{in}}, c, t, \rho, \sigma$ with the following tests.

(i) **Claim:** $y' = y + p_{\mathrm{in}}$.

(ii) **Incoming:** $\sum_v q_v m_v + y' = \sum_v q_v p_v + p_{\mathrm{in}}$.

(iii) **Outgoing:** $\sum_j a_j(\theta)\widehat{w}_j = B_m(\theta) + D_c(\theta)$.

(iv) **Trace:** $\mathrm{Tr}(U_k) = 0$ for every k; these use only public values.

(v) **First anchored equation:** $A + \rho + \sum_k W_k(z)t_k = 0$.

(vi) **Second anchored equation:**

$$A^2 + \sigma + \sum_k W_k(z)^2(U_k + V_k(p, c, \theta) + t_k) = 0.$$

A separate underlying commitment relation must link the hidden packed columns to the fixed Boolean witness table.

Lemma 9.25 (The claimed-value and incoming tests are affine)**.** *Once the public parameters and (m, y') are fixed, the claim and incoming equations are affine over L in the committed secret variables p, p_{in}.*

Proof. Rewrite the claim equation as $p_{\mathrm{in}} = y' + y$. Rewrite the incoming equation as $\sum_v q_v p_v + p_{\mathrm{in}} = \sum_v q_v m_v + y'$. In both expressions the coefficients and right sides are public, and each hidden element occurs with degree one, proving the assertion. □

Lemma 9.26 (The outgoing test is affine)**.** *Once the public (m, θ) and the high-axis weights are fixed, the outgoing test is affine over L in the committed packed columns $\widehat{w}$ and coefficients c.*

Proof. Expand $D_c(\theta) = \sum_u \lambda_u(\theta)c_u$. Then the outgoing relation reads

$$\sum_j a_j(\theta)\widehat{w}_j + \sum_u \lambda_u(\theta)c_u = B_m(\theta).$$

Every coefficient and the right side are publicly determined. The hidden values occur linearly. □

Lemma 9.27 (Affineness of the anchored equations)**.** *For fixed public (m, y', θ, U, z, A), the two anchored equations of Definition 9.24 are affine over L in (p, c, t, ρ, σ). No product of two committed secret elements appears.*

Proof. The first anchored equation is $A + \rho + \sum_k W_k(z)t_k = 0$: its coefficients $W_k(z)$ and A are public. For the second, A^2 and every $W_k(z)^2$ are public scalars. By Lemma 7.68, $V_k(p, c, \theta)$ is L-linear in the secret pad and correction variables. Thus $A^2 + \sigma + \sum_k W_k(z)^2(U_k + V_k + t_k) = 0$ is affine in (p, c, t, σ). In particular $\sigma = \rho^2$ does *not* appear among the affine verifier equations. □

Definition 9.28 (Algebraic residuals at the second challenge)**.** For arbitrary fixed hidden variables and public (θ, U), define

$$\delta := \sigma + \rho^2, \qquad e_k := U_k + V_k(p, c, \theta) + t_k^2 + t_k.$$

These values are *defined mathematically* for soundness analysis; the verifier need not compute them directly or ask for nonlinear proofs.

Lemma 9.29 (Accepted affine equations imply a residual root)**.** *If both anchored equations pass, then*

$$\boxed{\delta + \sum_k e_k W_k(z)^2 = 0.}$$

Proof. From the first equation, $A = \rho + \sum_k W_k(z) t_k$. Squaring in characteristic two gives $A^2 = \rho^2 + \sum_k W_k(z)^2 t_k^2$. Substitute this into the second affine equation and collect the scalar term $\rho^2 + \sigma = \delta$ and the summands $U_k + V_k + t_k^2 + t_k = e_k$. The remaining equality is the displayed equation. □

Theorem 9.30 (Anchored residual detection)**.** *Fix $\theta, p, c, t, \rho, \sigma, U$ before an independent uniform second challenge $z \in L$. If $(\delta, e) \neq (0, 0)$, then even for an arbitrary public A chosen after seeing z,*

$$\Pr_z[\textit{both anchored affine equations pass}] \leq \min\{1, n/q\}.$$

Proof. The fixed-before-z hypothesis makes (δ, e) independent of the challenge. By Lemma 9.29, acceptance implies $\delta + \sum_k e_k W_k(z)^2 = 0$. The anchored residual root bound Theorem 6.81 bounds this event by n/q for any nonzero residual tuple. The probability is also at most one. □

Corollary 9.31 (No secret squaring check is needed for this algebraic soundness)**.** *In the ideal algebraic setting, an adversarial choice $\sigma \neq \rho^2$ fixed before the second challenge is accepted by both anchored checks with probability at most* $\min\{1, n/q\}$.

Proof. The inequality $\sigma \neq \rho^2$ is equivalent to $\delta \neq 0$, so $(\delta, e) \neq (0, 0)$ regardless of the vector e. Apply Theorem 9.30. □

Lemma 9.32 (Public trace checks impose no secret multiplication)**.** *The trace constraints* $\mathrm{Tr}(U_k) = 0$ *can be checked from public field elements alone.*

Proof. Each U_k is explicitly published. The trace is the finite field expression $\sum_{i=0}^{n-1} U_k^{2^i}$ by Definition 6.34. Its computation accesses only the public value U_k and public field arithmetic. □

Remark 9.33 (What zero hidden-quadratic constraints means). Only the *six displayed algebraic checks* above are affine in their hidden field values; the trace check is public and the second equation involves public A^2. The bit-packing/commitment binding, an implementation's internal arithmetic, and the ability to prove these affine relations in zero knowledge are separate assumptions. Consequently this statement is not a count of all gates or all transmitted field elements in a concrete end-to-end PCS.

9.5 False-claim soundness and repeated-opening failures

Lemma 9.34 (False claim forces a partial-evaluation discrepancy)**.** *Fix a Boolean table w, packed columns $\widehat{w}$, and evaluation points $(r_{\mathrm{lo}}, r_{\mathrm{hi}})$. Suppose the public $y \neq y_w$. If the claim and incoming affine equations pass, then*

$$e := m + s^* + p \in L^I$$

is nonzero.

Proof. If $e = 0$ then $m = s^* + p$. The claim and incoming equations imply $y = \sum_v q_v(m_v + p_v)$ by Lemma 7.51. Hence $y = \sum_v q_v s_v^* = y_w$ by Lemma 7.27, contradicting the false claim. □

Lemma 9.35 (Correct-transposition discrepancy)**.** *Under Lemma 9.34, if $c = c^p$, then passing the outgoing equation is equivalent to $B_e(\theta) = 0$, where e is fixed and nonzero.*

Proof. By Lemma 7.56, the outgoing discrepancy is $B_e(\theta) + D_{c+c^p}(\theta)$. The assumed equality $c = c^p$ makes the second term zero. By Lemma 9.34, $e \neq 0$, and the outgoing equation holds precisely when the first term vanishes. □

Lemma 9.36 (Incorrect hidden transposition gives a degree-ℓ mismatch)**.** *If $c \neq c^p$ is fixed before a uniform $\theta \in L^\ell$, the probability that $D_c(\theta) = \Delta_p(\theta)$ is at most $\min\{1, \ell/q\}$.*

Proof. Apply Corollary 6.61. Its hypotheses are exactly the stated coefficient mismatch and independence from the challenge. □

Theorem 9.37 (Ideal masked ring-switching false-claim soundness)**.** *Fix a Boolean witness, its exact packed columns, public $(r_{\mathrm{lo}}, r_{\mathrm{hi}}, y)$, and the public m, y' and hidden p, p_{in}, c before the independent uniform first challenge θ. After θ, a malicious prover may choose the checker values (t, ρ, σ, U) adaptively, but must bind them before the independent uniform second challenge z. It may choose A after receiving z. Assume all ideal field-affine checks and public trace checks of Definition 9.24 are enforced exactly. If $y \neq y_w$, then*

$$\boxed{\Pr[\text{accept}] \leq \min\left\{1, \frac{\ell + n}{q}\right\}.}$$

Proof. Distinguish two mutually exclusive cases, which are already fixed before θ.

If $c = c^p$, whenever the claim and incoming equations pass the vector $e = m + s^* + p$ is nonzero by Lemma 9.34. By Lemma 9.35, outgoing acceptance requires $B_e(\theta) = 0$. The discrepancy polynomial has degree at most ℓ and is nonzero; Theorem 7.35 bounds this event by ℓ/q. Further checker constraints can only reduce acceptance.

If $c \neq c^p$, let E denote the exceptional first-challenge event $D_c(\theta) = \Delta_p(\theta)$. Lemma 9.36 gives $\Pr[E] \leq \ell/q$. On its complement, if the public trace checks pass then Lemma 6.69 guarantees a nonzero vector $e_k = U_k + V_k + t_k^2 + t_k$ for at least one coordinate. The auxiliary values are fixed before the fresh z, so the residual detection theorem (Theorem 9.30) bounds passage of the two anchored equations by n/q. Conditioning on the malicious prover's chosen state and private random tape and then averaging gives $\Pr[\text{accept}] \leq (\ell + n)/q$ in this case. Taking the maximum of the two bounds, and capping by one, proves the theorem. □

Definition 9.38 (Two-opening pad-reuse experiment)**.** Let a single witness w be evaluated at two distinct high-axis points $r_{\mathrm{hi}}^{(1)}, r_{\mathrm{hi}}^{(2)}$ with the same low-axis point. Let $s^{*(i)}$ be the associated partial-evaluation vectors. A *reused-pad* implementation chooses one uniform $p \in L^I$ and publishes $m^{(1)} = s^{*(1)} + p$ and $m^{(2)} = s^{*(2)} + p$.

Lemma 9.39 (Reusing a pad reveals the difference of partial evaluations)**.** *In the two-opening pad-reuse experiment,*

$$\boxed{m^{(1)} + m^{(2)} = s^{*(1)} + s^{*(2)}}$$

for every outcome of the pad.

Proof. Add the two definitions: $m^{(1)} + m^{(2)} = s^{*(1)} + p + s^{*(2)} + p$. Because $p + p = 0$ in characteristic two, the pad cancels. □

Theorem 9.40 (Perfect two-opening leakage under pad reuse)**.** *For any $\ell \geq 1$ and $h \geq 1$, there are two Boolean witnesses having the same authorized evaluation values at two public points, but whose reused-pad public pairs $(m^{(1)}, m^{(2)})$ have statistical distance* 1*.*

Proof. Take $r_{\mathrm{lo}} = 0_\ell$ and $r_{\mathrm{hi}}^{(1)} = 0_h$, $r_{\mathrm{hi}}^{(2)} = (1, 0, \ldots, 0)$. Let $w^{(0)}$ be identically zero. Choose $v_* \neq 0_\ell$ and define $w^{(1)}$ with exactly one nonzero entry at $(v_*, 0_h)$. Both public evaluations, at both chosen points, are zero for both witnesses: the low-axis equality weights select only the row $v = 0_\ell$, which is zero.

For $w^{(0)}$, both partial vectors vanish, so Lemma 9.39 gives $m^{(1)} + m^{(2)} = 0$ always. For $w^{(1)}$, the first high-axis point selects the Boolean column $j = 0_h$, producing a partial vector with a single 1 in row v_*. The second high-axis point selects the column $(1, 0, \ldots, 0)$, where the witness is zero, and hence $s^{*(2)} = 0$. Thus $m^{(1)} + m^{(2)}$ has a nonzero entry at v_* with probability one. The deterministic event $m^{(1)} + m^{(2)} = 0$ distinguishes the two distributions with probabilities 1 and 0. Lemma 3.5 makes the statistical distance 1. □

Corollary 9.41 (Fresh partial-evaluation pads are necessary for this implementation)**.** *Any multi-opening version that publishes raw masked partial vectors in the form $m = s^* + p$ must not reuse the same pad across arbitrary evaluation points if it is to satisfy witness privacy for the authorized evaluation values.*

Proof. The reuse construction of Definition 9.38 violates the two-witness closeness required by statistical zero knowledge, as shown in Theorem 9.40. Thus that reuse rule is not compatible with the stated privacy goal. □

Remark 9.42 (No universal lower bound)*.* Corollary 9.41 rules out this exact pad-reuse pattern; it does not prove an information-theoretic lower bound for all possible multi-opening ZK schemes. Correlated masks with additional zero-knowledge machinery would require their own proof.

9.6 Ideal complete-view privacy and modular cryptographic composition

The public core is not the complete verifier view. A real implementation includes commitments to packed witness elements, hidden pads, correction coefficients and auxiliary masks, plus the affine-proof transcripts and any authenticated oracle queries. We now state exactly what an external proof system must supply.

Definition 9.43 (Opaque commitment-and-affine ideal functionality)**.** An *opaque ideal proof interface* stores hidden witness data in a private registry, but its public transcript has only the following effects:

(i) On binding a hidden tuple, it emits a public registration token sampled from a specified distribution independent of the hidden tuple. The token is produced before the challenges that rely on its binding.

(ii) At each scheduled protocol stage, it accepts *only* the corresponding prescribed relation from Definition 9.24, with coefficients determined by the public instance, messages, and chosen challenges. It privately tests this relation on the registered tuple and releases only its accept/reject bit. It is *not* an oracle for arbitrary verifier-chosen affine predicates about the hidden tuple.

(iii) A request to test any extraneous predicate, or to obtain a hidden field value, an unmasked codeword symbol, or an authenticated opening of a hidden value, is rejected with a fixed public error response independent of the hidden tuple.

(iv) For honest admissible executions, every prescribed relation is valid for every chosen challenge, so all scheduled acceptance bits equal 1. For malicious executions it enforces those prescribed relations exactly. Public registration tokens may be reused or queried only through explicitly allowed witness-independent interface operations.

This is a *mathematical ideal functionality*, not an assertion that a concrete ordinary binding commitment is perfectly opaque.

Lemma 9.44 (Opaque tokens do not perturb the public-core law)**.** *Assume all opaque interface tokens and outputs for an honest execution are independent of the private witness and of the random masks used in the public core, except that the acceptance bits are the constant* 1*. Then, conditioned on any allowed visible history of tokens, the honest public-core transitions have the same uniform conditional laws as Lemmas 9.9, 9.10, and 9.11.*

Proof. Independence implies that conditioning on the tokens does not change the joint law of the private masks and the verifier's previous public-core messages. Constant acceptance bits reveal no new information. The input-pair translation argument, trace-zero pushforward argument, and final uniform-ρ translation argument therefore hold under the additional conditioning without change. These are exactly the three cited lemmas. □

Definition 9.45 (Complete ideal verifier view)**.** For a single admissible opening, the *complete ideal view* consists of the public commitment tokens, the verifier's own random tape, (m, y'), its challenges (θ, z), the public vector U, the final scalar A, the public trace checks, the acceptance bits of all required affine relations, and all interaction with the opaque ideal interface. The verifier may choose its challenges adaptively after all messages visible at that stage, but may not replace the prescribed affine verification equations with arbitrary secret-dependent predicates. Unsupported requests produce only the fixed error responses of Definition 9.43.

Theorem 9.46 (Perfect complete-view ZK in the opaque ideal model)**.** *For any causal verifier and any two admissible witnesses for the same fixed public evaluation instance, the complete ideal verifier views of Definition 9.45 are identically distributed. A simulator without the witness generates the opaque registration tokens from their specified distributions, uses Definition 9.12 for the public core, and returns the constant honest acceptance bits from the ideal checks.*

Proof. The simulator samples each token from the witness-independent distribution prescribed by Definition 9.43, and couples the verifier's random tape with the real experiment. By Lemma 9.44, conditioning on these tokens preserves the uniform laws of all public core messages. Theorem 9.13 then gives identical conditional distributions at every public-core round, including when the verifier chooses challenges from the earlier tokens and messages. The honest hidden state satisfies all *prescribed* ideal affine relations by Lemmas 9.6 and 9.7; hence their public accept/reject bits are the constant 1 in both experiments. All outputs are either the same coupled verifier actions, identically distributed tokens, identically distributed core messages, or these deterministic accept bits. Thus the complete finite joint view laws coincide. □

Definition 9.47 (Sequential external-interface simulation contract)**.** Fix a finite interactive implementation, a verifier class and a bound on authenticated queries. Resolve each protocol step into a finite transition on its *visible history*, which records the entire public prefix and the verifier's random tape. For a fixed admissible witness w, let $K_i^w(h)$ be the real conditional distribution of the next visible state from history h. Let $L_i(h)$ be a causal, *witness-independent* transition kernel producing an ideal opaque-interface view as in Definition 9.45. An *online sequential realization with local errors* $(\varepsilon_0, \dots, \varepsilon_N)$ means:

(i) initial real and ideal visible-state distributions have statistical distance at most ε_0;

(ii) for each stage $1 \le i \le N$ and each possible visible history h,

$$\Delta(K_i^w(h), L_i(h)) \le \varepsilon_i,$$

uniformly over all admissible witnesses and all verifiers in the chosen class;

(iii) the ideal kernels L_i faithfully implement the public-core simulator and opaque-interface simulator, including the same ordering and causal challenge dependence as the real protocol.

This requires checking correlations with any preceding commitments and authenticated openings: the perfect public-core law of Theorem 9.13 does *not* alone establish (ii) for histories containing real commitment roots. For histories that are unreachable under a particular process, one may choose a regular conditional kernel arbitrarily, but the uniform bound must still hold under the chosen extension.

Theorem 9.48 (Conditional real-world statistical ZK composition)**.** *Suppose a concrete single-opening implementation satisfies Definition 9.47 with local errors $(\varepsilon_0, \dots, \varepsilon_N)$ for a given query bound and verifier class. Then for every admissible witness, the implemented complete view has a witness-independent simulator with statistical distance at most*

$$\boxed{\varepsilon_{\mathrm{ZK}} \le \min\left\{1, \sum_{i=0}^{N} \varepsilon_i\right\}.}$$

If all external interfaces are perfectly simulatable, the complete protocol is perfect zero knowledge for that single-opening game.

Proof. Use the finite visible history spaces of the real and ideal processes, with the verifier's random tape incorporated into each history. Condition (ii) of Definition 9.47 bounds the total variation between the real and ideal transition kernels at every stage and every visible history. Condition (i) bounds the initial-state distance. The stepwise hybrid lemma, Lemma 3.10, therefore bounds the distance between the final visible-history distributions by $\sum_{i=0}^{N} \varepsilon_i$. Condition (iii) and Theorem 9.46 provide a witness-independent implementation of the ideal visible process. A deterministic projection from final histories to the actual verifier view cannot increase total variation, by Corollary 3.8. Finally, total variation is at most one for any two distributions. □

Definition 9.49 (External algebraic-soundness failure)**.** Let $\mathcal{B}$ be the event that an implemented verifier accepts despite some underlying commitment-binding, packed-witness consistency, or advertised field-affine constraint failing to hold for the fixed hidden tuple. Assume a separately established implementation-specific error bound $\Pr[\mathcal{B}] \le \varepsilon_{\mathrm{ext,snd}}$.

Theorem 9.50 (Conditional full implementation soundness)**.** *Under the causal and fixed-before-challenge hypotheses of Theorem 9.37, and assuming implementation acceptance outside $\mathcal{B}$ implies acceptance of all ideal algebraic checks on the same fixed hidden values, a false evaluation claim is accepted with probability at most*

$$\boxed{\varepsilon_{\mathrm{snd}} \le \min\left\{1, \frac{\ell + n}{q} + \varepsilon_{\mathrm{ext,snd}}\right\}.}$$

Proof. Outside $\mathcal{B}$, implementation acceptance entails acceptance of the ideal algebraic checker. Thus the implementation-accept event is contained in the union of the ideal-accept event and $\mathcal{B}$. Theorem 9.37 bounds the first event by $(\ell+n)/q$, and Definition 9.49 bounds the second by $\varepsilon_{\mathrm{ext,snd}}$. Apply the probability union bound and cap the result by one. □

Remark 9.51 (No unconditional real-world ZK claim). Theorem 9.48 is deliberately conditional: the implementation must actually satisfy a simulation contract for its *full* commitment and proof transcripts. Neither ordinary computational hiding of one commitment nor a proof that (m, y', U, A) has uniform marginals establishes that contract. A real online commitment chosen before the evaluation point also requires a simulator capable of maintaining consistency with later chosen evaluations.

9.7 Efficiency, assumptions, and the path to a genuine ZK-KBFold PCS

Definition 9.52 (Field-message accounting for the public core)**.** Count only publicly transmitted elements of L in the core of Definition 9.4, not commitments, field traces, challenge bits or the separate affine proofs. Let $n = |I| = 2^\ell$.

Lemma 9.53 (Exact single-opening public-core message count)**.** *The honest public core transmits $2n + 2$ elements of L: n entries of m, one y', n entries of U and one A. It also receives $\ell + 1$ challenge elements of L: the ℓ coordinates of θ and one z.*

Proof. Read the five ordered stages of Definition 9.4. The lengths of the prover-message vectors are $n, 1, n, 1$ respectively, whose sum is $2n + 2$. The verifier supplies a vector $\theta \in L^\ell$ and a scalar $z \in L$, contributing $\ell+1$ challenge elements. No other elements are counted by Definition 9.52. □

Lemma 9.54 (Naive affine-coefficient count)**.** *The family of public affine coefficients $H_{v,k}(\theta)$ from Definition 6.56 contains n^2 elements of L. A direct implementation of their defining sums uses n summands per coefficient, hence n^3 basis-coordinate terms.*

Proof. Both indices v, k range over I, a set of size n, so there are n^2 ordered pairs. By definition, each coefficient $H_{v,k}$ is a sum over a third index $u \in I$ with n possible values. Counting the summands gives $n \cdot n^2 = n^3$ terms. This is an operation-count upper bound for the naive formula, not a lower bound on optimized implementations. □

Lemma 9.55 (Exact private state field-element count)**.** *Excluding the already committed packed witness, the auxiliary private tuple $(p, p_{\mathrm{in}}, c, t, \rho, \sigma)$ has $3n + 3$ field slots: $n + 1 + n + n + 1 + 1$. In the honest sampler the genuinely independent randomness in that tuple consists of $2n + 2$ field elements: n for p, one for p_{in}, n for t and one for ρ.*

Proof. The displayed tuple has, in order, $n, 1, n, n, 1, 1$ field entries, summing to $3n+3$. The honest sampler chooses only $p, p_{\mathrm{in}}, t, \rho$ independently. Their entry counts sum to $n+1+n+1 = 2n+2$. The remaining entries satisfy $c = \mathcal{T}(p)$ and $\sigma = \rho^2$, so they are deterministic functions of the sampled entries. □

Definition 9.56 (Certification milestones)**.** The following are *separate* certification tasks rather than consequences of the preceding theorems:

(i) **Real commitment semantics:** a concrete binding and hiding scheme with a consistent simulator for commitments and authenticated queries.

(ii) **Affine proof realization:** an implementation that enforces every relation in Definition 9.24 on one fixed hidden tuple with explicit completeness, soundness and zero-knowledge error.

(iii) **Multi-opening ZK:** commitment simulation before evaluation points are known; fresh masks per opening; and a joint simulator for every adaptive challenge, proof transcript and opening.

(iv) **Cross-field integration:** the source KBFold PCS uses odd-characteristic multiplicative-subgroup word folding. The present binary-field checker does not inherit its proximity or QROM theorem.

(v) **Concrete optimization:** avoid or compress the naive n^2 public affine coefficients and verify the resulting circuit/prover work, rather than asserting asymptotic efficiency without implementation.

Proposition 9.57 (Binary-field folding is not directly interchangeable with KBFold's $\pm x$ fold)**.** *The characteristic-two ring-switching checker cannot reuse the original multiplicative domain pairing $x, -x$ as a two-point local fold over $L = \mathbb{F}_{2^n}$.*

Proof. In characteristic two, $-x = x$ for every $x \in L$. Thus the supposed two distinct evaluation positions coincide. Furthermore the standard even/odd word-fold formulas divide by $2x$, but $2 = 0$ in L, so this expression has no inverse for any x. The required field hypotheses of Definition 5.41 therefore fail, as shown in Lemma 6.92. □

Remark 9.58 (Historical and originality boundary). Diamond–Posen supply the original tensor-algebra compiler [DP24]. Kochergin's 2026 note [Koc26] and Frigo–shelat's sumcheck bit-packing paper [Fs26] give different zero-knowledge routes for binary-field packing. The reusable mathematical contribution developed here is the exact kernel-coordinate/transposition dictionary, its anchored affine checker and a rigorous separation between perfect public-core simulation and full proof-system simulation. This chapter does not claim that trace-zero masking, the general idea of replacing quadratic witness constraints by public squaring, or a single-challenge polynomial identity test originated here.

Remark 9.59 (The next theorem genuinely needed). For the *original* odd-characteristic KBFold PCS, the central outstanding theorem is a composable adaptive view simulator for committed masked fold oracles, including every sumcheck round polynomial, authenticated opening and the degree-aware linear final closure of Chapter 5. A public-core theorem for binary Ring Switching does not discharge that security obligation.

9.8 Summary and exact theorem boundary

For one admissible opening, sample independent uniform $p \in L^{I_\ell}$, $p_{\mathrm{in}} \in L$, $t \in L^{I_\ell}$ and $\rho \in L$ and define

$$m = s^* + p, \quad y' = y + p_{\mathrm{in}}, \quad c = \mathcal{T}(p), \quad U_k = V_k(p, c, \theta) + t_k^2 + t_k, \quad A = \rho + \sum_k W_k(z) t_k.$$

Theorem 9.13 establishes the *exact* witness-independent joint distribution of the public core for every causal verifier:

$$\boxed{\mathsf{Law}(\mathsf{Core}_{w,V^*}) = \mathsf{Law}(\mathsf{Sim}_{\mathrm{core}}^{V^*}).}$$

All enforced algebraic constraints are affine in hidden extension-field variables after the public challenges and messages are fixed, and a nonzero anchored residual is detected with probability

at least $1 - n/q$ (Theorem 9.30). For the combined ideal masked Ring Switching interface, a false committed evaluation claim is accepted with probability at most

$$\boxed{\varepsilon_{\mathrm{alg}} \leq \min\{1, (\ell + n)/q\}.}$$

Theorem 9.46 gives perfect complete-view zero knowledge only in a strictly opaque ideal interface. Theorem 9.48 states the precise additional requirement on a concrete external proof system: a sequential witness-independent simulator for its complete commitment, proof and authenticated-opening transcripts.

Neither result proves real-world zero knowledge for a fully deployed PCS. The multiple-opening pad-reuse pattern is insecure (Theorem 9.40), and an unmasked original KBFold oracle is not private (Theorem 3.50). The next stage must instantiate and prove the missing external-interface realization and the odd-characteristic KBFold-specific adaptive view simulator.

10 Commitment and Affine-Proof Realization

The masked Ring Switching interface of Chapter 9 requires proofs of field-affine relations on a fixed hidden tuple. This chapter provides an *explicit classical realization of those relations*: binary-coordinate compilation, fresh Pedersen commitments, and Schnorr-style OR proofs for parity constraints. The foundational primitives are classical [Ped92; Sch91; CDS94]. The result is a concrete perfect *honest-sigma-verifier* zero-knowledge theorem for one opening, with arbitrary causal choices of the outer Ring Switching challenges. The exact extraction timing is isolated as a separate knowledge-soundness contract; full malicious-verifier zero knowledge, succinctness, quantum security, and integration with KBFold's original authenticated oracle are *not* asserted.

10.1 Compiling binary-field affine equations into parity constraints

Throughout Chapter 10 let $L = \mathbb{F}_{2^n}$ with $n = 2^\ell$. For compatibility with the packed columns of Chapters 6–9, fix the *Boolean-kernel evaluation basis* $\gamma_i := \beta_i$ for $0 \leq i < n$, identifying I_ℓ with its binary integer indices, and let $d_k : L \to \mathbb{F}_2$ be its corresponding coordinate maps. The affine-to-parity equivalence itself holds for every public binary basis, but the packed-column identity later in the chapter uses the specific choice $\gamma_i = \beta_i$.

Definition 10.1 (Field-variable bit representation)**.** For field variables $x_1, \ldots, x_s \in L$ define their bits $x_{j,i} := d_i(x_j) \in \mathbb{F}_2$, so that

$$x_j = \sum_{i=0}^{n-1} \gamma_i x_{j,i}.$$

Let $a_1, \ldots, a_s, b \in L$ be public. The target affine equation is $\sum_{j=1}^{s} a_j x_j = b$.

Definition 10.2 (Public binary coefficient matrix)**.** For every output coordinate k and input coordinate (j, i) set

$$M_{k,j,i} := d_k(a_j \gamma_i) \in \mathbb{F}_2, \qquad b_k := d_k(b) \in \mathbb{F}_2.$$

The k-th associated parity constraint is

$$\sum_{j=1}^{s} \sum_{i=0}^{n-1} M_{k,j,i} x_{j,i} = b_k \quad \text{in } \mathbb{F}_2.$$

Lemma 10.3 (Coordinates of a publicly scaled element)**.** *For $a, x \in L$ and $0 \le k < n$,*

$$d_k(ax) = \sum_{i=0}^{n-1} d_k(a\gamma_i) d_i(x).$$

Proof. Expand $x = \sum_i \gamma_i d_i(x)$, multiply by a, and apply the $\mathbb{F}_2$-linear coordinate map d_k. Because $d_i(x) \in \mathbb{F}_2$, each scalar may be moved through d_k. The result is the displayed finite sum. □

Theorem 10.4 (Exact affine-to-parity equivalence)**.** *For any $x_1, \ldots, x_s \in L$,*

$$\boxed{\sum_j a_j x_j = b \quad \Longleftrightarrow \quad \sum_{j,i} M_{k,j,i} x_{j,i} = b_k \text{ for every } 0 \le k < n.}$$

The equivalence holds for arbitrary public coefficients chosen after the bit variables have been committed.

Proof. Apply d_k to $\sum_j a_j x_j$ and use its additivity. Apply Lemma 10.3 to each summand. This gives the k-th displayed parity expression. If the original field equality holds, all its coordinates agree. Conversely, if all n coordinates agree, then the field element $\sum_j a_j x_j + b$ has every binary coordinate zero, so it is zero by uniqueness of the fixed basis expansion. The coefficients are only parameters of the algebraic identity; their time of choice does not affect its validity. □

Corollary 10.5 (No multiplication of two hidden field elements)**.** *Every affine field equation with public coefficients reduces to n parity equations in committed Boolean coordinates, with no hidden-variable multiplication gates.*

Proof. Each row in Definition 10.2 is a sum of terms 0, $x_{j,i}$, or their $\mathbb{F}_2$-linear sum, equal to a public bit. No term contains a product of two hidden coordinates. □

Lemma 10.6 (Row size bound)**.** *For s field variables, each parity row has at most sn hidden bits, and there are exactly n output-coordinate rows.*

Proof. There are s input-variable indices and n coordinate indices. Each row uses only a subset of their sn pairs. The output-coordinate index takes exactly n values. □

Remark 10.7 (Distinct scalar characteristics)*.* The field L has characteristic 2; the prime-order group used for Pedersen commitments below has an *odd* prime order P. The reduction above is the required bridge. In particular, binary XOR may not be implemented by simply adding commitment exponents modulo P.

10.2 A concrete hiding and computationally binding bit commitment

We use Pedersen's group commitment [Ped92] for *binary coordinates*, not for raw field addition in L.

Definition 10.8 (Prime-order commitment parameters)**.** Let G be a cyclic multiplicative group of *odd prime* order P, with generators $g, h \in G$. Setup is trusted only in the following precise sense: no efficient participant knows $\alpha = \log_g h$. The discrete-logarithm assumption is that a probabilistic polynomial-time algorithm given the public pair (g, h) cannot compute α except with negligible probability. We require P to exceed every parity row's number of summands.

Definition 10.9 (Pedersen commitment to a binary coordinate)**.** For $b \in \{0,1\} \subset \mathbb{Z}_P$ and $r \in \mathbb{Z}_P$, define

$$\mathsf{Com}(b;r) := g^b h^r \in G.$$

An opening is the pair (b,r), with b restricted to $\{0,1\}$. Every bit commitment uses a fresh independent uniform r.

Lemma 10.10 (Exact perfect hiding)**.** *For each fixed bit b, if r is uniform in $\mathbb{Z}_P$, then $\mathsf{Com}(b;r)$ is uniform on G. Consequently, any vector of independently blinded bit commitments is jointly uniform on the appropriate power of G and independent of its binary message vector.*

Proof. Because h generates G, the map $r \mapsto h^r$ is a bijection from $\mathbb{Z}_P$ to G. Multiplication by the fixed g^b is another bijection. The composition sends uniform r to uniform group elements. For a vector of commitments, the maps act on independently sampled blindings coordinatewise, so the joint distribution is the product of the individual uniform distributions. □

Lemma 10.11 (Two openings recover the forbidden setup logarithm)**.** *If $g^b h^r = g^{b'} h^{r'}$ with $b \neq b'$ in $\mathbb{Z}_P$, then $r' \neq r$ and*

$$\boxed{\log_g h = (b-b')(r'-r)^{-1} \pmod{P}.}$$

Proof. Rearrange the commitment equality: $g^{b-b'} = h^{r'-r} = g^{\alpha(r'-r)}$. If $r' = r$, then $g^{b-b'} = 1$, implying $b = b'$ because g has order P. Thus $r' - r \neq 0$, and the inverse exists in the prime field $\mathbb{Z}_P$. Equate the exponents modulo P and divide to obtain the formula. □

Theorem 10.12 (Computational binding for bits)**.** *Under the discrete-logarithm assumption of Definition 10.8, no efficient committer can output a group element and valid openings to two distinct bits with non-negligible probability.*

Proof. Given such an algorithm, run it on public (G,P,g,h). Whenever it outputs two distinct openings, apply Lemma 10.11. This computes $\log_g h$ in polynomial time with the algorithm's success probability, contradicting the discrete-logarithm assumption. □

Definition 10.13 (Aggregating commitments)**.** For an index set T of m committed bits $C_i = \mathsf{Com}(b_i;r_i)$, put

$$C_T := \prod_{i \in T} C_i, \qquad S_T := \sum_{i \in T} b_i \in \{0,\dots,m\}, \qquad R_T := \sum_{i \in T} r_i \in \mathbb{Z}_P.$$

The sum S_T is an *ordinary integer*, subsequently embedded in $\mathbb{Z}_P$.

Lemma 10.14 (Exact aggregate opening)**.** *For each finite T,*

$$\boxed{C_T = g^{S_T} h^{R_T}.}$$

Proof. Multiply the defining equations $C_i = g^{b_i} h^{r_i}$ over all $i \in T$. Because G is abelian, powers of g and h may be collected separately. Their exponents add, yielding the identity. □

Lemma 10.15 (Why naive XOR homomorphism fails)**.** *For two committed bits $b_1 = b_2 = 1$, their aggregate opens to the integer $S_T = 2$, whereas $b_1 + b_2 = 0$ in $\mathbb{F}_2$. Consequently the natural product of two bit commitments is not automatically an opening of their XOR with the sum of blindings.*

Proof. The product is $g^2 h^{r_1+r_2}$ by Lemma 10.14. An opening to the XOR using the same blinding $r_1 + r_2$ would instead be $h^{r_1+r_2}$. As g has odd prime order P, we have $g^2 \neq 1$. Thus the elements are different. □

Remark 10.16 (Setup and post-quantum scope). The Pedersen instantiation in a conventional discrete-logarithm group is *not* post-quantum secure against general quantum computers. It is an explicit classical commitment/proof candidate for the affine checker, not a post-quantum implementation of the original KBFold PCS. A transparent post-quantum realization remains separate.

10.3 Schnorr knowledge proofs and explicit OR composition

The three-move proof of knowledge for discrete logarithms is due to Schnorr [Sch91]; its OR composition is a standard instance of the partial-knowledge methods of Cramer and coauthors [CDS94]. We give all protocol messages and their elementary proofs.

Definition 10.17 (Schnorr relation)**.** For a group element $Y \in G$, define the relation $\mathcal{R}_{\log}(Y; r)$ by $Y = h^r$, $r \in \mathbb{Z}_P$. The prover sends $T = h^u$ for uniform $u \in \mathbb{Z}_P$; the verifier responds with uniform challenge $e \in \mathbb{Z}_P$; the prover returns $z = u + er$. The verifier accepts if $h^z = TY^e$.

Lemma 10.18 (Schnorr completeness)**.** *For every valid witness r and all u, e, the Schnorr verifier accepts.*

Proof. Since $Y = h^r$ and $z = u + er$, $h^z = h^u h^{er} = TY^e$. □

Lemma 10.19 (Two Schnorr transcripts extract the logarithm)**.** *If two accepting Schnorr transcripts have the same Y, T but distinct challenges $e \neq e'$ and responses z, z', then*

$$\boxed{r = (z - z')(e - e')^{-1}}$$

satisfies $Y = h^r$.

Proof. Divide the two verifier equations to obtain $h^{z-z'} = Y^{e-e'}$. Since $e - e' \neq 0$ modulo the prime P, it has an inverse. Raise both sides to that inverse: $h^{(z-z')(e-e')^{-1}} = Y$. □

Lemma 10.20 (Perfect honest-verifier Schnorr simulation)**.** *For a uniformly sampled verifier challenge e, the joint transcript has exactly the distribution obtained without the witness by sampling independent uniform (e, z) and setting $T = h^z Y^{-e}$.*

Proof. For any fixed valid witness r, the honest challenge e is uniform and independent of uniform prover nonce u. Conditional on e, the response $z = u + er$ is uniform on $\mathbb{Z}_P$. The verifier equation uniquely determines $T = h^z Y^{-e}$ from (e, z). Therefore the simulated and honest distributions of the entire triple coincide. □

Definition 10.21 (Schnorr OR proof for a finite candidate set)**.** Fix nonempty $S = \{t_1, \ldots, t_d\} \subseteq \mathbb{Z}_P$ with distinct elements and a public commitment $C \in G$. For each branch set $Y_j = Cg^{-t_j}$. The prover knows an index j_* and a scalar r such that $Y_{j_*} = h^r$. Before receiving the global verifier challenge, it chooses independent uniform $e_j, z_j \in \mathbb{Z}_P$ for $j \neq j_*$, sets

$$T_j = h^{z_j} Y_j^{-e_j} \quad (j \neq j_*),$$

and chooses uniform u to set $T_{j_*} = h^u$. It sends $(T_1, \ldots, T_d)$. After the verifier draws uniform $e \in \mathbb{Z}_P$, the prover sets

$$e_{j_*} := e - \sum_{j \neq j_*} e_j, \qquad z_{j_*} := u + e_{j_*} r,$$

and sends all $(e_j, z_j)_{j=1}^d$. The verifier accepts if

$$\sum_j e_j = e \quad \text{and} \quad h^{z_j} = T_j Y_j^{e_j} \quad \forall j.$$

All challenge and response scalars belong to $\mathbb{Z}_P$.

Lemma 10.22 (OR-proof perfect completeness)**.** *For any candidate t_{j_*} with a known opening $C = g^{t_{j_*}} h^r$, every honest execution of Definition 10.21 accepts.*

Proof. By construction the local challenges sum to the verifier's global challenge. For $j \neq j_*$ the prover defined T_j precisely so that $h^{z_j} = T_j Y_j^{e_j}$. For $j = j_*$, $T_{j_*} Y_{j_*}^{e_{j_*}} = h^u h^{r e_{j_*}} = h^{u + e_{j_*} r} = h^{z_{j_*}}$. Thus all equations pass. □

Theorem 10.23 (OR-proof perfect honest-verifier zero knowledge)**.** *With uniform independent global challenge e, the full accepting OR-proof transcript can be simulated* exactly *without any opening of C: choose independent uniform $(e_j, z_j) \in \mathbb{Z}_P^2$ for every branch, put $e = \sum_j e_j$, and set*

$$\boxed{T_j = h^{z_j} (C g^{-t_j})^{-e_j}.}$$

The resulting joint distribution agrees with the honest prover/verifier distribution for every valid witness branch.

Proof. For the honest prover, the challenges e_j on false branches are independent uniform. The true-branch challenge $e_{j_*} = e - \sum_{j \neq j_*} e_j$ is uniform and independent of them because the global honest-verifier challenge is uniform. The false-branch responses z_j were drawn independent uniform; the true response $z_{j_*} = u + e_{j_*} r$ is also uniform conditioned on the entire challenge vector because u was uniform independent. Thus the complete vector $(e_j, z_j)_j$ is uniform on $(\mathbb{Z}_P^2)^d$. Every first message T_j is uniquely determined by the verifier equation as $h^{z_j} Y_j^{-e_j}$. The simulator uses exactly this deterministic expression from exactly the same distribution, so the complete transcript laws coincide. □

Theorem 10.24 (OR-proof special soundness)**.** *Two accepting OR-proof transcripts for the same statement C, S and the same first-message vector $(T_j)_j$ but different global challenges $e \neq e'$ determine an index j and an opening*

$$\boxed{C = g^{t_j} h^r, \qquad r = (z_j - z'_j)(e_j - e'_j)^{-1}.}$$

Proof. The sums of the two local challenge vectors are different, so for some branch j we have $e_j \neq e'_j$. Divide the two accepting verifier equations for that branch, obtaining $h^{z_j - z'_j} = Y_j^{e_j - e'_j}$. As in Lemma 10.19, inverting the nonzero challenge difference yields $Y_j = h^r$ for the displayed r. Since $Y_j = C g^{-t_j}$, multiplying by g^{t_j} gives $C = g^{t_j} h^r$. □

Remark 10.25 (Proof of knowledge versus a nontrivial language). Because h generates the entire group G, each group element Y_j has some logarithm mathematically. The OR protocol proves *efficient knowledge of an opening*, not mere set membership. Its computational knowledge-soundness guarantee requires the standard rewinding extractor and independent uniform global challenges; the two-transcript special-soundness theorem does not alone bound a malicious prover's success probability in the entire composed protocol.

Remark 10.26 (Honest-verifier scope). Theorem 10.23 assumes the sigma verifier chooses its challenge uniformly. It makes no assertion of straight-line zero knowledge against an arbitrary malicious verifier that chooses the sigma challenge as a function of its first message. A separate compiler is needed for that stronger notion.

10.4 Exact zero-knowledge parity proofs from bit commitments

Definition 10.27 (Parity-compatible integer openings)**.** For $m \geq 0$ and a public desired parity $b \in \{0,1\}$, define

$$\mathcal{S}_{m,b} := \{t \in \{0,\dots,m\} : t \equiv b \pmod 2\}.$$

If $\mathcal{S}_{m,b}$ is empty, the verifier rejects the parity statement deterministically. Assume $P > m$, so all candidates map injectively into $\mathbb{Z}_P$.

Lemma 10.28 (Correct integer sum lies in the permitted set)**.** *Let $b_1,\dots,b_m \in \{0,1\}$. If $\sum_i b_i = b$ in $\mathbb{F}_2$, then the ordinary integer $S = \sum_i b_i$ belongs to $\mathcal{S}_{m,b}$.*

Proof. As an integer, $0 \leq S \leq m$. Reduction of the sum modulo 2 is exactly addition of the bits in $\mathbb{F}_2$, so $S \equiv b \pmod 2$, giving membership. □

Definition 10.29 (Concrete parity proof)**.** Given public bit commitments $C_i = g^{b_i} h^{r_i}$, form $C_T = \prod_i C_i$. The prover proves:

(i) For each C_i it knows a bit opening, using Definition 10.21 with candidate set $\{0,1\}$.

(ii) It knows an opening of C_T to an integer $t \in \mathcal{S}_{m,b}$, using Definition 10.21 with candidate set $\mathcal{S}_{m,b}$.

All OR-proofs have independent fresh prover randomness and independent uniform sigma verifier challenges. The first bitness proofs can be executed when the commitments are bound; the aggregate parity proof is performed when the public row coefficients and target parity become known.

Theorem 10.30 (Perfect completeness of the parity compiler)**.** *For honest bit commitments and bits satisfying $\sum_i b_i = b$ in $\mathbb{F}_2$, the concrete parity proof accepts with probability 1.*

Proof. Each C_i has the known bit opening (b_i, r_i), so its two-branch OR proof is perfectly complete by Lemma 10.22. Lemma 10.28 places the true integer sum $S = \sum_i b_i$ in $\mathcal{S}_{m,b}$. Lemma 10.14 gives the known aggregate opening $C_T = g^S h^{R_T}$ with $R_T = \sum_i r_i$. The aggregate OR proof is therefore complete by the same lemma. All checks pass. □

Lemma 10.31 (An incorrect parity gives two different aggregate openings)**.** *Suppose each C_i has an extracted valid bit opening (b_i, r_i), and the aggregate OR proof has an extracted opening (t, r_T) with $t \in \mathcal{S}_{m,b}$. If $\sum_i b_i \neq b$ in $\mathbb{F}_2$, then*

$$t \not\equiv S := \sum_i b_i \pmod P.$$

Proof. The incorrect parity means $S \not\equiv b \pmod 2$, whereas every $t \in \mathcal{S}_{m,b}$ satisfies $t \equiv b \pmod 2$. Thus $S \neq t$ as integers. Both lie in $\{0,\dots,m\}$ and $P > m$, so they remain distinct modulo P. □

Theorem 10.32 (False parity reveals the setup logarithm)**.** *Under the hypotheses of Lemma 10.31, the bit openings and aggregate opening explicitly determine the setup logarithm:*

$$\boxed{\log_g h = (S - t)(r_T - R_T)^{-1} \pmod P, \quad R_T = \sum_i r_i.}$$

Therefore a polynomial-time adversary that yields extracted accepting proofs for a false parity relation with non-negligible probability contradicts the discrete-logarithm assumption.

Proof. From the extracted bit openings, Lemma 10.14 gives $C_T = g^S h^{R_T}$. The extracted aggregate opening gives $C_T = g^t h^{r_T}$. Hence $g^{S-t} = h^{r_T - R_T}$. The previous lemma gives $S - t \neq 0$ modulo P, so $r_T - R_T$ cannot vanish modulo P: otherwise the nonidentity element g^{S-t} would equal 1. The formula follows by exponent comparison and inversion, exactly as in Lemma 10.11. Computing it from the alleged proof extractor contradicts discrete-logarithm hardness whenever such adversaries succeed non-negligibly and extraction itself succeeds. □

Theorem 10.33 (Perfect honest-verifier privacy of parity proofs)**.** *For a fixed public commitment vector and public parity row, the full set of honest-verifier bitness and aggregate OR-proof transcripts has a perfect simulator using only the commitments and row coefficients. The simulator requires no committed bits or blinding scalars.*

Proof. For each bitness proof and the aggregate proof, use the witness-free perfect simulator from Theorem 10.23. The publicly determined aggregate commitment is the product of the given commitments, so the simulator can construct all branch statements. Honest provers choose independent fresh OR-protocol randomness and honest sigma verifiers choose independent uniform challenges. Conditional on the fixed public statements, each OR transcript therefore has exactly its independently simulated law. Multiplying these conditional laws yields equality of the complete joint transcript distribution. □

Corollary 10.34 (Exact empty-row handling)**.** *If a row contains no committed bits, its parity constraint is true exactly when its public target is* 0*, and it can be checked without any cryptographic proof.*

Proof. The empty sum in $\mathbb{F}_2$ equals 0. Thus its equality to the target holds precisely for target 0. The candidate sets are $\mathcal{S}_{0,0} = \{0\}$ and $\mathcal{S}_{0,1} = \varnothing$, in agreement with direct checking. □

Remark 10.35 (Extractor scope). The parity binding theorem is an *extracted-witness* argument: it assumes the standard knowledge extractor succeeds on each accepted Schnorr OR proof. Its explicit computational error must be added when this extractor is composed with the outer ring-switching challenges. It is not a claim of unconditional information-theoretic soundness.

10.5 A complete field-affine proof compiler

Definition 10.36 (Bit-committed field variables)**.** To register a hidden field element $x \in L$, commit independently to its n fixed-basis coordinates

$$C_i(x) := \mathsf{Com}(d_i(x); r_i), \qquad 0 \le i < n, \quad r_i \xleftarrow{\$} \mathbb{Z}_P.$$

Write $\mathbf{C}(x)$ for the n-tuple. The n bitness OR proofs establish that every registered group element has a known binary opening. Registration occurs before the outer random challenge whose soundness argument relies on the field value.

Lemma 10.37 (Registered coordinates reconstruct one field element)**.** *Given extracted binary openings (b_i, r_i) of every $C_i(x)$, there is a unique associated field value*

$$x^{\mathrm{ext}} := \sum_{i=0}^{n-1} \gamma_i b_i \in L.$$

Proof. The fixed basis coefficients $b_i \in \mathbb{F}_2$ determine the displayed field element. Uniqueness follows from the uniqueness of basis expansions; no second field element can have the same coordinate vector. □

Definition 10.38 (Affine proof compiler)**.** For registered hidden field variables $x_1, \ldots, x_s$ and a public affine equation $\sum_j a_j x_j = b$, the compiler:

(i) computes all public binary coefficients $M_{k,j,i} = d_k(a_j \gamma_i)$ and $b_k = d_k(b)$;

(ii) for each row k, selects the set $T_k = \{(j,i) : M_{k,j,i} = 1\}$;

(iii) forms the public group product $C_{T_k} = \prod_{(j,i)\in T_k} C_i(x_j)$;

(iv) proves the integer opening of C_{T_k} lies in $\mathcal{S}_{|T_k|,b_k}$ using the OR protocol of Definition 10.21.

Each registered bit must additionally have a valid bitness proof from Definition 10.29. No secret field element is transmitted or opened.

Lemma 10.39 (All affine row statements are publicly computable)**.** *Given the public group commitments, public basis, coefficients $(a_j)_j$ and right side b, a verifier can compute every C_{T_k} and candidate set $\mathcal{S}_{|T_k|,b_k}$ without knowing a hidden field element.*

Proof. Every $M_{k,j,i}$ equals the publicly computable coordinate $d_k(a_j\gamma_i)$. The set T_k and $b_k = d_k(b)$ therefore depend only on public data. Its aggregate commitment is the product of public commitments indexed by T_k, and the candidate set depends only on the publicly known row cardinality and target bit. □

Theorem 10.40 (Perfect completeness of the field-affine compiler)**.** *If every registered commitment opens to the correct coordinate of x_j and $\sum_j a_j x_j = b$ holds in L, then all bitness and compiled parity proofs accept for any public coefficient choices.*

Proof. Every bitness proof has a known binary opening and is complete by Lemma 10.22. By Theorem 10.4, each of the n compiled parity rows is true for these coordinate bits. Theorem 10.30 makes every corresponding parity OR proof complete. All checks pass. □

Theorem 10.41 (Extracted field-affine correctness)**.** *Suppose all registered commitments have extracted binary openings, and each compiled parity OR proof has an extracted opening to one of its allowed candidates. If the public field equation is false for the extracted field elements x_j^{ext}, then the extracted openings give a polynomial-time solution to the setup discrete-logarithm problem.*

Proof. If the field equation is false, Theorem 10.4 implies that at least one output-coordinate parity row is false for the extracted bits. Apply Theorem 10.32 to that row's extracted bit openings and allowed aggregate opening. It computes $\log_g h$ in polynomial time, contradicting setup hardness. □

Lemma 10.42 (Faithful compilation of the Chapter 9 checker)**.** *Each hidden-field equation in Definition 9.24 can be expressed in the affine input format of Definition 10.38 after its stipulated public messages and challenges are fixed. The trace constraints are public and require no hidden-field proof.*

Proof. Lemma 9.25 handles the claimed-value and incoming equations. Lemma 9.26 handles the outgoing equation, and Lemma 9.27 handles both anchored equations. In each case, all coefficients multiplying hidden field variables belong to the publicly known field L once the respective public messages and challenges have been received. Lemma 9.32 makes every trace-zero test directly computable from public values. □

Lemma 10.43 (Packed-column equality is also field affine in bits)**.** *For a binary witness column $(w_{v,j})_{v\in I}$, its packed field value $\widehat{w}_j = \sum_v \beta_v w_{v,j}$ is equivalent to n binary linear relations between the bits of $\widehat{w}_j$ and the witness bits $w_{v,j}$.*

Proof. Apply each coordinate map d_k to $\widehat{w}_j = \sum_v \beta_v w_{v,j}$. Since every $w_{v,j} \in \mathbb{F}_2$, linearity gives

$$d_k(\widehat{w}_j) = \sum_v d_k(\beta_v) w_{v,j} = w_{k,j},$$

using $d_k(\beta_v) = \delta_{kv}$. Thus the n equalities are simply $d_k(\widehat{w}_j) = w_{k,j}$. Conversely these n coordinate equalities imply the field equality by unique basis reconstruction. □

Remark 10.44 (Separate global witness binding)*.* The last lemma allows commitment-level linking between the packed table and a separately committed binary table if both are exposed to the concrete bit-proof system. It does not automatically link those commitments to an *external KBFold polynomial oracle* whose authentication structure is different. That link needs an explicitly specified cross-commitment statement and a proof.

10.6 A concrete perfect honest-verifier simulator

Definition 10.45 (Restricted honest-sigma-verifier model)**.** Consider a single opening of the Chapter 9 public-core protocol with the bit commitments of Definition 10.36. Every commitment uses fresh independent uniform Pedersen blinding, and every prescribed bitness or affine-parity proof uses fresh independent Schnorr OR-prover randomness. Each *sigma-proof challenge* is an independent uniform element of $\mathbb{Z}_P$, sampled after the corresponding OR first message. The outer Ring Switching challenges θ, z may still be arbitrary causal functions of the entire visible history, including commitments and earlier sigma transcripts. The verifier is not permitted to request extraneous secret-dependent predicates or actual commitment openings. This is the *honest-sigma-verifier model*.

Lemma 10.46 (Fresh commitments are conditionally uniform)**.** *At every registration stage, conditional on the previous visible history and on any actual private bit vector chosen at that stage, the fresh Pedersen commitments are jointly uniform on the corresponding power of G. This conditional law is independent of the committed bit values.*

Proof. The fresh commitment blindings are chosen independently uniformly from $\mathbb{Z}_P$ after the prior history. Once the bits are fixed, Lemma 10.10 makes every commitment uniform on G and the full vector independently uniform. Because the law is the same for every bit vector, averaging over any distribution of private bits conditional on the history does not change the uniform law. □

Lemma 10.47 (A prescribed OR proof has a witness-free conditional simulator)**.** *For a public OR-proof statement determined by the current visible history and any fixed valid hidden witness, if the new global sigma challenge is independent uniform, then the conditional transcript law equals that of Theorem 10.23, regardless of the witness and preceding proof transcripts.*

Proof. All OR-protocol randomizers for the current proof are fresh and independent of the previous history. The prover's witness may be a deterministic function of previously registered openings and current public coefficients, but the proof of Theorem 10.23 is pointwise in every fixed witness and every fixed public statement. Thus the same perfect simulation equality holds after conditioning on the current history and witness. Since its conditional law depends only on the statement, averaging over the witness preserves it. □

Definition 10.48 (Concrete honest-verifier simulator)**.** The simulator operates at the same causal protocol stages as the real implementation:

(i) at each hidden-bit registration, it samples independent uniform elements of G as the public commitments without choosing openings;

(ii) at each prescribed bitness or affine-parity OR proof, it uses the *witness-free* transcript simulator of Theorem 10.23 on the publicly known commitments and candidates;

(iii) at public-core stages it uses the causal simulator of Definition 9.12;

(iv) it forwards the verifier's allowed adaptive outer challenges and returns only the honest constant acceptance bits.

The simulator is defined for the honest-sigma-verifier model of Definition 10.45.

Theorem 10.49 (Perfect honest-verifier ZK of the concrete affine transcript)**.** *For one admissible Ring Switching opening and every causal verifier obeying Definition 10.45, the joint visible transcript of all fresh Pedersen bit commitments, prescribed Schnorr OR bitness and affine-parity proofs, and the public core (m, y', θ, U, z, A) has exactly the same law as the transcript generated by Definition 10.48. The equality holds for any two admissible witnesses with the same public instance. It excludes an external polynomial-commitment scheme and its authenticated openings.*

Proof. Include the verifier's random tape in the finite visible state and induct over the ordered message stages. For the initial hidden-variable registrations, Lemma 10.46 makes each fresh commitment vector conditionally uniform independent of the hidden bit values, exactly as in the simulator. For each prescribed OR proof, Lemma 10.47 identifies the complete conditional transcript law with the simulator's witness-free OR-proof law, even when the public statement depends on earlier messages or outer challenges. For the intervening public core, Theorem 9.13 gives the witness-free conditional distributions for (m, y'), U, and A. Conditioning on previous fresh perfectly hiding commitments and their perfectly honest-verifier-simulated proof transcripts cannot disclose the underlying private masks: the conditional laws established in the two preceding lemmas do not depend on those masks. Therefore the three public-core transition laws remain unchanged under that additional conditioning. At stages where the verifier selects an outer challenge, the same causal verifier transition kernel is applied to identically distributed visible histories in both experiments. Every prescribed honest affine statement is valid by Lemmas 9.6 and 9.7, and the concrete proofs are perfectly complete by Theorem 10.40; hence the associated acceptance bits are always 1. Thus every transition in the real and simulated finite transcript processes has the same conditional law. Induction on their length establishes exact equality of the complete visible joint laws. □

Corollary 10.50 (No witness-dependent commitment-token distribution)**.** *Under the hypotheses of Theorem 10.49, the complete vector of registered Pedersen commitment tokens is jointly independent of the hidden witness.*

Proof. It is a deterministic projection of the perfectly simulated visible transcript of that theorem. Alternatively, iterate Lemma 10.46 over all registration stages. □

Remark 10.51 (Exactly what has been realized). The theorem gives *perfect honest-verifier zero knowledge* for the explicit bit-commitment and affine-proof transcripts, with adaptive *outer* challenges. It is not full malicious-verifier ZK, because the sigma-proof verifier challenges are assumed independently uniform. It is not an end-to-end KBFold PCS theorem, because the original folded oracle commitment, Merkle authentication paths, and masked sumcheck transcript are absent. The commitment is classical discrete-logarithm based and is not post-quantum binding.

10.7 Temporal extraction, soundness composition, and remaining real-world gaps

Definition 10.52 (Extraction-before-challenge contract)**.** A concrete execution satisfies the *early extraction contract* if there is a computational coupling with an ideal run in which, except with probability at most $\varepsilon_{\mathrm{ext}}$:

(i) every bit commitment whose value is used in the first-challenge discrepancy is assigned a unique extracted binary opening *before* the verifier samples θ;

(ii) every auxiliary bit commitment whose value enters the anchored residual is assigned a unique extracted binary opening *before* the verifier samples z;

(iii) every accepted compiled parity OR proof has an extracted aggregate opening (t, R) with t in the public permitted candidate set; every bitness proof has an extracted binary opening; and every packed-column witness used by the outer protocol is linked to its registered binary coordinates by the separately checked packing relation;

(iv) the resulting coupled ideal protocol has the same public messages and the same fresh independent θ, z as the real execution outside the bad event.

The contract is a *knowledge and timing property to be proved for an actual composition of the OR extractors*, not a consequence of bare computational binding alone.

Lemma 10.53 (Accepted parity equations are valid under extracted consistency)**.** *Suppose all bit and aggregate OR proofs have successful extracted openings and the adversary has not solved the setup discrete logarithm. Then each accepted compiled affine equation holds for the extracted registered field values.*

Proof. If one accepted affine equation failed, Theorem 10.41 would turn the extracted openings for its nonzero parity row into a solution of the setup discrete-logarithm problem. This contradicts the stipulated non-solution event. Hence every such equation holds. □

Theorem 10.54 (Conditional soundness of the explicit bit-commitment compiler)**.** *Assume the false public evaluation instance and timing hypotheses of Theorem 9.37. Suppose the concrete bit-commitment and OR-proof implementation satisfies the early extraction contract of Definition 10.52 with bad-event probability at most $\varepsilon_{\mathrm{ext}}$, and the chance of breaking the discrete-logarithm binding assumption in this coupling is at most $\varepsilon_{\mathrm{DL}}$. Then*

$$\boxed{\Pr[\text{accept false claim}] \le \min\left\{1, \frac{\ell + n}{q} + \varepsilon_{\mathrm{ext}} + \varepsilon_{\mathrm{DL}}\right\}.}$$

Proof. Remove the event that the early extraction contract fails and the event that a forbidden discrete-logarithm opening collision is produced. Their total probability is at most $\varepsilon_{\mathrm{ext}} + \varepsilon_{\mathrm{DL}}$ by the union bound. On the complementary event, Lemma 10.53 makes every accepted compiled affine equation true on the fixed extracted field values, with the required order of commitments and random challenges. The coupled execution is therefore a legal instance of the ideal algebraic checker in Theorem 9.37. That theorem bounds false-claim acceptance on this good event by $(\ell + n)/q$. Add the two removed bad-event probabilities and cap by one. □

Lemma 10.55 (Multi-statement extraction cannot be silently assumed)**.** *The two-transcript special soundness of an individual Schnorr OR proof does not, as a matter of logical implication alone, establish the early extraction contract for an arbitrary adaptive multi-stage protocol using many proofs.*

Proof. Theorem 10.24 has hypotheses concerning two accepting transcripts of one fixed statement sharing exactly one first-message vector and carrying different global challenges. Definition 10.52 instead quantifies over many interleaved statements and their dependency order relative to two outer challenges and a fixed witness representation. The former statement contains neither the latter's challenge timing hypothesis nor a coupling of the extracted witnesses across distinct proofs. Hence the latter requires additional premises and a separate composition proof. □

Definition 10.56 (Malicious sigma-verifier security target)**.** The *malicious-sigma-verifier simulation target* removes the requirement that each global Schnorr OR challenge be independently uniform. The verifier may compute it as an arbitrary efficient function of its entire view, including the OR first messages, commitments, outer challenges and prior proofs. The target requires one efficient online witness-independent simulator for the full resulting interactive verifier view.

Proposition 10.57 (The proven HVZK simulator does not discharge the malicious target)**.** *The proof of Theorem 10.49 does not establish Definition 10.56.*

Proof. The simulator of Theorem 10.23 first chooses a uniform vector of branch challenges and defines the OR first messages from that choice. Consequently the global challenge it outputs is uniformly distributed and is not obtained by running an arbitrary verifier on the already determined first messages. Definition 10.56 permits the verifier to choose its challenge by such an arbitrary function. The simulator's argument therefore does not cover that quantifier. No impossibility of an alternative simulator is claimed. □

Definition 10.58 (Concrete ZK compiler obligation)**.** A *malicious-verifier zero-knowledge sigma compiler* for this chapter is an explicit transformation of the bitness and parity proof transcripts together with a proof that: (i) its complete visible transition kernels satisfy Definition 9.47 for all allowed malicious verifiers; (ii) its knowledge extraction respects the early challenge-timing contract; and (iii) it retains binding to the same committed witness. The transformation may use a designated-verifier technique, a proven malicious-verifier interactive compiler, or an appropriately modeled Fiat–Shamir transform [FS87], but neither its implementation nor its proof is supplied here.

Remark 10.59 (ROM and QROM are different proof obligations). A Fiat–Shamir transform of these Schnorr-style proofs is not automatically a quantum-random-oracle proof, and conventional discrete-log commitments are not quantum-resistant. To claim a post-quantum KBFold extension, the commitments, knowledge-soundness extractor, adaptive ZK simulator, authenticated fold-oracle openings, and original KBFold QROM analysis must all be revisited with quantum adversaries.

Remark 10.60 (The honest-verifier result is substantive but bounded). Chapter 10 constructs the concrete commitment and affine proof at the binary-coordinate level. The perfect honest-sigma-verifier simulation theorem is a real transcript result for this restricted model, not just an ideal-functionality restatement. It must not be labeled full zero knowledge without a malicious-verifier compiler and the external KBFold PCS commitment/opening simulator.

10.8 Cost accounting, historical placement, and formal certification

Definition 10.61 (Affine compilation parameters)**.** Let s be the number of distinct secret field elements registered in one stage, and let E be the number of field-affine equations verified after registration. For the e-th equation and k-th output coordinate, write $m_{e,k}$ for the number of nonzero binary coefficients in that parity row. Write $B := sn$ for the number of committed binary coordinates.

Lemma 10.62 (Exact bit commitment and parity proof counts)**.** *The compiler registers $B = sn$ Pedersen commitments and, if every coordinate is registered once, executes B binary bitness OR proofs. It executes exactly nE parity rows, except that a zero-length row may be checked directly and omitted. Each nonempty row's OR proof has*

$$d_{e,k} = |\mathcal{S}_{m_{e,k},b_{e,k}}| \le \left\lceil \frac{m_{e,k}+1}{2} \right\rceil$$

candidate branches.

Proof. Every field element has exactly n binary basis coordinates, so s elements supply $sn = B$ commitments and one bitness proof per coordinate. Theorem 10.4 produces one parity equation for each of the n output coordinates of each of the E field equations. For a row with m inputs, the allowed integers in $[0, m]$ of one specified parity form one of the two parity classes. The larger class has $\lceil (m+1)/2 \rceil$ elements, giving the branch bound. The empty-row handling follows Corollary 10.34. □

Lemma 10.63 (Linear branch-size upper bound)**.** *The total number of candidate branches across all nonempty parity OR proofs is at most*

$$\boxed{\sum_{e=1}^{E} \sum_{k=0}^{n-1} \left\lceil \frac{m_{e,k}+1}{2} \right\rceil \le nE \left\lceil \frac{sn+1}{2} \right\rceil .}$$

Proof. The first inequality follows from Lemma 10.62 applied rowwise, and zero rows contribute no actual proof branches. Every row weight is at most sn by Lemma 10.6. The ceiling function is monotone, so each term in the double sum is at most $\lceil (sn+1)/2 \rceil$. There are nE such terms, giving the second inequality. □

Lemma 10.64 (Two group elements per branch in a raw OR proof)**.** *An uncompressed d-branch Schnorr OR proof transmits d group elements in its first message and $2d$ elements of $\mathbb{Z}_P$ in its responses, in addition to one verifier challenge in $\mathbb{Z}_P$.*

Proof. Definition 10.21 sends the vector $(T_j)_{j=1}^{d}$ of d group elements, followed by the vector (e_j, z_j) of two scalars per branch. The verifier supplies one global challenge e. No compression or derived challenge optimization is assumed. □

Proposition 10.65 (Raw compiler overhead)**.** *The raw compiler commits to every secret field coordinate separately. Its parity proofs also grow with the number of binary row entries. Consequently the construction in this chapter* does not *establish a polylogarithmic-size zero-knowledge PCS, even if the original KBFold commitment scheme has succinct opening proofs.*

Proof. Lemma 10.62 requires sn separate bit commitments and bitness proofs in the direct implementation. Lemma 10.63 counts a candidate set for each parity row, and Lemma 10.64 requires a group message and scalar responses for every candidate branch. These costs are proportional to explicit field-coordinate and row representations; nothing in the construction compresses them to a polylogarithmic dependence on the witness length. □

Definition 10.66 (Chapter 10 Lean module contract)**.** The formalization splits into:

(i) exact finite-field affine-to-binary coordinate matrices;

(ii) prime-order cyclic group algebra and Pedersen hiding/binding;

(iii) Schnorr and OR-protocol message equations, HVZK simulators and two-transcript extraction;

(iv) integer-lift parity candidate sets and wrong-parity discrete-log reduction;

(v) compilation of all Chapter 9 affine equations and packing links;

(vi) finite adaptive public-core plus honest-sigma proof-transcript simulation;

(vii) a parameterized early-extraction contract and its conditional outer soundness implication.

Remark 10.67 (Prior work and originality). The commitment construction and its hiding/binding theorem come from Pedersen [Ped92]. The logarithm proof is Schnorr's [Sch91], while OR proofs of partial knowledge are classical [CDS94]. The affine-to-parity application here is an explicit composition in our Boolean-kernel Ring Switching notation, not a claim to have invented bit commitments or OR proofs. The target PCS and its post-quantum motivation are from the original KBFold manuscript [Mkh26].

Remark 10.68 (Main remaining theorem). The next goal is a malicious-verifier simulator and a sound extractor for the bound commitments. Both must remain secure under adaptive outer challenges. After that, one must bridge these characteristic-two bit commitments to KBFold's odd-characteristic, authenticated folding-oracle PCS.

10.9 Main theorem chain and certified scope

For public $a_j, b \in L = \mathbb{F}_{2^n}$ and hidden $x_j \in L$, Theorem 10.4 proves

$$\boxed{\sum_j a_j x_j = b \iff \sum_{j,i} d_k(a_j \gamma_i) d_i(x_j) = d_k(b) \quad \forall\, 0 \le k < n.}$$

Commit each hidden binary coordinate using the freshly blinded Pedersen equation

$$C_i = g^{b_i} h^{r_i}$$

in an odd-prime-order group. The commitment is perfectly hiding and computationally binding under discrete-log hardness (Theorems 10.12 and 10.32). A parity row is proved by a Schnorr OR proof that its product commitment opens to an integer with the correct parity. Those proofs have perfect completeness, perfect honest-verifier simulation and two-transcript special soundness (Theorems 10.30, 10.33, 10.24).

The concrete affine transcript, combined with the single-opening public core of Chapter 9, admits an exact simulator for verifiers whose *inner sigma challenges are honest uniform*, while their *outer* Ring Switching challenges may be adaptively chosen:

$$\boxed{\mathsf{Law}(\mathrm{FullView}_{\text{honest sigma}}) = \mathsf{Law}(\mathsf{Sim}_{\text{honest sigma}}).}$$

The associated computational soundness implication requires the early extraction contract:

$$\boxed{\Pr[\text{false evaluation accepted}] \leq \min\left\{1, \frac{\ell + n}{2^n} + \varepsilon_{\text{ext}} + \varepsilon_{\text{DL}}\right\}.}$$

The theorem is *conditional* until the multi-stage extractor and its exact errors are constructed and audited.

The raw compiler is not succinct, is not post-quantum binding, and is not a proof of malicious-verifier or online multiple-opening zero knowledge for the original KBFold PCS. These are subsequent, explicitly identified research obligations.

11 Malicious-Verifier Zero Knowledge and Temporal Extraction

Chapter 10 constructed a concrete classical commitment-and-affine proof with perfect honest-sigma-verifier simulation, but did not allow the inner sigma challenges to depend on the prover's first messages. This chapter removes that restriction for the *concrete binary-coordinate affine transcript*. The key tool is a sequentially repeated *one-bit* Schnorr OR proof [Sch91; CDS94] and classical verifier rewinding [Gol01]. We derive its exact fixed-challenge law, an abort-corrected malicious-verifier simulator, a quantitative two-run forking theorem, and the precise limit of local temporal knowledge extraction. The underlying KBFold polynomial oracle, Merkle paths and post-quantum model are *not* part of this chapter's concrete simulator.

11.1 One-bit OR proofs and their causal security model

The three-move Schnorr OR proof of Chapter 10 has a challenge in $\mathbb{Z}_P$. A generic simulator that guesses this entire challenge succeeds with probability $1/P$, which is unsuitable when P is cryptographically large. We use the classical *one-bit challenge and rewinding* technique [Sch91; CDS94] to obtain an explicit malicious-verifier simulator. A textbook treatment of the bit-challenge technique is given in [Gol01].

Definition 11.1 (One-bit OR mini-proof)**.** Fix a public group commitment C, a nonempty finite set $S = \{t_1, \dots, t_d\} \subset \mathbb{Z}_P$, and a witness (j_*, r) satisfying $C = g^{t_{j_*}} h^r$. The prover executes Definition 10.21, except that the *global* verifier challenge is a bit $e \in \{0, 1\} \subset \mathbb{Z}_P$. The internal branch challenges e_j and responses z_j remain arbitrary elements of $\mathbb{Z}_P$ and must obey $\sum_j e_j = e \pmod P$. An honest verifier samples the challenge bit uniformly *after* receiving the entire first-message vector T.

Lemma 11.2 (Perfect completeness of the mini-proof)**.** *For either global challenge bit, every honest execution of Definition 11.1 satisfies the OR verifier equations.*

Proof. The completeness proof of Lemma 10.22 is an identity for every $e \in \mathbb{Z}_P$, hence in particular for $e = 0$ and $e = 1$. □

Definition 11.3 (Schedule-respecting malicious verifier)**.** A malicious verifier V^* is a probabilistic algorithm with a private random tape and a rewindable state. At each OR mini-proof stage, after observing the statement, previous transcript, and first message T, it may either: (i) send either bit 0 or 1; or (ii) terminate with a distinguished abort symbol $\perp$. Any other response is normalized to $\perp$. The verifier may choose the bit as an arbitrary efficient function of its entire visible history and its private tape. All other public Ring Switching challenges may likewise be chosen causally from the current visible history. The verifier cannot demand an extraneous secret opening or change the prescribed protocol equations.

Definition 11.4 (One-step black-box rewinding)**.** A simulator is allowed to save the verifier state immediately before one OR first message, execute the verifier on a trial first message, inspect the bit challenge or abort, and restore the saved state with the *same* verifier random tape. Only the current mini-proof is rewound. After a trial is selected, the simulator continues from the selected verifier state without changing any earlier public message. This is the standard classical black-box rewinding model and does not grant rewinding access to a quantum verifier.

Lemma 11.5 (A fixed statement admits a known witness in the honest case)**.** *For every bitness or parity OR statement generated by an honest registration and a true affine relation in Chapter 10, the honest prover knows a candidate index j_* and blinding r required by Definition 11.1.*

Proof. For bitness, the actual bit and its fresh Pedersen blinding give $C = g^b h^r$ with candidate $b \in \{0, 1\}$. For an affine parity row, the true integer sum lies in its permitted parity set by Lemma 10.28. Lemma 10.14 supplies its known aggregate blinding. Both are valid OR witnesses. □

Remark 11.6 (Cost of choosing a shorter challenge). The mini-proof has challenge entropy only one bit. Its standalone knowledge error cannot be treated as $1/P$. It must be repeated with fresh prover randomness and fresh verifier challenge bits; the soundness analysis of this repetition appears in Section 11.5.

11.2 A simulator for any fixed one-bit challenge

The central technical fact is stronger than ordinary honest-verifier zero knowledge: for every *fixed* global challenge bit the Schnorr OR transcript admits a perfect witness-free simulator whose first message has the same distribution, independently of that bit.

Definition 11.7 (Fixed-bit OR transcript simulator)**.** Given only (C, S, e) for a specified bit $e \in \{0, 1\}$, write $Y_j = Cg^{-t_j}$. Independently sample $e_1, \ldots, e_{d-1}, z_1, \ldots, z_d$ uniformly in $\mathbb{Z}_P$; set

$$e_d := e - \sum_{j<d} e_j \pmod{P}, \qquad T_j := h^{z_j} Y_j^{-e_j}.$$

Output $(T, (e_j, z_j)_{j=1}^d)$. This produces a transcript accepted for the global challenge e, and requires no OR witness.

Lemma 11.8 (The simulator always satisfies the verifier equations)**.** *For every public input and bit e, the output of Definition 11.7 is accepting.*

Proof. By definition $\sum_j e_j = e$ in $\mathbb{Z}_P$. Multiplying $T_j = h^{z_j} Y_j^{-e_j}$ by $Y_j^{e_j}$ gives h^{z_j}. These are exactly the OR verifier equations. □

Lemma 11.9 (Honest local challenges are uniform subject to their sum)**.** *Fix an honest OR witness branch j_* and a global challenge $e \in \mathbb{Z}_P$. The complete local challenge vector in an honest execution is uniform over*

$$\mathcal{H}_e := \{(e_1, \ldots, e_d) \in \mathbb{Z}_P^d : \sum_j e_j = e\}.$$

Proof. The honest prover chooses the $d - 1$ local challenges outside branch j_* independently uniformly, and sets the remaining coordinate to e minus their sum. This is a bijection between $\mathbb{Z}_P^{d-1}$ and $\mathcal{H}_e$, so every vector in $\mathcal{H}_e$ occurs with the same probability $P^{-(d-1)}$. □

Lemma 11.10 (Honest responses are independent uniforms given challenges)**.** *For fixed (C, S), honest witness (j_*, r), and global challenge e, conditional on the complete local challenge vector, the responses $z_1, \ldots, z_d$ are independent uniform elements of $\mathbb{Z}_P$.*

Proof. The honest prover sampled all false-branch responses z_j independently uniformly. The true-branch response is $z_{j_*} = u + e_{j_*} r$ with independent uniform nonce u. Once the challenge vector and witness are fixed, the map $u \mapsto u + e_{j_*} r$ is a translation and hence preserves the uniform distribution. It is independent of all false-branch responses. □

Theorem 11.11 (Perfect simulation conditional on any fixed challenge)**.** *For every valid OR witness and every fixed $e \in \{0, 1\}$, the distribution of the entire honest transcript*

$$(T, (e_j, z_j)_j) \mid e$$

is identical *to the distribution produced by Definition 11.7.*

Proof. By Lemma 11.9, the honest challenge vector is uniform over $\mathcal{H}_e$. Definition 11.7 selects exactly this uniform law by freely choosing $d-1$ coordinates and solving for the last. By Lemma 11.10, the honest responses conditioned on the challenge vector are independent uniforms; the simulator samples the same conditional law. Finally every accepting transcript uniquely determines the first message as $T_j = h^{z_j} Y_j^{-e_j}$. The simulator and honest prover therefore use identical laws for all coordinates and the same deterministic relation for T. □

Lemma 11.12 (First message is challenge-independent)**.** *For either fixed challenge bit, the first message $T = (T_1, \ldots, T_d)$ of the fixed-challenge simulator is uniform on G^d. In particular its law is independent of the challenge and the OR witness.*

Proof. Condition on any local challenge vector. Each z_j is independent uniform in $\mathbb{Z}_P$. Since h generates G, the map $z_j \mapsto h^{z_j} Y_j^{-e_j}$ is a bijection onto G. The coordinates T_j are thus conditionally independent uniform elements of G, with a law independent of the selected local challenge vector. Averaging over that vector does not change the uniform product law. □

Corollary 11.13 (Fixed-bit response simulation conditioned on T)**.** *For every fixed valid witness, every $e \in \{0,1\}$ and every possible first message T, the conditional distribution of the honest OR response given (T, e) is the same as that of Definition 11.7 given (T, e).*

Proof. Theorem 11.11 identifies the joint laws of first messages and responses for the specified e. Lemma 11.12 assigns every possible T positive probability $|G|^{-d}$ in both experiments. Therefore their conditional response laws at each (T, e) are identical. □

11.3 Perfect black-box simulation for a malicious verifier

Definition 11.14 (Balanced one-bit rewinding simulator)**.** At the beginning of one prescribed OR mini-proof, save the complete verifier state, including its random tape. Repeat the following trial:

(i) Restore the saved verifier state. Independently guess $\widehat{e} \xleftarrow{\$} \{0,1\}$ and run the fixed-bit simulator of Definition 11.7 for $\widehat{e}$, obtaining first message T and response R.

(ii) Send T to V^* and read its reply. If its reply is a bit e, *select* the trial exactly when $e = \widehat{e}$. If its reply is $\perp$, independently flip an additional fair coin and select the trial exactly when this coin is 1.

(iii) If unselected, rewind and retry. If selected with a valid bit, send R as the prover response and continue from that verifier state. If selected with $\perp$, preserve that terminating verifier state and output its abort transcript.

The independent fair coin on abort is essential: it makes the trial-selection probability the same for aborts and valid challenges.

Lemma 11.15 (Every trial is selected with probability one half)**.** *Fix the entire visible history and verifier random tape at the beginning of a mini-proof. For every possible sampled first message T, the probability that a trial of Definition 11.14 is selected is exactly* $1/2$.

Proof. By Lemma 11.12, the guessed bit $\widehat{e}$ is independent of T and is uniform. For a fixed verifier state and T, the verifier's reply is determined. If the reply is 0 or 1, the independent uniform $\widehat{e}$ equals it with probability $1/2$. If the reply is $\perp$, the simulator's additional fair selection coin equals 1 with probability $1/2$. These exhaust the possible replies. □

Lemma 11.16 (Expected trial count)**.** *For every fixed verifier state, the balanced simulator selects a trial almost surely; the number N of trials satisfies*

$$\Pr[N > t] = 2^{-t} \quad (t \geq 0), \qquad \mathbb{E}[N] = 2.$$

Proof. Each retry uses fresh independent simulator randomness from the same saved verifier state. Lemma 11.15 gives conditional selection probability $1/2$ in every trial. Thus the probability of failing the first t trials is $(1/2)^t$. Finally $\mathbb{E}[N] = \sum_{t\geq 0} \Pr[N > t] = \sum_{t\geq 0} 2^{-t} = 2$. □

Lemma 11.17 (Balanced selection preserves the real first-message law)**.** *For a fixed verifier state, the selected first message T is uniform on G^d, exactly as in an honest real execution. The verifier's decision to abort and its chosen challenge bit are distributed in the same way in both experiments.*

Proof. Every trial samples T uniformly on G^d by Lemma 11.12. Lemma 11.15 states that, conditional on each T, it is selected with the same probability $1/2$. Consequently selection does not reweight the uniform distribution of T. The verifier's response is a fixed function of this selected T and its saved random tape; hence its response has the real-execution distribution. □

Lemma 11.18 (A selected accepting response has the real conditional law)**.** *Condition on any selected non-aborting first message T and challenge bit e. The balanced simulator's OR response has the same conditional distribution as that of the honest prover answering e after T.*

Proof. The trial is selected with a valid response exactly when $\widehat{e} = e$. Under this event, the trial's response comes from the fixed-e simulator. The selection event imposes no further condition on the sampled response once $(T, \widehat{e})$ are fixed, because the verifier chooses e based on the first message and its saved state. Corollary 11.13 identifies this response law with the honest conditional response law. □

Theorem 11.19 (Perfect malicious-verifier zero knowledge for one mini-proof)**.** *For every fixed valid OR statement and every rewindable, schedule-respecting malicious verifier V^*, the balanced simulator is witness free, terminates almost surely in expected two trials, and produces* exactly *the joint real distribution of the verifier random tape, first message, challenge or abort, and (when present) prover response.*

Proof. Couple the verifier's initially sampled random tape in the two experiments and fix it. By Lemma 11.17, the selected T and the verifier's reply have the honest real distribution. If the verifier aborts, the selected trial reports precisely that abort history; there is no prover response. If the verifier sends a challenge bit, Lemma 11.18 gives precisely the honest conditional prover-response law. These are all components of the visible mini-proof transcript, so their joint laws match for the fixed tape. Averaging over the tape gives exact equality of the joint distributions. Lemma 11.16 proves almost-sure termination and the expected trial count. □

Lemma 11.20 (Why unconditional acceptance of simulated aborts is biased)**.** *Suppose a trial were accepted whenever the verifier aborts, but accepted after a valid challenge only when the guessed bit matches. If the verifier aborts on half of the possible uniform first messages, the selected transcript would abort with probability $2/3$, although the honest prover's transcript aborts with probability $1/2$.*

Proof. Let A be the event that the verifier aborts on T. By assumption $\Pr[A] = 1/2$. Under the incorrect selection rule, the probability of selecting a trial with abort is $1/2$; that of selecting one with a valid challenge is $(1/2)(1/2) = 1/4$. Conditioned on selection, the probability of abort would therefore be $(1/2)/(1/2 + 1/4) = 2/3$. The honest prover sends a uniformly distributed T without selection, so it encounters abort with probability $1/2$. The balanced rule avoids this discrepancy. □

Remark 11.21 (Classical scope). The simulator rewinds the verifier to the same *classical* random tape and state. This proof does not authorize rewinding quantum verifier states or claim security in the quantum-random-oracle model.

11.4 Sequential repetition and full affine-transcript simulation

Definition 11.22 (Sequential amplification parameter)**.** Fix an integer $\kappa \geq 1$. For every prescribed Chapter 10 bitness or parity OR statement, replace its single proof by κ *sequential* fresh copies of Definition 11.1. The verifier sends the bit challenge for copy i before the prover sends the first message for copy $i+1$. All κ copies must accept. An honest verifier samples independent uniform challenge bits; a malicious verifier may choose them causally. All corresponding commitments and the statement are fixed before the first copy begins.

Lemma 11.23 (Sequential composition preserves honest completeness)**.** *Every valid OR witness causes all κ honest mini-proofs to accept for every possible verifier challenge sequence.*

Proof. Each mini-proof has perfect completeness for either challenge by Lemma 11.2. Sequential execution does not change the witness relation or any individual verifier identity, so all κ pass. □

Lemma 11.24 (Sequential simulator cost)**.** *For a fixed valid OR statement, its κ-copy malicious-verifier simulator has expected at most 2κ mini-proof trials. For M prescribed mini-proofs in one execution, the expected total number of trials is at most $2M$, where M counts the already repeated one-bit mini-proofs.*

Proof. Fix the visible history and verifier state at each mini-proof start. By Lemma 11.16, each copy needs two expected trials. Early verifier aborts only reduce the number of copies executed. Linearity of expectation over the at most κ copies per statement, or at most M copies in total, gives the bounds. □

Theorem 11.25 (Perfect simulation of sequential OR statements)**.** *For any causal rewindable verifier, the joint transcript of κ sequential honest OR mini-proofs of a fixed valid statement is identically distributed to the transcript obtained by invoking the balanced simulator separately at each copy. The claim includes aborts and arbitrary history-dependent verifier challenge bits.*

Proof. Induct over the sequential mini-proofs. At the first, Theorem 11.19 identifies the real and simulated conditional transition laws. Suppose the visible prefix laws agree through copy $i-1$. For any realized non-aborting prefix, the next OR statement is the same public statement with a fresh independent honest prover random tape. Theorem 11.19 holds pointwise for every such prefix and verifier state, giving identical conditional laws for copy i. The verifier's own next action is a function of that visible history and its private tape, so it is coupled identically. If a prefix aborts, both processes terminate with the same conditional distribution. Induction gives equality of the entire ordered transcript laws. □

Definition 11.26 (Chapter 11 concrete affine protocol)**.** The *sequential-bit affine protocol* uses the Pedersen bit-commitment register and field-affine parity compilation of Chapter 10, but replaces *every* Schnorr OR bitness and aggregate proof by the sequential repetition of Definition 11.22. The Chapter 9 public-core messages (m, y', U, A) and outer challenges (θ, z) are unchanged. Fresh Pedersen blindings and fresh private Ring Switching masks are used as in Chapters 9–10. This protocol's transcript does *not* include the original KBFold folded-oracle commitment or Merkle openings.

Theorem 11.27 (Perfect expected-time malicious-verifier ZK of the affine transcript)**.** *For any two admissible witnesses for the same single-opening public instance and any probabilistic polynomial-time, classically rewindable, schedule-respecting verifier V^*, the complete visible transcripts of Definition 11.26 have identical distributions. There is a witness-independent, expected-polynomial-time black-box simulator for those transcripts, allowing V^* to choose* all inner bit challenges *and the outer Ring Switching challenges adaptively.*

Proof. Simulate every registered Pedersen bit commitment by an independent uniform element of G, as justified by Lemma 10.46. Simulate the public-core messages using Definition 9.12. For each prescribed bitness or parity OR mini-proof, execute the balanced simulator of Definition 11.14 relative to the current public statement. The statement is computable from the public commitment vector, fixed basis, public affine coefficients and required parity target by Lemma 10.39.

To prove equality of transcript laws, condition on any current visible history and the verifier's private random tape. Fresh commitment blindings hide the underlying bits perfectly, even under this conditioning; the commitment message transition is therefore the same in the real and simulated processes. The exact conditional laws of the public core were proved in Lemmas 9.9, 9.10, and 9.11. Those laws remain unchanged by conditioning on the public proof messages: each proof message has a witness-independent conditional distribution by Theorem 11.19, and all honest prover coins for each new proof are fresh. For the next inner OR mini-proof, Theorem 11.19 matches the exact conditional transcript law for the current visible history, regardless of how the malicious verifier chooses the bit. Outer challenges are supplied by the same causal verifier kernel in both experiments and thus have identical conditional laws. All honest affine relations are valid by Lemmas 9.6 and 9.7; their compiled proof checks pass by Lemma 11.23. Induction over the finite ordered message stages therefore proves equality of full visible laws. Finally Lemma 11.24 bounds the expected number of replayed mini-proofs by twice their polynomial total count, so the simulator has expected polynomial running time. □

Definition 11.28 (Bounded-retry simulator)**.** Given a deterministic retry bound $R \geq 1$, modify the balanced simulator so that after R unselected trials of a mini-proof it outputs a distinguished simulation failure symbol and terminates. Otherwise it runs unchanged.

Theorem 11.29 (Strict polynomial time with negligible statistical error)**.** *Suppose a protocol execution uses at most M sequential one-bit mini-proofs. The bounded-retry full-transcript simulator of Definition 11.28 is strict polynomial time whenever MR is polynomial, and its output law has statistical distance at most*

$$\boxed{\min\{1, M2^{-R}\}}$$

from the honest real transcript law. In particular $R = \lambda + \lceil \log_2 M \rceil$ gives distance at most $2^{-\lambda}$.

Proof. By Lemma 11.16, the probability of R consecutive unselected trials at any reached mini-proof is 2^{-R}, conditional on its starting visible history. A union bound over at most M mini-proofs gives failure probability at most $M2^{-R}$. Couple the truncated simulator to the unbounded perfect simulator using identical trial coins until a truncation occurs. The views coincide on the complementary event, so their statistical distance is at most the truncation probability. Theorem 11.27 identifies the unbounded simulator's output law with the real transcript. Each mini-proof makes at most R verifier invocations, hence at most MR in total. For the stated choice of R, $M2^{-R} \leq 2^{-\lambda}$. □

Remark 11.30 (Sequentiality is essential for this simulator). A single parallel challenge containing κ unpredictable bits would require guessing all κ bits at once under the simple transcript-guessing approach, with acceptance probability $2^{-\kappa}$. Sequential repetition permits a separate one-bit rewind

after each first message, costing two expected trials per copy. The next section makes this contrast exact.

11.5 Two-run forking and quantitative knowledge error

For soundness the *honest* sigma verifier chooses the bit challenges uniformly. This is a different security game from the malicious-verifier ZK simulation above.

Definition 11.31 (Resettable fixed-statement prover)**.** Fix a public OR statement (C, S) before the first one-bit verifier challenge. A resettable prover has a fixed initial state and a complete fixed random tape for a sequence of κ fresh one-bit mini-proofs for that same statement. Given any verifier challenge string $\mathbf{e} = (e_1, \ldots, e_\kappa) \in \{0,1\}^\kappa$, the resulting messages and accepting or rejecting outcome are deterministic. The prover may adapt its next first message to earlier challenges, but not to the current challenge before sending that first message. A reset restarts it from the same initial state and random tape.

Definition 11.32 (Accepting challenge set)**.** For the fixed deterministic state of Definition 11.31, let

$$\mathcal{A} := \{\mathbf{e} \in \{0,1\}^\kappa : \text{every mini-proof accepts}\}, \quad m := |\mathcal{A}|, \quad p := m/2^\kappa.$$

The quantity p is the conditional success probability against independent uniform verifier challenge bits.

Lemma 11.33 (Earliest difference yields one OR fork)**.** *If two distinct challenge strings $\mathbf{e}, \mathbf{e}' \in \mathcal{A}$ are both accepted from the same reset prover state and random tape, then their transcripts contain two accepting responses to the* same *OR first message under the* different *global bits* 0 *and* 1*.*

Proof. Let i be the smallest index with $e_i \neq e'_i$. All earlier verifier challenges and prover responses are identical, because the starting state and random tape are fixed and the prover is causal. Immediately before the i-th challenge, both runs therefore have the same prover state and the same first message T_i. Their current challenges differ: one is 0 and the other 1. Both full challenge strings lie in $\mathcal{A}$, so the verifier accepts the i-th mini-proof in both runs. The two local accepting transcripts have the required common first message and distinct global challenges. □

Theorem 11.34 (Combinatorial knowledge extraction from two accepting runs)**.** *Under the hypotheses of Lemma 11.33, one can compute in polynomial time from the two transcripts an OR witness (t_j, r) such that*

$$\boxed{t_j \in S, \qquad C = g^{t_j} h^r.}$$

Proof. The preceding lemma supplies two accepting mini-proofs with the same statement and first message and two different global challenges. Apply the explicit extraction formula of Theorem 10.24. It produces a branch index j and its opening exponent r. Comparing the two transcripts to find the earliest challenge difference, and computing the one modular inverse, are polynomial-time operations. □

Lemma 11.35 (Exact two-run forking probability)**.** *For a fixed deterministic prover with success probability $p = m/2^\kappa$, sample two independent uniform challenge strings and reset the same prover state between runs. The probability that both runs accept and their challenge strings are distinct is exactly*

$$\boxed{p(p - 2^{-\kappa}).}$$

Proof. There are m accepting strings. The probability that both independent samples belong to $\mathcal{A}$ is $(m/2^\kappa)^2 = p^2$. Among those ordered pairs, there are exactly m diagonal pairs with identical challenge strings. Each ordered pair has probability $2^{-2\kappa}$, so the diagonal probability equals $m2^{-2\kappa} = p2^{-\kappa}$. Subtracting yields the formula. □

Theorem 11.36 (Randomized-prover two-run forking bound)**.** *Let a randomized causal prover have acceptance probability ε against κ independent uniform challenge bits, on a fixed statement. Run it twice from the* same *initial state and* same *private random tape but with independent verifier challenge strings. The probability of obtaining an extractable fork is at least*

$$\boxed{\max\{0, \varepsilon(\varepsilon - 2^{-\kappa})\}.}$$

Proof. Condition on the prover's initial private random tape R, and write p_R for its conditional acceptance probability. Lemma 11.35 gives conditional fork probability $p_R^2 - 2^{-\kappa}p_R$. Averaging over R yields

$$\mathbb{E}[p_R^2] - 2^{-\kappa}\mathbb{E}[p_R].$$

By the elementary nonnegativity of variance, $\mathbb{E}[p_R^2] \geq (\mathbb{E}[p_R])^2 = \varepsilon^2$. The expectation of p_R is ε by the law of total probability. The lower bound is therefore $\varepsilon^2 - 2^{-\kappa}\varepsilon$. A probability is also nonnegative, giving the stated maximum. Theorem 11.34 turns any successful fork into a valid OR witness. □

Corollary 11.37 (Knowledge-error dichotomy)**.** *Fix a resettable prover running in polynomial time. Either $\varepsilon \leq 2^{-\kappa} + \delta$, or a two-run black-box extractor finds an OR witness with probability at least $\varepsilon\delta$, where $\delta > 0$ is any chosen threshold.*

Proof. If the first alternative fails, then $\varepsilon - 2^{-\kappa} > \delta$. Theorem 11.36 makes the extraction probability at least $\varepsilon(\varepsilon - 2^{-\kappa}) > \varepsilon\delta$. □

Lemma 11.38 (Why a single global κ-bit guess is costly)**.** *Suppose a verifier independently samples a uniform bit vector $\mathbf{e} \in \{0,1\}^\kappa$ after receiving all first messages, and a simulator guesses one independent uniform $\widehat{\mathbf{e}}$ in advance. Then the probability of a complete match is exactly $2^{-\kappa}$, so the expected number of complete independent trials until a match is 2^κ.*

Proof. For every guessed vector, precisely one of the 2^κ equally likely verifier vectors matches. Independent repeated trials therefore have success probability $2^{-\kappa}$ and geometric expected trial count its reciprocal, namely 2^κ. □

Remark 11.39 (What this theorem actually extracts). Theorems 11.34 and 11.36 establish extraction for *one fixed OR statement* across resettable one-bit repetitions. They do not by themselves yield a single online extractor for every interleaved proof and a complete Ring Switching transcript. That multistage composition must respect the two outer challenge boundaries; we state it separately.

11.6 Extraction before the outer challenges

Definition 11.40 (Pre-challenge registration boundary)**.** The first outer challenge boundary is the instant immediately before θ is sampled. All commitments to the witness, packed columns, $m, p, p_{\text{in}}, c, y'$ that influence the first discrepancy must be registered before this instant. All bitness proofs required to bind their binary-coordinate interpretations are completed before this boundary. The second boundary is immediately before z is sampled. The post-θ variables t, ρ, σ and their

required bitness proofs are completed before this second boundary. The *registration prologue* at either boundary consists of the appropriate bit commitments followed by sequential one-bit OR proof repetitions, with no outer random challenge interleaved inside that prologue.

Lemma 11.41 (A local reset occurs before the protected challenge)**.** *If the entire OR prologue for a statement C, S is completed before an outer challenge is sampled, then the two-run forking experiment of Theorem 11.36 can be performed using only states and verifier challenge bits from* before *that outer challenge.*

Proof. The prologue begins with fixed public statement and a prover state prior to the protected outer challenge. Both forked runs replay the prologue from this saved state and use only new independent *inner* verifier bits. By the defined schedule, neither run needs or receives the future outer challenge. The extractor's messages, including the second branch challenge that creates the fork, are therefore all generated before the outer challenge. □

Theorem 11.42 (Stand-alone early extraction with quantitative success)**.** *Fix one registration OR statement and a resettable polynomial-time prover in a κ-copy prologue that ends before an outer challenge. If its probability of passing this prologue against independent uniform challenge bits is ε, then a two-run extractor* before that outer challenge *produces a valid permitted opening of the statement with probability at least*

$$\boxed{\max\{0, \varepsilon(\varepsilon - 2^{-\kappa})\}.}$$

Its two executions use no outer challenge value and therefore do not condition on one.

Proof. Lemma 11.41 places the fork entirely in the pre-challenge prologue. Theorem 11.36 gives the probability of two distinct accepting challenge strings, and Theorem 11.34 computes an allowed witness from the resulting fork. All actions happen before the protected outer challenge is sampled. □

Lemma 11.43 (At most one successful challenge path without a fork)**.** *For a fixed deterministic prover state and random tape, if no pair of distinct accepting challenge strings yields a valid fork, then at most one of the 2^κ challenge strings can be accepting; its acceptance probability is therefore at most $2^{-\kappa}$.*

Proof. If two distinct accepting challenge strings existed, Lemma 11.33 would yield a pair of accepting responses to one first message under different challenge bits. That is a valid fork, contrary to hypothesis. Thus the accepting set has cardinality at most one. Its uniform probability is at most $1/2^\kappa$. □

Lemma 11.44 (False parity plus a fork solves setup discrete log)**.** *Suppose the registered bits of an aggregate commitment are fixed and the extractor knows valid binary openings (b_i, r_i). Let $S = \sum_i b_i$ be its integer sum, and assume the claimed parity target is false for these bits. If an accepting κ-copy aggregate parity OR proof yields a fork, one can compute $\alpha = \log_g h$ in polynomial time.*

Proof. Theorem 11.34 extracts an opening (t, R) with t in the parity-compatible candidate set. The aggregate commitment also has the known true opening $(S, \sum_i r_i)$ by Lemma 10.14. Because the target parity is false and P exceeds the row length, Lemma 10.31 shows that t and S are distinct modulo P. The explicit formula of Theorem 10.32 then recovers the setup logarithm. □

Theorem 11.45 (Quantitative soundness for a false parity with known true opening)**.** *Suppose the correct binary openings of a fixed aggregate commitment are known to the reduction, but their*

parity disagrees with the public permitted candidate set. Let ε be the success probability of a causal prover against κ sequential honest one-bit challenges. There is a two-run discrete-log reduction whose success probability is at least

$$\boxed{\max\{0, \varepsilon(\varepsilon - 2^{-\kappa})\}.}$$

Consequently, if that reduction's discrete-log success probability is at most δ_{DL}, then

$$\boxed{\varepsilon \le 2^{-\kappa} + \sqrt{\delta_{\mathrm{DL}}}.}$$

Proof. Apply the two-run forking bound Theorem 11.36. For every successful fork, Lemma 11.44 computes the forbidden logarithm, giving the first inequality. Write $a = 2^{-\kappa}$. If $\varepsilon \le a$, the second bound holds trivially. Otherwise $\varepsilon(\varepsilon - a) \le \delta_{\mathrm{DL}}$. Since $\varepsilon \ge \varepsilon - a$, we have $(\varepsilon - a)^2 \le \delta_{\mathrm{DL}}$. Taking nonnegative square roots yields $\varepsilon \le a + \sqrt{\delta_{\mathrm{DL}}}$. □

Lemma 11.46 (Different first messages do not imply extractability)**.** *Two accepting OR mini-proof transcripts with the same statement but different first-message vectors are not sufficient to apply the special-soundness formula of Theorem 10.24.*

Proof. Special soundness explicitly assumes the same T in both transcripts. Indeed, for any public OR statement and either challenge bit, the witness-free algorithm of Definition 11.7 generates accepting transcripts. Two independent runs generally produce different first messages yet require no witness to generate. Thus the mere existence of two accepting transcripts with different first messages supplies none of the algebraic cancellation used by the extraction proof. □

Definition 11.47 (Global temporal coupling obligation)**.** The *global extraction obligation* is to build an efficient extractor for the actual adaptive interaction that:

(i) assigns a unique extracted binary value to *all* pre-θ committed field variables before sampling the real θ;

(ii) similarly extracts all required post-θ, pre-z values before the real z;

(iii) preserves a *single* consistent accepted visible transcript, including its already sampled outer challenge and authenticated openings;

(iv) keeps the real independent outer challenge laws intact in the coupled ideal run; and

(v) incurs an explicitly bounded bad-event probability efficient enough to combine with the algebraic soundness theorem of Chapter 9.

This is the stronger contract of Definition 10.52 with a fully instantiated, efficient multi-statement extractor.

Proposition 11.48 (Local extraction does not yet prove global coupling)**.** *The extraction theorem Theorem 11.42 does not by itself discharge the global obligation of Definition 11.47.*

Proof. The stand-alone theorem concerns one fixed statement, one saved initial prover state and two replayed *inner* challenge sequences. It yields a witness on the fork event but says nothing about selecting one accepted continuation that simultaneously matches all other registered bit statements, their dependent affine rows, and the later public history. The global obligation additionally requires consistent extraction of all statements, a coupling of the final real accepted transcript, and preservation of the independent outer challenges. None of these conclusions is among the hypotheses or statements proved for one isolated OR prologue. A separate multi-stage extractor is therefore required; no impossibility is claimed. □

11.7 The complete security statement and what remains conditional

Definition 11.49 (Per-proof conditional false-parity bound)**.** Let Q be an upper bound on the number of affine parity rows tested during one protocol execution. Assume every row commitment and its extracted individual bit openings are fixed before the *first inner bit challenge* of that row. For any row whose parity is false on these extracted bits, suppose the probability that its κ sequential OR mini-proofs all accept, conditional on the entire earlier visible history, is at most η_{row}. In the stand-alone resettable model with a uniform discrete-log reduction bound δ_{DL}, Theorem 11.45 allows the choice

$$\eta_{\mathrm{row}} := \min\{1,\, 2^{-\kappa} + \sqrt{\delta_{\mathrm{DL}}}\}.$$

Applying this bound uniformly over adaptive histories is a hypothesis of the composed security theorem.

Lemma 11.50 (Union bound for false accepted parity rows)**.** *Under Definition 11.49, the probability that any false parity row is accepted is at most* $\min\{1, Q\eta_{\mathrm{row}}\}$.

Proof. Order the at most Q row checks by their occurrence in the protocol. For each row, condition on its prior visible history and the already extracted bit openings. If the row is false, its conditional acceptance probability is at most η_{row} by hypothesis; if it is true, it contributes nothing to the false-row event. Averaging over histories bounds each false-row acceptance event by η_{row}. The probability union bound over at most Q rows gives $Q\eta_{\mathrm{row}}$, capped by one. □

Definition 11.51 (Global extraction error)**.** Write $\varepsilon_{\mathrm{couple}}$ for a proven failure bound of an actual multi-stage extractor satisfying Definition 11.47. Write $\varepsilon_{\mathrm{oracle}}$ for a separately proven probability of an accepted but inconsistent external polynomial-commitment or authenticated-opening relation. These quantities are *parameters* and are not assigned numerical values until the corresponding constructions and reductions are supplied.

Theorem 11.52 (Conditional soundness of the sequential-bit compiler)**.** *Assume the honest uniform outer challenges, field representation, packing consistency and fixed-value conditions of Theorem 9.37. Suppose the implemented protocol satisfies the global extraction and oracle assumptions of Definition 11.51, and each false accepted compiled parity row obeys Definition 11.49. For every false public evaluation claim,*

$$\boxed{\Pr[\mathrm{accept}] \le \min\left\{1,\, \frac{\ell+n}{2^n} + \varepsilon_{\mathrm{couple}} + Q\eta_{\mathrm{row}} + \varepsilon_{\mathrm{oracle}}\right\}.}$$

Proof. Let B_1 be failure of the global extraction coupling, B_2 the event that a false compiled parity row is accepted, and B_3 an external oracle or packing-link failure. Their probabilities are bounded by $\varepsilon_{\mathrm{couple}}$, $Q\eta_{\mathrm{row}}$, and $\varepsilon_{\mathrm{oracle}}$ by hypothesis and Lemma 11.50. Outside $B_1 \cup B_2 \cup B_3$, all registered field variables are fixed at their required outer challenge boundaries and every accepted affine constraint holds on the extracted field values. The coupled accepting run is then an accepting run of the ideal algebraic checker of Chapter 9, with the same independent uniform θ, z. Theorem 9.37 bounds the probability of accepting the false public claim by $(\ell+n)/2^n$. Add the probabilities of the three excluded events and cap by one. □

Lemma 11.53 (Outer challenge assumptions)**.** *The malicious-verifier simulation Theorem 11.27 allows arbitrary causal outer challenges θ, z. The soundness bound of Theorem 11.52 requires these outer challenges to be fresh independent uniforms.*

Proof. The simulation proof proceeds by comparing conditional transition laws at each public history. Lemmas 9.10 and 9.11 hold after fixing any causal outer challenge values, and the inner simulator matches arbitrary verifier challenge bits. By contrast, the algebraic root bounds invoked in Theorem 9.37 are Schwartz–Zippel bounds that count roots under uniform independent θ, z. Without that sampling condition their numerical probabilities do not follow. □

Theorem 11.54 (An unmasked authenticated opening defeats witness privacy)**.** *Suppose an otherwise perfectly simulated single-opening verifier view additionally publishes the entire unmasked partial vector s^*, or publishes the full transposed pad c^p together with the masked vector m. Then there exist two admissible witnesses with the same public evaluation instance whose resulting complete views are perfectly distinguishable.*

Proof. For the first disclosure rule, use the two equal-evaluation witnesses constructed in Theorem 8.52. Their partial vectors differ in one coordinate, which directly distinguishes the views with probability one. For the second rule, Theorem 7.71 gives the deterministic reconstruction $s^* = m + \mathcal{T}(c^p)$. Apply the same distinguishing test to the reconstructed partial vector. All other transcript messages can be ignored by the distinguisher. □

Proposition 11.55 (The Chapter 11 transcript result does not certify original KBFold)**.** *Theorem 11.27 does not imply an end-to-end zero-knowledge or quantum-soundness theorem for the original odd-characteristic KBFold PCS.*

Proof. The transcript modeled in Definition 11.26 contains only binary-field masked Ring Switching messages, Pedersen bit commitments and their affine proof transcripts. It does not contain KBFold's odd-characteristic folded codeword oracles, authenticated openings, or its proximity and final-closure checks. Moreover the simulator uses classical rewinding of the malicious verifier, which is not the quantum simulation model. The conclusion therefore has a strictly smaller transcript domain and different field and adversary hypotheses than the claimed PCS statement. □

Remark 11.56 (What Chapter 11 genuinely advances). The honest-sigma-verifier restriction of Chapter 10 is removed for the *concrete affine transcript* by sequential one-bit challenges and a perfect expected-time classical rewinding simulator. Local temporal extraction is proved with an exact two-run forking probability. The global extraction contract, malicious verifier access to the underlying oracle and the post-quantum KBFold composition are still independent obligations.

11.8 Complexity ledger and the next certification boundary

Definition 11.57 (Compiled proof counts)**.** Let $B = sn$ be the number of registered binary commitments for s field elements and let $Q \le nE$ count the nonempty affine-parity rows for E field-affine equations. Each bit commitment has one bitness OR statement and each nonempty row has one aggregate OR statement. Write $J := B + Q$ for this number of prescribed OR statements. Each statement is replaced by κ sequential mini-proofs.

Lemma 11.58 (Exact mini-proof count)**.** *The modified compiler performs*

$$\boxed{M = \kappa J = \kappa(B + Q)}$$

one-bit OR mini-proofs, provided none is skipped by early rejection. Its verifier sends M challenge bits, one after each OR first message, in addition to the outer field challenges.

Proof. There are B bitness statements and Q aggregate statements by Definition 11.57. Each runs exactly κ copies of Definition 11.1 under a non-rejecting execution. Thus their sum is $\kappa(B+Q)$. Each mini-proof receives exactly one bit challenge after sending its first message. □

Lemma 11.59 (Expected and bounded simulator costs)**.** *For the M mini-proofs of Lemma 11.58, the exact malicious-verifier simulator uses at most $2M$ expected mini-proof trials. A truncated simulator with*

$$R = \lambda + \lceil \log_2 M \rceil$$

uses at most MR trials and has statistical simulation error at most $2^{-\lambda}$.

Proof. Apply Lemma 11.24 to all M possible copies to obtain the expectation. Apply Theorem 11.29 with the displayed R for the strict upper bound and its error estimate. □

Proposition 11.60 (A security-efficiency tradeoff)**.** *Amplifying the inner challenge with κ sequential one-bit repetitions reduces the baseline challenge-guessing knowledge error to $2^{-\kappa}$ while increasing raw proof communication by a factor of κ. The resulting uncompressed affine compiler remains nonsuccinct.*

Proof. A deterministic prover with no two accepting challenge strings passes at most one of the 2^{κ} challenge sequences, by Lemma 11.43. Thus its baseline acceptance is at most $2^{-\kappa}$. Lemma 11.58 shows that each prescribed OR statement is transmitted κ times, using fresh first messages and responses. The size per copy remains proportional to its branch set by Lemma 10.64, so the raw communication is multiplied by κ. The direct compiler was already nonsuccinct by Proposition 10.65; sequential repetition does not add a compression step. □

Definition 11.61 (Lean certification graph)**.** The intended dependency order of Chapter 11 is:

(i) fixed-bit OR uniform hyperplane and conditional transcript distribution;

(ii) balanced malicious-verifier simulator and abort-preserving selection lemma;

(iii) sequential transcript composition and bounded retries;

(iv) challenge-tree forking, same-first-message extraction and the quantitative ε inequality;

(v) stand-alone temporal extraction and false-parity discrete-log reduction;

(vi) a separately constructed global temporal extractor for all dependent commitments;

(vii) adaptive KBFold oracle opening simulation and post-quantum soundness.

Items (vi)–(vii) are *unproved obligations*, not Lean declarations claimed to exist.

Remark 11.62 (Historical placement). The one-bit challenge rewinding strategy is classical zero-knowledge methodology [Gol01]; our concrete OR transcript uses the established Schnorr and Cramer–Damgård–Schoenmakers constructions [Sch91; CDS94]. The mathematical work here is to make the challenge range, abort conditioning, sequential composition, exact forking probability and extraction timing explicit for the kernel-coordinate affine Ring Switching interface. No claim that one-bit rewinding itself is new is made.

Remark 11.63 (Next substantive mathematical problem). To close the Chapter 10 early extraction obligation, the multistage extractor must produce *one* fixed hidden field tuple before each outer challenge, with the same accepted public history and independent outer challenge law. Only after that can the resulting malicious-verifier ZK affine proof be safely composed with KBFold's original authenticated folding-oracle transcript, and its odd- characteristic proximity theorem.

11.9 The theorem chain and exact achieved security

A single one-bit OR transcript has a perfect witness-free simulator for each *fixed* verifier challenge $e \in \{0,1\}$. Its first message is uniform on G^d, independent of the challenge and witness (Theorem 11.11 and Lemma 11.12). The balanced rewinding simulator accepts one trial with probability $1/2$, even when the verifier aborts, and perfectly simulates any classical schedule-respecting verifier in two expected trials (Theorem 11.19). Sequential composition gives perfect expected-time *malicious-verifier zero knowledge* for the full single-opening *affine commitment and proof transcript*, including adaptive outer Ring Switching challenges:

$$\boxed{\mathrm{Law}(\mathsf{View}_{\mathrm{affine}}^{P_w,V^*}) = \mathrm{Law}(\mathsf{View}_{\mathrm{affine}}^{\mathsf{Sim},V^*}).}$$

Theorem 11.27 does not cover the original KBFold oracle or its authenticated openings. A strict polynomial-time simulator with M mini-proofs and R trials per mini-proof has statistical error $\leq M2^{-R}$ (Theorem 11.29).

For knowledge extraction, two accepted *distinct* κ-bit challenge strings against one reset deterministic prover expose a common first-message fork. If its acceptance probability is ε, the two-run extractor succeeds with probability at least

$$\boxed{\max\{0, \varepsilon(\varepsilon - 2^{-\kappa})\}.}$$

Theorem 11.45 turns an extracted false parity proof into a discrete-log solver and yields $\varepsilon \leq 2^{-\kappa} + \sqrt{\delta_{\mathrm{DL}}}$ for a fixed statement with known true bit openings. Its extraction can occur *before* the protected outer challenge when the complete OR registration prologue runs before that challenge (Theorem 11.42).

The stronger *global temporal extraction coupling* for all interleaved commitments remains an open formal obligation: the conditional full false-claim bound is

$$\boxed{\Pr[\text{false claim accepted}] \leq \min\left\{1,\ \frac{\ell + n}{2^n} + \varepsilon_{\mathrm{couple}} + Q\eta_{\mathrm{row}} + \varepsilon_{\mathrm{oracle}}\right\}.}$$

This statement is conditional on the explicit coupling and authenticated-oracle hypotheses of Theorem 11.52. The direct bit proof is nonsuccinct, discrete-log based, and not post-quantum; the original odd-characteristic KBFold PCS requires a separate end-to-end adaptive zero-knowledge and proximity proof.

12 Global Temporal Extraction and Classical Soundness

The chronological extraction contract left open in Chapters 10–11 is addressed here *for the explicit sequential one-bit Schnorr OR affine-proof compiler*. We retain the same accepted main transcript while privately replaying each accepted OR statement from its saved pre-statement classical state. The resulting *bounded-replay* extractor assigns one consistent binary-coordinate registry before each protected outer challenge, or computes the forbidden Pedersen setup discrete logarithm. The construction and its proof rely on classical checkpoint access, fixed public statements, a strict epoch schedule and an independent public-coin verifier. The underlying authenticating KBFold oracle remains a separate interface for Chapter 13; our result does not silently assume its extractability or quantum soundness.

12.1 One accepted main path with protected challenge boundaries

The only new proof mechanism in this chapter is an *extract-and-restore* operation at the end of an already accepted sequential OR proof. It does not change the verifier messages on the main execution path. All extra executions run from saved classical prover states, with the *same* prover random tape, and their messages are discarded before continuing the main path. This is a stand-alone, classical interactive reduction; the external KBFold oracle is explicitly separated.

Definition 12.1 (Three outer epochs)**.** A *two-challenge execution* is divided into three ordered epochs:

(i) *Epoch 0*, before the first fresh independent outer challenge $\theta \leftarrow L^{\ell}$: the hidden witness/packed-column bit coordinates and the hidden field variables p, p_{in}, c are registered with mandatory bitness proofs; the public masked values m, y' are published and fixed. Every variable entering the first discrepancy is thus fixed before θ.

(ii) *Epoch 1*, between θ and the second fresh independent outer challenge $z \leftarrow L$: all additional hidden variables t, ρ, σ and the public trace vector U are fixed, with their mandatory bitness proofs completed.

(iii) *Epoch 2*, after z: the verifier finishes the remaining parity proofs and checks the external authenticated oracle. A parity proof may occur earlier as soon as its public coefficients are fixed.

No commitment or field variable from a later epoch may replace an already registered value used in an earlier discrepancy polynomial.

Definition 12.2 (Sequential OR blocks)**.** An *OR block* consists of one public fixed statement (C, S) and $\kappa \geq 1$ sequential one-bit OR mini-proofs of Definition 11.22, with one fresh uniform verifier bit after each first message. The prover's behavior, including later first messages in this block, may depend on earlier bits in that same block. The verifier requires all κ mini-proofs to accept before continuing. There are at most J OR blocks on any accepting main path; J is a public polynomial bound.

Definition 12.3 (Checkpoint-replay access)**.** Immediately before each OR block, a classical black-box reduction may save the prover's complete machine state and random tape. It can execute alternative sequences of inner verifier bits from that state and then *restore the exact state at the end of the already sampled main block*. The reduction does not send alternative transcripts to the main verifier. It cannot rewind a quantum prover, cannot change an outer challenge already

sampled, and cannot reset an external stateful oracle unless that capability is included in the explicit external model.

Definition 12.4 (Main and side executions)**.** At every OR block the reduction first runs one *main* execution using the usual independent uniform verifier bits. If the main block rejects, the entire main execution rejects. If it accepts, the reduction may run *side* executions from the saved pre-block checkpoint, using new independent uniform inner bits, before restoring the accepted main post-block state and proceeding. Every later main inner or outer challenge is sampled exactly as in the unmodified honest-verifier protocol.

Lemma 12.5 (Replay preserves the main-path law)**.** *For every causal randomized classical prover satisfying Definition 12.3, the public main transcript, including both outer challenges and all inner verifier bits, has exactly the same distribution with or without any number of checkpoint-replay side executions.*

Proof. Fix the prover's random tape and induct over the ordered main protocol stages. At a stage with no OR block, both executions send the same messages and sample the same distribution of verifier challenges. At an OR block, the main challenge sequence and the resulting accepting or rejecting post-block state are sampled first, just as in the unmodified protocol. Every side execution begins from the saved pre-block state, and the reduction restores the fixed sampled main post-block state before the next main stage. Thus the side runs do not alter any main message, state or verifier coin. Induction gives equality of conditional main-transcript laws for the fixed prover tape. Averaging over that tape proves the lemma. □

Lemma 12.6 (Protected outer challenges stay independent)**.** *Suppose θ is sampled by the honest verifier only after all Epoch 0 extraction attempts, and z only after all Epoch 1 extraction attempts. Conditional on the entire state of the reduction immediately before either sampling, that outer challenge has its prescribed fresh uniform law.*

Proof. The verifier samples its outer challenge from fresh independent randomness, which has not been used in either the main preceding transcript or any side replay. The reduction's pre-challenge state is a function of earlier prover states, inner bits and replay coins only. Independence of the fresh random sample from that sigma-field yields the uniform conditional law. □

Remark 12.7 (Essential scheduling rule)*.* The bitness proof for a value that enters the first challenge-dependent discrepancy must finish in Epoch 0; a bitness proof for an auxiliary residual value must finish in Epoch 1. Simply committing to a Pedersen group element before a challenge and deferring its *knowledge proof* until afterward does not meet this extractor schedule.

12.2 A quantitative extract-and-restore lemma

Put $a_\kappa := 2^{-\kappa}$. The central improvement over Chapter 11 is to *condition on the already sampled accepting main path* and search only for an *alternative* accepted inner challenge string. The main path is never replaced by the successful fork. The elementary probability bound below makes this strategy work uniformly over all adaptive pre-block histories.

Definition 12.8 (Local accepted challenge tree)**.** Fix a pre-block public history, a public OR statement (C, S), and the entire saved private state and random tape of a causal prover. For each challenge string $e \in \{0,1\}^\kappa$, run the deterministic prover from that state. Let

$$\mathcal{A} := \{e : \text{all } \kappa \text{ mini-proofs accept}\}, \quad m := |\mathcal{A}|, \quad p := m/2^\kappa.$$

For an accepted main string $e \in \mathcal{A}$, a side run is a *fork* if it accepts on some string $e' \in \mathcal{A} \setminus \{e\}$.

Lemma 12.9 (Probability of a distinct accepting side string)**.** *For every fixed accepted main string e, one independently sampled uniform side challenge string gives a fork with probability*

$$\boxed{p - a_\kappa}.$$

Proof. The accepted challenge set has m elements, exactly one of which is the fixed main string e. Each of the other $m-1$ strings has probability $2^{-\kappa}$. Thus the success probability equals $(m-1)2^{-\kappa} = p - 2^{-\kappa} = p - a_\kappa$. □

Lemma 12.10 (Exact failure probability after bounded replay)**.** *Fix the deterministic pre-block state. Run a fresh uniform main OR block and, only if it accepts, make $R \geq 1$ independent side attempts for a* distinct *accepted challenge string. The joint probability that the main block accepts but no fork is found is*

$$\boxed{p(1 - p + a_\kappa)^R}.$$

Proof. The main string is accepted with probability p. Conditioned on any accepted main string, Lemma 12.9 makes the failure probability of one side attempt $1 - p + a_\kappa$. The side challenge strings are mutually independent and also independent of the main sample. Thus all R attempts fail with probability $(1 - p + a_\kappa)^R$. This expression does not depend on the particular accepted main string. Multiplying by p gives the formula. □

Lemma 12.11 (Uniform elementary replay bound)**.** *For every admissible $p = m2^{-\kappa}$ and every integer $R \geq 1$,*

$$\boxed{p(1 - p + a_\kappa)^R \leq a_\kappa + \frac{1}{R+1}}.$$

Proof. If $m = 0$, the left side is zero. Otherwise $p \geq a_\kappa$. Put $u := p - a_\kappa \in [0, 1]$. Then

$$p(1 - p + a_\kappa)^R = (u + a_\kappa)(1 - u)^R \leq u(1 - u)^R + a_\kappa.$$

The real-valued function $f(u) = u(1-u)^R$ on $[0, 1]$ is zero at the endpoints. For $0 < u < 1$ its derivative is

$$f'(u) = (1 - u)^{R-1}(1 - (R + 1)u).$$

Hence the maximum is attained at $u = 1/(R+1)$ and equals $R^R/(R+1)^{R+1}$, which is at most $1/(R+1)$. Substitution gives the required bound. □

Theorem 12.12 (Per-block online extraction with a preserved main path)**.** *Fix a public OR statement before its first one-bit challenge and let a causal randomized classical prover be checkpoint-replayable. There is a black-box procedure that:*

(i) *runs the unmodified main OR block against κ independent uniform challenge bits;*

(ii) *if it accepts, makes at most R independent side replays and attempts the explicit special-soundness extraction of Chapter 11;*

(iii) *restores the exact main post-block state before the next protocol stage.*

It produces a valid OR opening on every successful fork, and

$$\boxed{\Pr[\textit{main accepts but extraction fails}] \leq 2^{-\kappa} + \frac{1}{R+1}}.$$

The bound holds for every pre-block history, including adaptively chosen public statements.

Proof. Condition on the complete saved prover state and tape at the start of the block. The statement is then fixed and Definition 12.8 gives the deterministic accepted challenge set. By Lemma 12.11, the probability that main acceptance is followed by R failed side attempts is at most $2^{-\kappa} + 1/(R+1)$.

If a distinct accepting side string is found, the two runs start from the identical saved state and tape. Lemma 11.33 supplies two accepting transcripts with the same OR first message and different one-bit challenges at the earliest differing index. Theorem 11.34 computes a valid permitted opening. The procedure then restores the main post-block state by Definition 12.3. The conditional bound is uniform over saved states, so averaging over the prover tape and all earlier adaptive public histories preserves it. □

Corollary 12.13 (Uniform extraction error over sequential blocks)**.** *For at most J sequential OR blocks, suppose every accepted block is followed by the extract-and-restore procedure of Theorem 12.12. The probability that the* main protocol accepts *yet at least one accepted OR block fails to yield an extracted opening is at most*

$$\boxed{\varepsilon_{\mathrm{fork}}(J, \kappa, R) := \min\left\{1, J\left(2^{-\kappa} + \frac{1}{R+1}\right)\right\}.}$$

Proof. At the start of the j-th reached block, condition on the entire history and saved prover state. Theorem 12.12 bounds the event that this block accepts but extraction fails. If the final main protocol accepts, every reached mandatory OR block accepted, so failure at any such block lies in one of these events. A union bound over at most J block positions gives the displayed bound, capped by one. □

Lemma 12.14 (Expected and worst-case replay overhead)**.** *The J-block procedure uses at most JR side block executions, and at most $J(R+1)$ total main-plus-side block executions. If a block contains κ one-bit mini-proofs, the number of executed mini-proof instances is at most $\kappa J(R+1)$.*

Proof. At each reached block the procedure makes exactly one main execution and at most R side executions. There are at most J blocks. Multiply the resulting number of blocks by the maximum of κ mini-proofs per block. □

Remark 12.15 (Why this is stronger than isolated forking). Chapter 11 obtained two accepting runs for one fixed statement. Theorem 12.12 instead *retains the actual main accepted run*, constructs a fork only as a private side computation, and restores the precise state that will receive the next outer challenge. The failure bound is unconditional over histories; no lower bound on the prover's conditional acceptance probability needs to be assumed.

12.3 A single extracted registry across all protocol epochs

We now combine the per-block extractor into one chronological procedure. The registry is an *analysis object*: it never appears in the honest public protocol, and the extractor does not open any commitment to the verifier.

Definition 12.16 (Tagged bit-commitment registry)**.** A registered hidden field element is represented by n tagged Pedersen bit commitments $(C_{x,0}, \dots, C_{x,n-1})$ and their prescribed bitness OR blocks. Each tag uniquely identifies a group element in the commitment record; subsequent affine rows must reference the *same* tag and group element, not a new value chosen under the same name. A canonical extracted entry is

$$(C_{x,i};\, b_{x,i}, r_{x,i}), \qquad C_{x,i} = g^{b_{x,i}} h^{r_{x,i}}, \quad b_{x,i} \in \{0,1\}.$$

The reconstructed field value is $x^{\mathrm{ext}} = \sum_i \beta_i b_{x,i}$.

Definition 12.17 (Global checkpoint extractor)**.** The extractor $\mathcal{E}_R$ follows exactly the *main* execution of the prescribed protocol in Definition 12.1. Whenever a mandatory OR block accepts, it immediately applies Theorem 12.12 with at most R side attempts, then restores the main state. For a successful bitness extraction it records the branch bit and blinding in the registry. For a successful aggregate-parity extraction it records a permitted integer candidate and its aggregate blinding, and verifies its consistency with the already extracted constituent bit entries. If any repeated commitment tag has a different extracted bit opening, or any aggregate opening is inconsistent with the true integer sum of its extracted constituent bits, it *records* the derived value $\log_g h$ using the reduction of Chapter 10. It does *not* terminate or change the main execution upon discovering a collision or an extraction failure: it restores the main state and continues the verifier to its natural completion. Afterward it returns the complete main transcript together with a consistent registry, a recorded setup discrete logarithm, or an extraction failure flag.

Lemma 12.18 (A group element has a unique blinding for a fixed bit)**.** *If* $\mathsf{Com}(b; r) = \mathsf{Com}(b; r')$*, then* $r = r'$ *in* $\mathbb{Z}_P$*.*

Proof. Cancel the common factor g^b. The equality becomes $h^r = h^{r'}$. Because h has prime order P, this implies $r = r' \pmod P$. □

Lemma 12.19 (Conflicting extracted bits reveal the setup logarithm)**.** *If two valid extracted openings* (b, r) *and* (b', r') *of the same tagged commitment have* $b \neq b'$*, then* $\mathcal{E}_R$ *computes* $\log_g h$*.*

Proof. The openings satisfy $g^b h^r = g^{b'} h^{r'}$. Lemma 10.11 provides the explicit modular inverse formula for $\log_g h$. □

Lemma 12.20 (False aggregate parity reveals the setup logarithm)**.** *Suppose an accepted aggregate parity block has extracted opening* (t, R_T)*, where* t *lies in the public permitted candidate set, and all of its tagged constituent bit commitments have canonical extracted openings* (b_i, r_i)*. If the parity of* $\sum_i b_i$ *differs from the public parity target, then* $\mathcal{E}_R$ *computes* $\log_g h$*.*

Proof. Let $S = \sum_i b_i$ and $R_{\mathrm{true}} = \sum_i r_i$. Lemma 10.14 gives the valid aggregate opening (S, R_{true}). The OR fork gives the second valid aggregate opening (t, R_T). Their parities differ, so $S \neq t$ as ordinary integers. The setup condition $P > |T|$ makes them distinct modulo P. Theorem 10.32 computes the forbidden logarithm. □

Lemma 12.21 (Correct parity of every row gives all field-affine equations)**.** *Suppose all registered bit openings are consistent, every prescribed affine parity row is accepted and extracted, and no aggregate row yields a setup logarithm under Lemma 12.20. Then each prescribed field-affine equation is true for the single field tuple reconstructed from the canonical bit registry.*

Proof. Every accepted parity OR block extracts an integer candidate t with the required target parity. If the true integer sum of its constituent canonical bits had the opposite parity, then Lemma 12.20 would return the setup logarithm, contrary to assumption. Thus every compiled row holds over $\mathbb{F}_2$ for the canonical bits. The exact affine-to-parity equivalence Theorem 10.4 then makes every original field-affine equation true for the uniquely reconstructed field values. □

Theorem 12.22 (Global chronological extraction)**.** *Assume the scheduling and checkpoint model of Definitions 12.1 and 12.3; every hidden bit appears under one stable commitment tag, all required bitness blocks are completed before the outer challenge that uses that bit, and all later compiled parity rows refer to those existing tags. For a main protocol with at most* J *mandatory OR blocks,* $\mathcal{E}_R$ *has the following properties:*

(i) *The main visible transcript has* exactly *the law of the unmodified honest-verifier interaction.*

(ii) *With probability at least* $1 - \varepsilon_{\text{fork}}$ *on the event of main acceptance, understood in the unconditioned joint-event sense*

$$\Pr[\text{main accepts} \ \wedge \ \text{any block extraction fails}] \leq \varepsilon_{\text{fork}},$$

every mandatory accepted block yields a valid opening.

(iii) *Outside that failure event, either* $\mathcal{E}_R$ *computes the setup discrete logarithm, or the extracted registry is consistent and every accepted compiled field-affine equation holds on one fixed field tuple.*

(iv) *The field variables registered in Epoch 0 are extracted* before *the actual sampling of* θ*; those registered in Epoch 1 are extracted* before *the actual sampling of* z*. The fresh outer challenges retain their prescribed independent uniform distributions.*

(v) *The worst-case replay count is at most* $J(R+1)$ *OR block executions.*

Here $\varepsilon_{\text{fork}} = \min\{1, J(2^{-\kappa} + 1/(R+1))\}$.

Proof. Apply the per-block procedure of Theorem 12.12 at each accepted mandatory OR block. Lemma 12.5 proves (i), because every side replay is discarded and the accepted main post-block state is restored. The union bound Corollary 12.13 proves the joint-event estimate (ii). On its complement, every accepted block has an extracted permitted opening. If two openings for the same tag disagree on their bits, Lemma 12.19 computes the setup logarithm; otherwise Lemma 12.18 makes every tagged opening unambiguous. If any extracted aggregate candidate disagrees with the bits of the same registered commitments, Lemma 12.20 computes the logarithm. If neither happens, Lemma 12.21 establishes all prescribed field-affine equations on the canonical tuple. This proves (iii).

For (iv), the protocol's Epoch 0 schedule finishes every required bitness block and its immediate side replay before the first outer sampling step. Therefore all Epoch 0 canonical entries are already fixed when θ is drawn. The identical argument applies to the Epoch 1 bitness blocks before z. Lemma 12.6 shows that the outer challenges remain independent uniforms. Finally Lemma 12.14 gives (v). □

Corollary 12.23 (No separately assumed global extractor for this affine compiler)**.** *Within the* classical checkpoint-replay model *and the explicit registration schedule of Theorem 12.22, the abstract global extraction obligation of Definition 11.47 is realized by* $\mathcal{E}_R$ *with the stated finite failure bound, up to a discrete-log-breaking event. This conclusion applies to the bit-commitment/affine-proof component, not to an unmodeled external KBFold authenticated oracle or to quantum provers.*

Proof. The five required properties are precisely the output, chronological consistency, challenge independence and same-main-path assertions of Theorem 12.22. The only exceptional events are a failed side extraction or a conflicting pair of valid openings, which yields the setup logarithm. The oracle exclusion is part of the checkpoint access model and the declared transcript domain, so it cannot be eliminated by this argument. □

Remark 12.24 (Conditioning is not being hidden). The extraction error bound controls the *joint event* that an accepted main run lacks a witness. It does not claim a negligible conditional failure probability given acceptance if the adversary's acceptance probability is itself tiny. This distinction is needed for the final soundness reduction.

12.4 Computational false-claim soundness with the constructed extractor

Definition 12.25 (External authenticated-oracle error)**.** Let $\mathcal{B}_{\text{oracle}}$ be the event that a concrete external PCS, Merkle/FRI authentication layer or packed-witness link accepts, even though its exposed field words or authenticated evaluations do not correspond to the one registered binary witness table extracted in Epoch 0. Assume a *separately proven* bound

$$\Pr[\mathcal{B}_{\text{oracle}}] \le \varepsilon_{\text{oracle}}.$$

The present chapter does not instantiate the original KBFold oracle or prove this bound. When the external oracle is replaced by an exact ideal functionality, put $\varepsilon_{\text{oracle}} = 0$.

Definition 12.26 (Forking reduction against the group setup)**.** For any fixed PPT classical cheating prover $\mathcal{P}^*$ and integer $R \ge 1$, define $\mathcal{D}_R^{\mathcal{P}^*}(g,h)$: it emulates the honest public-coin verifier, runs the global checkpoint extractor $\mathcal{E}_R$ at every accepted OR block, and outputs $\log_g h$ if a pair of inconsistent extracted openings is found. It otherwise outputs failure. Write

$$\varepsilon_{\text{DL}}(R) := \Pr[\mathcal{D}_R^{\mathcal{P}^*}(g,h) \text{ solves } \log_g h].$$

For polynomial J, κ, R and polynomial prover runtime, this is a PPT algorithm in the checkpoint-replay model.

Lemma 12.27 (A good accepting false claim is an algebraic exception)**.** *Suppose a false public evaluation claim is accepted by the concrete affine verifier. Outside the events of (i) a missing fork for an accepted OR block, (ii) an extracted opening collision revealing* $\log_g h$*, and (iii)* $\mathcal{B}_{\text{oracle}}$*, the main transcript satisfies all ideal algebraic checks of Definition 9.24 on one extracted field tuple, with the first-challenge variables fixed before* θ *and second-challenge variables fixed before* z*.*

Proof. Theorem 12.22 gives valid consistent openings for all accepted bitness and aggregate OR blocks outside the first two events. Lemma 12.21 then makes the claim, incoming, outgoing and anchored field-affine relations true on the extracted tuple. Public trace constraints are checked directly and do not involve a hidden extraction. Outside $\mathcal{B}_{\text{oracle}}$, the same registered packed bit table defines the committed witness underlying the claimed evaluation, so the claim is false for that very table. The epoch boundary assertion in Theorem 12.22 fixes the appropriate tuple coordinates before the two respective challenges, and Lemma 12.6 preserves their uniform independence. These are exactly the hypotheses of the ideal algebraic checker. □

Theorem 12.28 (Explicit finite-replay soundness reduction)**.** *Let* $\mathcal{P}^*$ *be a PPT classical prover in the checkpoint-replay model attempting a false public evaluation claim. Suppose the prescribed affine compiler and commitment timing satisfy Definitions 12.1 and 12.16, with at most* J *sequential OR blocks, each using* κ *one-bit challenges, and let the authenticated-oracle error be as in Definition 12.25. For every integer* $R \ge 1$*,*

$$\boxed{\Pr[\text{false claim accepted}] \le \min\Big\{1,\ \frac{\ell+n}{2^n} + \varepsilon_{\text{oracle}} + \varepsilon_{\text{DL}}(R) + J\Big(2^{-\kappa} + \frac{1}{R+1}\Big)\Big\}.}$$

No abstract $\varepsilon_{\text{couple}}$ *is assumed for the concrete affine compiler: its extraction error is the explicit last term.*

Proof. Run $\mathcal{D}_R^{\mathcal{P}^*}$ and focus on its unchanged main protocol transcript, which has the honest-verifier distribution by Lemma 12.5. Let F be the event that some accepted mandatory block lacks a successful fork. By Corollary 12.13, $\Pr[\text{main accepts} \cap F] \le J(2^{-\kappa} + 1/(R+1))$. Let D denote a successful output of the setup logarithm by $\mathcal{D}_R$ and let $O = \mathcal{B}_{\text{oracle}}$. On the event of main acceptance outside $F \cup D \cup O$, Lemma 12.27 places the main transcript in the ideal algebraic accepting event with an unchanged false public claim. By Theorem 9.37, the probability of that event is at most $(\ell + n)/2^n$. The union bound therefore gives

$$\Pr[\text{accept}] \le \frac{\ell + n}{2^n} + \Pr[O] + \Pr[D] + \Pr[\text{accept} \cap F].$$

Substitute $\Pr[O] \le \varepsilon_{\text{oracle}}$, $\Pr[D] = \varepsilon_{\text{DL}}(R)$ and the bound on F. Every probability is at most one, giving the displayed result. □

Corollary 12.29 (Choose replay budget from a claimed cheating advantage)**.** *Suppose a cheating prover's false-claim acceptance probability is $\varepsilon > 0$, and write*

$$\varepsilon_0 := \frac{\ell + n}{2^n} + \varepsilon_{\text{oracle}} + J2^{-\kappa}.$$

For $R = \max\{1, \lceil 4J/\varepsilon \rceil\}$, the explicit reduction satisfies

$$\boxed{\varepsilon_{\text{DL}}(R) \ge \frac{3}{4}\varepsilon - \varepsilon_0.}$$

In particular, if $\varepsilon_0 \le \varepsilon/4$, then $\varepsilon_{\text{DL}}(R) \ge \varepsilon/2$.

Proof. Rearrange Theorem 12.28 before its cap at one:

$$\varepsilon_{\text{DL}}(R) \ge \varepsilon - \varepsilon_0 - \frac{J}{R+1}.$$

The choice $R \ge 4J/\varepsilon$ implies $J/(R+1) \le \varepsilon/4$. Substitution gives the first claimed bound. Under the additional hypothesis $\varepsilon_0 \le \varepsilon/4$, it is at least $\varepsilon/2$. □

Theorem 12.30 (Classical computational soundness of the affine component)**.** *Let λ be a security parameter. Assume J, κ and the cheating prover's runtime are polynomial in λ, the setup discrete logarithm is hard for PPT algorithms, and*

$$\frac{\ell + n}{2^n}, \quad \varepsilon_{\text{oracle}}, \quad J2^{-\kappa}$$

are negligible in λ. Assume checkpoint-replay access and the exact pre-challenge registration schedule of Chapter 12. Then any PPT cheating prover has negligible probability of accepting a false evaluation claim in the concrete sequential-bit affine verification protocol.

Proof. Assume toward a contradiction that some PPT prover has a non-negligible false-claim acceptance function $\varepsilon(\lambda)$. Then there exist a polynomial $p(\lambda)$ and infinitely many security parameters such that $\varepsilon(\lambda) \ge 1/p(\lambda)$. Choose a polynomial replay budget $R(\lambda) = \max\{1, \lceil 4J(\lambda)p(\lambda) \rceil\}$. The runtime bound of Lemma 12.14 makes $\mathcal{D}_R$ PPT. For all sufficiently large parameters in the infinite subsequence, negligibility of the three error terms implies $\varepsilon_0 \le \varepsilon/4$. Corollary 12.29 then gives $\varepsilon_{\text{DL}}(R) \ge \varepsilon/2 \ge 1/(2p(\lambda))$. This is a non-negligible discrete-logarithm advantage, contradicting the assumed hardness. □

Remark 12.31 (Scope of the result). The theorem discharges the *classical global temporal extraction* obligation for the registered Pedersen/Schnorr affine component under a strict sequencing and reset model. It does *not* certify the external KBFold oracle: its error $\varepsilon_{\text{oracle}}$ is an explicit additional assumption that Chapter 13 must instantiate. Nor does it establish a quantum rewinding extractor or a quantum-hard commitment. Those belong to Chapters 13–14 within the locked plan.

12.5 Executable extractor contract, cost, and handoff to KBFold

The proof above replaces the abstract $\varepsilon_{\text{couple}}$ of Chapter 11 by an explicit bounded-replay operation on the prescribed classical OR blocks. The following pseudocode fixes which messages are visible to the *main* verifier and which are private to the reduction.

Definition 12.32 (Checkpoint extractor pseudocode)**.** For each mandatory public OR statement $Q = (C, S)$, the reduction executes:

```
saved = snapshot(prover_before_Q)
(main_bits, main_messages, main_end) =
    run_one_block_with_fresh_uniform_bits(Q)

if not verify_all(main_messages):
    reject_main_and_stop

restore(saved)
repeat at most R times:
    (side_bits, side_messages, _) =
        run_one_block_with_fresh_uniform_bits(Q)
    if verify_all(side_messages) and side_bits != main_bits:
        opening = extract_at_earliest_difference(
                    main_messages, side_messages)
        record_or_compare_with_canonical_registry(opening)
        break
restore(main_end)
continue_main_verifier()
```

The snapshot includes the private random tape and complete prover state. The side executions neither change the group commitments in the statement nor publish messages to the main verifier. The extractor invokes no irreversible external or authenticated oracle during a replayed block.

Lemma 12.33 (Every successful replay extracts from the original statement)**.** *If the extractor reaches the* `extract_at_earliest_difference` *line, the extracted opening is for the original main-path public statement* $Q = (C, S)$.

Proof. Every side run begins at the same pre-block checkpoint and receives the same fixed public statement Q. The challenge paths differ only in fresh inner verifier bits. By causality, their first messages are identical up to the earliest different challenge bit, and the verifier accepts both mini-proofs at that index. The OR special soundness theorem (Theorem 10.24) returns an opening of the unchanged group element C to one of the unchanged candidates in S. □

Lemma 12.34 (All extracted bit values are registered before use)**.** *On an accepted main run with successful extraction of every bitness block, no first-challenge field variable is defined by an*

extraction occurring after θ, and no second-challenge residual variable is defined by an extraction occurring after z.

Proof. By Definition 12.1, every mandatory bitness block for the first-challenge variables completes in Epoch 0, and all auxiliary second-challenge bitness blocks complete in Epoch 1. The pseudocode always performs its side replays immediately after the main block and *before* continuing to the next stage. Thus the registered field values are extracted before the respective outer sampling boundaries, as also quantified in Theorem 12.22. □

Proposition 12.35 (Polynomial overhead for non-negligible cheating advantage)**.** *Let the cheating advantage be at least $\varepsilon(\lambda) \geq 1/p(\lambda)$ for some polynomial p. Choosing $R = \lceil 4Jp \rceil$ makes the reduction execute at most*

$$\boxed{O(J^2 \kappa p)}$$

one-bit mini-proof instances, apart from the cost of restoring and executing the prover state.

Proof. Lemma 12.14 bounds the count by $\kappa J(R+1)$. With $R = \lceil 4Jp \rceil$, this is at most $\kappa J(4Jp+2)$, which has the displayed polynomial order. □

Definition 12.36 (Chapter 13 external interface)**.** The next chapter receives the following *fixed contract*: a binary-field affine verifier whose public messages and sequential OR proofs satisfy Theorem 12.22, with internal algebraic error $(\ell + n)/2^n$ and an explicit computational discrete-log reduction. It must add a *separately proven* authenticated polynomial-oracle interface that:

(i) binds the original committed witness/codeword to the extracted canonical bit registry before the corresponding challenge;

(ii) keeps masked opening responses and Merkle/FRI authentication data compatible with one globally simulatable verifier view;

(iii) preserves KBFold's original odd-characteristic local fold/degree and proximity checks, without transferring a binary-field characteristic-two identity to an odd-characteristic word fold.

Its total failure is recorded as $\varepsilon_{\text{oracle}}$ until a concrete theorem proves it.

Remark 12.37 (No fifth workstream). All remaining work belongs inside the locked final structure: Chapter 13 constructs the full masked KBFold prover/verifier and external oracle; Chapter 14 proves or precisely bounds its end-to-end security; Chapter 15 completes Lean and Rust certification, complexity and the final manuscript audit. No additional numbered chapter is introduced.

12.6 Theorem chain and exact error budget

For a sequential OR block with κ honest one-bit challenges, let p be the accepted challenge-string fraction in a fixed saved classical prover state. If the main block accepts, R independent side attempts fail to find an accepted *different* challenge string with joint probability

$$\boxed{p(1 - p + 2^{-\kappa})^R \leq 2^{-\kappa} + \frac{1}{R+1}.}$$

Theorem 12.22 runs this extraction at every accepted block and restores the original main post-block state. Its probability of an accepted main execution with some missing opening is at most

$$\boxed{\varepsilon_{\text{fork}} \leq J\left(2^{-\kappa} + \frac{1}{R+1}\right).}$$

The canonical extracted bit values are fixed before θ or z according to their epoch, and all accepted parity checks agree with one extracted field tuple unless their conflicting openings compute $\log_g h$. The fresh outer challenges retain their independent uniform distributions.

Consequently the explicit classical false-claim soundness reduction is

$$\boxed{\varepsilon_{\text{false}} \le \min\Bigg\{1,\ \frac{\ell+n}{2^n} + \varepsilon_{\text{oracle}} + \varepsilon_{\text{DL}}(R) + J\left(2^{-\kappa} + \frac{1}{R+1}\right)\Bigg\}.}$$

For any non-negligible false-claim probability, choosing polynomial R inversely proportional to that probability gives a non-negligible discrete-log solver (Corollary 12.29). Thus, when the field root error, the external oracle error and $J2^{-\kappa}$ are negligible, the *classical affine component is computationally sound* (Theorem 12.30). This is the quantitative replacement for the formerly unspecified global temporal extraction error.

The remaining cryptographic boundary is *one concrete authenticated KBFold oracle* with a proven binding and joint zero-knowledge simulation contract. That is the whole scope of Chapter 13; the locked manuscript concludes with Chapter 14 (security) and Chapter 15 (certification).

13 Complete Masked KBFold Protocol and Authenticated Oracles

This chapter works in the *original odd-characteristic field* of KBFold [Mkh26]. It assembles the already proved normalized fold dictionary and degree-aware high-degree masking (Chapter 5), the genuine KBFold Sumcheck transcript (Chapter 3), and an authenticated *hiding* oracle interface. The public Merkle leaves are fresh Pedersen commitments to field symbols, rather than the field symbols themselves. Sumcheck round coefficients and final folded values remain hidden behind prescribed relation proofs; only the authorized evaluation $v = \widetilde{f}(z)$ is public.

The protocol defined here uses a *full-relation* verification mode: it checks the committed polynomial words and source-to-fold identities in a polynomial-size NP circuit. This is a nonsuccinct reference protocol with fully specified algebraic predicates. A future *sparse* optimization would retain KBFold's local-query architecture, but requires a new adversarial proximity proof for the enlarged degree budgets. The relation proof engine is an explicit cryptographic interface with circuit-level statements; its real-world knowledge extraction and adaptive ZK simulation are the final security obligations of Chapter 14. No other research chapter is introduced.

13.1 Parameters, hiding leaves, and authenticated oracle semantics

Throughout this chapter $\mathbb{F}$ is a finite field of *odd* characteristic; $N = 2^m$, $M = 2^{m+s}$, $s \geq 1$, and Ω_j is the subgroup of size $M_j = M/2^j$ from Section 5.5. Fix $1 \leq B \leq N$, $B_j = \lceil B/2^j \rceil$, $N_j = 2^{m-j}$ and $D_j = N_j + B_j \leq M_j$. No characteristic-two trace or Artin–Schreier identity is invoked in this chapter.

Definition 13.1 (Canonical field encoding)**.** Fix a public injective map $\mathsf{enc}_{\mathbb{F}} : \mathbb{F} \hookrightarrow \{0,1\}^b$ with $b = \lceil \log_2 |\mathbb{F}| \rceil$. The image of this map is the set of *valid* encodings; a relation circuit must check validity, rather than interpreting every b-bit string as a field element. All field arithmetic in the relation is arithmetic of $\mathbb{F}$, not arithmetic of the commitment group.

Definition 13.2 (Hiding commitment to a field symbol)**.** Let $G = \langle g \rangle = \langle h \rangle$ be an odd-prime-order group as in Definition 10.8. For $a \in \mathbb{F}$ with encoding $(a_1, \ldots, a_b)$, sample independent $r_1, \ldots, r_b \leftarrow \mathbb{Z}_P$ and form

$$\mathsf{VC}(a; \mathbf{r}) := (g^{a_1} h^{r_1}, \ldots, g^{a_b} h^{r_b}) \in G^b.$$

Each component is a Pedersen bit commitment. The map VC is *not* assumed to be additive over $\mathbb{F}$.

Lemma 13.3 (Perfect uniformity of a committed field symbol)**.** *For every fixed $a \in \mathbb{F}$ and fresh independent uniform $\mathbf{r}$, the public vector $\mathsf{VC}(a; \mathbf{r})$ is uniform on G^b. For any finite collection of possibly correlated field symbols, their independently blinded commitment vectors are jointly uniform on the corresponding power of G, independently of all symbol values.*

Proof. Apply Lemma 10.10 to each of the b Pedersen bit commitments. Their blindings are independent, so their joint law is uniform on G^b, for every fixed encoding. Conditioning on any fixed collection of field symbols, the entire collection of commitments is thus a product uniform law independent of those symbols. Averaging over any distribution on the symbols preserves the same product law. □

Definition 13.4 (Domain-separated Merkle trees)**.** Fix a deterministic hash $H : \{0,1\}^* \to \{0,1\}^{\lambda_H}$ and a canonical injective serialization ser for group elements, indices and domain identifiers. For each level j and ordered index $x \in \Omega_j$, put

$$L_{j,x} := H(\mathsf{ser}(\texttt{leaf}, j, x, \mathsf{VC}(Q_j(x); \mathbf{r}_{j,x}))).$$

Internal binary nodes hash $\mathsf{ser}(\mathtt{node}, j, \text{left}, \text{right})$. Pad non-power-of-two levels with a publicly fixed convention if needed. Write root_j for the resulting root and $\mathsf{Path}_{j,x}$ for the leaf's sibling hashes and orientation bits. A *public opening shell* consists of the index x, the group commitment vector $\mathsf{VC}(Q_j(x); \mathbf{r}_{j,x})$ and its Merkle authentication path. It does *not* reveal $Q_j(x)$ or $\mathbf{r}_{j,x}$.

Lemma 13.5 (Perfect privacy of the authenticated shell)**.** *Fix any finite collection of field words $w_j : \Omega_j \to \mathbb{F}$. Build all the Merkle trees with independent uniform Pedersen blindings at every (j, x). Then the joint distribution of all roots, all commitment vectors and* all *their Merkle paths is independent of the words $(w_j)_j$. The result holds for any deterministic hash H.*

Proof. By Lemma 13.3, the complete public array of bit-commitment group elements is a product uniform distribution independent of the words. All leaf hashes, internal hashes, roots and paths are deterministic functions of that public array and fixed indices. Applying the same deterministic function to equal input distributions yields equal output distributions. □

Definition 13.6 (Authenticated shell opening)**.** For a public root root_j, the authenticated shell at (j, x) reveals a *commitment vector* $\mathbf{C}_{j,x}$ and its Merkle path, but not its field-symbol opening. The verifier recomputes the leaf and root and checks equality with root_j. Any assertion about the hidden symbol $a_{j,x} \in \mathbb{F}$ and the blinding $\mathbf{r}_{j,x}$ is part of a separate zero-knowledge *relation proof*, not of the public opening shell.

Lemma 13.7 (Merkle opening conflict implies a hash collision)**.** *Fix one root and index, and assume canonical injective serialization and proper domain separation. If two valid authentication shells for that root and index contain distinct commitment vectors or disagree at a node of the authenticated path, then these shells yield two distinct inputs with the same hash output, unless their entire authenticated leaf-to-root sequences are identical.*

Proof. Compare the two purported authenticated chains from the root downward. They share the same terminal root hash. If the underlying child encodings at the highest differing node are different, but their parent hashes agree, the two distinct serialized parent inputs collide under H. If no internal node differs, then at the leaf the distinct commitment vectors give distinct injectively serialized leaf inputs with the same leaf hash, again a collision. Domain separation prevents a leaf encoding from coinciding with a node encoding. □

Remark 13.8 (Exact authentication and privacy distinction)*.* Collision resistance bounds malicious changes to authenticated commitment vectors. Discrete-log binding bounds changes to their hidden bit values. Neither property makes a *raw* unmasked field-symbol opening private. Accordingly the interface above never sends the field symbol itself; a zero-knowledge relation proof must enforce its opening and all arithmetic constraints.

13.2 A masked sumcheck with hidden offsets

Let $f : I_m \to \mathbb{F}$ be the private witness, $z \in \mathbb{F}^m$ the public evaluation point, and $v = \widetilde{f}(z)$ the authorized public value. Put $\chi_0(a) = \mathrm{eq}_m(a, z)$ and

$$\sigma_0 := \sum_{a \in I_m} f(a)\chi_0(a) = v.$$

For each round $1 \le j \le m$ use the unmasked degree-two polynomial $s_j(T)$ from (3.7). Its challenge $r_{j-1} \in \mathbb{F}$ is sampled only after the round-message commitment has been bound.

Definition 13.9 (True sumcheck states)**.** Starting with $(f_0, \chi_0) = (f, \chi_0)$, put

$$f_j := \mathsf{Res}_{r_{j-1}} f_{j-1}, \qquad \chi_j := \mathsf{Res}_{r_{j-1}} \chi_{j-1}, \qquad \sigma_j := \sum_{a \in I_{m-j}} f_j(a)\chi_j(a).$$

The prover computes $s_j(T)$ from f_{j-1}, χ_{j-1} by Definition 3.68.

Lemma 13.10 (Exact sumcheck transition)**.** *For every honest round $1 \le j \le m$,*

$$\boxed{s_j(0) + s_j(1) = \sigma_{j-1}, \qquad s_j(r_{j-1}) = \sigma_j.}$$

Proof. The first equality is Lemma 3.69. For the second, each summand of $s_j(r_{j-1})$ is the product of the two affine restrictions $\mathsf{Res}_{r_{j-1}}(f_{j-1})(b)$ and $\mathsf{Res}_{r_{j-1}}(\chi_{j-1})(b)$. By Definition 13.9 these are $f_j(b)$ and $\chi_j(b)$. Summing their products over $b \in I_{m-j}$ gives σ_j. □

Definition 13.11 (Hidden-offset Sumcheck)**.** Initialize $b_0 := 0$. At the beginning of round j, sample a fresh independent $c_j \leftarrow \mathbb{F}$ and define

$$\begin{aligned} H_j(T) &:= \frac{b_{j-1}}{2} + c_j(2T - 1), \\ \mu_j(T) &:= s_j(T) + H_j(T), \\ b_j &:= H_j(r_{j-1}) = \frac{b_{j-1}}{2} + c_j(2r_{j-1} - 1). \end{aligned}$$

The prover sends *only* a fresh, perfectly hiding commitment $\mathbf{C}_{\mu_j}$ to the three coefficients of the degree-≤ 2 polynomial μ_j. The coefficients, c_j, b_j, and s_j remain private. The public challenge r_{j-1} is chosen after $\mathbf{C}_{\mu_j}$ is fixed.

Lemma 13.12 (Endpoint carry for the round mask)**.** *For every j,*

$$\boxed{H_j(0) + H_j(1) = b_{j-1}.}$$

Proof. From the defining formula, $H_j(0) = b_{j-1}/2 - c_j$ and $H_j(1) = b_{j-1}/2 + c_j$. Their sum is b_{j-1}. The division by two is valid because $\mathbb{F}$ has odd characteristic. □

Theorem 13.13 (Exact masked sumcheck invariant)**.** *For every $1 \le j \le m$, the honest hidden round polynomial satisfies*

$$\boxed{\begin{aligned} \mu_j(0) + \mu_j(1) &= \sigma_{j-1} + b_{j-1}, \\ \mu_j(r_{j-1}) &= \sigma_j + b_j, \end{aligned}}$$

and $\deg \mu_j \le 2$. Thus the publicly authorized initial claim is $\sigma_0 + b_0 = v$ and the private running claim after round j is $\sigma_j + b_j$.

Proof. Lemma 13.10 gives the two unmasked sumcheck equalities. Lemma 13.12 gives the mask's endpoint sum, and Definition 13.11 gives its evaluation at the fresh challenge. Adding the respective equalities proves both displayed identities. The degree bound holds because s_j has degree at most two by Lemma 3.70 and H_j has degree at most one. The starting relation follows from $\sigma_0 = v$ and $b_0 = 0$. □

Lemma 13.14 (Public evaluation weight at the last round)**.** *At the end of all m rounds, the zero-variable weight is the public scalar*

$$\boxed{\chi_m = \prod_{i=0}^{m-1} ((1-r_i)(1-z_i) + r_i z_i).}$$

Moreover $\sigma_m = \widetilde{f}(r)\chi_m$.

Proof. The function χ_0 is the Boolean equality table $b \mapsto \mathrm{eq}_m(b, z)$. Successive affine restrictions at $r_0, \ldots, r_{m-1}$ produce the multilinear extension $\widetilde{\chi_0}(r)$ by Lemma 5.7. Since the equality table is the product of coordinate-wise Boolean weights, its multilinear extension is the displayed product. The final table f_m has one entry, equal to $\widetilde{f}(r)$ by Corollary 5.8. Its product with the single entry χ_m is σ_m by definition. □

Corollary 13.15 (Masked sumcheck to linear fold closure)**.** *Let Q_m be the honestly masked fold polynomial of Theorem 5.37, and put $a := [X^0]Q_m$. Then*

$$\boxed{\mu_m(r_{m-1}) = \chi_m a + b_m.}$$

Proof. By Theorem 13.13, $\mu_m(r_{m-1}) = \sigma_m + b_m$. Lemma 13.14 makes $\sigma_m = \widetilde{f}(r)\chi_m$. The constant-term closure identity Corollary 5.38 gives $a = \widetilde{f}(r)$. Substitution proves the equality. □

Lemma 13.16 (Publicly revealing the masked round coefficients is unsafe)**.** *The mask H_j does not alter the quadratic coefficient of s_j. In particular,*

$$[T^2]\mu_j = [T^2]s_j.$$

Therefore the prescribed protocol must not publish the entire μ_j coefficient vector and infer privacy merely from the uniformity of c_j.

Proof. By Definition 13.11, H_j has degree at most one. Consequently its T^2 coefficient is zero. Taking the coefficient of T^2 in $\mu_j = s_j + H_j$ proves the displayed identity. A public quadratic coefficient would hence reveal the unmasked quadratic coefficient exactly. □

Remark 13.17 (Relation-proof requirement). The verifier must certify the endpoint sum, the masked challenge transition, the updated private offset b_j, and the final equation of Corollary 13.15 *without revealing* their secret values. A generic zero-knowledge proof of the explicit finite-field relations will perform these checks; ordinary raw sumcheck message verification cannot.

13.3 The masked fold chain and confidential local checks

Definition 13.18 (Honest masked fold-oracle chain)**.** Before the evaluation point z is known, sample $R \xleftarrow{\$} \mathbb{F}[X]_{<B}$ and set

$$Q_0 = U_f^{(m)} + X^N R.$$

After the sumcheck round challenge r_j, define $Q_{j+1} = \mathsf{pfold}_{r_j}(Q_j)$ for $0 \le j < m$. At each level set

$$w_j(x) := Q_j(x) \quad (x \in \Omega_j),$$

and commit to *every* $w_j(x)$ with fresh independent field-symbol commitment $\mathsf{VC}(w_j(x); \mathbf{r}_{j,x})$. The resulting Merkle root root_j is published *before* that level's local authentication-query indices are selected.

Lemma 13.19 (Exact degree and word-fold invariants)**.** *For an honest masked chain,*

$$\deg Q_j < D_j, \qquad w_{j+1} = \mathsf{WFold}_{r_j,j}(w_j)$$

for every $0 \le j < m$*. At the final level,* $Q_m = a + \rho X$ *with* $a = \widetilde{f}(r)$*.*

Proof. The degree and codeword statements are respectively Theorem 5.48 and Corollary 5.45. The final linear form and its constant coefficient follow from Theorem 5.37. □

Definition 13.20 (Confidential authenticated fold query)**.** For a query (j, y) with $0 \le j < m$ and $y \in \Omega_{j+1}$, choose either public square root $x \in \Omega_j$ satisfying $x^2 = y$. The prover gives public Merkle authentication shells for (j, x), $(j, -x)$ and $(j + 1, y)$. It then proves in zero knowledge that the three authenticated commitment vectors possess *valid field-symbol openings* (a_+, a_-, b) satisfying

$$\boxed{b = (2r_j - 1)\frac{a_+ + a_-}{2} + (1 - r_j)\frac{a_+ - a_-}{2x}.}$$

The proof uses the same hidden openings as the corresponding registered word, not independently chosen values for each query. No a_+, a_-, b or commitment blindings are disclosed.

Lemma 13.21 (The query equation is independent of square-root choice)**.** *Replacing* x *by* $-x$ *and interchanging the two level-*j *openings leaves the equation of Definition 13.20 unchanged.*

Proof. The sum $a_+ + a_-$ is unchanged. The difference $a_+ - a_-$ and the denominator $2x$ both change sign, so their quotient is unchanged. The coefficients $(2r_j - 1)$ and $(1 - r_j)$ do not depend on the root. □

Lemma 13.22 (Honest confidential fold queries pass)**.** *Every honest authenticated fold query has valid committed field openings satisfying its hidden fold equation.*

Proof. By Definition 13.18, the three openings are the true values $Q_j(x), Q_j(-x), Q_{j+1}(y)$. The word-fold identity of Lemma 13.19 is exactly the query equation at $y = x^2$. The blinding values are those generated for the registered leaves. Therefore all queried statements have valid witnesses. □

Definition 13.23 (Confidential final closure)**.** Let $Q_m(X) = a + \rho X$ be the last hidden polynomial. The public verifier requests authenticated commitment shells at two publicly fixed, distinct points $x_0, x_1 \in \Omega_m$. The prover supplies a zero-knowledge relation proof that:

(i) the two hidden authenticated field values u_0, u_1 lie on a common degree-< 2 polynomial;

(ii) the constant a of that line satisfies $x_1 u_0 - x_0 u_1 = (x_1 - x_0)a$;

(iii) a agrees with the hidden sumcheck terminal relation $\mu_m(r_{m-1}) = \chi_m a + b_m$.

The proof binds a to the same final degree-< 2 word as the registered final root.

Lemma 13.24 (Honest final closure is valid)**.** *The final closure relation of Definition 13.23 has a witness in every honest execution, provided* $|\Omega_m| \ge 2$*.*

Proof. The group domain has $M_m = 2^s \ge 2$ elements by Lemma 5.40, so distinct x_0, x_1 exist. By Lemma 13.19, $Q_m = a + \rho X$. Its two evaluations lie on this line and Lemma 5.57 gives the required constant coefficient equation. Corollary 13.15 gives the final hidden sumcheck equation, proving every item. □

Lemma 13.25 (Degree-certified final word is uniquely determined by two values)**.** *If $P, Q \in \mathbb{F}[X]_{<2}$ agree at two distinct points of $\mathbb{F}$, then $P = Q$.*

Proof. The difference $P - Q$ has degree at most one and two distinct roots. The univariate root bound Lemma 2.7 forces $P - Q = 0$. □

Remark 13.26 (The proximity gap)*.* A zero-knowledge proof of the *entire* degree-$< D_j$ word and each fold relation is a valid but nonsuccinct baseline. Replacing that full relation by sparse local checks requires a proximity theorem for *arbitrary adversarial words* in the enlarged degree budgets D_j, not merely a root-count lemma for two known polynomials. Chapter 14 keeps those two verification modes separate.

13.4 The complete interactive prover and verifier

Definition 13.27 (Explicit relation-proof interface)**.** Fix an interactive zero-knowledge *proof-of-relation* engine Π_{rel} for polynomial-size arithmetic and Boolean circuits over $\mathbb{F}$, including: valid field-symbol encodings, Pedersen bit openings, canonical Merkle-tree reconstruction, polynomial arithmetic and equality, and all public constraints specified below. Its public input is a tuple of public commitments, roots, points and claimed scalars. Its private witness is the collection of field values, blinding scalars, word evaluations and paths required to evaluate the circuit. The verifier checks the *entire* relation without receiving any private circuit input.

The *implemented relation predicate* in this chapter is Π_{full}: it certifies the entire committed word at every level, its degree bound, source relation and fold identity. This defines one nonsuccinct reference protocol. A proposed *sparse optimization* Π_{sparse} would replace these complete-word proofs with authenticated local statements and a separately proved proximity and initial-word binding theorem. It is a target of Chapter 14, not an alternative instantiated verifier in the present chapter. The original KBFold proximity analysis cannot simply be reused with the enlarged degree budgets.

Definition 13.28 (Public commitment registry)**.** For each named hidden field variable, keep a single public vector of Pedersen bit commitments. An invocation of Π_{rel} must refer to that same vector whenever it uses the variable. At preprocessing the registered variables include all entries $f(a)$, all coefficients of R, and every symbol of the level-zero masked word. At each Sumcheck round the registered variables include the three coefficients of μ_j, the round mask c_j and, after r_{j-1}, the offset b_j and every newly committed level-j oracle symbol. The equality of a named field variable across two circuit invocations is checked through its fixed public commitment vector.

Definition 13.29 (Complete preprocessing relation)**.** The stage-zero public statement consists of the initial Merkle root root_0, the public source and mask commitment vectors $\mathbf{C}_f, \mathbf{C}_R$, and parameters. The private witness of $\mathcal{R}_{\text{pre}}$ consists of (f, R), every related blinding vector and the full level-zero Merkle tree. The relation checks:

(i) every source/mask commitment opens to the correctly encoded field value, with $\deg R < B$;

(ii) $Q_0 = U_f^{(m)} + X^N R$;

(iii) every level-zero leaf commitment opens to $Q_0(x)$ and all its hashes reconstruct the announced root_0;

(iv) $\deg Q_0 < D_0$.

The preprocessing proof is a *knowledge-binding registration proof* completed before z is chosen. Its extraction property is a named cryptographic interface contract of Chapter 14.

Definition 13.30 (Round relations)**.** For round $1 \le j \le m$, the *pre-challenge relation* $\mathcal{R}_{j,\mathrm{pre}}$, proved after commitment of μ_j but before sampling r_{j-1}, binds the hidden c_j, the three coefficients of μ_j, and earlier offset b_{j-1} to

$$\mu_j(T) = s_j(T) + b_{j-1}/2 + c_j(2T-1),$$

where s_j is reconstructed from the same committed source table f and already fixed earlier challenges.

The *post-challenge relation* $\mathcal{R}_{j,\mathrm{post}}$, proved after r_{j-1} but before any level-j query, checks

$$\begin{aligned} b_j &= b_{j-1}/2 + c_j(2r_{j-1}-1), \\ \mu_j(0) + \mu_j(1) &= \sigma_{j-1} + b_{j-1}, \\ \mu_j(r_{j-1}) &= \sigma_j + b_j, \\ Q_j &= \mathsf{pfold}_{r_{j-1}}(Q_{j-1}), \\ \deg Q_j &< D_j, \end{aligned}$$

and, in Π_{full} mode, checks all openings of the level-j Merkle leaf commitments against the coefficients of Q_j. The public challenge is not allowed to modify any earlier commitment or named hidden value.

Definition 13.31 (Full protocol $\Pi^{\mathrm{hid}}_{\mathrm{KBFold}}$)**.** The interactive opening protocol is:

(i) **Before evaluation:** the prover samples R and fresh bit-commitment blindings, builds Q_0, and publishes $(\mathbf{C}_f, \mathbf{C}_R, \mathsf{root}_0)$. It proves $\mathcal{R}_{\mathrm{pre}}$. The verifier verifies this proof. No evaluation point z has yet been chosen.

(ii) **Evaluation claim:** after preprocessing, the verifier chooses $z \in \mathbb{F}^m$, and the prover publishes a claimed scalar v. An honest prover sends $v = \widetilde{f}(z)$. The prover initializes χ_0, the claimed running sum $\sigma_0 = v$, and $b_0 = 0$. The relation $v = \sum_a f(a)\chi_0(a)$ against the *preprocessing-bound* source table is included in the first round's proof. A cheating prover may choose a false v; the verifier must still enforce the relation.

(iii) **For $j = 1, \ldots, m$:** the prover computes s_j and samples fresh c_j, commits to the three coefficients of μ_j and c_j, and proves $\mathcal{R}_{j,\mathrm{pre}}$. Only then the honest verifier samples the fresh fold/Sumcheck challenge $r_{j-1} \in \mathbb{F}$. The prover computes b_j, Q_j, publishes their fresh hiding commitments and root_j, and proves $\mathcal{R}_{j,\mathrm{post}}$.

(iv) **Authenticated queries:** after root_j is bound, the verifier may choose up to q_j adaptive indices $y \in \Omega_j$ for the previous fold $j-1$. For each it receives the three shells of Definition 13.20 and a zero-knowledge proof of their hidden fold equation and registered leaf openings. It checks every Merkle path and proof. No field-symbol value is publicly revealed.

(v) **Final closure:** after all m rounds, the verifier requests two distinct final-level authenticated shells and a proof of the degree-< 2 final relation and the terminal Sumcheck link of Definition 13.23. It accepts if and only if all public Merkle checks and all required relation proofs have accepted.

The full verifier view records every commitment vector and root, all inner proof messages, z, v, r, adaptive query indices, all authenticated shells and all final proof messages.

Lemma 13.32 (All protocol relations are finite arithmetic circuits)**.** *For fixed finite field representation, m, B, M and a bounded number of queries, each relation in Definitions 13.29, 13.30, 13.20 and 13.23 is decidable by an explicit finite Boolean/arithmetic circuit of size polynomial in the expanded input and committed-word lengths and the hash-circuit complexity.*

Proof. A field element has a finite canonical b-bit encoding. Every field sum, product, inverse of a fixed nonzero public scalar and basis expansion has a finite arithmetic circuit over the fixed representation. The polynomials $U_f, Q_j, s_j, H_j, \mu_j$ have publicly bounded degrees and are computed by finitely many sums and products with indices bounded by N, B, M. A Merkle leaf hash and its internal hash circuits are finite by assumption on the efficiently computable hash H; the bounded Merkle reconstruction is a composition of such circuits. Group scalar exponentiation to check a Pedersen commitment is computable by repeated squaring in $O(\log P)$ group multiplications. The tests in the displayed definitions are equality comparisons of these circuit outputs. Combining finitely many polynomial-size circuits gives the asserted finite relation circuit. □

Theorem 13.33 (Perfect algebraic completeness of the full protocol)**.** *For every f, z, $v = \widetilde{f}(z)$, uniform mask R and honest prover random tape, every relation statement submitted by the honest prover of Definition 13.31 has a valid hidden witness. Consequently the complete protocol accepts with probability one whenever Π_{rel} has perfect completeness and the honest verifier follows the stated authenticated-query schedule.*

Proof. The initial relation is satisfied by construction of Q_0 and the Merkle tree. Lemma 13.10 and Theorem 13.13 establish every Sumcheck pre- and post-challenge relation, including the initial claim. Lemma 13.19 proves the fold, degree and oracle-word equalities at every level. Lemma 13.22 gives valid authenticated hidden witnesses for every queried fold triple. Lemma 13.24 gives the terminal witness. Every Pedersen commitment opens using its actual honest blinding and every Merkle path reconstructs the announced root. Thus all circuit relations are true; perfect completeness of the relation engine and deterministic correctness of the hash checks imply acceptance with probability one. □

Remark 13.34 (Two verification modes). The defined protocol runs the full-relation circuit checks over all committed word symbols and degree budgets. This is a nonsuccinct finite reference protocol. Replacing those checks with sparse KBFold proximity queries requires new source binding and proximity theorems for the degree budgets D_j; that optimization belongs to Chapter 14.

13.5 Online verifier view and the exact simulation boundary

Definition 13.35 (Public authenticated shell transcript)**.** The *shell projection* of a full verifier view consists of the source/mask/round/fold bit commitments, all Merkle roots, z, v, the outer challenge vector r, selected query indices and all publicly authenticated commitment vectors and paths. It omits all Π_{rel} proof messages and *never* contains unblinded field-symbol values, masked round coefficients or mask offsets.

Lemma 13.36 (Hiding roots can be sampled without any witness)**.** *For each prescribed commitment stage, the conditional public distribution of its fresh commitment vectors and the Merkle roots computed from them is produced exactly by sampling uniform vectors in G^b at all new leaf positions and hashing the resulting arrays. The sampler does not need to know f, R, Q_j, s_j, μ_j or any committed secret.*

Proof. Condition on any fixed hidden field values that the honest prover intends to commit at this stage. By Lemma 13.3, the new independently blinded bit-commitment vectors are product-uniform on G^b, independently of those values. The Merkle roots are deterministic functions of these vectors, and the remaining named field-variable commitments use the same independent blinding argument. The conditional law therefore equals the output of the stated witness-free sampler. Averaging over the hidden values leaves this law unchanged. □

Lemma 13.37 (Authenticated shell replies are deterministic after root construction)**.** *After the simulator has sampled and stored an entire leaf commitment array and built its Merkle tree, every public authentication shell for a subsequently queried index is computable without a hidden field opening. Repeated queries to the same indexed leaf obtain the same commitment vector and path.*

Proof. The shell is defined only by the public commitment vector at that index and sibling hashes on the unique canonical tree path. Both are determined by the stored array and the public hash function. Reading the stored elements again reproduces the same values for a repeated index. No opening bit or Pedersen blinding is required. □

Theorem 13.38 (Perfect online privacy of the public shell)**.** *Let a causal verifier choose its evaluation point after the preprocessing shell and later choose its outer challenges and authentication-query indices as functions only of the previous shell messages, its own random tape and the authorized public evaluation value v when available. For any fixed witness f, the public-shell process of the honest prover is* exactly *simulatable by a witness-free online algorithm with access only to the authorized evaluation functionality $\mathcal{F}_{\mathrm{eval}}^{f}$. The simulator samples its first commitment shell* before *the verifier chooses z, and answers all later queries consistently.*

Proof. The simulator first draws the public commitments and initial Merkle tree with Lemma 13.36; their law is independent of f. It forwards the resulting shell to the verifier, which chooses z as a causal function of the same visible history in the real and simulated processes. The simulator then obtains the authorized scalar v from $\mathcal{F}_{\mathrm{eval}}^{f}(z)$. At every later registration or word-commitment stage, Lemma 13.36 gives the same conditional distribution of fresh public group commitments and roots in both processes, independent of the new private values. At every authenticated query stage, Lemma 13.37 constructs the same deterministic reply from the already stored public array. The verifier's next causal choice therefore has the same conditional law in both experiments. Induction over the finite stage and query sequence proves equality of the joint online shell views, including the verifier's random tape. The theorem does not include relation-proof messages, which have been explicitly excluded from the shell view. □

Lemma 13.39 (Any group commitment is mathematically openable to any fixed bit)**.** *Let $C \in G$ and $b \in \{0,1\}$. There exists a unique $r \in \mathbb{Z}_P$ such that $C = g^b h^r$. The assertion is an existence statement, not an efficient opening algorithm.*

Proof. The element Cg^{-b} belongs to the cyclic group generated by h. Hence it equals h^r for some unique $r \in \mathbb{Z}_P$. Rearranging gives the required commitment equation. Finding r from public (C, g, h) may require solving a discrete logarithm. □

Corollary 13.40 (Simulated public trees admit algebraically consistent hidden interpretations)**.** *Fix any arrays of public commitment vectors with valid deterministic Merkle roots, any public point $z \in \mathbb{F}^m$, and any authorized scalar $v \in \mathbb{F}$. There exists a* mathematical *hidden assignment satisfying all honest algebraic relationships of Definitions 13.29 and 13.30, with a table f for which $\widetilde{f}(z) = v$. The assignment need not be efficiently computable from the public arrays.*

Proof. Choose the constant table $f(a) = v$ for every $a \in I_m$, and choose any $R \in \mathbb{F}[X]_{<B}$, for example $R = 0$. Define Q_j, s_j, μ_j, c_j, b_j as in the honest protocol, using the already selected public challenges and any fixed choices of the round masks. Their algebraic relations hold by Theorems 5.33 and 13.13. For each existing public bit commitment vector, apply Lemma 13.39 coordinatewise to obtain some blinding exponents that open it to the required valid field-symbol encoding. Use the provided deterministic Merkle trees as the hash witnesses. The hidden interpretations therefore satisfy all specified relations. The table's multilinear extension is constant v, so its authorized evaluation is v. No algorithm for computing the potentially hard opening exponents has been supplied. □

Definition 13.41 (Full-proof simulation contract)**.** A concrete Π_{rel} satisfies the *online sequential relation-simulation contract* if, for every allowed classical malicious verifier and query budget, its pre-evaluation knowledge-binding proof and all later adaptively scheduled relation proofs admit one causal witness-independent simulator, with a bound $\varepsilon_{\mathrm{proofZK}}$ on the statistical distance or computational distinguishing advantage of their *joint* messages, conditioned on the entire previous visible shell and proof history. The simulator must operate on the true public statements built from simulated shells, including the commitments registered before z, and must not require efficient knowledge of the hidden openings in Corollary 13.40. This is a required property of the chosen proof engine; it is not inferred from Pedersen hiding alone.

Theorem 13.42 (Conditional complete online-view simulation)**.** *Suppose the relation-proof engine satisfies Definition 13.41 and preserves the exact causal protocol schedule. Then the complete single-opening verifier view, including all public roots, authenticated shells and proof transcripts, has an online simulator whose advantage is at most $\varepsilon_{\mathrm{proofZK}}$, in the corresponding statistical or computational model. The simulator publishes its initial commitment before learning z and uses only the authorized ideal evaluation value v afterward.*

Proof. First use the witness-free public-shell simulator from Theorem 13.38. It samples and retains the commitment trees and can answer every future adaptive shell query. By Corollary 13.40, the resulting adaptively constructed public proof statements have consistent mathematical witnesses. At each scheduled proof stage, invoke the online simulator guaranteed by Definition 13.41, using the current public statement and verifier state. The contract controls the complete joint proof-message distribution, including its correlation with the shell and previous proofs, by $\varepsilon_{\mathrm{proofZK}}$. All other transitions are identical conditional distributions by Theorem 13.38. The stepwise hybrid bound Lemma 3.67 (or its computational analogue) gives at most the contracted error for the complete verifier view. □

Remark 13.43 (Critical security boundary). The shell simulator and its mathematical witness existence theorem are *proved*. The complete online-view statement is conditional on an actual *adaptive* zero-knowledge argument-of- knowledge realization for the stage circuits. The general binary-field Pedersen/Schnorr affine compiler of Chapter 12 does *not* prove all odd-characteristic field multiplications, Merkle hash gates and source-word polynomial constraints needed by Π_{rel}. Those circuits and their cryptographic security are the Chapter 14 task.

13.6 Binding, cost, and the Chapter 14 handoff

Definition 13.44 (Nonsuccinct full-relation soundness events)**.** For a cheating classical prover in the *full-relation mode*, introduce:

(i) $\mathcal{B}_{\text{pre}}$: the preprocessing proof accepts but its knowledge extractor cannot assign one source table f and mask R to the commitments *before* the evaluation point;

(ii) $\mathcal{B}_{\text{rel}}$: an accepted relation proof has a false circuit statement for the fixed registered values;

(iii) $\mathcal{B}_{\text{bind}}$: one Pedersen bit commitment has two distinct accepted openings in the protocol;

(iv) $\mathcal{B}_{\text{hash}}$: one Merkle root has two conflicting authenticated commitment shells for the same indexed position.

Their probabilities are parameters ε_{pre}, ε_{rel}, $\varepsilon_{\text{bind}}$, $\varepsilon_{\text{hash}}$, to be bounded by separate cryptographic reductions in Chapter 14.

Lemma 13.45 (Sound full-relation checks fix the original evaluation)**.** *Assume none of the events in Definition 13.44 occurs and that the initial claim relation $\sigma_0 = \sum_a f(a)\chi_0(a) = v$ is enforced using the* same *pre-evaluation extracted source table f. Then*

$$\boxed{v = \widetilde{f}(z).}$$

Proof. The initial claim relation identifies v with $\sum_a f(a)\chi_0(a)$. By construction $\chi_0(a) = \mathrm{eq}_m(a, z)$. The multilinear interpolation identity Lemma 1.9 says exactly that this sum equals $\widetilde{f}(z)$. The no-bad-event hypotheses ensure that the source table in this relation is the same one bound before z. □

Theorem 13.46 (Conditional full-relation false-evaluation bound)**.** *For any adversary producing an accepted claimed value $v \neq \widetilde{f}(z)$ relative to the source table extracted at preprocessing, the acceptance probability in full-relation mode is bounded by*

$$\boxed{\varepsilon_{\text{full,snd}} \leq \min\{1,\ \varepsilon_{\text{pre}} + \varepsilon_{\text{rel}} + \varepsilon_{\text{bind}} + \varepsilon_{\text{hash}}\}.}$$

This theorem does not use a proximity reduction: the entire word relation is checked in the nonsuccinct NP circuit.

Proof. By Lemma 13.45, an accepted transcript outside the four bad events must satisfy $v = \widetilde{f}(z)$ for the preprocessing-extracted table. Thus acceptance of a false evaluation implies at least one of the four bad events. The union bound gives their total error, capped at one. No sparse-query proximity event has been invoked. □

Definition 13.47 (Sparse KBFold proximity obligation)**.** For the *sparse* mode, define $\varepsilon_{\text{prox}}$ as a claimed bound on the event that an adversary supplies committed words outside one consistent degree-bounded masked fold chain but still passes the authenticated local checks, degree/proximity tests, and hidden Sumcheck relations. The event and bound must be stated for the actual sequence of enlarged Reed–Solomon degree budgets

$$D_j = 2^{m-j} + \lceil B/2^j \rceil$$

and the correlation of its adaptive fold and query challenges. No numerical value is assigned to $\varepsilon_{\text{prox}}$ in this chapter.

Lemma 13.48 (Exact total fold-word size)**.** *The total number of committed field-symbol positions in the entire $(m+1)$-level word chain is*

$$\boxed{\sum_{j=0}^{m} M_j = M(2 - 2^{-m}) < 2M.}$$

Proof. By Lemma 5.40, $M_j = M2^{-j}$. Summing the finite geometric series gives $M\sum_{j=0}^{m} 2^{-j} = M(2-2^{-m})$. Because $m \geq 1$, the last factor is strictly less than two. □

Lemma 13.49 (Bit-commitment count for fold oracles)**.** *With the field encoding length* $b = \lceil \log_2 |\mathbb{F}| \rceil$*, the full word chain uses exactly*

$$\boxed{bM(2-2^{-m})}$$

Pedersen group elements for its field-symbol commitments, excluding separate source, mask, round-polynomial and offset commitments.

Proof. Every committed field symbol has exactly b bit commitments by Definition 13.2. Multiply the position count of Lemma 13.48 by b. □

Lemma 13.50 (Authenticated query-shell complexity)**.** *In a balanced binary Merkle tree with* M_j *leaves, an authenticated commitment shell contains* b *group elements and* $\log_2 M_j$ *sibling hashes, apart from its public index. Thus a local fold query opens at most three such shells and uses at most*

$$\boxed{3b \text{ group elements} + 3\log_2 M \text{ sibling hashes.}}$$

The zero-knowledge relation-proof cost is additional.

Proof. The tree height is $\log_2 M_j$ because M_j is a power of two. A path consists of one sibling hash at each tree level. The leaf holds b group elements. The fold query requests two symbols at level j and one at $j+1$, for three such shells. Since $M_j \leq M$, each individual path has length at most $\log_2 M$, giving the stated upper bound. □

Proposition 13.51 (Full-relation mode has linear committed-word overhead)**.** *The reference full-relation mode requires* $\Theta(bM)$ *public group elements for its authenticated word commitments and at least the full word arrays as private relation witnesses. It does not preserve the polylogarithmic opening-size objective of the source KBFold PCS.*

Proof. Lemma 13.49 gives a count between bM and $2bM$. Consequently the public commitment and input representation alone are linear in the expanded word size M, up to the encoding factor b. A relation that checks the entire word and its Merkle commitments must access each committed leaf and its corresponding hidden opening, so its uncompressed witness includes all these entries. No compression theorem has been supplied by this mode. □

Definition 13.52 (Locked Chapter 14 security inputs)**.** The following objects, and no additional chapter, are the Chapter 14 inputs:

(i) complete prover/verifier $\Pi^{\mathrm{hid}}_{\mathrm{KBFold}}$, its circuit relations and online shell simulator;

(ii) the pre-evaluation knowledge-binding reduction, the Pedersen/hash consistency errors and the online adaptive relation-proof simulation contract;

(iii) the enlarged-budget sparse proximity event $\varepsilon_{\mathrm{prox}}$, or the nonsuccinct full-relation alternative;

(iv) the odd-characteristic Sumcheck and linear fold-closure identities;

(v) a separate treatment of classical interactive, Fiat–Shamir ROM and post-quantum/QROM models.

Chapter 15 remains the final Lean/Rust certification and document audit block.

Remark 13.53 (Exact current status). The algebraic prover/verifier and its confidential authentication checks are fully defined and have perfect completeness. The Merkle/commitment shell has exact witness- independent online distribution. A complete cryptographic security theorem still needs a concrete relation proof engine with the precise online simulation and pre-evaluation extraction guarantees, and the sparse mode additionally needs a proximity theorem for the changed RS rates. Those obligations are confined to Chapter 14.

13.7 The exact achieved theorem chain

For a private table f and mask $R \in \mathbb{F}[X]_{<B}$, the honest prover commits to the entire masked fold chain

$$Q_0 = U_f^{(m)} + X^{2^m} R, \qquad Q_{j+1} = \mathsf{pfold}_{r_j}(Q_j), \qquad \deg Q_j < 2^{m-j} + \left\lceil B/2^j \right\rceil .$$

The full masked Sumcheck keeps its round coefficients and running offset hidden:

$$\boxed{\begin{aligned} \mu_j(0) + \mu_j(1) &= \sigma_{j-1} + b_{j-1}, \\ \mu_j(r_{j-1}) &= \sigma_j + b_j, \\ \mu_m(r_{m-1}) &= \chi_m[X^0] Q_m + b_m. \end{aligned}}$$

The final word has degree at most one and its hidden constant coefficient is authenticated by two distinct final leaves. Theorem 13.33 proves perfect algebraic completeness of the *entire* declared prover/verifier protocol.

Every public oracle leaf is a freshly blinded Pedersen bit-commitment vector. Theorem 13.38 establishes exact *online witness independence* of the public commitments, roots, adaptive authenticated shells and their paths, even when the commitment is sent before the evaluation point is chosen. The complete view also contains interactive proof transcripts: Theorem 13.42 gives their conditional simulation reduction with an explicit $\varepsilon_{\text{proofZK}}$ contract.

In full-relation mode, accepted false claims are confined to the precise pre-knowledge, relation-soundness, Pedersen binding and hash-collision events:

$$\boxed{\varepsilon_{\text{full,snd}} \le \varepsilon_{\text{pre}} + \varepsilon_{\text{rel}} + \varepsilon_{\text{bind}} + \varepsilon_{\text{hash}}.}$$

The direct construction uses $bM(2 - 2^{-m})$ public group elements for fold-word symbols, and therefore is not a succinct PCS. The sparse mode must replace the full circuit checks using a proved $\varepsilon_{\text{prox}}$ for the new RS rates. Chapter 14 closes the security theorem statements in the chosen classical/ROM/quantum model, and Chapter 15 is reserved exclusively for certification and final manuscript audit.

14 Final Security Theorems: Classical Reference Protocol, Sparse Obstruction, and Quantum Boundary

Chapters 1–13 established the odd-characteristic kernel/folding algebra, the masked Sumcheck recurrence and a fully specified authenticated-commitment reference protocol. This chapter closes their *end-to-end security statement* without confusing that nonsuccinct reference with a proved succinct zero-knowledge KBFold PCS.

There are three distinct results. First, complete-word circuit checks reduce classical false-evaluation acceptance to prefix-preserving knowledge extraction, discrete-log binding and Merkle collision resistance. Second, a *constructive* online simulator follows from equivocal Pedersen commitments and adaptive multi-proof witness indistinguishability of the chosen NP proof engine. Its source may be changed *inside the simulation* after an authorized evaluation is known, because the simulator retains the generator-relation trapdoor. Third, we give *negative* theorems: sparse folding does not certify initial Reed–Solomon membership, and the Pedersen instantiation is broken by a quantum discrete-log algorithm [Sho97]. The exact remaining proof-engine, sparse proximity and QROM obligations are contained in this chapter and passed directly to the locked final Chapter 15.

14.1 Single-opening games, setup, and proof-engine hypotheses

We work throughout over the *odd-characteristic* field $\mathbb{F}$, with $N = 2^m$, $M = 2^{m+s}$, $1 \le B \le N$ and the full-relation protocol of Definition 13.31. Its basic claim is the evaluation of *one* preprocessing-bound table $f : I_m \to \mathbb{F}$ at *one* point z chosen after the first commitment. The classical security parameter is λ; statistical distance is denoted Δ, and computational distinguishing advantage is denoted $\mathrm{Adv}^{\mathrm{dist}}$.

Definition 14.1 (Classical commitment setup)**.** Sample a cyclic prime-order group $G = \langle g \rangle$ of order $P > 2$ and $\alpha \xleftarrow{\$} \mathbb{Z}_P^\times$. Publish $h = g^\alpha$, but withhold α in the *real* setup. The relation-proof engine has a separate reference string $\mathsf{crs}_{\mathrm{rel}}$, generated by its specified setup algorithm. The field-symbol commitment of Definition 13.2 uses independent uniform blindings for *every* encoded bit of every named hidden variable, including the masked words. The adversary receives $(G, P, g, h, \mathsf{crs}_{\mathrm{rel}})$.

Definition 14.2 (Evaluation-binding game)**.** The cheating prover emits a pre-evaluation commitment registry, the initial word root, and an accepted preprocessing relation proof. Only *after* this transcript is fixed does the honest verifier sample and reveal $z \in \mathbb{F}^m$. The prover then announces $v \in \mathbb{F}$ and completes all masked Sumcheck, fold-oracle, authentication and relation proof stages of Definition 13.31. The game is won if all verifier checks accept but

$$v \neq \widetilde{f}_{\mathrm{pre}}(z),$$

where f_{pre} is the single source table extracted by the pre-evaluation knowledge extractor. Extraction failure is counted as a separate bad event, not as an arbitrary choice of f_{pre} after observing z.

Definition 14.3 (Authorized one-opening zero-knowledge game)**.** A classical malicious verifier chooses its point z *after* receiving the pre-evaluation commitment and relation proof. It may choose later field challenges and authenticated query indices causally from its complete view. The authorized functionality $\mathcal{F}_{\mathrm{eval}}^{f}$ reveals $\widetilde{f}(z)$ to the simulator when z has been selected. The full view includes $\mathsf{crs}_{\mathrm{rel}}$, the verifier's random tape, all commitment vectors, Merkle roots and paths, all relation-proof transcripts and the public evaluation claim v. The privacy experiment makes one opening; no repeated evaluation query to the same source commitment is included.

Definition 14.4 (Online relation-engine extraction contract)**.** For the fixed finite family of circuit relations in Chapter 13, let K be the maximum number of accepted proof invocations on any opening transcript. A *prefix-preserving extraction implementation* of Π_{rel} supplies a reduction with an extraction method for each accepted circuit proof. Except for an event of probability at most $\varepsilon_{\mathrm{KE}}$ over the *whole* adaptive execution, these extractions:

(i) give actual openings of every registered Pedersen bit commitment and valid witnesses to every accepted circuit statement;

(ii) extract one source table and mask from the preprocessing proof *before* the real point z is sampled;

(iii) complete the extractions of all pre-challenge registered values before their respective fresh outer challenges;

(iv) leave the accepted main transcript and its future honest outer-challenge distributions unchanged; and

(v) return all witnesses with the same named public commitment vectors and Merkle roots seen by the verifier.

For a suitable trapdoor-extractable NIZK argument of knowledge this can be sought via its independent extraction key; for a rewindable interactive argument it requires a separate checkpoint theorem. *Chapter 12's one-bit affine extractor alone does not discharge this contract for arbitrary hash and field circuits.*

Definition 14.5 (Adaptive witness-indistinguishability contract)**.** The engine Π_{rel} has $\varepsilon_{\mathrm{WI}}$-*adaptive multi-proof witness indistinguishability* if, for every classical PPT verifier and every sequence of at most K adaptively selected *true* public statements, its entire joint transcript when proofs use one sequence of valid witnesses is computationally indistinguishable, with advantage at most $\varepsilon_{\mathrm{WI}}$, from its transcript when they use another sequence of valid witnesses for the same public statements. The verifier's auxiliary information, previous proof transcripts, causal challenge choices and any retained public commitment roots are included. This is a *joint* contract; no marginal-message hiding shortcut is permitted. If each of K adaptive proof stages contributes at most δ_{WI} under a valid conditional hybrid, one may set $\varepsilon_{\mathrm{WI}} \le K\delta_{\mathrm{WI}}$.

Definition 14.6 (Reduction error terms)**.** Write $\varepsilon_{\mathrm{DL}}$ for the success probability of the concrete reduction that derives $\log_g h$ from inconsistent extracted bit openings, and $\varepsilon_{\mathrm{CRH}}$ for the success probability of its collision finder against the Merkle hash. Write $\varepsilon_{\mathrm{KE}}$ for Definition 14.4 and $\varepsilon_{\mathrm{WI}}$ for Definition 14.5. These are *assumption-dependent advantages*, not unconditional numerical constants.

Lemma 14.7 (Completeness carries to the final security game)**.** *If the relation engine has perfect completeness and the honest prover uses the field, masks and query schedule of Chapter 13, the full-relation verifier accepts every correct single evaluation with probability one.*

Proof. The game uses exactly the honest algorithm and verifier predicates of Definition 13.31. Theorem 13.33 supplies a valid hidden witness for every statement and proves all algebraic and Merkle checks. The stipulated proof-engine completeness turns those valid witnesses into accepted proofs. □

Remark 14.8 (Not yet an instantiated NP argument). The commitment group and full arithmetic circuits are concrete. The contracts of Definitions 14.4 and 14.5 specify the *precise* remaining cryptographic properties of the chosen NP relation-proof engine. A concrete scheme, CRS, proof format and reduction meeting both contracts must be supplied before labeling the whole construction a deployed secure PCS.

14.2 A final classical evaluation-binding theorem

Lemma 14.9 (Canonical bit opening is unique except under discrete-log failure)**.** *Fix the public Pedersen key (g, h) and a collection of tagged bit commitments. If two accepted extracted witnesses assign different bits to the same tagged group element, then from the two witnesses one can compute $\alpha = \log_g h$. If their bit values agree, their blinding exponents agree as well.*

Proof. For distinct bits, apply the explicit collision formula of Lemma 10.11. For equal bits, cancel the shared factor g^b from the two valid openings, giving $h^r = h^{r'}$. Since h has prime order P, the blindings agree modulo P by Lemma 12.18. This is pointwise for every tagged element, hence for the whole registered bit vector. □

Lemma 14.10 (A changed authenticated leaf gives a hash collision)**.** *Suppose the preprocessing extractor obtains one valid indexed Merkle commitment tree under root root_j, and a later accepted relation proof or authentication shell assigns a different commitment vector to an existing position under the* same *root. Then a collision in the underlying hash function can be computed from the two authenticated paths.*

Proof. The preprocessing and later witnesses provide the different indexed leaf vectors and their paths to the identical root. The domain-separated Merkle conflict reduction of Lemma 13.7 finds the first incompatible serialized node inputs whose hash outputs agree. The differing inputs form an explicit hash collision. □

Lemma 14.11 (The preprocessing witness fixes one source table)**.** *On the good event of the prefix-preserving extraction contract, the accepted pre-evaluation transcript determines one canonical source table $f_{\mathrm{pre}} : I_m \to \mathbb{F}$ before the point z is chosen. Every later accepted relation that references the registered source commitments uses the same source table, unless a discrete-log opening conflict is obtained.*

Proof. Definition 14.4 extracts actual canonical bit openings for every source field variable in the preprocessing relation, prior to revealing z. Definition 13.1 makes the source field value uniquely determined by its valid bit encoding. Every subsequent use of that source value refers to its same named group commitments, by Definition 13.28. A different extracted source value would change at least one encoding bit and therefore give two valid openings of the same tagged group element. Lemma 14.9 then extracts the setup discrete logarithm. □

Theorem 14.12 (End-to-end classical evaluation binding: full-relation mode)**.** *Under Definitions 14.1 and 14.4, for any classical PPT malicious prover in the one-opening game, the full-relation protocol of Chapter 13 satisfies*

$$\boxed{\Pr[\text{accept} \ \wedge \ v \neq \widetilde{f}_{\mathrm{pre}}(z)] \leq \min\{1, \varepsilon_{\mathrm{KE}} + \varepsilon_{\mathrm{DL}} + \varepsilon_{\mathrm{CRH}}\}.}$$

In particular, when the three underlying advantages are negligible, the protocol is classically computationally evaluation binding. There is no additional univariate root-test or Reed–Solomon proximity error in this full-relation security bound.

Proof. Run the prescribed extractor along the unchanged accepted main path. Let E be the event of a failed extraction, D an extracted Pedersen bit-opening conflict, and H an authenticated leaf conflict for a fixed root. Their probabilities are bounded by the three displayed terms, respectively. Outside $E \cup D \cup H$, all accepted proofs have valid witnesses on one registered source table f_{pre}: Lemma 14.11 fixes the source before z, Lemma 14.9 prevents conflicting registered field values, and Lemma 14.10 prevents conflicting authenticated leaf vectors. The initial claimed-evaluation relation in Definition 13.31 is an *exact* circuit equality for those same registered source bits:

$$v = \sum_{a \in I_m} f_{\mathrm{pre}}(a)\, \mathrm{eq}_m(a, z).$$

By the multilinear interpolation identity Lemma 1.9, the right side equals $\widetilde{f}_{\mathrm{pre}}(z)$. Thus an accepted false claim cannot occur outside $E \cup D \cup H$. Apply the union bound and cap by one. The proof never replaces a complete-word relation by a sparse-query degree test; hence there is no Schwartz–Zippel or proximity term to add. □

Corollary 14.13 (Knowledge-soundness of the expanded witness relation)**.** *On the complement of the extraction, discrete-log and hash-collision events, an accepted transcript determines a single registered source table f_{pre} and witnesses for all accepted NP relation statements that verify the original masked fold chain and terminal Sumcheck relation.*

Proof. Definition 14.4 returns a valid circuit witness for every accepted proof. The canonical commitment and Merkle consistency lemmas make their common registered values agree. The circuit relations in Definitions 13.29 and 13.30 then give the fold chain, degree budgets and masked Sumcheck equalities. The terminal relation is verified by the final extracted circuit witness. All values refer to the same source table by Lemma 14.11. □

Proposition 14.14 (Random challenges are unnecessary for full-relation algebraic binding)**.** *Once the source has been extracted before z and all complete relations are enforced, the implication*

$$\mathrm{accept} \Longrightarrow v = \widetilde{f}_{\mathrm{pre}}(z)$$

outside the named cryptographic bad events holds for every fixed z and every sequence of legal fold challenges.

Proof. The proof of Theorem 14.12 uses only the exact registered-source identity and multilinear interpolation, not a probabilistic root bound. Both are valid for every field input. Therefore the conclusion holds pointwise for any legal challenge sequence. □

Remark 14.15 (Why this is not yet an unconditional deployed theorem). Theorem 14.12 is a *rigorous reduction under an explicit NP argument-of-knowledge contract.* Chapter 13 provides the circuits; Chapters 10–12 provide a concrete classical affine component. But an implemented proof system for the *full odd-field circuits* with the prefix-preserving extraction property has not yet been supplied or machine audited. Assigning $\varepsilon_{\mathrm{KE}} = 0$ without such an instantiation would be unjustified.

14.3 Constructive online simulation using equivocal Pedersen openings

The privacy proof can be made more explicit than the abstract online proof-simulation condition of Chapter 13. The zero-knowledge simulator need not solve discrete logarithms of arbitrary sampled group elements. Instead it samples each public commitment *together with a known opening to zero* in a setup where it knows the relation between the generators, then *equivocates* those same group elements to any later required field values. The public group distribution is unchanged.

Definition 14.16 (Simulated commitment key)**.** The simulator samples the same prime-order group G, generator g, and $\alpha \xleftarrow{\$} \mathbb{Z}_P^\times$ as in Definition 14.1. It sets $h = g^\alpha$, publishes (G, P, g, h) and retains the *simulation trapdoor* α. It also samples the relation-engine CRS from its normal setup distribution; it is not assumed to know any proof-engine extraction key.

Lemma 14.17 (Simulated and real commitment keys have identical public laws)**.** *The public generator pair (g, h) produced by Definition 14.16 has* exactly *the real-setup distribution of Definition 14.1.*

Proof. Both algorithms sample α uniformly among the nonzero elements of $\mathbb{Z}_P$ and publish $h = g^\alpha$, withholding or retaining α only as local private state. The published group and generator pairs therefore have identical distributions. The relation-engine CRS is independently sampled in both cases. □

Definition 14.18 (Trapdoor equivocation algorithm)**.** To publish a bit commitment, the simulator samples $\rho \xleftarrow{\$} \mathbb{Z}_P$ and outputs $C = h^\rho$ with a private record of ρ. Later, for any desired $b \in \{0, 1\}$, define

$$\boxed{r_b := \rho - b\alpha^{-1} \pmod P.}$$

To commit to a field symbol, repeat this for all bits of its canonical encoding. No equivocated bit or blinding is publicly opened in the target protocol.

Theorem 14.19 (Exact trapdoor equivocation with uniform public commitments)**.** *For every sampled pair (C, ρ) and every bit b, the algorithm of Definition 14.18 computes a valid opening*

$$\boxed{C = g^b h^{r_b}.}$$

The public C is uniform on G, independent of the chosen b. Moreover, for each fixed *target bit, the produced opening blinding r_b is uniform in $\mathbb{Z}_P$.*

Proof. As $h = g^\alpha$,

$$g^b h^{r_b} = g^b g^{\alpha(\rho - b\alpha^{-1})} = g^{\alpha\rho} = h^\rho = C.$$

The map $\rho \mapsto h^\rho$ is a bijection from $\mathbb{Z}_P$ onto G, so C is uniform. For a fixed bit b, the map $\rho \mapsto \rho - b\alpha^{-1}$ is a translation of $\mathbb{Z}_P$, so r_b is uniform as well. The assertion holds coordinatewise for field-symbol encodings. □

Definition 14.20 (Online equivocal full-protocol simulator)**.** The simulator $\mathsf{Sim}_{\mathrm{eq}}$ has the public parameter and the authorized evaluation functionality but no access to the real witness f. It proceeds in the *actual* protocol order:

(i) **Before z:** sample the public bit commitments and level-zero Merkle tree by the zero-opening method of Definition 14.18. For the hidden preprocessing witness, choose the *dummy* table $f^{\mathrm{pre}} \equiv 0$ and mask $R^{\mathrm{pre}} = 0$. Use the trapdoor to produce valid openings to those dummy values; run the honest prover of Π_{rel} on that true preprocessing statement. Publish the root and its proof before receiving z.

(ii) **After z:** obtain the authorized scalar $v = \mathcal{F}_{\mathrm{eval}}^f(z)$. Choose the *virtual* table

$$\boxed{f^z(a) := v \quad (a \in I_m)}$$

and virtual mask $R^z = 0$. Reinterpret the *same* already published source, mask and level-zero commitments by equivocation to these new hidden values.

(iii) **Each fold round:** run the honest hidden Sumcheck and polynomial-fold algebra for f^z, R^z and the actual verifier challenges received so far. Sample every new public commitment with known zero openings, then equivocate to the current correct hidden coefficients, offsets and word values. Publish each Merkle root at its required stage and use the actual chosen witnesses to produce the prescribed relation proofs.

(iv) **Oracle queries:** answer every authenticated shell query from the stored commitment trees, preserving all repeated-leaf consistency. Produce honest relation proofs for the corresponding virtual hidden field values. Finish with the honest two-point terminal relation for the virtual fold word.

The *internal* source used in the pre-evaluation proof may differ from the virtual source used after z. This is permitted only in the *simulator's* equivocal setup. Honest real provers and malicious provers without α cannot rely on such a change without violating computational binding.

Lemma 14.21 (The virtual source matches every authorized one-point claim)**.** *For the constant virtual table $f^z(a) = v$,*

$$\boxed{\widetilde{f^z}(z) = v}$$

for every $z \in \mathbb{F}^m$.

Proof. By multilinear interpolation,

$$\widetilde{f^z}(z) = v \sum_{a \in I_m} \mathrm{eq}_m(a, z).$$

The sum of all Boolean equality weights is $\prod_{i=0}^{m-1}((1 - z_i) + z_i) = 1$. Thus the expression equals v identically. □

Lemma 14.22 (Every simulated circuit statement has a known valid witness)**.** *The algorithm $\mathsf{Sim}_{\mathrm{eq}}$ can compute in polynomial time a valid private witness for each* individual *public relation statement it submits, including the pre-evaluation statement before learning z.*

Proof. Before z, the dummy source $f^{\mathrm{pre}} = 0$ and mask $R^{\mathrm{pre}} = 0$ satisfy the preprocessing relation by honest construction. The simulator knows the blinding exponents for all its public commitments by Theorem 14.19, and it stores the full Merkle tree. Thus it knows a preprocessing witness. After z, the constant source $f^z = v$ has the correct public evaluation by Lemma 14.21. The simulator computes the genuine masked Sumcheck and fold chain from that source, plus all the offsets and low-degree witnesses of Chapter 13. Theorem 14.19 opens all previously or newly sampled Pedersen commitments to the desired bits. Lemmas 13.22 and 13.24 show the authenticated query and terminal constraints hold. Therefore every submitted statement has a known valid witness. The pre-evaluation and post-evaluation witnesses need not be the same *joint* witness; the real protocol's binding prevents that behavior for efficient ordinary provers. □

Lemma 14.23 (The simulator's public authenticated shells have the honest law)**.** *At every stage, conditioned on the preceding visible shell transcript and actual causal verifier actions, the new public Pedersen commitments and their Merkle roots in $\mathsf{Sim}_{\mathrm{eq}}$ have exactly the honest distribution. Authenticated path replies use the same deterministic stored-array rule as the real protocol.*

Proof. For each newly committed field-symbol bit, the simulator samples $C = h^\rho$, which is uniform on G by Theorem 14.19. The real prover samples $C = g^b h^r$ with uniform r, which is also uniform

by Lemma 10.10. All fresh blindings are independent on both sides, so their entire conditional public arrays have the same product distribution regardless of the hidden word and its correlations. Merkle roots and paths are deterministic functions of those arrays. Repeated queries read the stored values, as in Lemma 13.37. Thus the conditional shell transition laws agree. □

Theorem 14.24 (Constructive classical online zero knowledge: full-relation mode)**.** *Assume the real and simulated commitment setup of Definitions 14.1 and 14.16, and assume* Π_{rel} *meets the adaptive multi-proof witness-indistinguishability contract of Definition 14.5. Then* $\mathsf{Sim}_{\mathrm{eq}}$ *is an* online, witness-independent, classical PPT *simulator for the complete verifier view of one correct opening, including all relation proofs, Merkle paths and adaptive query choices. Its distinguishing advantage is bounded by*

$$\boxed{\mathrm{Adv}^{\mathrm{ZK}}_{\mathrm{full}} \le \varepsilon_{\mathrm{WI}}.}$$

The simulator sends the preprocessing commitment before *learning the evaluation point and receives only the authorized evaluation value afterward.*

Proof. Compare a real opening of an arbitrary fixed private source table f with the proposed simulation.

First couple the public commitment setup by Lemma 14.17; both distributions are identical, and the simulator merely retains the secret exponent α locally. For every private bit value in the real execution, choose a fresh zero-opening record ρ and set $r_b = \rho - b\alpha^{-1}$. Theorem 14.19 shows that this gives the same public commitment and the same uniform honest blinding law as the real protocol. The entire shell law, including all root-before-query timing, is therefore identical by Lemma 14.23.

Next consider the sequence of at most K relation proofs. In the real execution, their witnesses come from the actual source table f and its honest fold chain. In the simulated execution, the pre-evaluation proof uses $f^{\mathrm{pre}} = 0$, while every subsequent relation proof uses the virtual source $f^z = v$, the actual outer challenges and its corresponding honest masked fold chain. Lemma 14.22 makes all these *individual* witnesses valid for their respective public statements. The public commitment arrays and roots can be coupled identically, with different private openings computed using α. The allowed public value v is identical in both processes because the simulator obtains $\widetilde{f}(z)$ from the authorized functionality, while Lemma 14.21 makes the virtual source evaluate to the same v. The adaptive multi-proof witness-indistinguishability contract then changes the real proof-witness sequence into the simulator's proof-witness sequence on these same public statements with advantage at most $\varepsilon_{\mathrm{WI}}$, including the verifier's responses to earlier proof messages. All non-proof transitions, Merkle query responses and causal verifier challenge kernels have identical conditional laws under this comparison, by the shell coupling. Thus the final complete-view distinguishing advantage is at most $\varepsilon_{\mathrm{WI}}$. Every simulated witness and opening is computed by finite polynomial-time operations from α, stored blinding records, the public challenges and the authorized value. This proves the claimed online PPT bound. □

Corollary 14.25 (Perfect zero knowledge under perfect adaptive WI)**.** *If the stipulated adaptive multi-proof witness indistinguishability holds with* $\varepsilon_{\mathrm{WI}} = 0$*, then the complete classical one-opening view is* perfectly *zero knowledge in the simulatable commitment setup model.*

Proof. Set $\varepsilon_{\mathrm{WI}} = 0$ in Theorem 14.24. The commitment setup and all shell transitions are already exactly distribution preserving. □

Remark 14.26 (Why the trapdoor is not a soundness backdoor). The real adversary receives only (g, h), not α. The simulator can equivocate because it retained α when generating an *identically*

distributed public key. The discrete-log reduction in the soundness theorem uses the distinct real experiment where no prover is given this exponent. The proof does not require one physical setup party to simultaneously keep and reveal α.

Remark 14.27 (Model and engine restriction). This is a *simulatable CRS* security theorem, not an assertion of plain-model simulation for a fixed uncontrolled commitment key. The concrete NP proof engine remains to be chosen and verified against the stated adaptive WI contract. The theorem does not cover multiple openings, quantum verifiers, or Fiat–Shamir transformations.

14.4 Sparse verification: exact rates and a decisive counterexample

The completed full-relation protocol verifies the degree and source relations explicitly in its NP circuit. A sparse-query replacement is not secured by the algebraic fold dictionary alone. The following quantitative statements identify precisely which piece of a new proximity theorem is missing.

Definition 14.28 (Enlarged masked Reed–Solomon codes)**.** At fold level j put

$$M_j = 2^{m+s-j}, \quad N_j = 2^{m-j}, \quad B_j = \lceil B/2^j \rceil, \quad D_j = N_j + B_j,$$

and define the code

$$\mathcal{C}_j := \mathrm{RS}(\Omega_j, D_j) = \{(P(x))_{x\in\Omega_j} : \deg P < D_j\} \subseteq \mathbb{F}^{M_j}.$$

Its Hamming distance is the ordinary number of unequal evaluation coordinates, and δ_j denotes its relative minimum distance.

Lemma 14.29 (Exact dimension and minimum distance)**.** *For $1 \le D_j \le M_j$, the code $\mathcal{C}_j$ has dimension D_j, minimum Hamming distance*

$$\boxed{d_j = M_j - D_j + 1}$$

and relative distance $\delta_j = (M_j - D_j + 1)/M_j$.

Proof. Evaluation at M_j distinct field points is injective on the space of polynomials of degree below D_j, because a nonzero polynomial of degree at most $D_j - 1 < M_j$ cannot vanish at all domain points. Thus the dimension is D_j. The difference of two distinct codewords is the evaluation of a nonzero polynomial of degree at most $D_j - 1$, with at most $D_j - 1$ roots on Ω_j. Its Hamming weight is therefore at least $M_j - (D_j - 1)$. Equality is attained by a polynomial which is the product of $D_j - 1$ distinct linear factors at evaluation points (use the constant polynomial 1 when $D_j = 1$). Hence the minimum distance is exactly $M_j - D_j + 1$. □

Lemma 14.30 (Degree-rate bounds and required blow-up)**.** *For $1 \le B \le N$ and $0 \le j \le m$,*

$$\boxed{\frac{D_j}{M_j} = 2^{-s}\left(1 + \frac{\lceil B/2^j \rceil}{N_j}\right) \le 2^{1-s}.}$$

In particular, $s \ge 2$ guarantees rate at most $1/2$ at every level. When $s = 1, B = N$, the code is the entire ambient space at every level.

Proof. Divide $D_j = N_j + B_j$ by $M_j = 2^s N_j$. Since $B \le N = 2^m$, we have $B_j = \lceil B/2^j \rceil \le \lceil 2^{m-j} \rceil = N_j$. This gives the inequality. For $s = 1, B = N$, $B_j = N_j$ and hence $D_j = 2N_j = M_j$, so evaluation polynomials of degree below M_j give every word in $\mathbb{F}^{M_j}$ by interpolation. □

Lemma 14.31 (Folding a monomial halves its exponent)**.** *For every integer $e \geq 0$ and fold parameter $r \in \mathbb{F}$, the polynomial $\mathsf{pfold}_r(X^e)$ has degree at most $\lfloor e/2 \rfloor$. Consequently for any m fold parameters,*

$$\boxed{\deg(\mathsf{pfold}_{r_{m-1}} \circ \cdots \circ \mathsf{pfold}_{r_0}(X^e)) \leq \lfloor e/2^m \rfloor.}$$

Proof. The defining monomial rules of Lemmas 5.10 and 5.11 send X^{2a} and X^{2a+1} to public scalar multiples of X^a, respectively. The scalar may vanish, which can only reduce the output degree. Thus the degree after one fold is at most $\lfloor e/2 \rfloor$. Iterate this inequality m times and use $\lfloor \lfloor e/2 \rfloor /2 \rfloor = \lfloor e/4 \rfloor$. □

Theorem 14.32 (Sparse local folds and final linearity do not certify initial degree)**.** *Assume $1 \leq B < N$ and $M \geq 2N$. Let*

$$\boxed{P_0(X) := X^{N+B}.}$$

The evaluation word of P_0 is not *in $\mathcal{C}_0 = \mathrm{RS}(\Omega_0, N + B)$. Nevertheless, for* every *sequence of fold challenges $r_0, \ldots, r_{m-1} \in \mathbb{F}$, the chain $P_{j+1} = \mathsf{pfold}_{r_j}(P_j)$ passes* every *polynomial/local-word fold consistency check, and its final polynomial satisfies $\deg P_m < 2$. Therefore even exhaustive local fold consistency and final degree-one closure do not imply the required initial Reed–Solomon membership.*

Proof. The polynomial P_0 has degree $N + B = D_0$. Because $B < N$,

$$D_0 = N + B < 2N \leq M.$$

If its evaluation word were in $\mathcal{C}_0$, some polynomial Q of degree below D_0 would agree with P_0 at all M domain points. Then $P_0 - Q$ would be nonzero of degree $D_0 < M$ with M distinct roots, contradicting the univariate root bound. Thus its word is outside $\mathcal{C}_0$.

Define all later polynomials by the *actual* normalized folding rule. For each pair $x, -x$ and square $y = x^2$, the evaluation fold identity Theorem 5.44 holds identically. Hence every local consistency check passes, regardless of the chosen challenges or the number of queries. By Lemma 14.31,

$$\deg P_m \leq \left\lfloor \frac{N + B}{N} \right\rfloor = 1,$$

since $0 < B < N$. Thus the final degree-one check also passes. This proves the stated non-implication. □

Remark 14.33 (Precise interpretation of the counterexample). The word P_0 above does *not* satisfy the full protocol's preprocessing source/mask relation. The theorem is a counterexample to the *proposed sparse replacement of that relation by local folding and a final line check*, not a forgery against the fully checked Chapter 13 protocol. An optimized sparse protocol must enforce both initial source/degree binding and robust proximity of the adversarial word chain.

Lemma 14.34 (Exact fixed-discrepancy sampling error)**.** *Fix two words on a domain of size M_j which disagree at exactly e coordinates, and sample q positions uniformly and independently* with replacement *after both words are fixed. The probability that all sampled coordinates miss the discrepancy is*

$$\boxed{(1 - e/M_j)^q.}$$

Proof. The complement of the discrepancy set has $M_j - e$ points. Each independent uniform sample avoids the discrepancy with probability $(M_j - e)/M_j$. The probability of all q misses is the product of those q equal factors. □

Definition 14.35 (Actual sparse proximity contract)**.** A valid sparse KBFold security theorem must prove a quantitative bound $\varepsilon_{\text{prox}}$ for the following full adaptive experiment: all committed word shells and their associated codeword commitments are fixed before their corresponding query indices; the adversary may supply arbitrary non-codeword words, may correlate words across rounds, and may adapt them to preceding public fold challenges, but not to the fresh query coins after a root is bound. The verifier's local fold checks, initial source/degree test, final line test and claim consistency must be specified. The reduction must bound acceptance of a chain not representing one valid initial degree-$< D_0$ masked word and its legal folds. The bound must use the *actual* budgets D_j/M_j from Lemma 14.30.

Proposition 14.36 (A fixed discrepancy bound is not a proximity theorem)**.** *Lemma 14.34 does not imply Definition 14.35 without an additional lower bound on discrepancy for every invalid adversarial chain.*

Proof. The fixed-discrepancy lemma assumes that the two words and the number e of their disagreements are already given. A proximity theorem must derive a positive discrepancy lower bound from non-membership in the appropriate degree-bounded code under correlated adaptive folds. The polynomial in Theorem 14.32 is outside the initial code but has *zero* local fold discrepancy at every location. Thus local disagreement cannot be inferred from initial non-membership alone. The missing implication requires separate code/commitment structure. □

Theorem 14.37 (Conditional sparse soundness composition)**.** *Suppose a particular sparse replacement of Chapter 13's full word-relation checks satisfies Definition 14.35 with error $\varepsilon_{\text{prox}}$, and retains the prefix-preserving source extraction, Pedersen binding, Merkle collision resistance and remaining valid relation proofs of Theorem 14.12. Then its false-evaluation acceptance probability is at most*

$$\boxed{\min\{1,\ \varepsilon_{\text{KE}} + \varepsilon_{\text{DL}} + \varepsilon_{\text{CRH}} + \varepsilon_{\text{prox}}\}.}$$

Proof. As in Theorem 14.12, exclude preprocessing extraction failures, conflicting registered field openings and conflicting authenticated Merkle shells. Their union probability is bounded by the first three errors. Outside these events, the sparse proximity contract ensures except with probability $\varepsilon_{\text{prox}}$ that all accepted sparse checks correspond to a valid initial source-bound masked word and its legal fold chain. On that good event the initial claimed evaluation is the true multilinear evaluation of the extracted source table by the same exact algebraic argument as in Lemma 13.45. Thus false acceptance belongs to the union of the four bad events. Apply the union bound. □

Remark 14.38 (Locked research boundary). No value of $\varepsilon_{\text{prox}}$ is supplied or implied by the current Chapter 13 reference implementation. The sparse optimization is not declared secure. Chapter 15 records this unresolved component inside its final certification ledger; no extra research chapter is added.

14.5 Security-model separation and a concrete quantum obstruction

The soundness and privacy results above are *classical interactive/CRS* results. Random-oracle compilation and quantum adversaries change both the proof model and the hardness assumptions. In particular, the present Pedersen commitment instantiation is *not post-quantum binding.*

Definition 14.39 (Four distinct claims)**.** We distinguish:

(i) *Classical interactive soundness:* the verifier samples every required outer challenge after the prescribed commitment/proof prefix, and the relation engine satisfies its knowledge-extraction contract against classical PPT provers.

(ii) *Classical online ZK in a simulatable CRS:* the commitment setup may be generated by a simulator retaining α; malicious verifiers and query indices are arbitrary classical causal algorithms, and relation proofs satisfy adaptive multi-proof WI.

(iii) *Fiat–Shamir ROM security:* the interactive challenges are derived from a domain-separated transcript hash, and a separate classical random-oracle proof justifies every extraction and simulation step in that noninteractive experiment.

(iv) *QROM/post-quantum security:* the adversary may perform quantum computation and quantum oracle queries. All commitment, proof-engine, hash and folding-oracle reductions must be valid against that adversary class.

Only (i)–(ii) are addressed by the proved conditional theorems of this chapter.

Lemma 14.40 (A known Pedersen setup logarithm gives two openings)**.** *If an adversary knows $\alpha = \log_g h$, then for every bit commitment $C = g^b h^r$ it can efficiently produce an opening to any desired bit $b' \in \{0, 1\}$:*

$$\boxed{r' = r + (b - b')\alpha^{-1} \pmod{P}.}$$

If $b' \neq b$ these are two different valid bit openings.

Proof. Compute

$$g^{b'} h^{r'} = g^{b'} g^{\alpha r + b - b'} = g^b h^r = C.$$

All operations are in the public prime-order scalar field; α is nonzero, so its inverse exists. If $b' \neq b$, the two openings have different bit values although the public commitment is unchanged. □

Theorem 14.41 (Quantum break of classical Pedersen-based binding)**.** *Assume the commitment group is a standard finite cyclic group whose discrete logarithm can be computed by a polynomial-time quantum algorithm [Sho97]. Then the field-symbol commitments of Chapter 13 are* not *quantum computationally binding. Indeed, a quantum adversary can run the following false-claim attack on the full-relation protocol: register a valid preprocessing proof for $f_{\mathrm{pre}} \equiv 0$, wait for the evaluation point z, and subsequently produce valid proofs for an equivocated source $f^{\mathrm{post}} \equiv 1$ and public claim $v = 1$. All public commitment vectors and the initial Merkle root remain unchanged. Whenever the underlying relation engine is perfectly complete and allows honest proofs of its true statements, this attack passes the algebraic and authentication checks while $1 \neq \widetilde{f}_{\mathrm{pre}}(z) = 0$.*

Proof. First sample the honest preprocessing commitment arrays and all required blinding scalars for the zero source and zero mask. Construct $Q_0 = 0$, its full authenticated commitment tree, and an honest proof of the true pre-evaluation relation. Submit the result before learning z. The preprocessing source is therefore the zero table, so $\widetilde{f}_{\mathrm{pre}}(z) = 0$ for every z.

After learning z, use the assumed quantum discrete-log algorithm to obtain $\alpha = \log_g h$ from the already published key, unless it was computed earlier. For every previously registered Pedersen bit commitment, use Lemma 14.40 to compute an opening to the bits of the desired constant-one source, zero mask and corresponding masked Q_0. The public commitment vectors, their hashes, the first Merkle root and its paths do not change. The pre-evaluation proof remains a valid proof of its original true statement. All subsequent statements are also true: take the virtual constant-one table, generate its honest Sumcheck masks and fold chain, equivocate the original level-zero leaf commitments to that chain, and build the later levels honestly. The identities and completeness of Theorem 13.33 make every post-evaluation NP relation and authenticated query valid. The

adversary possesses every required witness and blinding and can therefore produce honest proofs without violating their proof-system soundness.

The verifier accepts by perfect completeness, yet the claimed value is $v = 1$ while the preprocessing source had evaluation 0. Finally, Shor's quantum discrete-logarithm algorithm [Sho97] runs in quantum polynomial time on the stated cyclic-group families. Thus neither the binding property nor the protocol's evaluation binding survives this quantum adversary. □

Corollary 14.42 (No post-quantum theorem for the present Pedersen instantiation)**.** *A claimed QROM or post-quantum knowledge-soundness theorem for the* unchanged *Pedersen-based full protocol would be false under the quantum group model of Theorem 14.41.*

Proof. The theorem explicitly constructs a polynomial-time quantum false-claim adversary accepted by the honest verifier. Therefore the required quantum evaluation-binding game does not have negligible adversarial success. A proof model cannot restore a binding property already broken by the group's quantum discrete-log algorithm. □

Proposition 14.43 (Original KBFold QROM soundness does not automatically transfer)**.** *The original KBFold knowledge-soundness/QROM theorems cannot be applied as black-box end-to-end security proofs of the Chapter 13 Pedersen-hiding protocol.*

Proof. The original scheme authenticates field codeword values with a transparent Merkle/FRI oracle and proves proximity for the specified unmasked code and challenge schedule. Chapter 13 changes the leaves to Pedersen bit-commitment vectors, hides every field opening behind independent NP proofs, changes the source relation and degree budgets to D_j, and introduces a distinct simulator and extraction interface. The assumptions and transcript distribution of the old theorem therefore do not match those of the new protocol. Moreover Theorem 14.41 exhibits a quantum binding failure in the new leaf scheme, independently of the original hash/FRI proximity argument. Consequently no automatic theorem transfer is valid. □

Definition 14.44 (Required post-quantum replacement contract)**.** A future QROM secure version of the *same Chapter 13 protocol structure* must replace the Pedersen binding layer by a quantum-resistant commitment with a suitable equivocal simulation setup or another proved online privacy method; instantiate a quantum-secure relation argument with compatible extraction and adaptive simulation; and re-prove authentication, proximity and noninteractive challenge-sampling bounds for the complete altered transcript. These are *Chapter 15 certification/open-obligation entries*, not a new Chapter 16.

Remark 14.45 (Fiat–Shamir is not a missing algebraic lemma). Hashing an interactive verifier message into a Fiat–Shamir challenge changes the prover's ability to correlate its commitments and challenge oracle queries. Without an appropriate ROM theorem for the exact engine and transcript order, the classical interactive proofs of Theorems 14.12 and 14.24 do not imply noninteractive security. No such reduction is claimed in this chapter.

14.6 Final security theorem, concrete status, and certification handoff

Definition 14.46 (Classical reference-protocol assumption package)**.** For the single-opening full-relation protocol $\Pi^{\mathrm{hid}}_{\mathrm{KBFold}}$, let $\mathsf{ClassicalRef}(\lambda)$ consist of:

(i) the odd-characteristic KBFold algebra and finite relation circuits of Chapter 13;

(ii) freshly blinded Pedersen bit commitments in the setup of Definition 14.1, with a discrete-logarithm hard group for classical PPT adversaries;

(iii) a collision-resistant, domain-separated Merkle hash;

(iv) a perfectly complete NP relation engine with prefix-preserving knowledge extraction (Definition 14.4) and adaptive multi-proof witness indistinguishability (Definition 14.5);

(v) a single authorized evaluation opening with commitment and pre-evaluation proof issued before the chosen point z;

(vi) a verifier enforcing every full-word, claimed-value, fold, degree, authenticated opening and terminal closure relation from Definition 13.31.

The source-word representation, bit encoding and commitment tags must be fixed across all relation invocations.

Theorem 14.47 (Final classical reference-protocol theorem)**.** *Under* $\mathsf{ClassicalRef}(\lambda)$ *the Chapter 13 protocol has the following properties for a* single *opening:*

(i) ***Perfect completeness:*** *honest true evaluations are accepted with probability one.*

(ii) ***Classical computational evaluation binding:***

$$\boxed{\varepsilon_{\mathrm{snd}} \leq \min\{1, \varepsilon_{\mathrm{KE}} + \varepsilon_{\mathrm{DL}} + \varepsilon_{\mathrm{CRH}}\}.}$$

(iii) ***Classical online computational zero knowledge:*** *a witness-free simulator publishes the initial commitment before z, uses only the authorized value v after z, and satisfies*

$$\boxed{\mathrm{Adv}^{\mathrm{ZK}} \leq \varepsilon_{\mathrm{WI}}.}$$

(iv) ***No inherited sparse succinctness:*** *the described full-relation protocol uses $bM(2 - 2^{-m})$ public fold-word commitment group elements and does not establish a succinct KBFold opening proof.*

These conclusions are classical/CRS and conditional *on the explicitly stated NP proof-engine contracts. No ROM or post-quantum claim is part of this theorem.*

Proof. Completeness is Lemma 14.7. The evaluation-binding error follows directly from Theorem 14.12. The online ZK simulator and bound follow from Theorem 14.24. The public commitment count and nonsuccinctness are proved by Lemma 13.49 and Proposition 13.51. All required assumptions are explicitly included in Definition 14.46, so these results may be composed for the same single-opening interactive protocol. □

Corollary 14.48 (Asymptotic conditional classical security)**.** *If the concrete relation-engine implementation makes $\varepsilon_{\mathrm{KE}}$ and $\varepsilon_{\mathrm{WI}}$ negligible, and discrete logarithms and collision finding are classically hard so that $\varepsilon_{\mathrm{DL}}$ and $\varepsilon_{\mathrm{CRH}}$ are negligible, the reference protocol has negligible false-evaluation acceptance and negligible complete-view distinguishing advantage.*

Proof. A sum of finitely many negligible functions is negligible. Apply the error bounds in Theorem 14.47. □

Definition 14.49 (Outstanding proof ledger for Chapter 15)**.** The final certification chapter records exactly the following unresolved cryptographic requirements, without creating additional research chapters:

(i) **NP engine:** a concrete odd-field circuit prover/verifier, its CRS and prefix-preserving extractor, and a verified adaptive joint WI bound;

(ii) **Sparse proximity:** an actual initial source binding/degree test and adversarial correlated Reed–Solomon proximity bound with budgets $D_j = N_j + B_j$;

(iii) **ROM/QROM:** an argument for the modified interactive transcript under the selected hash model and, if quantum security is required, a post-quantum replacement for Pedersen commitments and proof machinery;

(iv) **Certification:** Lean 4 statements/proofs for the new exact algebraic identities and reductions, Rust executable sanity tests, complexity accounting and full manuscript/source audit.

Proposition 14.50 (The four-block scope is closed)**.** *Every remaining verification or instantiation task of this manuscript is covered by the Chapter 15 certification and unresolved-obligation ledger above. No additional numbered chapter is required to state the proved final security theorem and its limits.*

Proof. The exact algebra and chronological extraction are developed in Chapters 1–12. Chapter 13 provides the full odd-field prover/verifier, authenticated shells and circuit relations. The present chapter gives the final conditional security theorems and precise negative results for sparse folding and quantum Pedersen binding. The only missing work is the concrete instantiation, stronger model security and formal/software certification enumerated in Definition 14.49, all of which are assigned to the final Chapter 15 by the locked roadmap. □

Remark 14.51 (Status, not a probability of mathematical correctness)*.* The statements Theorems 14.12, 14.24, 14.32 and 14.41 have paper proofs with their displayed hypotheses. Their new Lean declarations are not yet certified. The full-reference end-to-end theorem is conditional on the actual NP proof engine; the sparse mode and post-quantum mode are *not* claimed secure.

14.7 Final theorem chain and exact logical status

Let $\Pi^{\mathrm{hid}}_{\mathrm{KBFold}}$ be the Chapter 13 *full-relation*, single-opening, classical reference protocol. Under its explicit source-extractable, adaptive-WI NP proof-engine and Pedersen/Merkle assumptions, Theorem 14.47 proves

$$\boxed{\begin{array}{c}\Pr[\text{accept} \wedge v \neq \widetilde{f}_{\mathrm{pre}}(z)] \leq \varepsilon_{\mathrm{KE}} + \varepsilon_{\mathrm{DL}} + \varepsilon_{\mathrm{CRH}}, \\ \mathrm{Adv}^{\mathrm{ZK}}_{\mathrm{online}} \leq \varepsilon_{\mathrm{WI}}, \qquad \Pr[\text{accept true}] = 1.\end{array}}$$

The ZK simulator makes its first commitments *before* learning z. For a source-bit commitment initially published as $C = h^{\rho}$, and the trapdoor $h = g^{\alpha}$, the simulator can open that very same public commitment to any bit b using

$$\boxed{r_b = \rho - b\alpha^{-1}.}$$

It constructs genuine post-evaluation witnesses for the virtual table $f^z(a) = v$, which satisfies $\widetilde{f^z}(z) = v$, and uses the prescribed adaptive-WI guarantee to hide the difference from the real pre-committed source. The complete proof-engine WI and extraction contracts are *not yet instantiated* by a concrete audited implementation.

For the sparse version, Theorem 14.32 shows that the invalid initial polynomial $P_0 = X^{N+B}$ passes every local fold equation and the final linear check when $1 \leq B < N$. Thus no sparse proximity error may be inferred from the algebraic fold identities. Moreover Theorem 14.41 gives an explicit quantum false-evaluation strategy against the current Pedersen instantiation: *the present protocol is not post-quantum secure.*

The manuscript remains strictly locked: Chapter 15 is the *final* block, covering Lean 4 and Rust certification, proof-engine/sparse open-obligation ledger and the final consolidated audit.

15 Final certification ledger and manuscript audit

This is the fifteenth and last chapter of the same KBFold ZK manuscript. It records the exact theorem chain, the formal and executable verification obligations, and the final limitations. It does not claim unexecuted Lean proofs or an uninstantiated NP proof engine have been certified.

15.1 The final reference theorem

Definition 15.1 (Scope)**.** The reference protocol is the fully checked, odd-characteristic, single-opening, classical full-relation protocol $\Pi^{\mathrm{hid}}_{\mathrm{KBFold}}$ of Chapter 13. Its assumptions are exactly those of Definition 14.46: prefix-preserving pre-point knowledge extraction, adaptive joint witness indistinguishability for the complete proof transcript, classical Pedersen binding, collision-resistant authenticated commitments, and an independent evaluation challenge. The source encoding and circuit relations must agree across all invocations.

Theorem 15.2 (Final conditional classical guarantee)**.** *Under Definition 15.1, for the source $\widetilde{f}_{\mathrm{pre}}$ committed before z,*

$$\Pr[\mathrm{accept} \wedge v \neq \widetilde{f}_{\mathrm{pre}}(z)] \leq \min\{1, \varepsilon_{\mathrm{KE}} + \varepsilon_{\mathrm{DL}} + \varepsilon_{\mathrm{CRH}}\}, \qquad \mathrm{Adv}^{\mathrm{ZK}}_{\mathrm{online}} \leq \varepsilon_{\mathrm{WI}}.$$

Honest true openings have acceptance probability one. This is a classical CRS theorem, conditional on the explicitly quantified NP proof-engine contracts; the full-relation protocol is not claimed succinct.

Proof. This is precisely Theorem 14.47, preserving all its assumptions and experiment order. Completeness follows from Lemma 14.7, binding from Theorem 14.12, and online simulation from Theorem 14.24. The group-element count for fold-word commitments is $bM(2-2^{-m})$ by Lemma 13.49. □

Remark 15.3 (Chronology). Extraction after learning z is not a substitute for a prefix-preserving extraction before z. Witness indistinguishability of a single relation proof does not automatically imply adaptive joint witness indistinguishability of the transcript.

15.2 Algebraic and negative-results checks

Lemma 15.4 (Pedersen equivocation identity)**.** *Let $h = g^{\alpha}$ in a prime-order cyclic group, $\alpha \neq 0$, and let $C = h^{\rho}$. For each desired bit b set $r_b = \rho - b\alpha^{-1}$. Then $C = g^b h^{r_b}$.*

Proof. $g^b h^{r_b} = g^{b+\alpha(\rho - b\alpha^{-1})} = g^{\alpha\rho} = C$. □

The identity alone does not simulate every NP proof message; adaptive joint witness indistinguishability remains a separate hypothesis.

Proposition 15.5 (Sparse local checks are insufficient)**.** *The sparse local fold equations and the final linear check alone do not enforce the enlarged Reed–Solomon degree requirement.*

Proof. The polynomial $P_0(X) = X^N + B$ in the parameter regime of Theorem 14.32 passes those checks while violating the degree condition. Hence no universal sparse proximity conclusion follows from those checks alone. □

Proposition 15.6 (No post-quantum claim)**.** *The unchanged Pedersen-based reference instantiation is not post-quantum evaluation-binding.*

Proof. Theorem 14.41 constructs a quantum false-evaluation strategy. It defeats the classical discrete-logarithm binding premise when transferred unchanged to the quantum setting. □

15.3 Lean 4 certification targets and acceptance rule

A theorem is *Lean certified* only when its exact declaration and dependencies compile under the recorded Lean/Mathlib toolchain, without `sorry`, hidden axioms masquerading as proofs, or missing imports. Cryptographic assumptions must remain explicit parameters.

L1. Normalize the odd-field fold recurrence, degree invariant, masked chain, and hidden final linear closure (Chapters 4–5, 13).

L2. Formalize source encoding, authenticated relation circuits, and the masked Sumcheck running-offset identities (Chapter 13).

L3. Formalize the checkpoint/replay prefix-preserving chronological extractor and its extraction-or-binding-failure disjunction (Chapter 12).

L4. Encode the pre-point source-binding event split and adaptive joint-WI simulation interface, retaining assumptions as theorem parameters (Chapter 14).

L5. Certify the explicit sparse high-degree counterexample and the algebraic equivocation identity above.

The manuscript's paper proofs must not be relabeled as machine-checked Lean proofs. Each certified item needs a source path, theorem identifier, pinned toolchain, build command and successful exit status.

15.4 Rust executable regression targets

Rust tests can catch counterexamples and implementation errors but cannot replace universal proof reductions. The required integration tests are:

R1. Odd-field masked fold and final-closure identities with degree budgets $D_j = N_j + B_j$.

R2. Malformed authentication paths, inconsistent source prefixes, reordered messages, correlated challenges and mask reuse.

R3. The sparse high-degree counterexample with a rejection check once an explicit initial degree test is implemented.

R4. Reproducible end-to-end prover/verifier/extractor/simulator transcript fixtures after the NP proof engine is instantiated.

Every executable claim must identify a commit, command, fixed seed where relevant, test result and exit code. These are obligations, not assertions that tests were executed for this chapter.

15.5 Open obligations: four locked categories

The outstanding list is the same as Definition 14.49.

O1. NP engine: instantiate an odd-field relation proof engine with pre-point prefix-preserving extraction and adaptive joint WI. Until then the final classical security theorem is conditional.

O2. Sparse proximity: add source/degree binding and prove an adversarial correlated Reed–Solomon proximity bound at enlarged rates. Sparse succinctness and soundness are not established.

O3. ROM/QROM: analyze any Fiat–Shamir transcript separately; replace Pedersen and proof machinery before asserting post-quantum security.

O4. Certification: compile Lean declarations, execute Rust integration checks, record communication/runtime and cross-check bibliography, labels, and source paths.

15.6 Final verdict and closure

Proposition 15.7 (No further chapter required)**.** *Chapters 1–15 constitute the closed manuscript scope. The positive endpoint is the explicitly conditional classical full-relation theorem; the sparse and quantum limitations are explicit negative results. The four obligations above remain engineering and proof-certification tasks, not additional manuscript chapters.*

Proof. Chapters 1–12 establish algebra, interfaces and chronological extraction; Chapter 13 specifies the full relation; Chapter 14 proves conditional security and gives both counterexamples; this chapter audits those dependencies and separates their evidentiary status. □

Remark 15.8 (Certification status). Paper theorem chain: established under its named assumptions. Concrete NP engine and complete Lean end-to-end verification: outstanding. Sparse and post-quantum guarantees: not established. This is a final manuscript audit, not a claim of formal/software certification.

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